

Theoretical quantum optics

Lecture notes

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Preamble

These lecture notes are still in the making, and updated versions will appear in the course of the lecture series. If you have suggestions for improvement or want to report an error, please don't hesitate to contact me in person or by email:

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Introduction

Literature

We begin with listing some literature.

Quantum mechanics For an introduction and/or recap of quantum mechanics and its fundamental principles there are many books. Let me mention some that I find useful:

- [1] C. Cohen-Tannoudji et al., *Quantum mechanics, volume 1: basic concepts, tools, and applications* (Wiley, 2019)

This is a standard reference every theoretician should have in his book shelf. There is also Volume 2 and 3, covering more advanced topics, some of which are relevant for this lecture.

- [2] M. Bartelmann et al., *Theoretische physik* (Springer Berlin Heidelberg, 2014)

This is a very detailed and extensive book about theoretical physics in general. It's a very useful reference book.

- [3] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information: 10th anniversary edition*, 10th (Cambridge University Press, New York, NY, USA, 2011)

This book focuses on quantum information, however, it provides a solid introduction to the formalism.

Quantum optics There is also a large number of books on quantum optics. Let me state some of them:

- [4] C. Cohen-Tannoudji et al., *Photons and atoms: introduction to quantum electrodynamics*, Physics textbook (Wiley-VCH, Weinheim, 2004)

This book provides a detailed discussion of classical electrodynamics and its quantization. It's the key reference for Chapters 2 and 3, and for the first parts of Chapters 7 and 8.

- [5] C. Gerry and P. Knight, *Introductory quantum optics* (Cambridge University Press, 2005)

- [6] D. Walls and G. Milburn, *Quantum optics* (Springer Berlin Heidelberg, 2008)

- [7] U. Leonhardt, *Essential quantum optics: from quantum measurements to black holes* (Cambridge University Press, 2010)

These three books provide a well structured and concise introduction to quantum optics.

- [8] R. Loudon, *The quantum theory of light*, Third (Oxford University Press, Oxford, USA, 2000)

This book is also one of the standard references for theoretical quantum optics. It is more on the theory side.

- [9] G. Grynberg et al., *Introduction to quantum optics: from the semi-classical approach to quantized light* (Cambridge University Press, Cambridge, UK; New York, NY, 2010)

This book discusses a large number of topics. The style is accessible without becoming overly theoretical.

- [10] W. Vogel and D.-G. Welsch, *Quantum optics*, 3rd revised and extended edition (Wiley-VCH, Weinheim, Germany, 2006)

One more book on the introduction to quantum optics. It's more on the theory side.

- [11] C. Cohen-Tannoudji et al., *Atom-photon interactions: basic processes and applications* (Wiley, New York, 1992)

Finally, I also would like to mention this book, which is build on Ref. [4]. It is not intended as an introduction, but serves as a key reference for all kinds of phenomena on light-matter interaction.

What is quantum optics

Quantum optics is the field of physics that studies the quantum mechanical properties of the electromagnetic field and its interaction with non-relativistic quantum matter, such as atoms, molecules, or electrons in a solid. It is the non-relativistic limit of quantum electrodynamics (QED), which is a fully relativistic field theory of light and matter. Typical table-top experiments working in the non-relativistic, low-energy limit fall into the scope of quantum optics. Quantum optics is thus central to understand how to trap, manipulate, probe, and measure (microscopic) quantum systems, such as ultracold atoms, single and multiple photons, molecules, or nano particles. It finds applications and overlaps with many fields in modern physics, ranging from atomic and molecular physics, to solid-state physics and quantum information.

A brief history of quantum optics

The early development of quantum optics

1901 Planck phenomenologically derives a formula that accurately describes the spectral energy density of black-body radiation by proposing that the energy of resonant modes of a cavity with frequency ν can only take values that are integer multiples of $h\nu$.

- [12] M. Planck, “Über das Gesetz der Energieverteilung im Normalspectrum”, [Annalen der Physik](#) **309**, 553–563 (1901)

1905 Einstein proposes a heuristic model of light in which energy is quantized into discrete “light quanta” (nowadays called photons). He uses this idea to explain the photoelectric effect.

- [13] A. Einstein, “Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt”, [Annalen der Physik](#) **322**, 132–148 (1905)

1909 Taylor demonstrates in a double-slit experiment that interference patterns could still be observed even when the light intensity was extremely low, approaching the level where individual photons are involved. This experiment provided early experimental evidence for the particle-like behavior of light, supporting the emerging quantum concept of light quanta.

- [14] G. I. Taylor, “Interference fringes with feeble light”, *Proceedings of the Cambridge Philosophical Society* **15**, 114–115 (1909)
- 1916 Einstein develops a quantum-theoretical description of how atoms emit and absorb radiation, introducing the concepts of spontaneous emission, stimulated emission, and absorption.
- [15] A. Einstein, “Strahlungs-Emission und Absorption nach der Quantentheorie”, *Verhandlungen der Deutschen Physikalischen Gesellschaft* **18**, 318–323 (1916)
- 1927 Dirac formulates the first fully quantum mechanical theory of light–matter interaction, treating both the electromagnetic field and atomic systems quantum mechanically. He introduces the quantization of the radiation field, the creation and annihilation operators, and derives the rates of spontaneous and stimulated emission.
- [16] P. A. M. Dirac, “The quantum theory of the emission and absorption of radiation”, *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* **114**, 243–265 (1927)
- 1940s The development of quantum electrodynamics (QED): Feynman, Tomonaga, and Schwinger independently formulated a fully relativistic, renormalized theory of QED, introducing tools like Feynman diagrams (Feynman) and covariant operator methods (Tomonaga and Schwinger) to calculate observable quantities such as the Lamb shift with unprecedented precision.
- [17] S. Tomonaga, “On a relativistically invariant formulation of the quantum theory of wave fields”, *Progress of Theoretical Physics* **1**, 27–42 (1946)
- [18] J. Schwinger, “On quantum-electrodynamics and the magnetic moment of the electron”, *Physical Review* **73**, 416–417 (1948)
- [19] R. P. Feynman, “Space-time approach to quantum electrodynamics”, *Physical Review* **76**, 769–789 (1949)
- 1954 Hanbury Brown and Twiss introduce the intensity interferometer, a method that measures correlations between intensity fluctuations at separate detectors to determine the angular size of radio sources. In 1956, they extended this concept to visible light, demonstrating photon correlations (photon bunching).
- [20] R. Hanbury Brown and R. Twiss, “A new type of interferometer for use in radio astronomy”, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* **45**, 663–682 (1954)
- [21] R. Hanbury Brown and R. Q. Twiss, “Correlation between Photons in two Coherent Beams of Light”, *Nature* **177**, 27–29 (1956)
- 1960 Maiman reports the first successful demonstration of a laser, using a synthetic ruby crystal to produce coherent red light through stimulated emission.
- [22] T. H. Maiman, “Stimulated Optical Radiation in Ruby”, *Nature* **187**, 493–494 (1960)
- 1963 Glauber formulates a quantum theory of optical coherence, describing light’s classical and nonclassical properties, while Sudarshan introduces the P-representation, unifying semiclassical and quantum descriptions and identifying truly nonclassical states.
- [23] R. J. Glauber, “The Quantum Theory of Optical Coherence”, *Physical Review* **130**, 2529–2539 (1963)

- [24] R. J. Glauber, “Coherent and Incoherent States of the Radiation Field”, [Physical Review](#) **131**, 2766–2788 (1963)
- [25] E. C. G. Sudarshan, “Equivalence of Semiclassical and Quantum Mechanical Descriptions of Statistical Light Beams”, [Physical Review Letters](#) **10**, 277–279 (1963)
- 1972 Clauser and Freedman measure polarization correlations of entangled photons produced in atomic calcium transitions and show results that violate Bell’s inequality, in agreement with quantum mechanics and inconsistent with local hidden-variable theories.
- [26] S. J. Freedman and J. F. Clauser, “Experimental test of local hidden-variable theories”, [Physical Review Letters](#) **28**, 938–941 (1972)
- 1977 Kimble, Dagenais, and Mandel report the first experimental observation of photon antibunching, showing that photons emitted by a single atom are never detected simultaneously. This result provided direct evidence of the quantum nature of light emission, demonstrating behavior that cannot be explained by any classical wave theory.
- [27] H. J. Kimble et al., “Photon Antibunching in Resonance Fluorescence”, [Physical Review Letters](#) **39**, 691–695 (1977)
- 1985 Slusher et al. report the first experimental observation of squeezed light, where quantum noise in one field quadrature is reduced below the shot-noise limit.
- [28] R. E. Slusher et al., “Observation of Squeezed States Generated by Four-Wave Mixing in an Optical Cavity”, [Physical Review Letters](#) **55**, 2409–2412 (1985)
- 1986 Grangier, Roger, and Aspect show that single photons incident on a beam splitter are never detected simultaneously in both output ports, providing experimental proof of the particle nature of light. They also repeat Taylor’s experiment with single photons in a Mach–Zehnder interferometer, where they observe interference with single photons, confirming that light exhibits both wave- and particle-like behavior within the same experimental framework.
- [29] P. Grangier et al., “Experimental Evidence for a Photon Anticorrelation Effect on a Beam Splitter: A New Light on Single-Photon Interferences”, [Europhysics Letters](#) **1**, 173 (1986)
- 1987 Hong, Ou, and Mandel demonstrate two-particle interference of identical photons incident on a beam splitter, showing that indistinguishable photons always exit together in the same output port.
- [30] C. K. Hong et al., “Measurement of subpicosecond time intervals between two photons by interference”, [Physical Review Letters](#) **59**, 2044–2046 (1987)
- 1992 Thompson, Rempe, and Kimble observe vacuum Rabi splitting for single atoms in a high-finesse optical cavity, demonstrating strong coherent coupling between light and matter.
- [31] R. J. Thompson et al., “Observation of normal-mode splitting for an atom in an optical cavity”, [Physical Review Letters](#) **68**, 1132–1135 (1992)

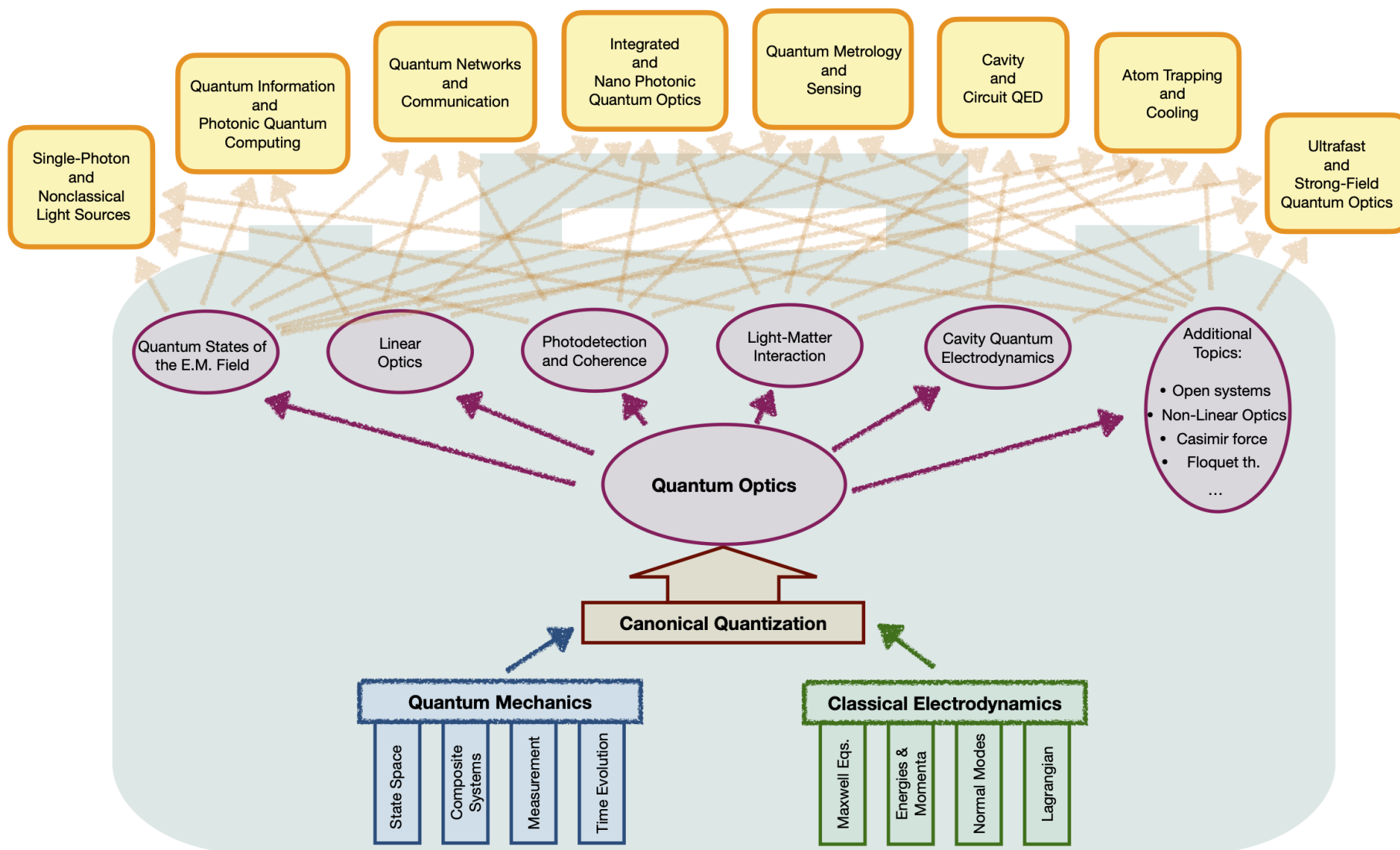
Branches of modern quantum optics Since then, many branches developed, some of which are listed in the following:

1. **Single-photon and nonclassical light sources** focus on generating and controlling individual photons and quantum states of light such as squeezed, entangled, or Fock states. These sources are essential for quantum communication, precision measurements, and quantum information processing.

2. **Quantum information and photonic quantum computing** use photons as qubits to encode, process, and transmit quantum information via linear-optical gates and photonic networks. This field enables quantum computing, teleportation, and entanglement distribution.
3. **Quantum networks and communication** use entangled photons and quantum repeaters to transmit information securely over long distances. This enables quantum key distribution, entanglement swapping, and scalable quantum networks for reliable quantum information transfer.
4. **Integrated and nanophotonic quantum optics** focuses on manipulating quantum light on chip-scale devices, integrating sources, waveguides, cavities, and detectors. This approach enables scalable, compact, and efficient quantum technologies for computing, communication, and sensing.
5. **Quantum metrology and sensing** exploit nonclassical light – such as squeezed or entangled states – to measure physical quantities with precision beyond classical limits. These techniques enhance applications like gravitational-wave detection, atomic clocks, and high-resolution spectroscopy.
6. **Cavity and circuit QED** studies the strong coupling between single atoms or artificial atoms (qubits) and a confined electromagnetic mode, enabling coherent exchange of energy between light and matter. This framework allows photon-mediated interactions, entanglement generation, and the implementation of quantum logic operations for quantum information processing.
7. **Atom trapping and cooling** techniques allow the preparation of ultracold atoms or ions, enabling precise control of their motion and quantum states. These methods enable Bose–Einstein condensates, quantum simulation, precision measurements, and quantum information processing.
8. **Ultrafast and strong-field quantum optics** explores light–matter interactions on femtosecond to attosecond timescales and at high intensities, probing electron dynamics and photoionization. It enables real-time control of quantum processes, steering electrons, manipulating states, and generating nonclassical light.

Aim of the lecture

The aim of this lecture is to set up the formal framework for the description of the quantized electromagnetic field and its interaction with matter (see the overview diagram below): We first recap the basic pillars of quantum mechanics (Chapter 1) as well as classical electrodynamics (Chapter 2). This will then allow us to quantize the electromagnetic field (Chapter 3) and dive into quantum optics. We will discuss diverse topics in quantum optics, ranging from quantum states of the electromagnetic field (Chapter 4), linear optics (Chapter 5), photodetection and coherence (Chapter 6), light-matter interaction (Chapters 7 and 8), cavity quantum electrodynamics (Chapter 9) to further topics which we will specify in the course of the lecture (note that these lecture notes contain complements, which possibly won't be discussed in the lecture). This will equip you with the fundamental toolbox needed to explore and conduct research on advanced topics in quantum optics.



Chapter 1

Quantum theory toolbox

We start with a very brief summary of the formalism of quantum mechanics. For details, please refer to the literature, e.g. [1–3].

1.1 Hilbert space and state vectors

Postulate 1. *Every isolated quantum system can be associated with a Hilbert space $\mathcal{H}$ (aka state space), which is a complex vector space together with an inner product. The system is completely described by its state vector $\psi \in \mathcal{H}$ (density operator ρ on $\mathcal{H}$).*

The inner product maps two state vectors (often simply called states, or vectors) $\psi, \phi \in \mathcal{H}$ to the complex numbers,

$$\mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}, \quad \psi, \phi \mapsto \langle \psi, \phi \rangle, \quad (1.1)$$

and satisfies, for all $c \in \mathbb{C}$,

$$\langle \psi, \phi \rangle = \langle \phi, \psi \rangle^* \quad (1.2a)$$

$$\langle \psi, c\phi \rangle = c \langle \psi, \phi \rangle \quad (1.2b)$$

$$\langle \psi, \psi \rangle \geq 0, \text{ with equality if and only if } \psi = 0. \quad (1.2c)$$

Note that the choice of the Hilbert space $\mathcal{H}$ depends on the physical scenario that is to be described. For example,

- (i) The Hilbert space of a two-level system is isomorphic to the complex vector space $\mathbb{C}^2$ together with the inner product $\langle \psi, \phi \rangle = \sum_{j=1}^2 \psi_j^* \phi_j$, where $\psi, \phi \in \mathbb{C}^2$. We typically write $\mathcal{H} \cong \mathbb{C}^2$ (in physics it is also common to write $\mathcal{H} = \mathbb{C}^2$), without explicitly mentioning the chosen inner product.
- (ii) For the Hilbert space of a particle in one spatial dimension, we choose the vector space $L_2(\mathbb{R})$ of square-integrable functions on $\mathbb{R}$, together with the inner product $\langle \psi, \phi \rangle = \int_{-\infty}^{\infty} dx \psi^*(x) \phi(x)$, where $\psi, \phi \in L_2(\mathbb{R})$. Accordingly, state vectors are represented by square-integrable functions,

$$\psi : x \mapsto \psi(x), \quad (1.3)$$

satisfying $\int_{-\infty}^{\infty} dx |\psi(x)|^2 < \infty$. Again, we typically write $\mathcal{H} \cong L_2(\mathbb{R})$, without mentioning the inner product.

Review Questions 1.1.

- What defines a Hilbert space? What is it good for?
- Consider the inner product $\langle \psi, \phi \rangle = \int_{-\infty}^{\infty} dx \psi^*(x) \phi(x) e^{-x^2}$, where $\psi, \phi : \mathbb{R} \rightarrow \mathbb{C}$. Can you come up with a definition of a complex vector space, which, together with the mentioned inner product, forms a physically relevant Hilbert space for a single continuous degree of freedom?

1.2 Operators and Dirac notation

1.2.1 Linear operators

An operator $A : \mathcal{H} \rightarrow \mathcal{H}$ is called a linear operator if it is linear in its inputs. For all $c \in \mathbb{C}$ and $\psi, \phi \in \mathcal{H}$, it satisfies

$$A(c\psi + \phi) = cA(\psi) + A(\phi). \quad (1.4)$$

In the following we only consider linear operators. For simplicity, we call a *linear operator* simply an *operator*, and write $A(\psi) = A\psi$, i.e., we drop the parentheses of the argument and suppose that the operator acts on the right.

The adjoint (often also called Hermitian conjugate or Hermitian adjoint) $A^\dagger$ of an operator A is defined by

$$\langle \psi, A\phi \rangle = \langle A^\dagger \psi, \phi \rangle. \quad (1.5)$$

Note that together with Eq. (1.2a), this yields

$$\langle \psi, A\phi \rangle = \langle \phi, A^\dagger \psi \rangle^*. \quad (1.6)$$

Accordingly, the elements $A_{\phi, \psi}^\dagger = \langle \phi, A^\dagger \psi \rangle$ of the adjoint $A^\dagger$ can thus be written in terms of the elements $A_{\psi, \phi} = \langle \psi, A\phi \rangle$ of A as

$$A_{\phi, \psi}^\dagger = A_{\psi, \phi}^*. \quad (1.7)$$

Consequently, in the case of a countable basis, the matrix representation of $A^\dagger$ (which we discuss further below) is obtained via transposition and complex conjugation of the matrix representation of A .

Consider now a composition of operators AB . Applying the definition (1.5) yields $\langle \psi, AB\phi \rangle = \langle A^\dagger \psi, B\phi \rangle = \langle B^\dagger A^\dagger \psi, \phi \rangle$. The adjoint of AB is therefore

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (1.8)$$

We further define the following operators:

- An operator A with domain $\text{dom}(A)$ is called Hermitian iff $\langle \psi, A\phi \rangle = \langle A\psi, \phi \rangle$ for all $\psi, \phi \in \text{dom}(A)$.
- An operator A on $\mathcal{H}$ with adjoint $A^\dagger = A$ is called self-adjoint.
- An operator A on $\mathcal{H}$ satisfying $A^\dagger A = AA^\dagger$ is called normal.
- An operator U on $\mathcal{H}$ satisfying $U^\dagger U = UU^\dagger = \mathbb{1}$ is called unitary.

- (v) An operator A on $\mathcal{H}$ is called positive (or positive semi-definite) if $\langle \psi, A\psi \rangle \geq 0$ for all $\psi \in \mathcal{H}$. If A is positive, we write $A \geq 0$.

Note that in *infinite-dimensional* Hilbert spaces, one has to be careful with the domain of linear operators. That is, an operator A may only be properly defined on a dense subset of $\mathcal{H}$. These operators are called unbounded. For example, the differential operator d/dx on the Hilbert space of continuous functions on the interval $[0, L]$ is unbounded, it is only properly defined for the subset of continuously *differentiable* functions on the interval $[0, L]$. In the definition (1.5) of the adjoint operators, this must be taken into account. One then finds that the domain of the adjoint is $\text{dom}(A^\dagger) \supset \text{dom}(A)$. Accordingly, in general, Hermiticity [see (i)] is weaker than self-adjointness [see (ii)], since the latter implies $\text{dom}(A^\dagger) = \text{dom}(A)$. In physics, this particularity is often overlooked, and Hermitian operators are often assumed to be self-adjoint. For example, a Hamiltonian is required to be Hermitian, however, we usually write $H^\dagger = H$, assuming that H is self-adjoint, which is a stronger requirement than Hermiticity. For the present lecture, this particularity is not important. Note that in *finite dimensional* Hilbert spaces, Hermiticity and self-adjointness are equivalent.

1.2.2 Dirac notation

The linearity (1.4) of operators as well as the inner product (1.2) allows us to apply Dirac's notation, which we use in the following. In particular, state vectors $\psi \in \mathcal{H}$ are written as kets $|\psi\rangle$, dual vectors ¹ from the dual space $\mathcal{H}^*$ are written as bras $\langle\phi|$, and the inner product $\langle\phi, \psi\rangle$ is written $\langle\phi|\psi\rangle$. Linear operators act on the kets to their right, that is, $A(\psi) = A\psi$ becomes $A|\psi\rangle$.

1.2.3 Operators in Dirac notation

If not otherwise mentioned, for the rest of this chapter we suppose that the Hilbert space $\mathcal{H}$ (which is a complex vector space) has a *countable* basis. Suppose that $\{|v_1\rangle, |v_2\rangle, \dots\}$ forms an orthonormal basis (ON-basis). State vectors $|\psi\rangle \in \mathcal{H}$ can then be written as a linear superposition of these basis states,

$$|\psi\rangle = \sum_j a_j |v_j\rangle, \quad (1.9)$$

with $|\psi\rangle$ normalized if $\|\psi\|^2 = \langle\psi|\psi\rangle = \sum_j |a_j|^2 = 1$. Similarly, operators A can be written in the outer product representation

$$A = \sum_{j,k} A_{j,k} |v_j\rangle\langle v_k|, \quad (1.10)$$

with $A_{j,k} \in \mathbb{C}$. If A is normal, i.e., $A^\dagger A = A A^\dagger$, it is diagonal with respect to some ON-basis $\{|\lambda_j\rangle\}_j$. This is called the spectral theorem, and states that A can be written as

$$A = V D V^\dagger = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|, \quad (1.11)$$

with $V = \sum_j |\lambda_j\rangle\langle v_j|$ a unitary operator, and $D = \sum_j \lambda_j |v_j\rangle\langle v_j|$ diagonal. Note that the spectral theorem is highly relevant, since it decomposes an operator A into its eigenstates $|\lambda_k\rangle$ with corresponding eigenvalues λ_k (for example, the spectral decomposition of the Hamilton operator allows you to solve Schrödinger's equation; see Sec. 1.5 below).

¹With each state vector, we can associate a dual vector $\phi \in \mathcal{H}^*$ in the dual space $\mathcal{H}^* = \{\langle\phi, \cdot\rangle \mid \phi \in \mathcal{H}\}$, which is the set of linear maps $\phi : \mathcal{H} \rightarrow \mathbb{C}$. That is, the dual vector $\phi \in \mathcal{H}^*$ is a linear map, $\phi : \mathcal{H} \rightarrow \mathbb{C}$ with the mapping $\psi \mapsto \langle\phi, \psi\rangle$, where $\langle\phi, \psi\rangle$ denotes the inner product of $\phi, \psi \in \mathcal{H}$.

A particular instance of the spectral theorem is the so-called completeness relation

$$\mathbb{1} = \sum_j |\lambda_j\rangle\langle\lambda_j|, \quad (1.12)$$

stating that the identity $\mathbb{1}$ is diagonal in every basis. Another important example is the projector on a m -dimensional subspace $W \subset \mathcal{H}$ with ON-basis $\{|w_1\rangle, \dots, |w_m\rangle\}$, defined by

$$P = \sum_{j=1}^m |w_j\rangle\langle w_j|. \quad (1.13)$$

Below, in Sec 1.4.2, projectors are used to describe projective measurements.

Finally, let us consider the trace of an operator A , which is central for the quantum mechanical description of measurements (see Sec. 1.4 below). It is defined, for any ON-basis $\{|v_l\rangle\}_l$, by

$$\text{Tr}(A) = \sum_l \langle v_l | A | v_l \rangle. \quad (1.14)$$

Inserting the outer product representation (1.10) and using the completeness relation (1.12), it is straightforward to see that $\text{Tr}(A) = \sum_j A_{j,j}$, which is the trace of the matrix representation of A . Important properties of the trace are as follows: For A, B operators, and $z \in \mathbb{C}$,

$$\text{Tr}(AB) = \text{Tr}(BA) \quad (1.15a)$$

$$\text{Tr}(A+B) = \text{Tr}(A) + \text{Tr}(B) \quad (1.15b)$$

$$\text{Tr}(zA) = z \text{Tr}(A). \quad (1.15c)$$

Review Questions 1.2.

- What is an operator? What defines a linear operator?
- What is the definition of the adjoint $A^\dagger$ of an operator A ?
- What is the adjoint of the composition of operators ABC ?
- Consider the operator $V = (|1\rangle\langle 1| - |1\rangle\langle 2| - |2\rangle\langle 1| + |2\rangle\langle 2|)/2$. Is V positive? unitary? self-adjoint? a projector? (hint: provide the spectral decomposition of V).
- Is the following correct? $\text{Tr}(ABC) = \text{Tr}(ACB)$. If not, why?

1.2.4 Density operators

In quantum theory, not all situations which we would like to consider as a possible *state* of the quantum system can be described by a single state vector $|\psi\rangle \in \mathcal{H}$. For example, if I through a coin and prepare $|0\rangle$ if it is heads and $|1\rangle$ if it is tails, and give you that state without telling which state I prepared, then your state of the system is $|0\rangle$ with classical probability $1/2$ and $|1\rangle$ with classical probability $1/2$. This is often called an ensemble (or statistical mixture), $\{(1/2, |0\rangle), (1/2, |1\rangle)\}$. To describe such an ensemble, we must therefore consider state operators instead of state vectors. The corresponding state operator for the ensemble that I prepared for you is $(|0\rangle\langle 0| + |1\rangle\langle 1|)/2$. Note that it includes no phase information between $|0\rangle$ and $|1\rangle$.

Note that the ensemble $\{(1/2, |0\rangle), (1/2, |1\rangle)\}$ with corresponding state operator $\rho = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2$ is fundamentally different from a superposition $|\psi\rangle = (|0\rangle + e^{i\varphi}|1\rangle)/\sqrt{2}$ with corresponding state operator $|\psi\rangle\langle\psi| = (|0\rangle\langle 0| + e^{-i\varphi}|0\rangle\langle 1| + e^{i\varphi}|1\rangle\langle 0| + |1\rangle\langle 1|)/2$. The superposition state has

a well-defined relative phase $e^{i\varphi}$ between $|0\rangle$ and $|1\rangle$, while the ensemble state that I prepared for you has no such well-defined phase.

In general, we describe the ensemble $\{p_j, |\psi_j\rangle\}_j$ of a quantum system, where the state vector $|\psi_j\rangle$ occurs with classical probability p_j , by the density operator (often also called density matrix, although the term *matrix* usually refers to the matrix representation of the density operator)

$$\rho = \sum_j p_j |\psi_j\rangle \langle \psi_j|, \quad \text{Tr}(\rho) = 1, \quad \rho \geq 0. \quad (1.16)$$

Note that the density operator is self-adjoint, i.e., $\rho = \rho^\dagger$. Further note that $\rho \geq 0$ implies $p_j \geq 0$, and $\text{Tr}(\rho) = 1$ implies $\sum_j p_j = 1$. Accordingly, p_j can be interpreted as the probability with which the state $|\psi_j\rangle$ occurs. Equation (1.16) is not necessarily the spectral decomposition of ρ , since the states $|\psi_j\rangle$ are not necessarily orthogonal. However, since ρ is self-adjoint, it is also normal, and, thus, possesses a spectral decomposition of the form (1.11),

$$\rho = \sum_j \lambda_j |\lambda_j\rangle \langle \lambda_j|, \quad (1.17)$$

with eigenvectors $|\lambda_j\rangle$, satisfying $\langle \lambda_j | \lambda_k \rangle = \delta_{j,k}$, and corresponding eigenvalues λ_j .

If ρ has a single non-vanishing eigenvalue, $\rho = |\psi\rangle \langle \psi|$, it is called pure. Otherwise it is called mixed. The degree of mixedness is often measured by the purity $\text{Tr}(\rho^2)$. It satisfies $\text{Tr}(\rho^2) \leq 1$, with equality iff ρ is pure.

1.2.5 Matrix representation

So far we considered state vectors as *abstract* objects living in the Hilbert space $\mathcal{H}$. In order to associate these abstract objects (which are elements of a complex vector space) with a set of complex numbers (which is useful for computer-based calculations), we have to choose a matrix representation, which corresponds to the choice of a coordinate system. Given an ON-basis $\{|v_j\rangle\}_j$, the most convenient choice is $|v_1\rangle = (1, 0, 0, \dots)^\top$, $|v_2\rangle = (0, 1, 0, \dots)^\top, \dots$. In this case, a state vector $|\psi\rangle = \sum_j \psi_j |v_j\rangle$ and its corresponding dual vector $\langle \psi|$ have the matrix representation

$$|\psi\rangle = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{pmatrix}, \quad \langle \psi| = (\psi_1^*, \psi_2^*, \dots). \quad (1.18)$$

Using the usual matrix multiplication, we get the inner product,

$$\langle \phi | \psi \rangle = (\phi_1^*, \phi_2^*, \dots) \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{pmatrix} = \sum_j \phi_j^* \psi_j. \quad (1.19)$$

and, similarly, the outer product representation,

$$|\psi\rangle \langle \phi| = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{pmatrix} (\phi_1^*, \phi_2^*, \dots) = \begin{pmatrix} \psi_1 \phi_1^* & \psi_1 \phi_2^* & \cdots \\ \psi_2 \phi_1^* & \psi_2 \phi_2^* & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \quad (1.20)$$

Note that the choice of the matrix representation of the ON-basis $\{|v_j\rangle\}_j$ is not unique. Instead of choosing the vectors $(0, \dots, 0, 1, 0, \dots, 0)^\top$ you could have chosen the representation $U(0, \dots, 0, 1, 0, \dots, 0)^\top$, with U any unitary matrix, which is, thus, unitarily equivalent to the

former representation. This freedom is due to the fact that unitary transformations preserve the inner product. That is, for $|w_j\rangle = U|v_j\rangle$, we have

$$\langle w_j | w_k \rangle = \langle v_j | \underbrace{U^\dagger U}_{\mathbb{I}} | v_k \rangle = \langle v_j | v_k \rangle. \quad (1.21)$$

Equation (1.21) essentially tells us that physics does not depend on the choice of your coordinate system. Unitary transformations of the whole system (i.e., transforming all elements such as states and operators) can therefore be seen as transformations between different coordinate systems. This is exactly what we discuss in Sec. 1.5 below, when we transform between the Schrödinger, Heisenberg, and interaction pictures.

Review Questions 1.3.

- What defines a density operator? Why are density operators needed?
- What are the density operators corresponding to the ensembles $\{(\frac{3}{4}, |0\rangle), (\frac{1}{4}, |1\rangle)\}$ and $\{(\frac{1}{2}, |a\rangle), (\frac{1}{2}, |b\rangle)\}$, where $|a\rangle = \sqrt{3/4}|0\rangle + \sqrt{1/4}|1\rangle$ and $|b\rangle = \sqrt{3/4}|0\rangle - \sqrt{1/4}|1\rangle$? Conclude that a density operator doesn't represent a unique ensemble. Instead, different ensembles can give rise to the same density operator.
- Consider the operator A with matrix representation

$$A = \begin{pmatrix} 1 & \frac{i}{2} \\ \frac{1+i}{4} & 3 \end{pmatrix}.$$

What is the matrix representation of its adjoint $A^\dagger$?

1.3 Composite systems

Postulate 2. The state space $\mathcal{H}$ of a composite physical system is the tensor product $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n$ of the state spaces $\mathcal{H}_j$ of the component physical systems. If system j is separately prepared in state ρ_j , then the joint state of the total system is $\rho_1 \otimes \rho_2 \otimes \cdots \otimes \rho_n$.

The tensor product

$$\otimes : \mathcal{H}_1 \times \mathcal{H}_2 \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2 \quad |v\rangle, |w\rangle \mapsto |v\rangle \otimes |w\rangle \quad (1.22)$$

gives rise to the composite Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$ of dimension $\dim(\mathcal{H}_1 \otimes \mathcal{H}_2) = \dim(\mathcal{H}_1) \dim(\mathcal{H}_2)$. It satisfies, for all $|v\rangle, |v_1\rangle, |v_2\rangle \in \mathcal{H}_1$, $|w\rangle, |w_1\rangle, |w_2\rangle \in \mathcal{H}_2$, and $z \in \mathbb{C}$,

$$z(|v\rangle \otimes |w\rangle) = (z|v\rangle) \otimes |w\rangle = |v\rangle \otimes (z|w\rangle) = z|v\rangle \otimes |w\rangle \quad (1.23a)$$

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle \quad (1.23b)$$

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle. \quad (1.23c)$$

Operators A on $\mathcal{H}_1$ and B on $\mathcal{H}_2$ act on their individual subspaces,

$$(A \otimes B)(|v\rangle \otimes |w\rangle) = A|v\rangle \otimes B|w\rangle, \quad (1.24)$$

and an ON-basis of $\mathcal{H}_1 \otimes \mathcal{H}_2$ can be obtained from the individual ON-bases $\{|v_j\rangle\}_j$ of $\mathcal{H}_1$ and $\{|w_k\rangle\}_k$ of $\mathcal{H}_2$ through $\{|v_j\rangle \otimes |w_k\rangle\}_{j,k}$. Note that there are many different ways to simplify notation. Usually we write

$$|v\rangle \otimes |w\rangle = |v\rangle |w\rangle = |v, w\rangle = |vw\rangle. \quad (1.25)$$

Suppose now we have a state ρ of a composite quantum system $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. Further suppose we have no access to system B, and we want to describe the state and measurement statistics of system A alone. That is, we want to describe the reduced density operator ρ_A of system A. This is obtained from ρ by tracing out system B via the partial trace,

$$\rho_A = \text{Tr}_B (\rho) = \sum_j \langle w_j | \rho | w_j \rangle, \quad (1.26)$$

where $\{|w_j\rangle\}_j$ is any ON-basis of $\mathcal{H}_B$ [c.f. Eq. (1.14)]. Note that for any states $|\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}_A$ and $|\phi_1\rangle, |\phi_2\rangle \in \mathcal{H}_B$, this yields

$$\begin{aligned} \text{Tr}_B (|\psi_1\rangle\langle\psi_2| \otimes |\phi_1\rangle\langle\phi_2|) &= |\psi_1\rangle\langle\psi_2| \sum_j \langle w_j | \phi_1 \rangle \langle \phi_2 | w_j \rangle \\ &= |\psi_1\rangle\langle\psi_2| \sum_j \langle \phi_2 | w_j \rangle \langle w_j | \phi_1 \rangle \\ &= |\psi_1\rangle\langle\psi_2| \langle \phi_2 | \underbrace{\left(\sum_j |w_j\rangle\langle w_j| \right)}_{\mathbb{1}} | \phi_1 \rangle \\ &= |\psi_1\rangle\langle\psi_2| \langle \phi_2 | \phi_1 \rangle. \end{aligned} \quad (1.27)$$

That is, we simply have to “swap” the kets and bras of subsystem B, i.e. $|\phi_1\rangle\langle\phi_2| \rightarrow \langle\phi_2|\phi_1\rangle$.

Review Questions 1.4.

- Calculate $\text{Tr}_B (A)$ for $A = \sum_{j,k,l,m} A_{j,k,l,m} |a_j, b_k\rangle\langle a_l, b_m|$, where $\{|a_j\rangle\}_j$ and $\{|b_j\rangle\}_j$ form an ON-basis of $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively.
- Show that $\text{Tr}_B (V \otimes W) = V \text{Tr} (W)$ for any operator $V \otimes W$ on $\mathcal{H}_A \otimes \mathcal{H}_B$.
- Calculate $\text{Tr}_B (\rho)$ and $\text{Tr}_A (\rho)$ for the density operator $\rho = \rho_A \otimes \rho_B$.
- Calculate $\text{Tr}_B (\rho)$ and $\text{Tr}_B (V \otimes W \rho)$ for $\rho = \sum_j \rho_A^{(j)} \otimes \rho_B^{(j)}$.

1.4 Measurements

1.4.1 General measurements

Postulate 3. Quantum measurements are described by a collection $\{M_m\}_m$ of measurement operators on $\mathcal{H}$, which satisfy the completeness equation $\sum_m M_m^\dagger M_m = \mathbb{1}$. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the system is ρ immediately before the measurement then the probability that result m occurs is given by

$$p(m) = \text{Tr} (M_m^\dagger M_m \rho), \quad (1.28)$$

and the state of the system after the measurement is

$$\rho|_m = \frac{M_m \rho M_m^\dagger}{\text{Tr} (M_m^\dagger M_m \rho)}. \quad (1.29)$$

1.4.2 Projective measurements

An important class of measurements are the so-called projective measurements (or projective-valued measures PVMs, or von Neumann measurements). In this case, the measurement operators are projectors, i.e., $M_m = P_m$ [c.f. Eq. (1.13)].

Projective measurement are often described through an observable M , which is a Hermitian operator on $\mathcal{H}$ with spectral decomposition

$$M = \sum_m m P_m. \quad (1.30)$$

The outcome m is thus associated with an eigenspace of M with eigenvalue m . The projectors P_m project on these eigenspaces and constitute the measurement operators. Since $P_m P_{m'} = \delta_{m,m'} P_m$, the probability to measure outcome m is

$$p(m) = \text{Tr}(P_m \rho), \quad (1.31)$$

and the post-measurement state is

$$\rho|_m = \frac{P_m \rho P_m}{\text{Tr}(P_m \rho)}. \quad (1.32)$$

Two important special cases are as follows:

- (i) If the observable M has non-degenerate eigenvalues, the measurement operators are one-dimensional projectors, i.e., $M_m = |\lambda_m\rangle\langle\lambda_m|$. Accordingly, $p(m) = \langle\lambda_m|\rho|\lambda_m\rangle$, and the post-measurement state is $\rho|_m = |\lambda_m\rangle\langle\lambda_m|$.
- (ii) If, additionally, ρ is pure, i.e., $\rho = |\psi\rangle\langle\psi|$, then, $p(m) = |\langle\lambda_m|\psi\rangle|^2$. This is often called Born's rule.

Note that beyond projective measurements, the measurement postulate 3 allows for more general quantum measurements which we don't discuss here.

1.4.3 Expectation values

Given an observable M , taking the average of the distribution of measurement results of M (resulting from measuring many copies of the state ρ) yields the expectation value

$$\langle M \rangle = \text{Tr}(M\rho) \stackrel{(1.30)}{=} \sum_m m \text{Tr}(P_m \rho) \stackrel{(1.31)}{=} \sum_m m p(m). \quad (1.33)$$

The n -th moment of the distribution of measurement results is obtained through $\langle M^n \rangle$, such that, for example, the variance of the distribution of measurement results is

$$\text{Var}(M) = \langle M^2 \rangle - \langle M \rangle^2. \quad (1.34)$$

Note that often the notation $\langle M \rangle = \langle M \rangle_\rho$ is used, to indicate that the expectation value is taken with respect to the density operator ρ .

Review Questions 1.5.

- What is a projective measurement?
- Consider the density operator $\rho_1 = |0\rangle\langle 0|$ of a two-level system, and the observable $M = -\frac{\hbar}{2}|0\rangle\langle 0| + \frac{\hbar}{2}|1\rangle\langle 1|$. What are the probabilities to find the outcomes $-\hbar/2$ and $\hbar/2$, respectively?
- What are the respective probabilities for $\rho_2 = |-\rangle\langle -|$ and $\rho_3 = (|-\rangle\langle -| + |+\rangle\langle +|)/2$, where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$?
- What are the expectation values $\langle M \rangle_\rho$ for the three cases from above? Discuss the result.

1.5 Time evolution

1.5.1 Schrödinger picture

Postulate 4. In the Schrödinger picture, the time evolution of a closed quantum system is due to the evolution of the system's state as described by the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle, \quad (1.35)$$

where $H(t)$ is a Hermitian^a operator known as the Hamiltonian of the closed system, and $\hbar$ is the reduced Planck's constant.

^aNote that in this lecture notes we consider a stronger criterion and suppose that the Hamiltonian is always self-adjoint, i.e., $H^\dagger(t) = H(t)$ [recall the discussion in Sec. 1.2.1]. If it is not, then we can restrict the domains of $H(t)$ and $H^\dagger(t)$, such that $H^\dagger(t) = H(t)$.

Recall that the state of a statistical mixture $\{p_j, |\psi_j\rangle\}_j$ is described by the density operator $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ [see Eq. (1.16)]. In a closed quantum system, the probabilities p_j , which weight the state $|\psi_j\rangle$, must be independent of time. Hence, in the Schrödinger picture, the evolution equation of the density operator is obtained by considering the Schrödinger equation (1.35) for each component $|\psi_j(t)\rangle$. Using the Hermitian conjugate $-i\hbar \frac{d}{dt} \langle\psi(t)| = \langle\psi(t)| H(t)$ of Schrödinger's equation (1.35), we get

$$\begin{aligned} i\hbar \frac{d}{dt} \rho(t) &= i\hbar \frac{d}{dt} \left[\sum_j p_j |\psi_j(t)\rangle\langle\psi_j(t)| \right] \\ &= \sum_j p_j [H(t) |\psi_j(t)\rangle\langle\psi_j(t)| - |\psi_j(t)\rangle\langle\psi_j(t)| H(t)] \\ &= [H(t), \rho(t)]. \end{aligned} \quad (1.36)$$

That is, we find that in a closed quantum system, the time evolution of the density operator $\rho(t)$ is described by the von Neumann equation (or Liouville-von Neumann equation)

$$\frac{d}{dt} \rho(t) = -\frac{i}{\hbar} [H(t), \rho(t)]. \quad (1.37)$$

1.5.2 Unitary time evolution operator

The solutions $|\psi(t)\rangle$ to the Schrödinger equation (1.35) and $\rho(t)$ to the von-Neumann equation (1.37) define the unitary time evolution operator $U(t, t_0)$ via

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle, \quad \rho(t) = U(t, t_0) \rho(t_0) U^\dagger(t, t_0), \quad (1.38)$$

which must satisfy $U(t, t) = \mathbb{1}$. By plugging $|\psi(t)\rangle$ from (1.38) into the Schrödinger equation (1.35), we see that the unitary time evolution operator obeys the differential equation

$$i\hbar \frac{d}{dt} U(t, t_0) = H(t) U(t, t_0), \quad (1.39)$$

with initial condition $U(t_0, t_0) = \mathbb{1}$. In order to find a solution for $U(t, t_0)$, let us integrate this equation, yielding

$$U(t, t_0) = \mathbb{1} - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1) U(t_1, t_0). \quad (1.40)$$

We see that this defines an iterative sequence. Hence, we can insert this expression for $U(t_1, t_0)$ on the right hand side, yielding

$$\begin{aligned} U(t, t_0) &= \mathbb{1} - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1) \left[\mathbb{1} - \frac{i}{\hbar} \int_{t_0}^{t_1} dt_2 H(t_2) U(t_2, t_0) \right] \\ &= \mathbb{1} - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1) + \left(-\frac{i}{\hbar} \right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H(t_1) H(t_2) U(t_2, t_0). \end{aligned} \quad (1.41)$$

By continuing with plugging (1.40) into the right hand side of (1.41), after an infinite number of iterations, we obtain the exact solution

$$U(t, t_0) = \mathbb{1} + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n H(t_1) H(t_2) \cdots H(t_n), \quad (1.42)$$

which is known as the Dyson series. We recognise that the limits of all integrals are different. If all integrals had the same limit, we see that Eq. (1.42) would simplify significantly, since then the integrals are decoupled. We therefore want to find a way of rewriting (1.42) such that all integrals have the same limits. To do so, we define the time-ordering operator $\mathcal{T}$, which takes n operators and orders them such that their time arguments are decreasing from left to right,

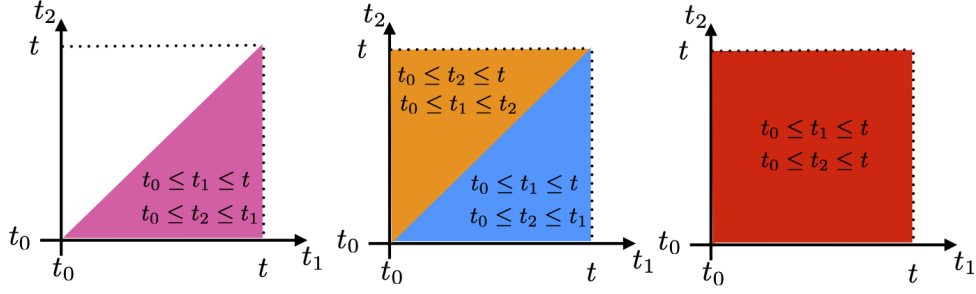
$$\mathcal{T}[H(t_1)H(t_2)\cdots H(t_n)] = H(t_j)H(t_k)\cdots H(t_l), \quad \text{where } t_j > t_k > \cdots > t_l. \quad (1.43)$$

Note that $\mathcal{T}$ only acts non-trivially if its arguments don't commute.

In order to see how we can simplify (1.42) using the time-ordering operator, consider the integral for $n = 2$:

$$\begin{aligned} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H(t_1) H(t_2) &= \frac{1}{2} \left(\int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H(t_1) H(t_2) + \int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 H(t_2) H(t_1) \right) \\ &= \frac{1}{2} \left(\int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \mathcal{T}[H(t_1)H(t_2)] + \int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \mathcal{T}[H(t_1)H(t_2)] \right) \\ &= \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \mathcal{T}[H(t_1)H(t_2)]. \end{aligned} \quad (1.44)$$

The last equality can easily be seen graphically:



This strategy to extend the integration over the whole parameter space $t_0 \leq t_1, t_2 \leq t$ can be generalized for the n -dimensional integral in (1.42): Here, the integral is defined in the space of an n -dimensional hypercube (which is the n -dimensional generalization of a cube) defined by the region $t_0 \leq t_j \leq t$ for all $j = 1, \dots, n$. Similar to the above illustrated case of $n = 2$, the integral merely runs over a slice of the hypercube, namely the region $t_0 \leq t_n \leq t_{n-1} \dots \leq t_1$, with a volume of $1/n!$ of the whole hypercube (recall that for $n = 2$ we integrated over the lower triangle of a square). It is now easy to see that when applying the time-ordering operator to $H(t_1)H(t_2) \dots H(t_n)$ in (1.42) and integrating over the whole hypercube, i.e., $t_0 \leq t_j \leq t$ for all $j = 1, \dots, n$, then we get $n!$ times the original integral. We must therefore compensate by a factor of $1/n!$, and get

$$U(t, t_0) = \mathbb{1} + \sum_{n=1}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \dots \int_{t_0}^t dt_n \mathcal{T} [H(t_1)H(t_2) \dots H(t_n)]. \quad (1.45)$$

Up to the time-ordering operator, this corresponds to the exponential of the operator $-\frac{i}{\hbar} \int_{t_0}^t ds H(s)$. We therefore define the time-ordered exponential

$$\mathcal{T} \exp \left(-\frac{i}{\hbar} \int_{t_0}^t ds H(s) \right) = \mathbb{1} + \sum_{n=1}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \dots \int_{t_0}^t dt_n \mathcal{T} [H(t_1)H(t_2) \dots H(t_n)], \quad (1.46)$$

such that the exact solution of the unitary time evolution operator can be written in the compact form

$$U(t, t_0) = \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_{t_0}^t ds H(s) \right), \quad (1.47)$$

with the time-ordering operator only acting non-trivially if $[H(t_1), H(t_2)] \neq 0$ for $t_1 \neq t_2$. We can immediately recognize two important special cases:

- (i) If $[H(t_1), H(t_2)] = 0$ for all $t_0 \leq t_1, t_2 \leq t$, the solution of the unitary time evolution operator is

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} \int_{t_0}^t ds H(s) \right). \quad (1.48)$$

- (ii) If the Hamiltonian is time independent, i.e., $H(t) = H$, the solution is

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} (t - t_0) H \right). \quad (1.49)$$

Let us finally note that the general solution (1.47) can also be written in the form of a Magnus expansion, reading

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1) - \frac{1}{2\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 [H(t_1), H(t_2)] + \dots \right), \quad (1.50)$$

where the higher order terms (not shown here) contain stacked commutators of three or more operators. This expansion is particularly fruitful if these stacked commutators vanish, i.e., if $[H(t_1), [H(t_2), H(t_3)]] = 0$ for all $t_0 \leq t_1, t_2, t_3 \leq t$.

Note that when t_0 is assumed to be known, we often simply write

$$U(t, t_0) = U(t), \quad |\psi(t_0)\rangle = |\psi\rangle, \quad \text{and} \quad \rho(t_0) = \rho. \quad (1.51)$$

1.5.3 Heisenberg picture

Recall the measurement postulate (3), stating that the measurement probabilities $p(m) = \text{Tr}(E_m \rho)$ are given by the expectation value of a positive operator $E_m = M_m^\dagger M_m$ [c.f. Eq. (1.28)]. Using the time-evolved state $\rho(t)$ in the Schrödinger picture from Eq. (1.38), as well as the cyclic property of the trace, the time dependent probabilities can be written as

$$p(m, t) = \text{Tr}(E_m \rho(t)) \stackrel{(1.38)}{=} \text{Tr}(E_m U(t) \rho U^\dagger(t)) \stackrel{(1.15a)}{=} \text{Tr}\left(\underbrace{U^\dagger(t) E_m U(t)}_{E_{H,m}(t)} \rho\right) = \text{Tr}(E_{H,m}(t) \rho). \quad (1.52)$$

Accordingly, instead of describing the time evolution in the Schrödinger picture through the evolution of quantum states $\rho(t)$, we can equally well consider a picture where the measurement operators $E_{H,m}(t)$ evolve in time and the quantum states ρ are independent of time. This is called the Heisenberg picture.

In general, it can happen that the measurement operator already has some time dependence in the Schrödinger picture, i.e., we consider $E(t)$ in the Schrödinger picture. The corresponding Heisenberg operator is then

$$E_H(t) = U^\dagger(t) E(t) U(t). \quad (1.53)$$

It evolves according to

$$\begin{aligned} i\hbar \frac{d}{dt} E_H(t) &\stackrel{(1.53)}{=} i\hbar \frac{d}{dt} (U^\dagger(t) E(t) U(t)) \\ &= U^\dagger(t) E(t) \underbrace{\left(i\hbar \frac{d}{dt} U(t)\right)}_{H(t)U(t)} + \underbrace{\left(i\hbar \frac{d}{dt} U^\dagger(t)\right)}_{-U^\dagger(t)H(t)} E(t) U(t) + U^\dagger(t) \left(i\hbar \frac{d}{dt} E(t)\right) U(t) \\ &\stackrel{(1.39)}{=} \underbrace{U^\dagger(t) E(t) U(t)}_{E_H(t)} \underbrace{U^\dagger(t) H(t) U(t)}_{H_H(t)} - \underbrace{U^\dagger(t) H(t) U(t)}_{H_H(t)} \underbrace{U^\dagger(t) E(t) U(t)}_{E_H(t)} + \left(i\hbar \frac{\partial}{\partial t} E(t)\right)_H \\ &= [E_H(t), H_H(t)] + \left(i\hbar \frac{\partial}{\partial t} E(t)\right)_H. \end{aligned} \quad (1.54)$$

That is, the Heisenberg equation of motion is

$$\boxed{\frac{d}{dt} E_H(t) = \frac{i}{\hbar} [H_H(t), E_H(t)] + \left(\frac{\partial}{\partial t} E(t)\right)_H}. \quad (1.55)$$

Note that here the Hamiltonian appears in the Heisenberg picture, i.e., $H_H(t) = U^\dagger(t) H(t) U(t)$. In the special case where $H(t) = H$ is time independent, the unitary $U(t) = \exp[-i(t - t_0)H/\hbar]$ commutes with H , and, hence, $H_H = H$. The Heisenberg equation of motion then simplifies to

$$\frac{d}{dt} E_H(t) = \frac{i}{\hbar} [H, E_H(t)] + e^{\frac{i}{\hbar}(t-t_0)H} \left(\frac{\partial}{\partial t} E(t)\right) e^{-\frac{i}{\hbar}(t-t_0)H}. \quad (1.56)$$

Note that the Heisenberg picture simply results from the Schrödinger picture by transforming your whole system via the unitary transformation $U^\dagger(t)$. That is, any operator $A(t)$ is transformed via $U^\dagger(t)A(t)U(t)$, and any state vector $|\psi(t)\rangle$ is transformed via $U^\dagger(t)|\psi(t)\rangle$. This includes the density operator, which reads in the Heisenberg picture $\rho_H(t) = U^\dagger(t)\rho(t)U(t) = U^\dagger(t)U(t)\rho U^\dagger(t)U(t) = \rho$. Accordingly, the transformation between the Schrödinger picture and the Heisenberg picture simply corresponds to a transformation of the coordinate system [recall our discussion around Eq. (1.21)], since there is a unitary freedom in choosing the coordinate system. And since you have this freedom at any instance of time, *any* time dependent unitary transformation of your whole system doesn't change the physics (similar to your freedom in choosing the coordinate system in the space of three spatial dimensions).

This discussion tells us that we can perform any unitary transformation, and not only stick to the Heisenberg or Schrödinger picture. Indeed, often the art is to find a unitary transformation (i.e., a change of the coordinate system) that significantly simplifies the equations of motion. One of these important transformations is the transformation into the interaction picture.

1.5.4 Interaction picture

Often one deals with a Schrödinger equation of the form

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = [H_0 + V(t)] |\psi(t)\rangle, \quad (1.57)$$

where the Hamiltonian decomposes into a time independent part H_0 whose solutions are known, and an additional, possibly time dependent part $V(t)$, which usually corresponds to an interaction or perturbation term. While solving the Schrödinger equation (1.57) for the full Hamiltonian $H = H_0 + V(t)$ is usually difficult, we can make use of the fact that solutions to H_0 are known, which allows us to transform into the interaction picture (or Dirac picture) with respect to H_0 via the unitary transformation $U_0^\dagger(t) = e^{\frac{i}{\hbar}(t-t_0)H_0}$. The state vector and operators in the interaction picture therefore become

$$|\psi_I(t)\rangle = U_0^\dagger(t) |\psi(t)\rangle, \quad A_I(t) = U_0^\dagger(t) A(t) U_0(t), \quad (1.58)$$

and the Schrödinger equation (1.57) modifies to

$$\begin{aligned} i\hbar \frac{d}{dt} \left(U_0(t) \underbrace{U_0^\dagger(t) |\psi(t)\rangle}_{|\psi_I(t)\rangle} \right) &= U_0(t) U_0^\dagger(t) [H_0 + V(t)] U_0(t) \underbrace{U_0^\dagger(t) |\psi(t)\rangle}_{|\psi_I(t)\rangle}, \\ \underbrace{\left(i\hbar \frac{d}{dt} U_0(t) \right)}_{H_0 U_0(t)} |\psi_I(t)\rangle + U_0(t) \left(i\hbar \frac{d}{dt} |\psi_I(t)\rangle \right) &= U_0(t) H_0 |\psi_I(t)\rangle + U_0(t) V_I(t) |\psi_I(t)\rangle \\ U_0(t) \left(i\hbar \frac{d}{dt} |\psi_I(t)\rangle \right) &= U_0(t) V_I(t) |\psi_I(t)\rangle. \end{aligned} \quad (1.59)$$

After multiplying both sides with $U_0^\dagger(t)$ from the left, the Schrödinger equation in the interaction picture becomes

$$\boxed{i\hbar \frac{d}{dt} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle}. \quad (1.60)$$

In short, knowing the solutions to H_0 allowed us to perform the transformation (1.58) into the interaction picture, where the Schrödinger equation simplifies significantly, and only contains the additional (interaction or perturbation) term $V_I(t) = U_0^\dagger(t) V(t) U_0(t)$.

Suppose now we can find the solution $|\phi_I(t)\rangle$ to the Schrödinger equation (1.60) in the interaction picture. To get the solution to the Schrödinger equation (1.57) in the original picture,

we simply have to transform back, yielding $|\phi(t)\rangle = U_0(t) |\phi_I(t)\rangle$ [this is what is usually done]. Alternatively we can stay in the interaction picture and [besides the solution $|\phi_I(t)\rangle$] consider also the measurement operators in the interaction picture through the transformation (1.58) [remember that this is simply a change of coordinates].

Review Questions 1.6.

- *What is the relation between the Schrödinger equation and the von Neumann equation?*
- *What is the time-ordered exponential? Why is it important for quantum mechanics?*
- *Suppose you know the spectral decomposition of a time-independent Hamiltonian, $H = \sum_j E_j |E_j\rangle \langle E_j|$. How does the eigenstate $|E_j\rangle$ evolve in time? Further suppose you have the initial state $|\phi(0)\rangle$. What is the time evolved state in the eigenbasis of H ? Conclude that diagonalizing the Hamiltonian essentially solves the Schrödinger equation.*
- *What is the difference between the Heisenberg and Schrödinger picture?*
- *Why is the interaction picture important?*
- *Are there any other “pictures” than the Heisenberg, Schrödinger, and interaction picture?*

Chapter 2

Classical electrodynamics

Our aim is to understand the dynamics of the electromagnetic field in the presence of charged particles. We will see that the classical electromagnetic field can be described as a set of classical harmonic oscillators. In the next chapter, this will allow us to develop a quantum mechanical description of the electromagnetic field through the quantization of these harmonic oscillators.

2.1 Lorentz force and Maxwell's equations

Consider a set of particles with masses m_α and charges q_α , and time dependent positions $\mathbf{r}_\alpha(t)$ and velocities $\mathbf{v}_\alpha(t) = \dot{\mathbf{r}}_\alpha(t)$. Under the influence of an electric field $\mathbf{E}(\mathbf{r}, t)$ and magnetic field $\mathbf{B}(\mathbf{r}, t)$, the particles' non-relativistic dynamics is described by the Newton-Lorentz equations

$$m_\alpha \frac{d^2}{dt^2} \mathbf{r}_\alpha(t) = q_\alpha [\mathbf{E}(\mathbf{r}_\alpha(t), t) + \mathbf{v}_\alpha(t) \times \mathbf{B}(\mathbf{r}_\alpha(t), t)]. \quad (2.1)$$

The set of particles gives rise to a charge density $\rho(\mathbf{r}, t)$ and current density $\mathbf{j}(\mathbf{r}, t)$,

$$\rho(\mathbf{r}, t) = \sum_\alpha q_\alpha \delta(\mathbf{r} - \mathbf{r}_\alpha(t)), \quad (2.2a)$$

$$\mathbf{j}(\mathbf{r}, t) = \sum_\alpha q_\alpha \mathbf{v}_\alpha(t) \delta(\mathbf{r} - \mathbf{r}_\alpha(t)), \quad (2.2b)$$

which, in turn, act as a source of the electric and magnetic field, as described by Maxwell's equations

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = \frac{\rho(\mathbf{r}, t)}{\epsilon_0}, \quad (2.3a)$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \quad (2.3b)$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t}, \quad (2.3c)$$

$$\nabla \times \mathbf{B}(\mathbf{r}, t) = \mu_0 \left(\mathbf{j}(\mathbf{r}, t) + \epsilon_0 \frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t} \right), \quad (2.3d)$$

where μ_0 is the vacuum permeability, and ϵ_0 the vacuum permittivity. Equations (2.1) and (2.3) form a set of coupled differential equations for the system composed of the fields+particles, i.e., they describe at any instance of time t_0 the fields $\mathbf{E}(\mathbf{r}, t_0)$ and $\mathbf{B}(\mathbf{r}, t_0)$ at all positions $\mathbf{r}$ of space, and the positions $\mathbf{r}_\alpha(t_0)$ and velocities $\mathbf{v}_\alpha(t_0)$ of all particles.

2.1.1 Maxwell's equations in reciprocal space

Our next aim is to express Maxwell's equations in reciprocal space (or $\mathbf{k}$ -space). To this end, consider any vector field $\mathbf{F}(\mathbf{r})$, whose Fourier transform $\tilde{\mathbf{F}}(\mathbf{k})$ is a function of the wavevector $\mathbf{k}$:

$$\tilde{\mathbf{F}}(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int d^3r \mathbf{F}(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}, \quad (2.4a)$$

$$\mathbf{F}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k \tilde{\mathbf{F}}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (2.4b)$$

Before we continue, let us list some important Fourier transforms, which will be used further below:

$$\frac{1}{4\pi r} = \frac{1}{(2\pi)^{3/2}} \int d^3k \frac{1}{(2\pi)^{3/2} k^2} e^{i\mathbf{k}\cdot\mathbf{r}} \quad \leftrightarrow \quad \frac{1}{(2\pi)^{3/2} k^2} \quad (2.5a)$$

$$\frac{\mathbf{r}}{4\pi r^3} = \frac{1}{(2\pi)^{3/2}} \int d^3k \frac{-i\mathbf{k}}{(2\pi)^{3/2} k^2} e^{i\mathbf{k}\cdot\mathbf{r}} \quad \leftrightarrow \quad \frac{-i\mathbf{k}}{(2\pi)^{3/2} k^2} \quad (2.5b)$$

$$\delta(\mathbf{r} - \mathbf{r}_\alpha) = \frac{1}{(2\pi)^{3/2}} \int d^3k \frac{e^{-i\mathbf{k}\cdot\mathbf{r}_\alpha}}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}} \quad \leftrightarrow \quad \frac{e^{-i\mathbf{k}\cdot\mathbf{r}_\alpha}}{(2\pi)^{3/2}}. \quad (2.5c)$$

Moreover, we will make use of the Parseval-Plancherel identity

$$\int d^3r \mathbf{F}^*(\mathbf{r}) G(\mathbf{r}) = \int d^3k \tilde{\mathbf{F}}^*(\mathbf{k}) \tilde{G}(\mathbf{k}), \quad (2.6)$$

and of the convolution theorem,

$$\frac{1}{(2\pi)^{3/2}} \int d^3r' \mathbf{F}(\mathbf{r}) G(\mathbf{r} - \mathbf{r}') \leftrightarrow \tilde{\mathbf{F}}(\mathbf{k}) \tilde{G}(\mathbf{k}), \quad (2.7)$$

where $\tilde{F}(\mathbf{k})$ and $\tilde{G}(\mathbf{k})$ are the Fourier transforms of $F(\mathbf{r})$ and $G(\mathbf{r})$, respectively.

Recall that Maxwell's equations (2.3) provide the divergence and curl of the electric and magnetic field. To this end, consider the divergence and curl of any vector field $\mathbf{F}(\mathbf{r})$, expressed in terms of its Fourier transform,

$$\nabla \cdot \mathbf{F}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k e^{i\mathbf{k}\cdot\mathbf{r}} i\mathbf{k} \cdot \tilde{\mathbf{F}}(\mathbf{k}) \quad (2.8a)$$

$$\nabla \times \mathbf{F}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k e^{i\mathbf{k}\cdot\mathbf{r}} i\mathbf{k} \times \tilde{\mathbf{F}}(\mathbf{k}). \quad (2.8b)$$

From this we see that taking the Fourier transform on both sides of Maxwell's equations (2.3) in real space, we obtain Maxwell's equations in reciprocal space to be

$$i\mathbf{k} \cdot \tilde{\mathbf{E}}(\mathbf{k}, t) = \frac{\tilde{\rho}(\mathbf{k}, t)}{\epsilon_0}, \quad (2.9a)$$

$$i\mathbf{k} \cdot \tilde{\mathbf{B}}(\mathbf{k}, t) = 0, \quad (2.9b)$$

$$i\mathbf{k} \times \tilde{\mathbf{E}}(\mathbf{k}, t) = -\frac{\partial \tilde{\mathbf{B}}(\mathbf{k}, t)}{\partial t}, \quad (2.9c)$$

$$i\mathbf{k} \times \tilde{\mathbf{B}}(\mathbf{k}, t) = \mu_0 \left(\tilde{\mathbf{j}}(\mathbf{k}, t) + \epsilon_0 \frac{\partial \tilde{\mathbf{E}}(\mathbf{k}, t)}{\partial t} \right). \quad (2.9d)$$

2.1.2 Transverse and longitudinal components

Given any vector field $\tilde{\mathbf{F}}(\mathbf{k})$ in reciprocal space, we can decompose it into a longitudinal component $\tilde{\mathbf{F}}_{\parallel}(\mathbf{k})$ and a transverse component $\tilde{\mathbf{F}}_{\perp}(\mathbf{k})$,

$$\tilde{\mathbf{F}}(\mathbf{k}) = \tilde{\mathbf{F}}_{\parallel}(\mathbf{k}) + \tilde{\mathbf{F}}_{\perp}(\mathbf{k}), \quad (2.10)$$

with the longitudinal component being parallel to the wavevector $\mathbf{k}$, reading

$$\tilde{\mathbf{F}}_{\parallel}(\mathbf{k}) = \left(\mathbf{k} \cdot \tilde{\mathbf{F}}(\mathbf{k}) \right) \frac{\mathbf{k}}{k^2}, \quad (2.11)$$

and the transverse component being orthogonal to $\mathbf{k}$, reading

$$\tilde{\mathbf{F}}_{\perp}(\mathbf{k}) = \tilde{\mathbf{F}}(\mathbf{k}) - \tilde{\mathbf{F}}_{\parallel}(\mathbf{k}) = \left(\mathbf{k} \times \tilde{\mathbf{F}}(\mathbf{k}) \right) \times \frac{\mathbf{k}}{k^2}. \quad (2.12)$$

As a direct consequence of Eqs. (2.11) and (2.12), we have

$$\mathbf{k} \cdot \tilde{\mathbf{F}}_{\perp}(\mathbf{k}) = 0 \quad (2.13a)$$

$$\mathbf{k} \times \tilde{\mathbf{F}}_{\parallel}(\mathbf{k}) = 0. \quad (2.13b)$$

In order to translate this property of the transverse and longitudinal field components to real space, let us transform the decomposition (2.10) to real space:

$$\mathbf{F}(\mathbf{r}) = \mathbf{F}_{\parallel}(\mathbf{r}) + \mathbf{F}_{\perp}(\mathbf{r}). \quad (2.14)$$

Here

$$\mathbf{F}_{\parallel}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k \tilde{\mathbf{F}}_{\parallel}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (2.15a)$$

$$\mathbf{F}_{\perp}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k \tilde{\mathbf{F}}_{\perp}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} \quad (2.15b)$$

are the inverse Fourier transforms of the longitudinal and transverse components of the vector field in reciprocal space. By taking the divergence and curl of any vector field $\mathbf{F}(\mathbf{r})$, we can use (2.13), such that (2.8) becomes

$$\nabla \cdot \mathbf{F}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k e^{i\mathbf{k} \cdot \mathbf{r}} i\mathbf{k} \cdot \tilde{\mathbf{F}}_{\parallel}(\mathbf{k}) = \nabla \cdot \mathbf{F}_{\parallel}(\mathbf{r}) \quad (2.16a)$$

$$\nabla \times \mathbf{F}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int d^3k e^{i\mathbf{k} \cdot \mathbf{r}} i\mathbf{k} \times \tilde{\mathbf{F}}_{\perp}(\mathbf{k}) = \nabla \times \mathbf{F}_{\perp}(\mathbf{r}). \quad (2.16b)$$

We thus conclude that

$$\nabla \cdot \mathbf{F}_{\perp}(\mathbf{r}) = 0, \quad (2.17a)$$

$$\nabla \times \mathbf{F}_{\parallel}(\mathbf{r}) = 0, \quad (2.17b)$$

which is the counterpart to (2.13) in real space.

Review Questions 2.1.

- Interpret Maxwell's equations.
- What is a wave vector? What does it describe?
- What are the properties of a transverse (longitudinal) vector field?
- Compare Eqs. (2.17) and (2.3) and conclude why we introduced the decomposition of a vector field in its transverse and longitudinal component.
- Prove the Parseval-Plancherel identity (2.6) with the help of the identity (2.5c)

2.1.3 Maxwell's equations in reciprocal space revisited

Using (2.13) in Maxwell's equations in reciprocal space (2.9), and multiplying (2.9a) and (2.9b) by $-\mathbf{i}\mathbf{k}/k^2$, Maxwell's equations read

$\tilde{\mathbf{E}}_{\parallel}(\mathbf{k}, t) = -\mathbf{i} \frac{\tilde{\rho}(\mathbf{k}, t)}{\epsilon_0} \frac{\mathbf{k}}{k^2},$	(2.18a)
$\tilde{\mathbf{B}}_{\parallel}(\mathbf{k}, t) = 0,$	(2.18b)
$\mathbf{i}\mathbf{k} \times \tilde{\mathbf{E}}_{\perp}(\mathbf{k}, t) = -\frac{\partial \tilde{\mathbf{B}}_{\perp}(\mathbf{k}, t)}{\partial t},$	(2.18c)
$\mathbf{i}\mathbf{k} \times \tilde{\mathbf{B}}_{\perp}(\mathbf{k}, t) = \mu_0 \left(\tilde{\mathbf{j}}_{\perp}(\mathbf{k}, t) + \epsilon_0 \frac{\partial \tilde{\mathbf{E}}_{\perp}(\mathbf{k}, t)}{\partial t} \right)$	(2.18d)

Note that by Eq. (2.18b), we directly see that the longitudinal magnetic field vanishes uniformly, i.e., the magnetic field is purely transverse,

$$\tilde{\mathbf{B}}(\mathbf{k}, t) = \tilde{\mathbf{B}}_{\perp}(\mathbf{k}, t) \quad \Leftrightarrow \quad \mathbf{B}(\mathbf{r}, t) = \mathbf{B}_{\perp}(\mathbf{r}, t). \quad (2.19)$$

Note that in the following we always write $\mathbf{B}(\mathbf{r}, t)$ and $\tilde{\mathbf{B}}(\mathbf{k}, t)$, keeping in mind that the magnetic field is purely transverse.

The longitudinal electric field, on the other hand, is in general, non-vanishing. Taking the inverse Fourier transform of (2.18a) by using Eq. (2.5b) and the convolution theorem (2.7), we get

$$\begin{aligned} \mathbf{E}_{\parallel}(\mathbf{r}, t) &= \frac{1}{(2\pi)^{3/2}} \int d^3k \, (2\pi)^{3/2} \frac{\tilde{\rho}(\mathbf{k}, t)}{\epsilon_0} \frac{-\mathbf{i}\mathbf{k}}{(2\pi)^{3/2}k^2} e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= \frac{1}{4\pi\epsilon_0} \int d^3r' \, \rho(\mathbf{r}', t) \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}. \end{aligned} \quad (2.20)$$

We thus see that the longitudinal electric field is fully governed by the charge density, i.e., by the positions of the charged particles. It also appears in the longitudinal part of Eq. (2.18d), which, using (2.18a), reads

$$\tilde{\mathbf{j}}_{\parallel}(\mathbf{k}, t) - \mathbf{i} \frac{\mathbf{k}}{k^2} \frac{\partial \tilde{\rho}(\mathbf{k}, t)}{\partial t} = \mathbf{0}. \quad (2.21)$$

Projecting this on $\mathbf{i}\mathbf{k}$ and using that $\mathbf{k} \cdot \tilde{\mathbf{j}}_{\parallel}(\mathbf{k}, t) = \mathbf{k} \cdot \tilde{\mathbf{j}}(\mathbf{k}, t)$, we obtain the continuity equation

$$\mathbf{i}\mathbf{k} \cdot \tilde{\mathbf{j}}(\mathbf{k}, t) + \frac{\partial \tilde{\rho}(\mathbf{k}, t)}{\partial t} = 0 \quad \stackrel{(2.8a)}{\Leftrightarrow} \quad \nabla \cdot \mathbf{j}(\mathbf{r}, t) + \frac{\partial \rho(\mathbf{r}, t)}{\partial t} = 0, \quad (2.22)$$

which expresses the conservation of charges, thus, contributing nothing new. We conclude that the longitudinal electric field $\mathbf{E}_{\parallel}(\mathbf{r}, t)$ does not constitute an independent dynamical degree of freedom. In consideration of the set of Maxwell equations in reciprocal space (2.18), we are thus left with only two coupled differential equations, the transverse parts of Eqs. (2.18c) and (2.18d),

$\mathbf{i}\mathbf{k} \times \tilde{\mathbf{E}}_{\perp}(\mathbf{k}, t) + \frac{\partial \tilde{\mathbf{B}}_{\perp}(\mathbf{k}, t)}{\partial t} = \mathbf{0},$	(2.23a)
$\mathbf{i}\mathbf{k} \times \tilde{\mathbf{B}}_{\perp}(\mathbf{k}, t) - \mu_0 \epsilon_0 \frac{\partial \tilde{\mathbf{E}}_{\perp}(\mathbf{k}, t)}{\partial t} = \mu_0 \tilde{\mathbf{j}}_{\perp}(\mathbf{k}, t).$	(2.23b)

In order to further simplify these differential equations, we continue with introducing the vector and scalar potential.

Review Questions 2.2.

- What does the continuity equation describe?
- What does it mean that the magnetic field is purely transverse?
- Use Eq. (2.2a) in (2.20) to show that $\mathbf{E}_{\parallel}(\mathbf{r}, t) = -\nabla \left(\frac{1}{4\pi\epsilon_0} \sum_{\alpha} \frac{q_{\alpha}}{|\mathbf{r} - \mathbf{r}_{\alpha}(t)|} \right)$. What does the term in parentheses correspond to?

2.1.4 Vector and scalar potential

Let us now come back for the moment to Maxwell's equations (2.3) in real space, and consider Eq. (2.3b), i.e., $\nabla \cdot \mathbf{B} = 0$: Since $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for any continuously twice-differentiable vector field $\mathbf{A}$, the magnetic field can be expressed as the curl of a vector potential $\mathbf{A}(\mathbf{r}, t)$, i.e.,

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (2.24)$$

Using this in Eq. (2.3c), i.e., $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$, we find $\nabla \times \mathbf{E} = -\nabla \times \partial \mathbf{A} / \partial t \Leftrightarrow \nabla \times (\mathbf{E} + \partial \mathbf{A} / \partial t) = 0$. And since $\nabla \times (\nabla \phi) = 0$ for any continuously twice-differentiable scalar field ϕ , we can write $\mathbf{E} + \partial \mathbf{A} / \partial t$ as the gradient of a scalar potential $-\phi(\mathbf{r}, t)$, such that the electric field reads

$$\mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} - \nabla \phi(\mathbf{r}, t). \quad (2.25)$$

We thus expressed the electric and magnetic field in terms of a vector potential $\mathbf{A}(\mathbf{r}, t)$ and scalar potential $\phi(\mathbf{r}, t)$. That is, instead of describing the electromagnetic field via the electric field $\mathbf{E}(\mathbf{r}, t)$ and magnetic field $\mathbf{B}(\mathbf{r}, t)$, we can equally well describe it via the vector potential $\mathbf{A}(\mathbf{r}, t)$ and scalar potential $\phi(\mathbf{r}, t)$. However, as we will see next, the choice of $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$ is not unique.

2.1.5 Gauge freedom

In order to find out how much freedom we have in choosing the potentials $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$, let us first note that by Eq. (2.17b), we have $\nabla \times \mathbf{A}_{\parallel}(\mathbf{r}, t) = 0$. The expression (2.24) of the magnetic field in terms of the vector potential thus becomes

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}_{\perp}(\mathbf{r}, t). \quad (2.26)$$

Next, note that since the Fourier transform of $\nabla \phi(\mathbf{r}, t)$ is $i\mathbf{k}\tilde{\phi}(\mathbf{k}, t)$, it only has a component parallel to $\mathbf{k}$, i.e., $\nabla \phi(\mathbf{r}, t) = (\nabla \phi(\mathbf{r}, t))_{\parallel}$. The components of Eq. (2.25) thus read

$$\mathbf{E}_{\perp}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}_{\perp}(\mathbf{r}, t)}{\partial t}, \quad (2.27)$$

and

$$\mathbf{E}_{\parallel}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}_{\parallel}(\mathbf{r}, t)}{\partial t} - \nabla \phi(\mathbf{r}, t). \quad (2.28)$$

While Eqs. (2.26) and (2.27) fix the transverse component $\mathbf{A}_{\perp}(\mathbf{r}, t)$ of the vector potential, Eq. (2.28) allows for the so-called gauge freedom in the definition of $\mathbf{A}_{\parallel}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$. That is, we can add an arbitrary longitudinal component $\nabla \chi(\mathbf{r}, t) = (\nabla \chi(\mathbf{r}, t))_{\parallel}$ to the vector potential, and compensate for that by adding $-\partial \chi(\mathbf{r}, t) / \partial t$ to the scalar potential. Accordingly, for any

sufficiently differentiable function $\chi(\mathbf{r}, t)$, the electric and magnetic fields are invariant under the transformation

$$\mathbf{A}(\mathbf{r}, t) \rightarrow \mathbf{A}(\mathbf{r}, t) + \nabla\chi(\mathbf{r}, t) \quad (2.29a)$$

$$\phi(\mathbf{r}, t) \rightarrow \phi(\mathbf{r}, t) - \frac{\partial\chi(\mathbf{r}, t)}{\partial t}. \quad (2.29b)$$

Note that in Eq. (2.29a) only the longitudinal component $\mathbf{A}_{\parallel}(\mathbf{r}, t)$ is modified. The transverse component $\mathbf{A}_{\perp}(\mathbf{r}, t)$ is gauge invariant. That is, gauge freedom implies that we can freely choose $\mathbf{A}_{\parallel}(\mathbf{r}, t)$ with the premise that we have to modify the scalar potential accordingly.

A common choice of gauge is the so-called Coulomb gauge (sometimes also called transverse gauge or radiation gauge),

$$\mathbf{A}_{\parallel}(\mathbf{r}, t) = \mathbf{0}, \quad \stackrel{(2.17a)}{\Rightarrow} \nabla \cdot \mathbf{A}(\mathbf{r}, t) = 0. \quad (2.30)$$

The name comes from the fact that, in the Coulomb gauge, the scalar potential becomes the Coulomb potential. To see this, consider Eq. (2.28), which becomes $\mathbf{E}_{\parallel}(\mathbf{r}, t) = -\nabla\phi(\mathbf{r}, t)$ in the Coulomb gauge. Using the longitudinal component of the electric field from Eq. (2.20), we get

$$\nabla\phi(\mathbf{r}, t) = -\frac{1}{4\pi\epsilon_0} \int d^3r' \rho(\mathbf{r}', t) \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} \quad (2.31)$$

and, thus,

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \rho(\mathbf{r}', t) \frac{1}{|\mathbf{r} - \mathbf{r}'|}, \quad (2.32)$$

which is just the Coulomb potential. Inserting the charge density $\rho(\mathbf{r}, t) = \sum_{\alpha} q_{\alpha} \delta(\mathbf{r} - \mathbf{r}_{\alpha}(t))$ from Eq. (2.2a), this turns into the Coulomb potential of charged particles,

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \sum_{\alpha} \frac{q_{\alpha}}{|\mathbf{r} - \mathbf{r}_{\alpha}(t)|}. \quad (2.33)$$

Note that for the time being we don't fix the gauge since we work with gauge-invariant quantities such as the transverse vector potential $\mathbf{A}_{\perp}(\mathbf{r}, t)$. **Further note that to ease notation in the following we stop stating the arguments $\mathbf{r}$, k , and t if there is no risk of confusion.**

Review Questions 2.3.

- What do the scalar and vector potential describe?
- What is the gauge freedom?
- Insert (2.24) and (2.25) into Maxwell's equations (2.3a) and (2.3d), and use $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \Delta \mathbf{A}$ and $c^2 = 1/\epsilon_0\mu_0$ to show that they can be written as

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta \mathbf{A} = \frac{\mathbf{j}}{\epsilon_0 c^2} - \nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) \quad (2.34a)$$

$$\Delta \phi = -\frac{\rho}{\epsilon_0} - \nabla \cdot \frac{\partial \mathbf{A}}{\partial t}. \quad (2.34b)$$

- Consider the so-called Lorentz gauge defined by

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0, \quad (2.35)$$

and show that the above equations become

$$\square \mathbf{A} = \frac{\mathbf{j}}{\epsilon_0 c^2} \quad (2.36a)$$

$$\square \phi = \frac{\rho}{\epsilon_0}, \quad (2.36b)$$

with $\square = \partial^2/\partial t^2 - \Delta$ the d'Alembert operator. These equations can be written more compactly using covariant notation with $\partial_\mu = (\frac{1}{c} \frac{\partial}{\partial t}, \nabla)$, $A^\mu = (\frac{\phi}{c}, \mathbf{A})$, and $j^\mu = (c\rho, \mathbf{j})$. The Lorentz gauge (2.35) then reads $\partial_\mu A^\mu = 0$, and Maxwell's equations (2.36a) become $\partial_\nu \partial^\nu A^\mu = j^\mu/\epsilon_0 c^2$.

- Discuss why it is always possible to find a gauge transformation such that (2.35) is satisfied.

2.1.6 Maxwell's equations in reciprocal space once more revisited

Recall that we found that Maxwell's equations in reciprocal space (2.18) reduce to the continuity equation (2.22) (expressing the conservation of charges; nothing new) and to the two coupled differential equations from Eq. (2.23). Let us now inspect these two differential equations by expressing the electric and magnetic fields in terms of the vector potential and scalar potential. In particular, using $\mathbf{B} = \nabla \times \mathbf{A}_\perp$ from (2.26), we have, by Eq. (2.8b), $\tilde{\mathbf{B}} = i\mathbf{k} \times \tilde{\mathbf{A}}_\perp$, and by Eq. (2.27), we have $\tilde{\mathbf{E}}_\perp = -\partial \tilde{\mathbf{A}}_\perp / \partial t$.

We start with Eq. (2.23a): Plugging the above expressions of $\tilde{\mathbf{B}}$ and $\tilde{\mathbf{E}}_\perp$ in Eq. (2.23a) gives

$$\mathbf{0} = i\mathbf{k} \times \tilde{\mathbf{E}}_\perp + \frac{\partial \tilde{\mathbf{B}}}{\partial t} = -i\mathbf{k} \times \frac{\partial \tilde{\mathbf{A}}_\perp}{\partial t} + i\mathbf{k} \times \frac{\partial \tilde{\mathbf{A}}_\perp}{\partial t} = \mathbf{0}. \quad (2.37)$$

We thus see that Eq. (2.23a) is trivially fulfilled. Accordingly, we can simply disregard this differential equation.

Next, consider Eq. (2.23b), for which we get

$$\mu_0 \tilde{\mathbf{j}}_\perp = i\mathbf{k} \times \tilde{\mathbf{B}} - \mu_0 \epsilon_0 \frac{\partial \tilde{\mathbf{E}}_\perp}{\partial t} = -\mathbf{k} \times (\mathbf{k} \times \tilde{\mathbf{A}}_\perp) + \mu_0 \epsilon_0 \frac{\partial^2 \tilde{\mathbf{A}}_\perp}{\partial t^2}. \quad (2.38)$$

Using $\mathbf{k} \times (\mathbf{k} \times \tilde{\mathbf{A}}_\perp) = (\mathbf{k} \cdot \tilde{\mathbf{A}}_\perp)\mathbf{k} - (\mathbf{k} \cdot \mathbf{k})\tilde{\mathbf{A}}_\perp = -k^2\tilde{\mathbf{A}}_\perp$, and introducing the speed of light $c = 1/\sqrt{\mu_0\epsilon_0}$, this results in

$$\frac{1}{c^2} \frac{\partial^2 \tilde{\mathbf{A}}_\perp}{\partial t^2} + k^2 \tilde{\mathbf{A}}_\perp = \mu_0 \tilde{\mathbf{j}}_\perp. \quad (2.39)$$

Transforming back to real space, and using that the inverse Fourier transform of $k^2 \tilde{\mathbf{A}}_\perp$ is $-\Delta \mathbf{A}_\perp$, where $\Delta = \nabla^2$, we obtain the d'Alembert wave equation with source term $\mu_0 \mathbf{j}_\perp$,

$$\boxed{\frac{1}{c^2} \frac{\partial^2 \mathbf{A}_\perp}{\partial t^2} - \Delta \mathbf{A}_\perp = \mu_0 \mathbf{j}_\perp.} \quad (2.40)$$

Let us stress that we essentially reduced the non-trivial dynamics expressed through Maxwell's equations to be governed by a single differential equation, the inhomogenous d'Alembert wave equation of the transverse vector field potential. Before we continue with solving this equation, let us discuss some important quantities of the system composed of particles+fields.

2.2 Energies and momenta

2.2.1 Conserved quantities

The equations of motion of the particles+fields (i.e., the Newton-Lorentz equations (2.1) and Maxwell's equations) can be shown to conserve the following quantities:

- the total energy of the system, given by the sum of the particles' kinetic energy and the integral over the field's energy density,

$$H = \sum_\alpha \frac{1}{2} m_\alpha \mathbf{v}_\alpha^2 + \int \left(\frac{\epsilon_0 \mathbf{E}^2}{2} + \frac{\mathbf{B}^2}{2\mu_0} \right) d^3r, \quad (2.41)$$

due to the invariance of the equations of motion with respect to time translations,

- the total momentum (the sum of the particles' momenta and the spatial integral over the Poynting vector)

$$\mathbf{P} = \sum_\alpha m_\alpha \mathbf{v}_\alpha + \epsilon_0 \int (\mathbf{E} \times \mathbf{B}) d^3r, \quad (2.42)$$

due to the invariance with respect to spatial translations, and

- the total angular momentum

$$\mathbf{J} = \sum_\alpha m_\alpha \mathbf{r}_\alpha \times \mathbf{v}_\alpha + \epsilon_0 \int \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) d^3r, \quad (2.43)$$

due to the invariance with respect to rotations.

Let us now distinguish between the longitudinal and transverse field components in the expression of the total energy from Eq. (2.41), and write

$$H = \sum_\alpha \frac{1}{2} m_\alpha \mathbf{v}_\alpha^2 + H_\parallel + H_\perp, \quad (2.44)$$

where we define

$$H_\perp = \int \left(\frac{\epsilon_0 \mathbf{E}_\perp^2}{2} + \frac{\mathbf{B}^2}{2\mu_0} \right) d^3r, \quad \text{and} \quad H_\parallel = \frac{\epsilon_0}{2} \int \mathbf{E}_\parallel^2 d^3r. \quad (2.45)$$

Similarly, we can decompose the total momentum as $\mathbf{P} = \sum_{\alpha} m_{\alpha} \mathbf{v}_{\alpha} + \mathbf{P}_{\parallel} + \mathbf{P}_{\perp}$, where we define

$$\mathbf{P}_{\perp} = \epsilon_0 \int (\mathbf{E}_{\perp} \times \mathbf{B}) d^3r \quad \text{and} \quad \mathbf{P}_{\parallel} = \epsilon_0 \int (\mathbf{E}_{\parallel} \times \mathbf{B}) d^3r \quad (2.46)$$

as the contribution to the total momentum arising from the transverse and longitudinal component of the *electric field*, respectively. Note that $\mathbf{P}_{\perp}$ ($\mathbf{P}_{\parallel}$) is not the transverse (longitudinal) component of a field: the notation indicates that it is the momentum associated with the transverse (longitudinal) part of the electric field, $\mathbf{E}_{\perp}$ ($\mathbf{E}_{\parallel}$), as defined in (2.46).

2.2.2 Canonical momentum

The component $\mathbf{P}_{\parallel}$ from Eq. (2.46) can be rewritten as follows:

$$\begin{aligned} \mathbf{P}_{\parallel} &= \epsilon_0 \int d^3r (\mathbf{E}_{\parallel} \times \mathbf{B}) \\ &\stackrel{(2.6)}{=} \epsilon_0 \int d^3k (\tilde{\mathbf{E}}_{\parallel}^* \times \tilde{\mathbf{B}}) \\ &\stackrel{(2.18a),(2.24)}{=} \epsilon_0 \int d^3k \left(i \frac{\tilde{\rho}^*}{\epsilon_0} \frac{\mathbf{k}}{k^2} \times [\mathbf{i} \mathbf{k} \times \tilde{\mathbf{A}}_{\perp}] \right) \\ &= - \int d^3k \tilde{\rho}^* \left(\frac{\mathbf{k}}{k} \times \left[\frac{\mathbf{k}}{k} \times \tilde{\mathbf{A}}_{\perp} \right] \right), \end{aligned} \quad (2.47)$$

where we used $\tilde{\mathbf{E}}_{\parallel}^* = i \frac{\tilde{\rho}^*}{\epsilon_0} \frac{\mathbf{k}}{k^2}$ [see Eq. (2.18a)], and $\tilde{\mathbf{B}} = \mathbf{i} \mathbf{k} \times \tilde{\mathbf{A}}_{\perp}$ [see Eqs. (2.24) and (2.8b)]. Using that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$, as well as $\mathbf{k} \cdot \tilde{\mathbf{A}}_{\perp} = 0$, we have

$$\frac{\mathbf{k}}{k} \times \left[\frac{\mathbf{k}}{k} \times \tilde{\mathbf{A}}_{\perp} \right] = \underbrace{\left(\frac{\mathbf{k}}{k} \cdot \tilde{\mathbf{A}}_{\perp} \right)}_{=0} \frac{\mathbf{k}}{k} - \underbrace{\left(\frac{\mathbf{k}}{k} \cdot \frac{\mathbf{k}}{k} \right)}_{=1} \tilde{\mathbf{A}}_{\perp} = -\tilde{\mathbf{A}}_{\perp}, \quad (2.48)$$

and, thus,

$$\begin{aligned} \mathbf{P}_{\parallel} &= \int d^3k \tilde{\rho}^* \tilde{\mathbf{A}}_{\perp} \\ &\stackrel{(2.6)}{=} \int d^3r \rho \mathbf{A}_{\perp}. \end{aligned} \quad (2.49)$$

Finally, using $\rho = \sum_{\alpha} q_{\alpha} \delta(\mathbf{r} - \mathbf{r}_{\alpha}(t))$, we arrive at

$$\mathbf{P}_{\parallel} = \sum_{\alpha} q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha}). \quad (2.50)$$

This suggests that we can combine $\mathbf{P}_{\parallel}$ with the particles' momenta $\sum_{\alpha} m_{\alpha} \mathbf{v}_{\alpha}$, and associate the canonical momentum

$$\mathbf{p}_{\alpha} = m_{\alpha} \mathbf{v}_{\alpha} + q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha}) \quad (2.51)$$

with each particle. Note that the canonical momentum is gauge invariant, since $\mathbf{A}_{\perp}$ doesn't change under gauge transformations. It is composed of two contributions: the kinetic momentum $m_{\alpha} \mathbf{v}_{\alpha}$ and the momentum $q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha})$ associated with the longitudinal field of the α -th particle. The total momentum then becomes $\mathbf{P} = \sum_{\alpha} \mathbf{p}_{\alpha} + \mathbf{P}_{\perp}$, and the total energy (2.44) can be written in terms of the canonical momenta as

$$\boxed{H = \sum_{\alpha} \frac{1}{2m_{\alpha}} (\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha}))^2 + H_{\parallel} + H_{\perp}.} \quad (2.52)$$

Note that the naming *canonical momentum* comes from the fact that $\mathbf{p}_\alpha$ is the canonically conjugate momentum of the position $\mathbf{r}_\alpha$. We will explicitly show this in Sec. 2.5.4 below. That is, the Hamilton equations for $\mathbf{r}_\alpha$ and $\mathbf{p}_\alpha$ with respect to the corresponding Hamiltonian, or, equivalently, Lagrangian's equations with respect to the corresponding Lagrangian, result in the equations of motion of charged particles in an electromagnetic field, i.e., in the Newton-Lorentz equations (2.1). We will show this via Lagrange's equations in Sec. 2.5.2 below.

2.2.3 Total energy of the longitudinal field

Let us briefly consider the longitudinal component $H_\parallel$ of the total energy from Eq. (2.45). To this end, recall the longitudinal component of the electric field in reciprocal space from Eq. (2.18a), reading $\tilde{\mathbf{E}}_\parallel = -i\tilde{\rho}\mathbf{k}/(\epsilon_0 k^2)$. Using this, $H_\parallel$ from Eq. (2.45) becomes

$$\begin{aligned} H_\parallel &= \frac{\epsilon_0}{2} \int d^3r \mathbf{E}_\parallel^2 \\ &\stackrel{(2.6)}{=} \frac{\epsilon_0}{2} \int d^3k |\tilde{\mathbf{E}}_\parallel|^2 \\ &= \frac{1}{2\epsilon_0} \int d^3k \tilde{\rho}^* \frac{\tilde{\rho}}{k^2}. \end{aligned} \quad (2.53)$$

Defining $\tilde{G} = \tilde{\rho}/k^2$ with inverse Fourier transform $G = \int d^3r' \rho(\mathbf{r}') \frac{1}{4\pi|\mathbf{r}-\mathbf{r}'|}$ [obtained from the convolution theorem (2.7) and $\frac{1}{(2\pi)^{3/2}k^2} \leftrightarrow \frac{1}{4\pi r}$ from Eq. (2.5a)], we get

$$\begin{aligned} H_\parallel &\stackrel{(2.6)}{=} \frac{1}{2\epsilon_0} \int d^3r \rho(\mathbf{r}) G(\mathbf{r}) \\ &= \frac{1}{8\pi\epsilon_0} \int \frac{\rho(\mathbf{r}, t)\rho(\mathbf{r}', t)}{|\mathbf{r}-\mathbf{r}'|} d^3r d^3r', \end{aligned} \quad (2.54)$$

This shows that $H_\parallel$ only depends on the charge density, taking the form of a Coulomb energy (i.e., the integrated Coulomb potential (2.32) seen by all elements of the charge density). This is why $H_\parallel$ is sometimes called the Coulomb electrostatic energy of the system of charges.

For the charge density $\rho = \sum_\alpha q_\alpha \delta(\mathbf{r} - \mathbf{r}_\alpha)$ of point particles, with Fourier transform (cf. Eq. (2.5c))

$$\tilde{\rho}(\mathbf{k}, t) = \frac{1}{(2\pi)^{3/2}} \sum_\alpha q_\alpha e^{-i\mathbf{k}\cdot\mathbf{r}_\alpha(t)}, \quad (2.55)$$

the last line in Eq. (2.53) becomes

$$\begin{aligned} H_\parallel &= \underbrace{\sum_\alpha \frac{q_\alpha^2}{2\epsilon_0(2\pi)^3} \int \frac{d^3k}{k^2}}_{\epsilon_{\text{Coul}}} + \sum_{\alpha \neq \beta} \frac{q_\alpha q_\beta}{2\epsilon_0} \underbrace{\frac{1}{(2\pi)^{3/2}} \int d^3k \frac{e^{i\mathbf{k}\cdot(\mathbf{r}_\alpha - \mathbf{r}_\beta)}}{(2\pi)^{3/2}k^2}}_{1/4\pi|\mathbf{r}_\alpha - \mathbf{r}_\beta|} \\ &= \epsilon_{\text{Coul}} + \frac{1}{8\pi\epsilon_0} \sum_{\alpha \neq \beta} \frac{q_\alpha q_\beta}{|\mathbf{r}_\alpha - \mathbf{r}_\beta|}. \end{aligned} \quad (2.56)$$

Here, the second term describes the Coulomb interaction between pairs of particles. On the other hand, the first term is a self-interaction term, with ϵ_{Coul} called the Coulomb self energy. We therefore define

$$\boxed{V_{\text{Coul}} := \epsilon_{\text{Coul}} + \frac{1}{8\pi\epsilon_0} \sum_{\alpha \neq \beta} \frac{q_\alpha q_\beta}{|\mathbf{r}_\alpha - \mathbf{r}_\beta|} = H_\parallel.} \quad (2.57)$$

Note that, since the integral $\int d^3k/k^2$ diverges,¹ strictly speaking the Coulomb self energy ϵ_{Coul} is infinite. Therefore, one introduces a cutoff at large k , making ϵ_{Coul} constant. Often the Coulomb self energy is simply discarded since it merely contributes a constant term. Further note that (2.56) holds independently of the choice of gauge.

Review Questions 2.4.

- What is the difference between the kinetic momentum and canonical momentum of a particle?
- What is the total energy coming from the longitudinal field components.
- Interpret the Coulomb electrostatic energy.
- Why can we readily consider a cutoff at large k for the Coulomb self energy?

2.3 Normal mode expansion

2.3.1 Confined electromagnetic field

Recall that in Sec. 2.1 we started from Maxwell's equations and showed that the non-trivial dynamics of the electromagnetic field is described by a single differential equation for the transverse vector field potential, the inhomogeneous d'Alembert wave equation from Eq. (2.40). Now our aim is to solve this equation and obtain a simple description of the electromagnetic field in terms of field modes. To this end, we suppose that **the electromagnetic field is confined in space, such as in a cavity or in a cube**. While this captures real physical scenarios (the electromagnetic field in a cavity), we will lift this assumption further below through the consideration of an infinitely large cube, i.e., the free space limit.

In order to solve the inhomogeneous d'Alembert wave equation (2.40) for the transverse vector field potential, let us consider the homogeneous part by setting $\mathbf{j}_\perp = \mathbf{0}$,

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{A}_\perp}{\partial t^2} - \Delta \mathbf{A}_\perp = \mathbf{0}. \quad (2.58)$$

A basis of spatial functions (decomposing the transverse field) is obtained by separating the real vector field $\mathbf{A}_\perp$ into a real mode function $\mathbf{u}(\mathbf{r})$ and a real time dependence $x(t)$ via the ansatz

$$\mathbf{A}_\perp(\mathbf{r}, t) = \sqrt{\frac{m}{\epsilon_0}} \mathbf{u}(\mathbf{r}) x(t), \quad (2.59)$$

with m a normalization constant. Plugging the ansatz (2.59) into (2.40), we get

$$\frac{\Delta \mathbf{u}(\mathbf{r})}{\mathbf{u}(\mathbf{r})} = \frac{1}{c^2 x(t)} \frac{\partial^2 x(t)}{\partial t^2}. \quad (2.60)$$

We can thus separate the variables by setting both sides equal to a constant, for which we choose $-\omega^2/c^2$. This leads to the Helmholtz equation

$$\Delta \mathbf{u}(\mathbf{r}) + \frac{\omega^2}{c^2} \mathbf{u}(\mathbf{r}) = \mathbf{0} \quad (2.61)$$

for the mode function, and to the equation of motion of the simple harmonic oscillator

$$\frac{d^2}{dt^2} x(t) + \omega^2 x(t) = 0 \quad (2.62)$$

for the time dependence $x(t)$.

¹In particular, $\int d^3k \frac{1}{k^2} = \int_0^\infty dk \int_0^\pi d\theta \int_0^{2\pi} d\varphi k^2 \sin \theta \frac{1}{k^2} = 4\pi \int_0^\infty dk$, which diverges unless there is a cutoff at large energies for which it becomes constant.

Mode functions Since we consider the field to be confined in a given region of space with appropriate boundary conditions, the Helmholtz equation (2.61) is solved by an complete, infinitely large, countable set of real orthonormal eigenfunctions $\mathbf{u}_\kappa(\mathbf{r})$, which we index by κ . Note that orthonormality is expressed by

$$\int d^3r \mathbf{u}_\kappa^*(\mathbf{r}) \cdot \mathbf{u}_{\kappa'}(\mathbf{r}) = \delta_{\kappa, \kappa'}, \quad (2.63)$$

and completeness implies that any transverse vector field $\mathbf{F}_\perp(\mathbf{r})$ subject to the boundary conditions can be expanded in this basis,

$$\mathbf{F}_\perp(\mathbf{r}) = \sum_\kappa \left(\int d^3r' \mathbf{u}_\kappa^*(\mathbf{r}') \cdot \mathbf{F}_\perp(\mathbf{r}') \right) \mathbf{u}_\kappa(\mathbf{r}). \quad (2.64)$$

Time dependence Let us also index the eigenvalue [of the Laplacian in (2.61)] corresponding to the solution $\mathbf{u}_\kappa(\mathbf{r})$ by the index κ , and write it as $-\omega_\kappa^2/c^2$. The corresponding time dependence $x_\kappa(t)$ (also labeled by κ) must then obey the equation of motion of the simple harmonic oscillator (2.62) with frequency ω_κ ,

$$\frac{d^2}{dt^2} x_\kappa(t) + \omega_\kappa^2 x_\kappa(t) = 0. \quad (2.65)$$

By defining the complex function (sometimes called normal variable)

$$\alpha_\kappa(t) = \sqrt{\frac{m\omega_\kappa}{2\hbar}} \left[x_\kappa(t) + \frac{i}{\omega_\kappa m} p_\kappa(t) \right], \quad (2.66a)$$

$$\alpha_\kappa^*(t) = \sqrt{\frac{m\omega_\kappa}{2\hbar}} \left[x_\kappa(t) - \frac{i}{\omega_\kappa m} p_\kappa(t) \right], \quad (2.66b)$$

where $p_\kappa(t) = m \frac{dx_\kappa(t)}{dt}$, and $\hbar$ is introduced as a constant with dimension of action, ensuring that α_κ is dimensionless, we have

$$x_\kappa(t) = \sqrt{\frac{\hbar}{2m\omega_\kappa}} [\alpha_\kappa(t) + \alpha_\kappa^*(t)], \quad (2.67a)$$

$$p_\kappa(t) = \frac{1}{i} \sqrt{\frac{\hbar m \omega_\kappa}{2}} [\alpha_\kappa(t) - \alpha_\kappa^*(t)]. \quad (2.67b)$$

Using this in Eq. (2.65), we get

$$\frac{d^2}{dt^2} x_\kappa(t) = \frac{1}{m} \frac{d}{dt} p_\kappa(t) = \frac{1}{i} \sqrt{\frac{\hbar \omega_\kappa}{2m}} \left[\frac{d}{dt} \alpha_\kappa(t) - \frac{d}{dt} \alpha_\kappa^*(t) \right], \quad (2.68a)$$

$$\omega_\kappa^2 x_\kappa(t) = \omega_\kappa \sqrt{\frac{\hbar \omega_\kappa}{2m}} [\alpha_\kappa(t) + \alpha_\kappa^*(t)], \quad (2.68b)$$

which turns (2.65) into the first-order differential equation

$$\frac{d}{dt} \alpha_\kappa(t) + i\omega_\kappa \alpha_\kappa(t) = 0. \quad (2.69)$$

Solving (2.69) yields a solution for $\alpha_\kappa(t)$ for the free electromagnetic field (for which $\mathbf{j}_\perp = \mathbf{0}$),

$$\alpha_\kappa(t) = \alpha_\kappa e^{-i\omega_\kappa t}, \quad (2.70)$$

with $\alpha_\kappa = \alpha_\kappa(0)$ a complex constant.

Transverse electromagnetic field We can now combine the set of solutions [obtained from the ansatz (2.59)], given by the mode functions $\mathbf{u}_\kappa(\mathbf{r})$ together with their corresponding time dependencies $x_\kappa(t)$, yielding the transverse vector potential

$$\mathbf{A}_\perp(\mathbf{r}, t) = \sum_\kappa \sqrt{\frac{m}{\epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) x_\kappa(t), \quad (2.71)$$

$$\stackrel{(2.67a)}{=} \sum_\kappa \sqrt{\frac{\hbar}{2\omega_\kappa \epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) [\alpha_\kappa(t) + \alpha_\kappa^*(t)]. \quad (2.72)$$

Further below, we will also make use of the momentum conjugate of $\mathbf{A}_\perp$, reading [see Eq. (2.143)]

$$\mathbf{\Pi} = \epsilon_0 \dot{\mathbf{A}}_\perp. \quad (2.73)$$

With the help of Eq. (2.71), it becomes, using $p_\kappa(t) = m \frac{dx_\kappa(t)}{dt}$,

$$\mathbf{\Pi}(\mathbf{r}, t) = \sum_\kappa \sqrt{\frac{\epsilon_0}{m}} \mathbf{u}_\kappa(\mathbf{r}) p_\kappa(t) \quad (2.74)$$

$$\stackrel{(2.67b)}{=} - \sum_\kappa i \sqrt{\frac{\hbar \omega_\kappa \epsilon_0}{2}} \mathbf{u}_\kappa(\mathbf{r}) [\alpha_\kappa(t) - \alpha_\kappa^*(t)]. \quad (2.75)$$

Moreover, with the help of Eqs. (2.26) and (2.27), we can now express the magnetic and transverse electric field in terms of $\mathbf{u}_\kappa(\mathbf{r})$ and $\alpha_\kappa(t)$,

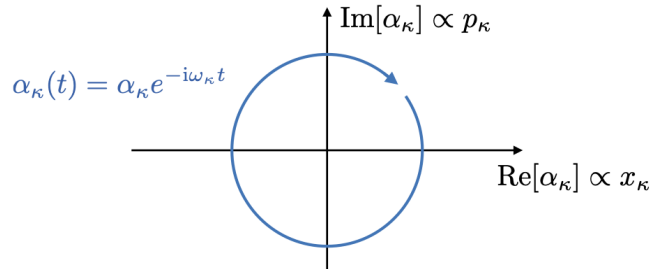
$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}_\perp(\mathbf{r}, t) = \sum_\kappa \sqrt{\frac{\hbar}{2\omega_\kappa \epsilon_0}} \nabla \times \mathbf{u}_\kappa(\mathbf{r}) [\alpha_\kappa(t) + \alpha_\kappa^*(t)], \quad (2.76)$$

$$\mathbf{E}_\perp(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathbf{A}_\perp(\mathbf{r}, t) = \sum_\kappa i \sqrt{\frac{\hbar \omega_\kappa}{2\epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) [\alpha_\kappa(t) - \alpha_\kappa^*(t)]. \quad (2.77)$$

Therewith, we can further calculate the energy $H_\perp = \int (\epsilon_0 \mathbf{E}_\perp^2 / 2 + \mathbf{B}^2 / 2\mu_0) d^3r$ of the transverse electromagnetic field from Eq. (2.45) in the cavity,² yielding

$$H_\perp = \sum_\kappa \frac{\hbar \omega_\kappa}{2} [\alpha_\kappa(t)^* \alpha_\kappa(t) + \alpha_\kappa(t) \alpha_\kappa(t)^*] \stackrel{(2.66)}{=} \sum_\kappa \left[\frac{p_\kappa^2(t)}{2m} + \frac{1}{2} m \omega_\kappa^2 x_\kappa^2(t) \right], \quad (2.78)$$

where $p_\kappa(t) = m \frac{d}{dt} x_\kappa(t)$. This shows, that each normal mode κ of the cavity can be associated with a classical harmonic oscillator whose state is described by the complex variable $\alpha_\kappa(t)$, defining a point in phase space with coordinates $\text{Re}[\alpha_\kappa(t)] \propto x_\kappa(t)$ and $\text{Im}[\alpha_\kappa(t)] \propto p_\kappa(t)$:



²For the calculation, we use the orthogonality of the mode functions and of their curls,

$$\int d^3r \mathbf{u}_\kappa^*(\mathbf{r}) \cdot \mathbf{u}_{\kappa'}(\mathbf{r}) = \delta_{\kappa, \kappa'},$$

$$\int d^3r (\nabla \times \mathbf{u}_\kappa(\mathbf{r})) \cdot (\nabla \times \mathbf{u}_{\kappa'}(\mathbf{r})) = \frac{\omega_\kappa^2}{c^2} \delta_{\kappa, \kappa'},$$

Including the inhomogeneity Note that so far we considered the free field, since we set $\mathbf{j}_\perp = \mathbf{0}$ in Eq. (2.58). The above expressions for the vector potential $\mathbf{A}_\perp$ and its momentum conjugate $\mathbf{\Pi}$ [see Eqs. (2.71) and (2.74)] are, however, also valid for a non-vanishing transverse current $\mathbf{j}_\perp$. To see this, note that we can expand the transverse current in the set of mode functions $\mathbf{u}_\kappa(\mathbf{r})$ as

$$\mathbf{j}_\perp(\mathbf{r}, t) = \epsilon_0 \sum_\kappa \sqrt{\frac{m}{\epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) b_\kappa(t), \quad (2.79)$$

with $b_\kappa(t)$ real. Using this together with the expansion $\mathbf{A}_\perp(\mathbf{r}, t) = \sum_\kappa \sqrt{\frac{m}{\epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) x_\kappa(t)$, the inhomogeneous d'Alembert wave equation (2.40) leads to the driven harmonic oscillator for the time dependence,

$$\frac{d^2}{dt^2} x_\kappa(t) + \omega_\kappa^2 x_\kappa(t) = b_\kappa(t). \quad (2.80)$$

By setting $b_\kappa(t) = 2\omega_\kappa \sqrt{\frac{\hbar\omega_\kappa}{2m}} \beta_\kappa(t)$, this results in

$$\frac{d}{dt} \alpha_\kappa(t) + i\omega_\kappa \alpha_\kappa(t) = i\omega_\kappa \beta_\kappa(t). \quad (2.81)$$

In this case, $\alpha_\kappa(t)$ describes a driven harmonic oscillator, whose dynamics may deviate from the undriven solution (2.70) of the free electromagnetic field.

Review Questions 2.5.

- *What is a normal mode? What is the origin of the name?*
- *What is $x(t)$? Reflect on the fact that $x(t)$ has nothing to do with “space”.*
- *Show the second equality of (2.78). If there is enough time, show also the first equality without commuting $\alpha_\kappa(t)$ and $\alpha_\kappa^*(t)$.*

2.3.2 The electromagnetic field in free space

In order to lift our assumption of a confined electromagnetic field and consider radiation in free space, we now consider a cubic cavity with side lengths L and periodic boundary conditions, solve the Helmholtz equation (2.61), and, thereafter, take the limit $L \rightarrow \infty$. Unlike earlier, we now consider complex mode functions $\boldsymbol{\mu}(\mathbf{r})$ and complex time dependencies $\xi(t)$ in the ansatz [c.f. Eq. (2.59)]

$$\mathbf{A}_\perp(\mathbf{r}, t) = \sqrt{\frac{m}{\epsilon_0}} \boldsymbol{\mu}(\mathbf{r}) \xi(t). \quad (2.82)$$

However, below we will assure that $\mathbf{A}_\perp$ is real.

For the cubic cavity with periodic boundary conditions, the Helmholtz equation (2.61) is solved by the complex mode functions

$$\boldsymbol{\mu}_{\mathbf{k},p}(\mathbf{r}) = \frac{1}{L^{3/2}} \exp(i\mathbf{k} \cdot \mathbf{r}) \mathbf{e}_{\mathbf{k},p}, \quad (2.83)$$

with $\omega_{\mathbf{k},p}^2/c^2 = k^2$. The mode functions are now indexed by the tuple $(\mathbf{k}, p)$, where

$$\mathbf{k} = \frac{2\pi}{L} (n_x \mathbf{e}_x + n_y \mathbf{e}_y + n_z \mathbf{e}_z), \quad n_x, n_y, n_z \in \mathbb{Z}, \quad (2.84)$$

is the discretised wave vector, and $p \in \{1, 2\}$ is a polarization index, referring to two orthonormal vectors $\mathbf{e}_{\mathbf{k},1}$ and $\mathbf{e}_{\mathbf{k},2}$ in the plane perpendicular to $\mathbf{k}$. These polarization vectors $\mathbf{e}_{\mathbf{k},p}$ are complex, allowing the description of elliptical polarization. Note that since $\omega_{\mathbf{k},p}^2/c^2 = k^2$, the frequency only depends on k , and we therefore write $\omega_{\mathbf{k},p} = \omega_{\mathbf{k}} = \omega_k$.

Using the mode functions (2.83), the ansatz (2.82) leads to the general solution

$$\mathbf{A}_\perp(\mathbf{r}, t) = \frac{1}{L^{3/2}} \sum_{\mathbf{k}} \underbrace{\left(\sum_p \sqrt{\frac{m}{\epsilon_0}} \xi_{\mathbf{k},p}(t) \mathbf{e}_{\mathbf{k},p} \right)}_{\tilde{\mathbf{A}}_\perp(\mathbf{k}, t)} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{L^{3/2}} \sum_{\mathbf{k}} \tilde{\mathbf{A}}_\perp(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (2.85)$$

which describes a Fourier series expansion. Since $\mathbf{A}_\perp$ is a real vector field, we thus must have $\tilde{\mathbf{A}}_{-\mathbf{k}}(t) = \tilde{\mathbf{A}}_\perp^*(\mathbf{k}, t)$, and, hence, $\xi_{-\mathbf{k},p}(t) = \xi_{\mathbf{k},p}^*(t)$ and $\mathbf{e}_{-\mathbf{k},p} = \mathbf{e}_{\mathbf{k},p}^*$. Analogously to (2.66), we can now set

$$\alpha_{\mathbf{k},p}(t) = \sqrt{\frac{m\omega_{\mathbf{k}}}{2\hbar}} \left(\xi_{\mathbf{k},p}(t) + \frac{i}{m\omega_{\mathbf{k}}} \pi_{\mathbf{k},p}(t) \right), \quad (2.86a)$$

$$\alpha_{-\mathbf{k},p}^*(t) = \sqrt{\frac{m\omega_{\mathbf{k}}}{2\hbar}} \left(\xi_{\mathbf{k},p}(t) - \frac{i}{m\omega_{\mathbf{k}}} \pi_{\mathbf{k},p}(t) \right), \quad (2.86b)$$

where $\pi_{\mathbf{k},p}(t) = m \frac{d\xi_{\mathbf{k},p}(t)}{dt}$, such that

$$\xi_{\mathbf{k},p}(t) = \sqrt{\frac{\hbar}{2m\omega_{\mathbf{k}}}} \left(\alpha_{\mathbf{k},p}(t) + \alpha_{-\mathbf{k},p}^*(t) \right), \quad (2.87a)$$

$$\pi_{\mathbf{k},p}(t) = \frac{1}{i} \sqrt{\frac{\hbar m \omega_{\mathbf{k}}}{2}} \left(\alpha_{\mathbf{k},p}(t) - \alpha_{-\mathbf{k},p}^*(t) \right). \quad (2.87b)$$

Plugging this into the general solution from Eq. (2.85), we obtain

$$\mathbf{A}_\perp(\mathbf{r}, t) = \frac{1}{L^{3/2}} \sum_{\mathbf{k},p} \sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}\epsilon_0}} \left[\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k} \cdot \mathbf{r}} \alpha_{\mathbf{k},p}(t) + \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k} \cdot \mathbf{r}} \alpha_{\mathbf{k},p}^*(t) \right], \quad (2.88)$$

where we changed the summation index from $\mathbf{k}$ to $-\mathbf{k}$ in the second term.

Finally, we take the limit $L \rightarrow \infty$ by converting the Fourier series in (2.88) into a Fourier integral: Given that by Eq. (2.84) neighbouring values of $\mathbf{k}$ are separated by $\Delta k = 2\pi/L$, we must replace

$$\sum_{\mathbf{k}} \Delta k^3 = \left(\frac{2\pi}{L} \right)^3 \sum_{\mathbf{k}} \rightarrow \int d^3k \quad (2.89)$$

in Eq. (2.85), leading to

$$\mathbf{A}_\perp(\mathbf{r}, t) = \frac{1}{(2\pi)^{3/2}} \int d^3k \underbrace{\left(\frac{L}{2\pi} \right)^{3/2} \tilde{\mathbf{A}}_\perp(\mathbf{k}, t)}_{\tilde{\mathbf{A}}_\perp(\mathbf{k}, t)} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{(2\pi)^{3/2}} \int d^3k \tilde{\mathbf{A}}_\perp(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (2.90)$$

with

$$\tilde{\mathbf{A}}_\perp(\mathbf{k}, t) = \left(\frac{L}{2\pi} \right)^{3/2} \tilde{\mathbf{A}}_\perp(\mathbf{k}, t) \stackrel{(2.85)}{=} \sum_p \underbrace{\sqrt{\frac{m}{\epsilon_0}} \left(\frac{L}{2\pi} \right)^{3/2} \xi_{\mathbf{k},p}(t)}_{\xi_p(\mathbf{k}, t)} \mathbf{e}_{\mathbf{k},p} = \sum_p \sqrt{\frac{m}{\epsilon_0}} \xi_p(\mathbf{k}, t) \mathbf{e}_{\mathbf{k},p}. \quad (2.91)$$

Here, the factor $(L/2\pi)^{3/2}$ rescales the coefficients to ensure that $\tilde{\mathbf{A}}_{\perp}(\mathbf{k}, t)$ doesn't vanish for large L (since $\tilde{\mathbf{A}}_{\perp}(\mathbf{k}, t) \rightarrow 0$ as $L \rightarrow \infty$). This rescaling defines

$$\xi_p(\mathbf{k}, t) = \left(\frac{L}{2\pi}\right)^{3/2} \xi_{\mathbf{k},p}(t), \quad (2.92a)$$

$$\pi_p(\mathbf{k}, t) = \left(\frac{L}{2\pi}\right)^{3/2} \pi_{\mathbf{k},p}(t), \quad (2.92b)$$

and, similarly, the coefficients $\alpha_p(\mathbf{k}, t) = (L/2\pi)^{3/2} \alpha_{\mathbf{k},p}(t)$, such that

$$\alpha_p(\mathbf{k}, t) \stackrel{(2.86a)}{=} \sqrt{\frac{m\omega_{\mathbf{k}}}{2\hbar}} \left(\xi_p(\mathbf{k}, t) + \frac{i}{m\omega_{\mathbf{k}}} \pi_p(\mathbf{k}, t) \right). \quad (2.93)$$

As a result, the transverse vector field potential from Eq. (2.88) turns into

$$\mathbf{A}_{\perp}(\mathbf{r}, t) = \sum_p \int d^3k \sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}\epsilon_0(2\pi)^3}} \left[\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k}\cdot\mathbf{r}} \alpha_p(\mathbf{k}, t) + \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k}\cdot\mathbf{r}} \alpha_p^*(\mathbf{k}, t) \right]. \quad (2.94)$$

Similar as above, using $\mathbf{\Pi} = \epsilon_0 \dot{\mathbf{A}}_{\perp}$, and Eqs. (2.26) and (2.27), the momentum conjugate of $\mathbf{A}_{\perp}$, and the magnetic and transverse electric field (for $\mathbf{j}_{\perp} = 0$) become

$$\mathbf{\Pi}(\mathbf{r}, t) = - \sum_p \int d^3k i \sqrt{\frac{\hbar\omega_{\mathbf{k}}\epsilon_0}{2(2\pi)^3}} \left[\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k}\cdot\mathbf{r}} \alpha_p(\mathbf{k}, t) - \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k}\cdot\mathbf{r}} \alpha_p^*(\mathbf{k}, t) \right], \quad (2.95)$$

$$\mathbf{B}(\mathbf{r}, t) = \sum_p \int d^3k i \sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}\epsilon_0(2\pi)^3}} \left[(\mathbf{k} \times \mathbf{e}_{\mathbf{k},p}) e^{i\mathbf{k}\cdot\mathbf{r}} \alpha_p(\mathbf{k}, t) - (\mathbf{k} \times \mathbf{e}_{\mathbf{k},p}^*) e^{-i\mathbf{k}\cdot\mathbf{r}} \alpha_p^*(\mathbf{k}, t) \right], \quad (2.96)$$

$$\mathbf{E}_{\perp}(\mathbf{r}, t) = \sum_p \int d^3k i \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3}} \left[\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k}\cdot\mathbf{r}} \alpha_p(\mathbf{k}, t) - \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k}\cdot\mathbf{r}} \alpha_p^*(\mathbf{k}, t) \right]. \quad (2.97)$$

Review Questions 2.6.

- How do we ensure that $\mathbf{A}_{\perp}(\mathbf{r}, t)$ describes a real field when allowing $\boldsymbol{\mu}(\mathbf{r})$ and $\xi(t)$ to be complex in the ansatz (2.82)?
- Calculate explicitly the step from Eq. (2.85) to Eq. (2.88)
- Show that the complex mode functions (2.83) are orthonormal, satisfying $\int d^3r \boldsymbol{\mu}_{\mathbf{k},p}^*(\mathbf{r}) \cdot \boldsymbol{\mu}_{\mathbf{k}',p'}(\mathbf{r}) = \delta_{\mathbf{k},\mathbf{k}'} \delta_{p,p'}$.

2.4 Our path to quantize the electromagnetic field

Above we have shown that the system of particles+fields corresponds to a set of mutually interacting particles and oscillators. If we were to quantize this system, the simplest idea would be to replace the particles' positions and momenta by their corresponding operators $\mathbf{r}_{\alpha}$ and $\mathbf{p}_{\alpha}$, whose commutator is

$$[r_{\alpha j}, r_{\alpha' j'}] = 0, \quad [p_{\alpha j}, p_{\alpha' j'}] = 0, \quad [r_{\alpha j}, p_{\alpha' j'}] = i\hbar \delta_{\alpha,\alpha'} \delta_{j,j'}, \quad (2.98)$$

and, similarly, the time dependencies $x_{\kappa}(t)$ and $p_{\kappa}(t) = m \frac{d}{dt} x_{\kappa}(t)$ of the κ -th field mode by their corresponding operators, satisfying

$$[x_{\kappa}, x_{\kappa'}] = 0, \quad [p_{\kappa}, p_{\kappa'}] = 0, \quad [x_{\kappa}, p_{\kappa'}] = i\hbar \delta_{\kappa,\kappa'}. \quad (2.99)$$

As a consequence of the latter, the normal variables α_κ and α_κ^* [see Eq. (2.66)] must then be replaced by the annihilation and creation operators a_κ and $a_\kappa^\dagger$, respectively, satisfying the commutation relations

$$[a_\kappa, a_{\kappa'}] = 0, \quad [a_\kappa^\dagger, a_{\kappa'}^\dagger] = 0, \quad [a_\kappa, a_{\kappa'}^\dagger] = \delta_{\kappa, \kappa'}. \quad (2.100)$$

Note that to simplify notation, we don't write hats on operators. We use the symbols $\mathbf{r}_\alpha$, $\mathbf{p}_\alpha$, x_κ , p_κ both for the variables as well as the corresponding operators. Further note that we consider the operators in the Schrödinger picture for $t = 0$, hence, there is no time argument.

Let us stress that such an approach to quantization is, however, heuristic. The reason is that we did not start from a Lagrangian or Hamiltonian formulation, such that we don't know yet whether $\mathbf{r}_\alpha$ and $\mathbf{p}_\alpha$ on the one hand, and x_κ and p_κ on the other hand, correspond to conjugate dynamical variables with respect to a Hamiltonian. That is, we don't know yet whether the above suggested quantization procedure is indeed correct. If we were to accept that it is correct, we can continue with the next chapter, the quantization of the electromagnetic field. If not, we must find a Lagrangian that gives rise to the correct classical dynamics (i.e., whose Lagrange equations correspond to Maxwell's equations and the Newton-Lorentz equations), and deduce the conjugate dynamical variables from it. This is briefly discussed in the next chapter, justifying that $\mathbf{r}_\alpha$ and $\mathbf{p}_\alpha$, as well as x_κ and p_κ are indeed conjugate variables.

2.5 Lagrangian and Hamiltonian approach

2.5.1 Standard Lagrangian

Let us consider the description of the electromagnetic field in terms of the vector potential $\mathbf{A}$ and scalar potential ϕ . The dynamical variables of the system of particles+fields are then given by

- (i) the three components $r_{\alpha j}$ of the position $\mathbf{r}_\alpha$ of the α -th particle, for all α ,
- (ii) the corresponding velocities $\dot{r}_{\alpha j}$,
- (iii) the three components $A_j(\mathbf{r})$ of the vector potential $\mathbf{A}(\mathbf{r})$, and the scalar potential $\phi(\mathbf{r})$,
- (iv) and the four corresponding velocities $\dot{A}_j(\mathbf{r})$ and $\dot{\phi}(\mathbf{r})$.

The standard Lagrangian in real space, from which the equations of motion can be derived turns out to be

$$L = \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 + \int d^3r \mathcal{L}(\mathbf{r}), \quad (2.101)$$

with the Lagrangian density

$$\mathcal{L}(\mathbf{r}) = \frac{\epsilon_0}{2} \left[(\dot{\mathbf{A}}(\mathbf{r}) + \nabla \phi(\mathbf{r}))^2 - c^2 (\nabla \times \mathbf{A}(\mathbf{r}))^2 \right] + \mathbf{j}(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}) - \rho(\mathbf{r}) \phi(\mathbf{r}). \quad (2.102)$$

Note that the interaction term in (2.102) is local, since the charge current $\mathbf{j}(\mathbf{r})$ and the vector field $\mathbf{A}(\mathbf{r})$ is evaluated at the same point $\mathbf{r}$, and the same holds for the charge density $\rho(\mathbf{r})$ and the scalar field $\phi(\mathbf{r})$. Moreover, the spatial derivatives of the vector field and scalar field expresses a coupling between the field variables from point to point. This coupling is the origin of the propagation of the free field.

As a next step, we want to get the Lagrangian with the fields in reciprocal space. This corresponds to a change of the dynamical variables from the components of $\mathbf{A}(\mathbf{r})$ and $\dot{\mathbf{A}}(\mathbf{r})$ to the components of $\tilde{\mathbf{A}}(\mathbf{k})$ and $\dot{\tilde{\mathbf{A}}}(\mathbf{k})$. In reciprocal space, however, the field is complex, such that

we now have twice as many dynamical variables as before. We must compensate for this by considering the constraint that the Fourier transform of a real vector field $\mathbf{F}(\mathbf{r})$ must satisfy

$$\tilde{\mathbf{F}}(\mathbf{k}) = \tilde{\mathbf{F}}^*(-\mathbf{k}). \quad (2.103)$$

We can, thus, include this constraint by considering only a half volume of the reciprocal space, since the other half volume is already accounted for by the fact that we consider real fields in real space. To this end, let us use the Parseval-Plancherel identity (2.6), which reads, for real fields $F(\mathbf{r})$ and $G(\mathbf{r})$,

$$\begin{aligned} \int d^3r F(\mathbf{r})G(\mathbf{r}) &= \int d^3k \tilde{F}^*(\mathbf{k})\tilde{G}(\mathbf{k}) \\ &= \oint d^3k \left[\tilde{F}^*(\mathbf{k})\tilde{G}(\mathbf{k}) + \tilde{F}(\mathbf{k})\tilde{G}^*(\mathbf{k}) \right], \end{aligned} \quad (2.104)$$

where $\oint d^3k$ denotes the integration over a half volume of the reciprocal space. Using this, we can transform the standard Lagrangian in the Coulomb gauge (2.101) into reciprocal space,

$$L = \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 + \oint d^3k \tilde{\mathcal{L}}(\mathbf{k}), \quad (2.105)$$

with the Lagrangian density in reciprocal space given by

$$\begin{aligned} \tilde{\mathcal{L}} &= \epsilon_0 \left[\left| \dot{\tilde{\mathbf{A}}}(\mathbf{k}) + i\mathbf{k}\tilde{\phi}(\mathbf{k}) \right|^2 - c^2 \left| \mathbf{k} \times \tilde{\mathbf{A}}(\mathbf{k}) \right|^2 \right] \\ &\quad + \tilde{\mathbf{j}}^*(\mathbf{k}) \cdot \tilde{\mathbf{A}}(\mathbf{k}) + \tilde{\mathbf{j}}(\mathbf{k}) \cdot \tilde{\mathbf{A}}^*(\mathbf{k}) - \tilde{\rho}^*(\mathbf{k})\tilde{\phi}(\mathbf{k}) - \tilde{\rho}(\mathbf{k})\tilde{\phi}^*(\mathbf{k}). \end{aligned} \quad (2.106)$$

To simplify notation, if there is no risk of confusion, we don't explicitly state the arguments $\mathbf{r}$ and $\mathbf{k}$ in the following

2.5.2 Equations of motion

In order to see that the standard Lagrangian (2.101) indeed leads to the correct equations of motion, i.e., the Newton-Lorentz equations (2.1) and Maxwell's equations (2.3), let us now derive them. To this end, we first recall Lagrange's equation for a Lagrangian $L(x_j(t), \dot{x}_j(t), t)$ of a system of N real dynamical variables $x_1, \dots, x_N$,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_j} \right) - \frac{\partial L}{\partial x_j} = 0. \quad (2.107)$$

Let us also recall the case of a Lagrangian $L = \int d^3k \mathcal{L}$, where the Lagrangian density

$$\mathcal{L}(\tilde{A}_j, \dot{\tilde{A}}_j, \partial_i \tilde{A}_j, \tilde{A}_j^*, \dot{\tilde{A}}_j^*, \partial_i \tilde{A}_j^*), \quad (2.108)$$

is due to a complex field $\tilde{A}_j$, where $\partial_i = \partial/\partial k_i$. The Lagrange equation with respect to $\tilde{A}_j^*$ is then

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\tilde{A}}_j^*} \right) - \frac{\partial \mathcal{L}}{\partial \tilde{A}_j^*} = - \sum_i \partial_i \frac{\partial \mathcal{L}}{\partial (\partial_i \tilde{A}_j^*)}. \quad (2.109)$$

Note that in our case, i.e., for the Lagrangian density (2.106), there is no dependence on $\partial_i \tilde{A}_j$ and $\partial_i \tilde{A}_j^*$, such that the right hand side in the Lagrange equation (2.109) becomes zero.

The Newton-Lorentz equation We start with the real dynamical variable $r_{\alpha j}$: Using the standard Lagrangian in real space (2.101), we get

$$\frac{\partial L}{\partial r_{\alpha j}} = q_{\alpha} \frac{\partial}{\partial r_{\alpha j}} [\dot{\mathbf{r}}_{\alpha} \cdot \mathbf{A}(\mathbf{r}_{\alpha}, t)] - q_{\alpha} \frac{\partial}{\partial r_{\alpha j}} \phi(\mathbf{r}_{\alpha}, t). \quad (2.110)$$

To calculate the first term, we utilize the vector identity

$$\nabla[\mathbf{A} \cdot \mathbf{B}] = [\mathbf{B} \cdot \nabla]\mathbf{A} + [\mathbf{A} \cdot \nabla]\mathbf{B} + \mathbf{B} \times [\nabla \times \mathbf{A}] + \mathbf{A} \times [\nabla \times \mathbf{B}], \quad (2.111)$$

such that

$$\begin{aligned} \nabla[\dot{\mathbf{r}}_{\alpha} \cdot \mathbf{A}(\mathbf{r}_{\alpha}, t)] &= \underbrace{[\mathbf{A}(\mathbf{r}_{\alpha}, t) \cdot \nabla]\dot{\mathbf{r}}_{\alpha}}_{=0} + [\dot{\mathbf{r}}_{\alpha} \cdot \nabla]\mathbf{A}(\mathbf{r}_{\alpha}, t) + \mathbf{A}(\mathbf{r}_{\alpha}, t) \times \underbrace{[\nabla \times \dot{\mathbf{r}}_{\alpha}]}_{=0} + \dot{\mathbf{r}}_{\alpha} \times [\nabla \times \mathbf{A}(\mathbf{r}_{\alpha}, t)] \\ &= [\dot{\mathbf{r}}_{\alpha} \cdot \nabla]\mathbf{A}(\mathbf{r}_{\alpha}, t) + \dot{\mathbf{r}}_{\alpha} \times [\nabla \times \mathbf{A}(\mathbf{r}_{\alpha}, t)]. \end{aligned}$$

Accordingly, for (2.110), we get

$$\frac{\partial L}{\partial r_{\alpha j}} = q_{\alpha} (\dot{\mathbf{r}}_{\alpha} \cdot \nabla) A_j(\mathbf{r}_{\alpha}, t) + q_{\alpha} [\dot{\mathbf{r}}_{\alpha} \times (\nabla \times \mathbf{A}(\mathbf{r}_{\alpha}, t))]_j - q_{\alpha} \frac{\partial}{\partial r_{\alpha j}} \phi(\mathbf{r}_{\alpha}, t). \quad (2.112)$$

On the other hand, we have

$$\frac{\partial L}{\partial \dot{r}_{\alpha j}} = m_{\alpha} \dot{r}_{\alpha j} + q_{\alpha} A_j(\mathbf{r}_{\alpha}, t), \quad (2.113)$$

and, thus,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}_{\alpha j}} = m_{\alpha} \ddot{r}_{\alpha j} + q_{\alpha} \frac{\partial}{\partial t} A_j(\mathbf{r}_{\alpha}, t) + q_{\alpha} (\dot{\mathbf{r}}_{\alpha} \cdot \nabla) A_j(\mathbf{r}_{\alpha}, t). \quad (2.114)$$

Combining (2.112) and (2.114) in accordance with Lagrange's equation (2.107) gives, for all three components $j = 1, 2, 3$,

$$\begin{aligned} m_{\alpha} \ddot{\mathbf{r}}_{\alpha} + q_{\alpha} \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}_{\alpha}, t) &= q_{\alpha} \dot{\mathbf{r}}_{\alpha} \times (\nabla \times \mathbf{A}(\mathbf{r}_{\alpha}, t)) - q_{\alpha} \nabla \phi(\mathbf{r}_{\alpha}, t) \\ \Leftrightarrow m_{\alpha} \ddot{\mathbf{r}}_{\alpha} &= q_{\alpha} \underbrace{\left(-\frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}_{\alpha}, t) - \nabla \phi(\mathbf{r}_{\alpha}, t) \right)}_{\mathbf{E}(\mathbf{r}_{\alpha}, t)} + q_{\alpha} \dot{\mathbf{r}}_{\alpha} \times \underbrace{(\nabla \times \mathbf{A}(\mathbf{r}_{\alpha}, t))}_{\mathbf{B}(\mathbf{r}_{\alpha}, t)} \\ \Leftrightarrow m_{\alpha} \ddot{\mathbf{r}}_{\alpha} &= q_{\alpha} [\mathbf{E}(\mathbf{r}_{\alpha}, t) + \dot{\mathbf{r}}_{\alpha} \times \mathbf{B}(\mathbf{r}_{\alpha}, t)], \end{aligned} \quad (2.115)$$

which is the Newton-Lorentz equation of the α -th particle (cf. Eq. (2.1)).

Maxwell's equations We continue with considering Lagrange's equation (2.109) for the dynamical variables of the field in reciprocal space. We begin with $\tilde{\phi}^*$. From the Lagrangian density in reciprocal space (2.106), we get

$$\frac{\partial \bar{\mathcal{L}}}{\partial \tilde{\phi}^*} = \epsilon_0 \frac{\partial}{\partial \tilde{\phi}^*} \left(\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi} \right) \cdot \left(\dot{\tilde{\mathbf{A}}}^* - i\mathbf{k}\tilde{\phi}^* \right) - \tilde{\rho} = -\epsilon_0 i\mathbf{k} \cdot \left(\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi} \right) - \tilde{\rho}, \quad (2.116)$$

and

$$\frac{\partial \bar{\mathcal{L}}}{\partial \dot{\tilde{\phi}^*}} = 0, \quad \text{and} \quad \frac{\partial \bar{\mathcal{L}}}{\partial (i\tilde{\phi}^*)} = 0. \quad (2.117)$$

Using this in Lagrange's equation (2.109), we get

$$\frac{d}{dt} \frac{\partial \bar{\mathcal{L}}}{\partial \dot{\tilde{\phi}}^*} - \frac{\partial \bar{\mathcal{L}}}{\partial \tilde{\phi}^*} = - \sum_i \partial_i \frac{\partial \bar{\mathcal{L}}}{\partial (\partial_i \tilde{\phi}^*)}. \quad \Leftrightarrow \quad -i\mathbf{k} \cdot (\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi}) = \frac{\tilde{\rho}}{\epsilon_0}. \quad (2.118)$$

Using $\tilde{\mathbf{E}} = -\dot{\tilde{\mathbf{A}}} - i\mathbf{k}\tilde{\phi}$, this becomes one of Maxwell's equation in reciprocal space (cf. Eq. (2.9a)),

$$i\mathbf{k} \cdot \tilde{\mathbf{E}} = \frac{\tilde{\rho}}{\epsilon_0}. \quad (2.119)$$

Next, we consider the Lagrange equation with respect to $\tilde{A}_j^*$. From the Lagrangian density (2.106), we get

$$\frac{\partial \bar{\mathcal{L}}}{\partial \tilde{A}_j^*} = -\epsilon_0 c^2 \frac{\partial}{\partial \tilde{A}_j^*} (\mathbf{k} \times \tilde{\mathbf{A}}) \cdot (\mathbf{k} \times \tilde{\mathbf{A}}^*) + \tilde{j}_j. \quad (2.120)$$

Using the identity³

$$(\mathbf{k} \times \tilde{\mathbf{A}}) \cdot (\mathbf{k} \times \tilde{\mathbf{A}}^*) = - \left[\mathbf{k} \times (\mathbf{k} \times \tilde{\mathbf{A}}) \right] \cdot \tilde{\mathbf{A}}^*, \quad (2.123)$$

we thus have

$$\frac{\partial \bar{\mathcal{L}}}{\partial \tilde{A}_j^*} = \epsilon_0 c^2 \left[\mathbf{k} \times (\mathbf{k} \times \tilde{\mathbf{A}}) \right]_j + \tilde{j}_j. \quad (2.124)$$

On the other hand, we have

$$\frac{\partial \bar{\mathcal{L}}}{\partial \dot{\tilde{A}}_j^*} = \epsilon_0 (\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi}) \frac{\partial}{\partial \dot{\tilde{A}}_j^*} (\dot{\tilde{\mathbf{A}}}^* - i\mathbf{k}\tilde{\phi}^*) = \epsilon_0 (\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi})_j = \epsilon_0 (\dot{\tilde{A}}_j + ik_j\tilde{\phi}), \quad (2.125)$$

and $\partial \bar{\mathcal{L}} / \partial (\partial_i \tilde{A}_j^*) = 0$. The Lagrange equation (2.109) then becomes

$$\begin{aligned} & \epsilon_0 (\ddot{\tilde{A}}_j + ik_j\dot{\tilde{\phi}}) - \epsilon_0 c^2 \left[\mathbf{k} \times (\mathbf{k} \times \tilde{\mathbf{A}}) \right]_j - \tilde{j}_j = 0 \\ \Leftrightarrow & \underbrace{-\epsilon_0 (\ddot{\tilde{A}}_j + ik_j\dot{\tilde{\phi}})}_{\dot{\tilde{E}}_j} + \epsilon_0 c^2 \underbrace{[\mathbf{k} \times (\mathbf{k} \times \tilde{\mathbf{A}})]_j}_{\tilde{B}} - \tilde{j}_j = 0 \\ & -\epsilon_0 \dot{\tilde{E}}_j + \epsilon_0 c^2 [\mathbf{k} \times \tilde{\mathbf{B}}]_j - \tilde{j}_j = 0. \end{aligned} \quad (2.126)$$

Finally, using $\mu_0 \epsilon_0 = 1/c^2$, this gives

$$i\mathbf{k} \times \tilde{\mathbf{B}} = \frac{1}{c^2} \dot{\tilde{\mathbf{E}}} + \frac{1}{\epsilon_0 c^2} \tilde{\mathbf{j}} = \mu_0 (\tilde{\mathbf{j}} + \epsilon_0 \dot{\tilde{\mathbf{E}}}), \quad (2.127)$$

which is another one of Maxwell's equations in reciprocal space (cf. Eq. (2.9d)).

³To see this, consider $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$, such that

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{d}) = a^2(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{d}) = [a^2 \mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{a}] \cdot \mathbf{d}. \quad (2.121)$$

On the other hand, from $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$, we have $\mathbf{a} \times (\mathbf{b} \times \mathbf{a}) = a^2 \mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{a}$. Combining this with the above yields

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{d}) = [\mathbf{a} \times (\mathbf{b} \times \mathbf{a})] \cdot \mathbf{d} = -[\mathbf{a} \times (\mathbf{a} \times \mathbf{b})] \cdot \mathbf{d}. \quad (2.122)$$

Note that the other two Maxwell equations simply result from the definitions $\tilde{\mathbf{E}} = -\dot{\tilde{\mathbf{A}}} - i\mathbf{k}\tilde{\phi}$ and $\tilde{\mathbf{B}} = i\mathbf{k} \times \tilde{\mathbf{A}}$: Using these definitions gives, on the one hand,

$$0 = i\mathbf{k} \cdot (i\mathbf{k} \times \tilde{\mathbf{A}}) = i\mathbf{k} \cdot \tilde{\mathbf{B}}, \quad (2.128)$$

which coincides with Maxwell's equation (2.9b), and, on the other hand,

$$-\dot{\tilde{\mathbf{B}}} = -i\mathbf{k} \times \dot{\tilde{\mathbf{A}}} = -i\mathbf{k} \times (\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi}) = i\mathbf{k} \times \dot{\tilde{\mathbf{E}}}, \quad (2.129)$$

which coincides with Maxwell's equation (2.9c).

In conclusion, we have seen that the standard Lagrangian (see Eqs. (2.101) and (2.105)) gives rise to the Newton-Lorentz equations for the particles' dynamics, and to Maxwell's equations for the field dynamics. We are now sure that this is the correct Lagrangian, and we can continue with finding the conjugate momenta of the dynamical variables $r_{\alpha j}$ and A_j , such that we can quantize the electromagnetic field in the end. However, before we can identify the conjugate momenta, we must get rid of redundancies in the set of all dynamical variables.

Review Questions 2.7.

- The action is given by the integral $S = \int_{t_1}^{t_2} dt L$. Discuss how the principle of least action is related to Lagrange's equations?
- Given a Lagrangian, how do we obtain the equations of motion?
- How do Lagrange's equations differ between discrete dynamical variables and continuous dynamical variables (i.e., fields)?

2.5.3 Elimination of redundant dynamical variables

Let us reconsider the standard Lagrangian density (2.102). There we directly see that $\dot{\phi}$ does not appear in the Lagrangian. This implies that the conjugate momentum $\partial\mathcal{L}/\partial\dot{\phi}$ associated with the variable ϕ is identically zero. When the velocity associated with a generalized coordinate does not appear in the Lagrangian, this coordinate can be eliminated by expressing it as a function of the other dynamical variables, giving a reduced Lagrangian. We can thus eliminate ϕ to get a Lagrangian where the dynamical variables of the field reduce to the three components of $\mathbf{A}$ and their time derivatives. To do so, consider Lagrange's equation with respect to $\tilde{\phi}$ from Eq. (2.118),

$$-i\mathbf{k} \cdot (\dot{\tilde{\mathbf{A}}} + i\mathbf{k}\tilde{\phi}) = \frac{\tilde{\rho}}{\epsilon_0} \quad \Leftrightarrow \quad -ik\dot{\tilde{A}}_{\parallel} + k^2\tilde{\phi} = \frac{\tilde{\rho}}{\epsilon_0} \quad (2.130)$$

where $\dot{\tilde{A}}_{\parallel} = \frac{\mathbf{k}}{k} \cdot \dot{\tilde{\mathbf{A}}}$, and, thus,

$$\tilde{\phi} = \frac{1}{k^2} \left(ik\dot{\tilde{A}}_{\parallel} + \frac{\tilde{\rho}}{\epsilon_0} \right). \quad (2.131)$$

We can now replace $\tilde{\phi}$ in the Lagrangian density (2.106) (and in all other expressions) with the above expression. A short calculation then results in the Lagrangian

$$\begin{aligned} L = & \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 + \epsilon_0 \int d^3k \left[\dot{\tilde{\mathbf{A}}}_{\perp}^* \cdot \dot{\tilde{\mathbf{A}}}_{\perp} - c^2 k^2 \tilde{\mathbf{A}}_{\perp}^* \cdot \tilde{\mathbf{A}}_{\perp} \right] + \int d^3k \left[\tilde{\mathbf{j}}^* \cdot \tilde{\mathbf{A}}_{\perp} + \tilde{\mathbf{j}} \cdot \tilde{\mathbf{A}}_{\perp}^* \right] \\ & - \underbrace{\int d^3k \frac{|\tilde{\rho}|^2}{\epsilon_0 k^2}}_{\stackrel{(2.53)}{=} H_{\parallel}} + \underbrace{\int d^3k \frac{i}{k} \left[(-ik\tilde{j}_{\parallel})\tilde{A}_{\parallel}^* - (-ik\tilde{j}_{\parallel})^* \tilde{A}_{\parallel} + \tilde{\rho}\dot{\tilde{A}}_{\parallel}^* - \tilde{\rho}^* \dot{\tilde{A}}_{\parallel} \right]}_{L_{\parallel}}. \end{aligned} \quad (2.132)$$

$\stackrel{(2.57)}{=} V_{\text{Coul}}$

Now consider that from the continuity equation (2.22) (i.e., the conservation of charges), we know that

$$\dot{\rho} = -\mathbf{i}\mathbf{k} \cdot \tilde{\mathbf{j}} = -\mathbf{i}k\tilde{j}_{\parallel}. \quad (2.133)$$

Accordingly, the last term in the Lagrangian (2.132), where the longitudinal component of $\tilde{A}_{\parallel}$ appears exclusively, can be written in terms of a total time derivative,

$$L_{\parallel} = \int d^3k \frac{\mathbf{i}}{k} [\dot{\rho}\tilde{A}_{\parallel}^* - \dot{\rho}^*\tilde{A}_{\parallel} + \tilde{\rho}\dot{\tilde{A}}_{\parallel}^* - \tilde{\rho}^*\dot{\tilde{A}}_{\parallel}] = \frac{d}{dt} \int d^3k \frac{\mathbf{i}}{k} [\tilde{\rho}\tilde{A}_{\parallel}^* - \tilde{\rho}^*\tilde{A}_{\parallel}]. \quad (2.134)$$

And since two Lagrangians which differ only in a total time derivative of a function of the coordinates are equivalent, we see that the evolution of the system of particles+fields must be independent of $\tilde{A}_{\parallel}$. That is, $\tilde{A}_{\parallel}$ is not a true dynamical variable: its value can be chosen without changing the dynamics of the system. This corresponds to the gauge freedom, since changing the gauge only affects $\tilde{A}_{\parallel}$ but leaves $\tilde{A}_{\perp}$ unaffected. We thus conclude that the dynamical variables of the field are given at each point by only four independent physical variables: the two transverse orthogonal components of the vector potential $\mathbf{A}_{\perp}$ and their time derivatives $\dot{\mathbf{A}}_{\perp}$, with the Lagrangian given by

$$L = \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 - V_{\text{Coul}} + \epsilon_0 \int d^3k \left[\dot{\tilde{\mathbf{A}}}_{\perp}^* \cdot \dot{\tilde{\mathbf{A}}}_{\perp} - c^2 k^2 \tilde{\mathbf{A}}_{\perp}^* \cdot \tilde{\mathbf{A}}_{\perp} \right] + \int d^3k \left[\tilde{\mathbf{j}}^* \cdot \tilde{\mathbf{A}}_{\perp} + \tilde{\mathbf{j}} \cdot \tilde{\mathbf{A}}_{\perp}^* \right]. \quad (2.135)$$

There are multiple possible choices of the gauge, which we are now fixing to the simplest one, the Coulomb gauge (2.30):

From now on we consider the Coulomb gauge (2.30), i.e., $\mathbf{A}_{\parallel} = 0$, and write $\mathbf{A}_{\perp} = \mathbf{A}$.

The Lagrangian in the Coulomb gauge then results from (2.135):

$$L = \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 - V_{\text{Coul}} + \int d^3k \bar{\mathcal{L}}_{\text{C}}, \quad (2.136a)$$

$$\bar{\mathcal{L}}_{\text{C}} = \epsilon_0 \left[\dot{\tilde{\mathbf{A}}}^* \cdot \dot{\tilde{\mathbf{A}}} - c^2 k^2 \tilde{\mathbf{A}}^* \cdot \tilde{\mathbf{A}} \right] + \tilde{\mathbf{j}}^* \cdot \tilde{\mathbf{A}} + \tilde{\mathbf{j}} \cdot \tilde{\mathbf{A}}^*, \quad (2.136b)$$

where $\tilde{\mathbf{A}} = \tilde{\mathbf{A}}_{\perp}$ (since we are in the Coulomb gauge). Transforming this into real space using (2.104), the Lagrangian in the Coulomb gauge in real space becomes

$$L = \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 - V_{\text{Coul}} + \int d^3r \mathcal{L}_{\text{C}}, \quad (2.137a)$$

$$\mathcal{L}_{\text{C}} = \frac{\epsilon_0}{2} \left[\dot{\mathbf{A}}^2 - c^2 (\nabla \times \mathbf{A})^2 \right] + \mathbf{j} \cdot \mathbf{A}. \quad (2.137b)$$

2.5.4 Conjugate variables

With the Lagrangians (2.136) and (2.137) at hand, we can now find the conjugate momenta associated with the particles' positions and the vector potential.

Conjugate particle momenta We start with $(\mathbf{r}_\alpha)_j = r_{\alpha j}$, whose conjugate momentum is $p_{\alpha j} = \partial L / \partial \dot{r}_{\alpha j}$. To this end, consider the Lagrangian (2.137), where $\dot{r}_{\alpha j}$ appears only in the terms

$$\frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 + \int d^3r \, \mathbf{j} \cdot \mathbf{A} \stackrel{(2.2b)}{=} \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 + \int d^3r \sum_{\alpha} q_{\alpha} \delta(\mathbf{r} - \mathbf{r}_{\alpha}(t)) \dot{\mathbf{r}}_{\alpha} \cdot \mathbf{A} \quad (2.138)$$

$$= \frac{1}{2} \sum_{\alpha} \left[m_{\alpha} \dot{\mathbf{r}}_{\alpha}^2 + q_{\alpha} \dot{\mathbf{r}}_{\alpha} \cdot \mathbf{A}(\mathbf{r}_{\alpha}) \right]. \quad (2.139)$$

Accordingly, the conjugate momentum becomes

$$p_{\alpha j} = \frac{\partial L}{\partial \dot{r}_{\alpha j}} = m_{\alpha} \dot{r}_{\alpha j} + q_{\alpha} A_j(\mathbf{r}_{\alpha}), \quad (2.140)$$

and when combining all three components $j = 1, 2, 3$ and using $\mathbf{A} = \mathbf{A}_{\perp}$ in the Coulomb gauge, we see that the conjugate momentum to the position $\mathbf{r}_{\alpha}$ of the α -th particle is the canonical momentum $\mathbf{p}_{\alpha}$ from Eq. (2.51),

$$\mathbf{p}_{\alpha} = m_{\alpha} \dot{\mathbf{r}}_{\alpha} + q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha}). \quad (2.141)$$

This now justifies why we called $\mathbf{p}_{\alpha}$ the canonical momentum in our initial definition (2.51). Let us stress that the canonical momentum, i.e., the conjugate momentum associated with $\mathbf{r}_{\alpha}$, is composed of the kinetic momentum $m_{\alpha} \dot{\mathbf{r}}_{\alpha}$ and the momentum $q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha})$ associated with the longitudinal field of the α -th particle.

Conjugate momenta for the field variables in real space First consider the Lagrangian from Eq. (2.137) in real space, where we have two independent field variables, $A_j(\mathbf{r})$ and $A_{j'}(\mathbf{r})$, at each point $\mathbf{r}$. The corresponding momentum to $A_j(\mathbf{r})$ is given by

$$\Pi_j(\mathbf{r}) = \frac{\partial \mathcal{L}_C}{\partial \dot{A}_j(\mathbf{r})} \stackrel{(2.137b)}{=} \epsilon_0 \dot{A}_j(\mathbf{r}). \quad (2.142)$$

Accordingly, the momentum conjugate with $\mathbf{A}$ is a vector with two orthogonal components,

$$\mathbf{\Pi}(\mathbf{r}) = \epsilon_0 \dot{\mathbf{A}}(\mathbf{r}). \quad (2.143)$$

Conjugate momenta for the field variables in reciprocal space In reciprocal space, the corresponding momentum to the complex variable $\tilde{A}_j(\mathbf{k})$ is obtained via

$$\tilde{\Pi}_j(\mathbf{k}) = \frac{\partial \bar{\mathcal{L}}_C}{\partial \dot{\tilde{A}}_j^*(\mathbf{k})} \stackrel{(2.136b)}{=} \epsilon_0 \dot{\tilde{A}}_j(\mathbf{k}), \quad (2.144)$$

such that the momentum conjugate with $\tilde{\mathbf{A}}$ is a complex vector with two orthogonal components,

$$\tilde{\mathbf{\Pi}}(\mathbf{k}) = \epsilon_0 \dot{\tilde{\mathbf{A}}}(\mathbf{k}), \quad (2.145)$$

which is simply the inverse Fourier transform of (2.143).

2.5.5 The Hamiltonian

Recall that for a set of real variables $\{x_j\}$ with conjugate momenta $\{p_j\}$, the Hamiltonian is obtained from the Lagrangian L via $H(x_j, p_j) = \sum_j \dot{x}_j p_j - L$. Accordingly, given the Lagrangian in the Coulomb gauge in real space, the Hamiltonian in the Coulomb gauge is

$$H = \sum_{\alpha} \mathbf{p}_{\alpha} \cdot \dot{\mathbf{r}}_{\alpha} + \int d^3r \, \mathbf{\Pi} \cdot \dot{\mathbf{A}} - L. \quad (2.146)$$

Using the Lagrangian from Eq. (2.137), and expressing $\dot{\mathbf{r}}_\alpha$ via Eq. (2.141) and $\dot{\mathbf{A}}$ via Eq. (2.143), a short calculation shows that the Hamiltonian in the Coulomb gauge reads

$$H = \sum_{\alpha} \frac{1}{2m_{\alpha}} [\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}(\mathbf{r}_{\alpha})]^2 + V_{\text{Coul}} + \frac{\epsilon_0}{2} \int d^3r \left[\left(\frac{\boldsymbol{\Pi}}{\epsilon_0} \right)^2 + c^2 (\boldsymbol{\nabla} \times \mathbf{A})^2 \right], \quad (2.147)$$

which is a function of conjugate dynamical variables. Similarly, in reciprocal space, the Hamiltonian is obtained via

$$H = \sum_{\alpha} \mathbf{p}_{\alpha} \cdot \dot{\mathbf{r}}_{\alpha} + \int d^3k \left[\tilde{\boldsymbol{\Pi}} \cdot \dot{\tilde{\mathbf{A}}}^* + \tilde{\boldsymbol{\Pi}}^* \cdot \dot{\tilde{\mathbf{A}}} \right] - L. \quad (2.148)$$

With the help of the Lagrangian (2.136) as well as Eqs (2.141) and (2.145), this results in the Hamiltonian in the Coulomb gauge in reciprocal space

$$H = \sum_{\alpha} \frac{1}{2m_{\alpha}} [\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}(\mathbf{r}_{\alpha})]^2 + V_{\text{Coul}} + \epsilon_0 \int d^3k \left[\frac{\tilde{\boldsymbol{\Pi}}^* \cdot \tilde{\boldsymbol{\Pi}}}{\epsilon_0^2} + c^2 k^2 \tilde{\mathbf{A}}^* \cdot \tilde{\mathbf{A}} \right]. \quad (2.149)$$

We are now ready to quantize the electromagnetic field.

Review Questions 2.8.

- *Why are the dynamics described by two Lagrangians differing by a total time derivative equivalent?*
- *What are the independent dynamical variables of the system particles+fields?*
- *What are the conjugate momenta of the dynamical variables?*

Chapter 3

Canonical quantization in the Coulomb gauge

So far, we studied the classical electromagnetic field in the presence of charged particles. We found that the electromagnetic field can be described via a set of fictitious harmonic oscillators and discussed the corresponding Lagrangian, which describes the dynamics of the system of particles+fields. What is most decisive is that we identified the conjugate momenta for the particle's positions [the canonical momentum (2.141)] and the conjugate momenta for the field variables in the Coulomb gauge [see Eqs. (2.143) and (2.145)]. This allows us in the following to quantize the system of particles+fields in the Coulomb gauge.

3.1 Quantization of the particles

We start with the quantization of the particles. To this end, we recall that the position and canonical momentum of the α -th particle are canonically conjugate variables. For their quantization, we can, thus, replace the position of the α -th particle by the quantum mechanical position operator $\mathbf{r}_\alpha$, and the momentum of the α -th particle by the quantum mechanical momentum operator $\mathbf{p}_\alpha$ (note that for the position and momentum operator we use the same notation as for the position and momentum, respectively; accordingly, in the following $\mathbf{r}_\alpha$ and $\mathbf{p}_\alpha$ always refer to operators. Further note that when writing an operator in bold, such as $\mathbf{r}_\alpha$, it is assumed to be a set of operators $r_{\alpha j}$). These operators must obey the commutation relation

$$\boxed{[r_{\alpha j}, p_{\alpha' j'}] = i\hbar\delta_{\alpha, \alpha'}\delta_{j, j'}}. \quad (3.1)$$

3.2 Quantization of the confined electromagnetic field

As justified in Sec. 2.5, for the quantization of the electromagnetic field, we replace the field variables and their conjugate momenta by operators. In particular, taking Eqs. (2.71) and (2.74), we have (at some initial time $t = 0$)

$$\mathbf{A}(\mathbf{r}) = \sum_{\kappa} \underbrace{\sqrt{\frac{m}{\epsilon_0}} x_{\kappa}}_{A_{\kappa}} \mathbf{u}_{\kappa}(\mathbf{r}) = \sum_{\kappa} A_{\kappa} \mathbf{u}_{\kappa}(\mathbf{r}), \quad (3.2)$$

and

$$\mathbf{\Pi}(\mathbf{r}) = \sum_{\kappa} \underbrace{\sqrt{\frac{\epsilon_0}{m}} p_{\kappa}}_{\Pi_{\kappa}} \mathbf{u}_{\kappa}(\mathbf{r}) = \sum_{\kappa} \Pi_{\kappa} \mathbf{u}_{\kappa}(\mathbf{r}), \quad (3.3)$$

where $A_\kappa = x_\kappa \sqrt{m/\epsilon_0}$ and $\Pi_\kappa = p_\kappa \sqrt{\epsilon_0/m}$ are now operators, satisfying the commutation relation

$$[A_\kappa, \Pi_{\kappa'}] = i\hbar \delta_{\kappa, \kappa'} \quad \Leftrightarrow \quad [x_\kappa, p_{\kappa'}] = i\hbar \delta_{\kappa, \kappa'}. \quad (3.4)$$

Note that to ease notation, we don't write hats on operators. Further note that we consider the operators x_κ and p_κ in the Schrödinger picture for $t = 0$, hence, there is no time argument.

Following the above described quantization procedure, we can now simply replace the complex variable α_κ from Eq. (2.66) by the so-called annihilation operator

$$a_\kappa = \sqrt{\frac{m\omega_\kappa}{2\hbar}} \left(x_\kappa + \frac{i}{m\omega_\kappa} p_\kappa \right), \quad (3.5)$$

and its complex conjugate α_κ^* by the so-called creation operator

$$a_\kappa^\dagger = \sqrt{\frac{m\omega_\kappa}{2\hbar}} \left(x_\kappa - \frac{i}{m\omega_\kappa} p_\kappa \right). \quad (3.6)$$

Since $[x_\kappa, p_{\kappa'}] = i\hbar \delta_{\kappa, \kappa'}$, these operators obey the commutation relations

$$\boxed{[a_\kappa, a_{\kappa'}^\dagger] = \delta_{\kappa, \kappa'}, \quad [a_\kappa, a_{\kappa'}] = 0, \quad [a_\kappa^\dagger, a_{\kappa'}^\dagger] = 0.} \quad (3.7)$$

The field operators for the vector fields in the Schrödinger picture at time $t = 0$ can then be obtained from Eqs. (2.71), (2.76), and (2.77) by setting $t = 0$ and applying the above replacement, yielding

$$\mathbf{A}(\mathbf{r}) = \sum_\kappa \sqrt{\frac{\hbar}{2\omega_\kappa \epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) [a_\kappa + a_\kappa^\dagger], \quad (3.8)$$

$$\mathbf{B}(\mathbf{r}) = \sum_\kappa \sqrt{\frac{\hbar}{2\omega_\kappa \epsilon_0}} \nabla \times \mathbf{u}_\kappa(\mathbf{r}) [a_\kappa + a_\kappa^\dagger], \quad (3.9)$$

$$\mathbf{E}_\perp(\mathbf{r}) = \sum_\kappa i \sqrt{\frac{\hbar\omega_\kappa}{2\epsilon_0}} \mathbf{u}_\kappa(\mathbf{r}) [a_\kappa - a_\kappa^\dagger]. \quad (3.10)$$

To express the Hamiltonian of the radiation, we must evaluate $H_\perp$ from Eq. (2.45) similar as in the classical case, now considering the field operators (3.9) and (3.10) as well as the commutation relations (3.7). This yields [c.f. Eq. (2.78)]

$$\boxed{H_\perp = \sum_\kappa \frac{\hbar\omega_\kappa}{2} [a_\kappa^\dagger a_\kappa + a_\kappa a_\kappa^\dagger] = \sum_\kappa \hbar\omega_\kappa \left(a_\kappa^\dagger a_\kappa + \frac{1}{2} \right),} \quad (3.11)$$

where the last term appears due to the commutation relation (3.7) (below we will see that it is associated with the vacuum energy).

In the Heisenberg picture, operators become time dependent and evolve according to the Heisenberg equation of motion (1.55). For the annihilation operators (3.5) it reads

$$\frac{d}{dt} a_\kappa(t) = -\frac{i}{\hbar} [a_\kappa(t), H_\perp] = -i\omega_\kappa a_\kappa(t). \quad (3.12)$$

By comparing with Eq. (2.69), we see that $a_\kappa(t)$ obeys the same evolution equation as its classical counterpart $\alpha_\kappa(t)$, with the solution

$$a_\kappa(t) = e^{iH_\perp t/\hbar} a_\kappa e^{-iH_\perp t/\hbar} = a_\kappa e^{-i\omega_\kappa t}, \quad (3.13)$$

and, accordingly,

$$a_{\kappa}^{\dagger}(t) = a_{\kappa}^{\dagger} e^{i\omega_{\kappa} t}. \quad (3.14)$$

The time dependent vector potential $\mathbf{A}(\mathbf{r}, t)$ in the Heisenberg picture then directly follows from Eq. (3.8),

$$\boxed{\mathbf{A}(\mathbf{r}, t) = \sum_{\kappa} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} \mathbf{u}_{\kappa}(\mathbf{r}) \left[a_{\kappa} e^{-i\omega_{\kappa} t} + a_{\kappa}^{\dagger} e^{i\omega_{\kappa} t} \right]}, \quad (3.15)$$

and $\mathbf{B}(\mathbf{r}, t)$ and $\mathbf{E}_{\perp}(\mathbf{r}, t)$ similarly follow from Eqs. (3.9) and (3.10). Note that these fields are often decomposed into a term with positive frequencies that involves only annihilation operators, and a term with negative frequencies that involves only creation operators. For example, for $\mathbf{A}(\mathbf{r}, t)$ from Eq. (3.15), the decomposition reads

$$\mathbf{A}(\mathbf{r}, t) = \mathbf{A}^{+}(\mathbf{r}, t) + \mathbf{A}^{-}(\mathbf{r}, t), \quad (3.16)$$

where

$$\mathbf{A}^{+}(\mathbf{r}, t) = \sum_{\kappa} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} \mathbf{u}_{\kappa}(\mathbf{r}) a_{\kappa} e^{-i\omega_{\kappa} t}, \quad \text{and} \quad \mathbf{A}^{-}(\mathbf{r}, t) = \left(\mathbf{A}^{+}(\mathbf{r}, t) \right)^{\dagger}. \quad (3.17)$$

3.3 Quantization of the electromagnetic field in free space

In Sec. 2.5 we found that the dynamical field variables, a two-component vector $\tilde{\mathbf{A}}(\mathbf{k})$ with components $\tilde{A}_j(\mathbf{k})$ and $\tilde{A}_{j'}(\mathbf{k})$, has conjugate momentum $\tilde{\Pi}(\mathbf{k}) = \epsilon_0 \dot{\tilde{\mathbf{A}}}(\mathbf{k})$ [see Eq. (2.145)]. We can therefore quantize the electromagnetic field by replacing the field variables and their conjugate momenta by operators satisfying the commutation relations¹

$$[\tilde{A}_j(\mathbf{k}), \tilde{\Pi}_{j'}(\mathbf{k}')] = 0 \quad [\tilde{A}_j(\mathbf{k}), \tilde{A}_{j'}(\mathbf{k}')] = 0 \quad [\tilde{\Pi}_j(\mathbf{k}), \tilde{\Pi}_{j'}(\mathbf{k}')] = 0 \quad (3.18a)$$

$$[\tilde{A}_j(\mathbf{k}), \tilde{\Pi}_j^{\dagger}(\mathbf{k}')] = i\hbar\delta_{j,j'}\delta(\mathbf{k} - \mathbf{k}') \quad [\tilde{A}_j(\mathbf{k}), \tilde{A}_j^{\dagger}(\mathbf{k}')] = 0 \quad [\tilde{\Pi}_j(\mathbf{k}), \tilde{\Pi}_j^{\dagger}(\mathbf{k}')] = 0. \quad (3.18b)$$

Now consider the expression of $\tilde{\mathbf{A}}(\mathbf{k})$ from Eq. (2.91) together with its conjugate momentum $\tilde{\Pi}(\mathbf{k}) = \epsilon_0 \dot{\tilde{\mathbf{A}}}(\mathbf{k})$ at some initial time $t = 0$,

$$\tilde{\mathbf{A}}(\mathbf{k}) = \sum_p \underbrace{\sqrt{\frac{m}{\epsilon_0}} \xi_p(\mathbf{k}) \mathbf{e}_{k,p}}_{\tilde{\mathbf{A}}_p(\mathbf{k})} = \sum_p \tilde{A}_p(\mathbf{k}) \mathbf{e}_{k,p} \quad (3.19)$$

$$\tilde{\Pi}(\mathbf{k}) = \sum_p \underbrace{\sqrt{\frac{\epsilon_0}{m}} \pi_p(\mathbf{k}) \mathbf{e}_{k,p}}_{\tilde{\Pi}_p(\mathbf{k})} = \sum_p \tilde{\Pi}_p(\mathbf{k}) \mathbf{e}_{k,p}. \quad (3.20)$$

The commutation relations (3.18) are, thus, equivalent to

$$[\xi_p(\mathbf{k}), \pi_{p'}(\mathbf{k}')] = 0 \quad [\xi_p(\mathbf{k}), \xi_{p'}(\mathbf{k}')] = 0 \quad [\pi_p(\mathbf{k}), \pi_{p'}(\mathbf{k}')] = 0, \quad (3.21a)$$

$$[\xi_p(\mathbf{k}), \pi_p^{\dagger}(\mathbf{k}')] = i\hbar\delta_{p,p'}\delta(\mathbf{k} - \mathbf{k}') \quad [\xi_p(\mathbf{k}), \xi_p^{\dagger}(\mathbf{k}')] = 0 \quad [\pi_p(\mathbf{k}), \pi_p^{\dagger}(\mathbf{k}')] = 0. \quad (3.21b)$$

¹Note that while for real fields, $A_j(\mathbf{r}), \Pi_{j'}(\mathbf{r}')$, the only non-vanishing commutation relation is $[A_j(\mathbf{r}), \Pi_{j'}(\mathbf{r}')] = i\hbar\delta_{j,j'}\delta(\mathbf{r} - \mathbf{r}')$, in the case of complex fields, $\tilde{A}_j(\mathbf{k}), \tilde{\Pi}_{j'}(\mathbf{k}')$, the only non-vanishing commutation relation is $[\tilde{A}_j(\mathbf{k}), \tilde{\Pi}_{j'}^{\dagger}(\mathbf{k}')] = i\hbar\delta_{j,j'}\delta(\mathbf{k} - \mathbf{k}')$. That is, the field operator and the **adjoint** of the operator associated with the momentum do not commute.

We can now consider the quantum counterpart to the complex coefficients $\alpha_p(\mathbf{k})$ [see Eq. (2.93)] at some initial time $t = 0$. According to our quantization procedure, these coefficients must be replaced by the annihilation operator

$$a_p(\mathbf{k}) = \sqrt{\frac{m\omega_{\mathbf{k}}}{2\hbar}} \left(\xi_p(\mathbf{k}) + \frac{i}{m\omega_{\mathbf{k}}} \pi_p(\mathbf{k}) \right), \quad (3.22)$$

and, similarly, $\alpha_p^*(\mathbf{k})$ by the creation operator

$$a_p^\dagger(\mathbf{k}) = \sqrt{\frac{m\omega_{\mathbf{k}}}{2\hbar}} \left(\xi_p^\dagger(\mathbf{k}) - \frac{i}{m\omega_{\mathbf{k}}} \pi_p^\dagger(\mathbf{k}) \right). \quad (3.23)$$

By the commutation relations (3.21), these operators satisfy

$$\left[a_p(\mathbf{k}), a_{p'}^\dagger(\mathbf{k}') \right] = \delta(\mathbf{k} - \mathbf{k}') \delta_{p,p'}, \quad [a_p(\mathbf{k}), a_{p'}(\mathbf{k}')] = 0, \quad [a_p^\dagger(\mathbf{k}), a_{p'}^\dagger(\mathbf{k}')] = 0. \quad (3.24)$$

The operator for the vector fields can now be obtained similar as in Eqs. (3.8)-(3.10). In particular, for the vector potential we get [cf. Eq. (2.94)]

$$\mathbf{A}(\mathbf{r}) = \sum_p \int d^3k \sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}\epsilon_0(2\pi)^3}} \left[\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k}\cdot\mathbf{r}} a_p(\mathbf{k}) + \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k}\cdot\mathbf{r}} a_p^\dagger(\mathbf{k}) \right], \quad (3.25)$$

Moreover, similar to Eq. (3.11), we can calculate the Hamiltonian of the transverse electromagnetic field using Eqs. (2.26), (2.27), (2.45), and the commutation relations (3.24), yielding

$$H_\perp = \sum_p \int d^3k \hbar\omega_{\mathbf{k}} \left(a_p^\dagger(\mathbf{k}) a_p(\mathbf{k}) + \frac{1}{2} \right). \quad (3.26)$$

Review Questions 3.1.

- How do we quantize the particles, and how do we quantize the electromagnetic field?
- What is the definition of the creation and annihilation operators? What are their commutation relations?
- Calculate the commutator $[a_\kappa^\dagger, a_\kappa^\dagger a_\kappa]$.

Chapter 4

Quantum states of the electromagnetic field

Recall that we decomposed the classical electromagnetic field into orthogonal modes (indexed by κ), which we further showed to correspond to classical harmonic oscillators. Due to the orthogonality of the mode functions, these harmonic oscillators are uncoupled in the absence of charges. They are only coupled indirectly through the matter degrees of freedom. After quantization, this is reflected in the fact that annihilation and creation operators corresponding to different modes commute. Accordingly, the quantized electromagnetic field corresponds to a composite system of infinitely many quantum harmonic oscillators. The corresponding Hilbert space is, thus, $\mathcal{H}_{\text{rad}} = \bigotimes_{\kappa} \mathcal{H}_{\kappa}$, where $\mathcal{H}_{\kappa} \simeq L_2(\mathbb{R})$, with $L_2(\mathbb{R})$ the square-integrable functions on $\mathbb{R}$. The Hilbert space $\mathcal{H}_{\text{rad}}$ is known as the (multi-mode) Fock space.

Suppose that the set $\{|\varphi_{\kappa}\rangle\}_{\varphi_{\kappa}}$ forms a basis of the single-mode Fock space $\mathcal{H}_{\kappa}$. Then states in the multi-mode Fock space $\mathcal{H}_{\text{rad}} = \bigotimes_{\kappa} \mathcal{H}_{\kappa}$ can be constructed from tensor products of these single-mode states,

$$\bigotimes_{\kappa} |\varphi_{\kappa}\rangle = |\varphi_1\rangle \otimes |\varphi_2\rangle \otimes |\varphi_3\rangle \otimes \cdots = |\varphi_1, \varphi_2, \varphi_3, \dots\rangle = |\{\varphi_{\kappa}\}\rangle, \quad (4.1)$$

with the set $\{|\varphi_1, \varphi_2, \varphi_3, \dots\rangle\}_{\varphi_1, \varphi_2, \varphi_3, \dots}$ forming a basis of $\mathcal{H}_{\text{rad}}$. To learn about the quantum states of the electromagnetic field, we start with considering quantum states of a single mode only. To ease notation we drop the mode index κ of the single mode under consideration.

4.1 Fock number states

The Fock states From the Hamiltonian (3.11) of the total, multi-mode radiation field, we directly see that the Hamiltonian of a single-mode is

$$H_{\text{single mode}} = \hbar\omega \left(a^{\dagger}a + \frac{1}{2} \right) = \hbar\omega \left(N + \frac{1}{2} \right), \quad (4.2)$$

where $N = a^{\dagger}a$ is the so-called number operator. A natural basis of the corresponding single-mode Fock space is therefore the eigenbasis of $H_{\text{single mode}}$, which is equivalent to the eigenbasis of N . That is, we consider the eigenstates $|n\rangle$ of N with corresponding eigenvalue n ,

$$N |n\rangle = n |n\rangle. \quad (4.3)$$

We now want to specify the eigenvalues n : To this end, we first note that for any state $|\psi\rangle$, we have $\langle\psi| N |\psi\rangle = \|a |\psi\rangle\|^2 \geq 0$. This states that N is a positive operator, and, thus, must have non-negative eigenvalues, i.e., $n \geq 0$. Next, we use the commutator $[N, a] = [a^{\dagger}a, a] = [a^{\dagger}, a]a = -a$, with which we find $[N, a] |n\rangle = -a |n\rangle \Leftrightarrow Na |n\rangle = (n-1)a |n\rangle$. After normalising

$a|n\rangle$ via $\langle n|a^\dagger a|n\rangle = n$, this states that the normalised state $\frac{1}{\sqrt{n}}a|n\rangle$ is an eigenstate of N with eigenvalue $n - 1$. Accordingly, by Eq. (4.3) we find $\frac{1}{\sqrt{n}}a|n\rangle = |n - 1\rangle$. Together with a similar calculation for the commutator $[N, a^\dagger]$, we, thus, have

$$a|n\rangle = \sqrt{n}|n - 1\rangle, \quad (4.4a)$$

$$a^\dagger|n\rangle = \sqrt{n + 1}|n + 1\rangle. \quad (4.4b)$$

Now recall that $n \geq 0$. In view of Eq. (4.4a), this can only be the case if there is an eigenstate $|0\rangle$ with eigenvalue 0, otherwise a repeated action of the annihilation operator a would lead to eigenstates with negative eigenvalues. From this we can also conclude that there can only be eigenstates $|n\rangle$ corresponding to eigenvalues $n \in \mathbb{N}_0$. This is why $|n\rangle$ is often called a number state. The eigenstate with corresponding eigenvalue 0 defines the vacuum state $|0\rangle$. With the help of Eq. (4.4b), we then obtain all other eigenstates via the repeated action of $a^\dagger$ on the vacuum state,

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle. \quad (4.5)$$

We have thus obtained the eigenstates $\{|n\rangle\}_{n=0}^\infty$ of N with corresponding eigenvalues $n \in \mathbb{N}_0$, which form an ON-basis of the single-mode Hilbert space, known as the Fock basis. Accordingly, the states $|n\rangle$ are also called Fock states.

Now that we found the eigenvalues n of the number operator N , we can conclude that the eigenvalues of the single-mode Hamiltonian (4.2) are $\hbar\omega(n + \frac{1}{2})$, with $n \in \mathbb{N}_0$. The eigenenergies of the single-mode field are thus separated by $\hbar\omega$. We therefore interpret n as the number of excitations in the electromagnetic field mode. These quantized excitations are called photons. Moreover, the names of the annihilation and creation operators now become clear: According to (4.4), annihilation operators remove a photon from the mode, and creation operators add a photon to the mode.

Fluctuations and quadrature operators Let us finally consider the expectation value of a product of creation and annihilation operators $f(a, a^\dagger)$ with respect to the number state $|n\rangle$, i.e., $\langle f(a, a^\dagger) \rangle_{|n\rangle} = \langle n|f(a, a^\dagger)|n\rangle$, for example $f(a, a^\dagger) = a^\dagger a a a^\dagger a$. From (4.5), we directly see that this expectation value must be identically zero if $f(a, a^\dagger)$ does not have as many creation as annihilation operators. Since this is the case for the field operators (3.8)-(3.10), the expectation value of the electromagnetic field with respect to a Fock state $|n\rangle$ vanishes. As an example, for [cf. Eq. (3.8)]

$$\mathbf{A}(\mathbf{r}) = \sqrt{\frac{\hbar}{2\omega\epsilon_0}}\mathbf{u}(\mathbf{r})[a + a^\dagger] = \mathbf{A}(\mathbf{r})^+ + \mathbf{A}(\mathbf{r})^-, \quad (4.6)$$

with $\mathbf{A}^+(\mathbf{r}) = \sqrt{\frac{\hbar}{2\omega\epsilon_0}}\mathbf{u}(\mathbf{r})a$, we get

$$\langle \mathbf{A}(\mathbf{r}) \rangle_{|n\rangle} = 0 \quad (4.7a)$$

$$\langle \mathbf{A}^2(\mathbf{r}) \rangle_{|n\rangle} = \langle n|[\mathbf{A}^+(\mathbf{r}) \cdot \mathbf{A}^-(\mathbf{r}) + \mathbf{A}^-(\mathbf{r}) \cdot \mathbf{A}^+(\mathbf{r})]|n\rangle = \frac{|\mathbf{u}(\mathbf{r})|^2}{\epsilon_0\omega^2}\hbar\omega\left(n + \frac{1}{2}\right). \quad (4.7b)$$

Therefore, the field in a number state is purely fluctuating with variance $(\Delta\mathbf{A})^2 = \langle \mathbf{A}^2(\mathbf{r}) \rangle_{|n\rangle}$ around zero. These fluctuations are also present for the vacuum state $|0\rangle$ associated with the zero point energy $\hbar\omega/2$ of the quantum harmonic oscillator. The vacuum fluctuations have physically observable consequences, such as the Casimir force or the Lamb shift of atomic levels.

The field fluctuations can be seen as uncertainty properties of the state, which have their origin in the creation and annihilation operators appearing in the expressions (3.8)-(3.10) of

the fields. For example, the vector field operators $\mathbf{A}(\mathbf{r})$ and $\mathbf{B}(\mathbf{r})$ [see Eqs. (3.8) and (3.9)] are determined by the operator $x \propto a + a^\dagger$, and, similarly, $\mathbf{E}_\perp(\mathbf{r})$ [see Eq. (3.10)] by $p \propto a - a^\dagger$. This motivates us to introduce rescaled versions of the canonically conjugate operators x and p , the so-called quadrature operators

$$X = \frac{a + a^\dagger}{2} = \sqrt{\frac{m\omega}{2\hbar}}x, \quad P = \frac{a - a^\dagger}{2i} = \frac{1}{\sqrt{2m\hbar\omega}}p, \quad (4.8)$$

whose commutator is $[X, P] = i/2$. The expectation values $\langle X \rangle, \langle P \rangle$ of these operators thus resemble those of the field operators. In particular, for the Fock state $|n\rangle$, we have

$$\langle X \rangle_{|n\rangle} = 0, \quad \langle X^2 \rangle_{|n\rangle} = \frac{n}{2} + \frac{1}{4}, \quad (4.9a)$$

$$\langle P \rangle_{|n\rangle} = 0, \quad \langle P^2 \rangle_{|n\rangle} = \frac{n}{2} + \frac{1}{4}. \quad (4.9b)$$

The uncertainties in the quadratures, e.g. $\Delta X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}$, are then seen to be

$$\Delta X = \Delta P = \sqrt{\frac{n}{2} + \frac{1}{4}}, \quad (4.10)$$

such that

$$\Delta X \Delta P = \frac{n}{2} + \frac{1}{4}. \quad (4.11)$$

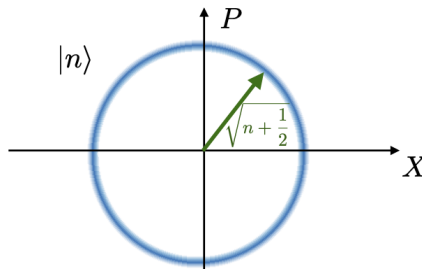
Fock number states thus exceed the Heisenberg uncertainty limit, except for the vacuum state $|0\rangle$, which is a state of minimal uncertainty, saturating the inequality of the uncertainty principle, given for the quadrature operators by

$$\Delta X \Delta P \geq \frac{1}{2} |\langle [X, P] \rangle| = \frac{1}{4}. \quad (4.12)$$

Finally, we note that due to $X^2 + P^2 = N + \frac{1}{2}$ and $N|n\rangle = n|n\rangle$, we have the eigenvalue equation

$$(X^2 + P^2)|n\rangle = \left(n + \frac{1}{2}\right)|n\rangle. \quad (4.13)$$

The number states may thus be qualitatively represented in the complex-plane representation of phase space by a circle of radius $\sqrt{n + \frac{1}{2}}$:



Review Questions 4.1.

- What is the Hilbert space of the free electromagnetic field?
- What do Fock states describe? What is the interpretation of the creation and annihilation operators?
- What is the consequence of ΔX and ΔP for a measurement of the electric and magnetic field? What does the uncertainty principle tell us about the measurement of the electric and magnetic field?
- Show that $\langle X^2 \rangle_{|n\rangle} = \langle P^2 \rangle_{|n\rangle} = \frac{n}{2} + \frac{1}{4}$ as provided in (4.9).

4.2 Coherent states

The coherent state Let us recall that in Sec. 2.3, we decomposed the classical electromagnetic field into uncoupled harmonic oscillators, whose evolution in phase space is described by the complex functions $\alpha_\kappa(t)$ from Eq. (2.66). Considering only a single mode, we are now looking for the quantum mechanical counterpart of this classical field description. To this end, we note that the quantization of the electromagnetic field was performed by replacing the complex function α with the corresponding annihilation operator a . We are therefore interested whether there exist eigenstates $|\alpha\rangle$ of a , whose eigenvalues correspond to the complex function α , i.e.,

$$a |\alpha\rangle = \alpha |\alpha\rangle. \quad (4.14)$$

Projecting this equation onto the Fock state $|n-1\rangle$ and using Eq. (4.4b) yields $\langle n-1|a|\alpha\rangle = \sqrt{n}\langle n|\alpha\rangle = \alpha\langle n-1|\alpha\rangle$. The last equality defines the recursion relation $\langle n|\alpha\rangle = \frac{\alpha}{\sqrt{n}}\langle n-1|\alpha\rangle$ through which the expansion of $|\alpha\rangle$ in the Fock basis becomes

$$|\alpha\rangle = \sum_{n=0}^{\infty} \langle n|\alpha\rangle |n\rangle = \langle 0|\alpha\rangle \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (4.15)$$

Normalising this state yields $1 = \langle \alpha|\alpha\rangle = |\langle 0|\alpha\rangle|^2 \sum_n \frac{|\alpha|^{2n}}{n!} = |\langle 0|\alpha\rangle|^2 e^{|\alpha|^2}$, which is satisfied if $\langle 0|\alpha\rangle = e^{-|\alpha|^2/2}$. By plugging this into Eq. (4.15), we finally find the coherent state

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \quad (4.16)$$

which obeys Eq. (4.14). From the expansion of the coherent state in the Fock basis, we directly see that the number of photons associated with $|\alpha\rangle$ follows a Poisson distribution,

$$p_n = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} \quad (4.17)$$

with expectation value and variance $|\alpha|^2$. Moreover, the variance of the photon number is equal to the average photon number,

$$(\Delta N)^2 = \langle N^2 \rangle_{|\alpha\rangle} - \langle N \rangle_{|\alpha\rangle}^2 = |\alpha|^2 = \langle N \rangle_{|\alpha\rangle}, \quad (4.18)$$

which is a characteristic of the Poisson distribution.

Displacement operator In view of Eq. (4.16), we now want to find an operator $D(\alpha)$ that creates the coherent state by acting on the vacuum, i.e., $D(\alpha)|0\rangle = |\alpha\rangle$ (similar as in Eq. (4.5) for Fock states). To this end, we define the displacement operator

$$D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a), \quad \alpha \in \mathbb{C}, \quad (4.19)$$

which is unitary, satisfying $D(-\alpha) = D^\dagger(\alpha) = D^{-1}(\alpha)$. In order to see that this operator satisfies the desired property, we use the operator identity

$$e^A B e^{-A} = B + \sum_{n=1}^{\infty} \frac{1}{n!} [A, B]_n, \quad (4.20)$$

with $[A, B]_1 = [A, B]$, and $[A, B]_{n+1} = [A, [A, B]_n]$.

For operators A and B both commuting with their commutator $[A, B]$, it can be shown that (4.20) implies that

$$e^{A+B} = e^A e^B e^{-[A, B]/2}. \quad (4.21)$$

With the help of (4.21), the displacement operator (4.19) can be rewritten as

$$D(\alpha) = e^{\alpha a^\dagger} e^{-\alpha^* a} e^{\frac{|\alpha|^2}{2} [a^\dagger, a]} = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a}, \quad (4.22)$$

and, therewith, shown to satisfy the desired property,

$$\begin{aligned} D(\alpha)|0\rangle &= e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a} |0\rangle \\ &= e^{-|\alpha|^2/2} e^{\alpha a^\dagger} |0\rangle \\ &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} (a^\dagger)^n |0\rangle \\ &\stackrel{(4.5)}{=} e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \\ &\stackrel{(4.16)}{=} |\alpha\rangle. \end{aligned} \quad (4.23)$$

Again, using (4.21), we find that the composition of displacement operators obeys

$$\begin{aligned} D(\alpha)D(\beta) &\stackrel{(4.19)}{=} e^{\alpha a^\dagger - \alpha^* a} e^{\beta a^\dagger - \beta^* a} \\ &= e^{(\alpha+\beta)a^\dagger - (\alpha+\beta)^* a} e^{[\alpha a^\dagger - \alpha^* a, \beta a^\dagger - \beta^* a]/2} \\ &= D(\alpha + \beta) e^{(\alpha\beta^* - \alpha^*\beta)/2} \\ &= D(\alpha + \beta) e^{i\text{Im}(\alpha\beta^*)}. \end{aligned} \quad (4.24)$$

The name of the displacement operator originates from the property that its action can be seen as a translation (or *displacement*) in phase-space,

$$D^\dagger(\alpha) a D(\alpha) = a + \alpha, \quad (4.25a)$$

$$D^\dagger(\alpha) a^\dagger D(\alpha) = a^\dagger + \alpha^*. \quad (4.25b)$$

These identities follow from (4.20).

Quadrature operators The naming of the displacement operator becomes more clear by considering the action of $D(\alpha)$ on the quadrature operators (4.8). Their expectation value with respect to the coherent state $|\beta\rangle$ is, using (4.14),

$$\langle X \rangle_{|\beta\rangle} = \langle \beta | X | \beta \rangle = \text{Re}(\beta), \quad \langle P \rangle_{|\beta\rangle} = \langle \beta | P | \beta \rangle = \text{Im}(\beta). \quad (4.26)$$

Now, considering (4.25), the action of $D(\alpha)$ on the quadrature operators,

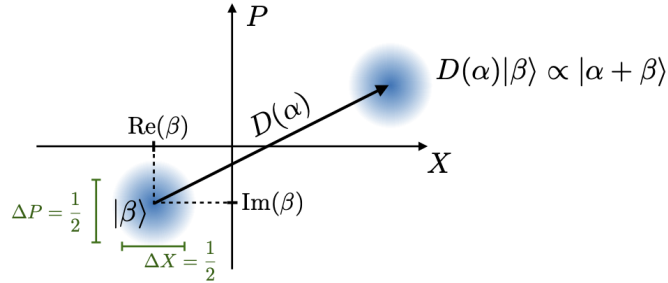
$$D^\dagger(\alpha) X D(\alpha) = X + \text{Re}(\alpha) \quad (4.27a)$$

$$D^\dagger(\alpha) P D(\alpha) = P + \text{Im}(\alpha), \quad (4.27b)$$

shows that

$$\begin{aligned} \langle X \rangle_{D(\alpha)|\beta\rangle} &= \langle \beta | D^\dagger(\alpha) X D(\alpha) | \beta \rangle = \text{Re}(\beta) + \text{Re}(\alpha) \\ \langle P \rangle_{D(\alpha)|\beta\rangle} &= \langle \beta | D^\dagger(\alpha) P D(\alpha) | \beta \rangle = \text{Im}(\beta) + \text{Im}(\alpha), \end{aligned} \quad (4.28)$$

which can be seen as a displacement in the complex-plane representation of phase-space:



Here, the clouds indicate uncertainties in the quadratures. In particular, from the expectation values

$$\langle X \rangle_{|\alpha\rangle} = \text{Re}(\alpha), \quad \langle X^2 \rangle_{|\alpha\rangle} = \text{Re}(\alpha)^2 + \frac{1}{4}, \quad (4.29a)$$

$$\langle P \rangle_{|\alpha\rangle} = \text{Im}(\alpha), \quad \langle P^2 \rangle_{|\alpha\rangle} = \text{Im}(\alpha)^2 + \frac{1}{4}, \quad (4.29b)$$

we find the uncertainties in the quadratures to be

$$\Delta X = \Delta P = \frac{1}{2}. \quad (4.30)$$

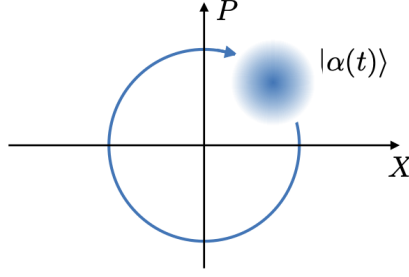
In the complex-plane representation of phase space, the coherent state $|\alpha\rangle$ is therefore centered around α , and corresponds to a state of minimum uncertainty:

$$\Delta X \Delta P = \frac{1}{2} |\langle [X, P] \rangle| = \frac{1}{4}. \quad (4.31)$$

Time evolution of coherent states Next, let us consider the time evolution of the coherent state (4.16) under the single-mode Hamiltonian (4.2), which is obtained through the unitary time-evolution operator $U(t) = \exp(-iH_{\text{single mode}}t/\hbar)$,

$$U(t) |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-i\omega(n+\frac{1}{2})t} |n\rangle = e^{-i\omega t/2} \underbrace{\{\alpha e^{-i\omega t}\}}_{\alpha(t)} = e^{-i\omega t/2} |\alpha(t)\rangle. \quad (4.32)$$

Accordingly, the coherent state $|\alpha\rangle$ remains in a coherent state under time evolution, $|\alpha(t)\rangle$, with the parameter $\alpha(t) = \alpha e^{-i\omega t}$ describing a circular motion in phase-space, similar to the classical dynamics from Eq. (2.70) [note that the global phase $e^{-i\omega t/2}$ in (4.32) appears due to the zero-point energy $\hbar\omega/2$ associated with the vacuum].

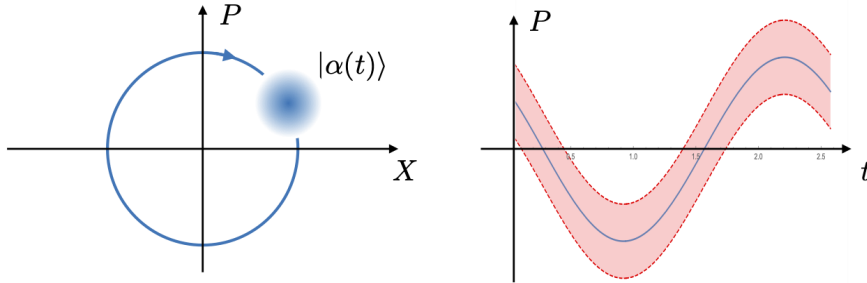


It is, thus, not surprising that the time-dependent expectation value of the (single-mode) vector potential from Eq. (3.8) resembles the classical vector potential from Eq. (2.71),

$$\langle \mathbf{A}(\mathbf{r}, t) \rangle_{|\alpha\rangle} = \langle \mathbf{A}(\mathbf{r}) \rangle_{|\alpha(t)\rangle} = \langle \alpha(t) | \mathbf{A}(\mathbf{r}) | \alpha(t) \rangle = \sqrt{\frac{\hbar}{2\omega\epsilon_0}} \mathbf{u}(\mathbf{r}) [\alpha(t) + \alpha^*(t)]. \quad (4.33)$$

Similar relations between the quantum mechanical expectation value and the classical field also hold for the magnetic and electric fields. The dynamics of the coherent state thus reproduces the classical dynamics of the electromagnetic field. Note that this is in strong contrast to the Fock number states for which the expectation values of the fields vanish.

Phasor diagram Now suppose that we measure, for the coherent state $|\alpha(t)\rangle$, the strength of the single-mode electric field, which, by Eq. (3.10), is $E_\perp = -\sqrt{\frac{2\hbar\omega}{\epsilon_0}}P$. That is, we essentially measure the quadrature P . Each measurement will give us a single value for $E_\perp \propto \langle P \rangle$. After a large number of measurements, we will then find the measurement outcomes to be gathered around the mean $\langle E_\perp \rangle \propto \langle P \rangle$ with standard deviation $\Delta E_\perp \propto \Delta P$. The outcome of the measurement can be illustrated by projecting the phase space diagram on the P -axis. As a result, one obtains the phasor diagram:



Here, the solid blue line corresponds to the mean $\langle P \rangle$, and the red shaded area to the standard deviation ΔP . Our measurement outcomes will thus be scattered around the reddish area.

The coherent state basis Coherent states form a non-orthogonal, overcomplete basis of the single-mode Fock space. Non-orthogonality is due to

$$\begin{aligned} \langle \alpha | \beta \rangle &= \langle 0 | D(-\alpha) D(\beta) | 0 \rangle \\ &\stackrel{(4.24)}{=} \langle 0 | D(-\alpha + \beta) e^{-i\text{Im}(\alpha\beta^*)} | 0 \rangle \\ &\stackrel{(4.23)}{=} \exp \left[-|\alpha - \beta|^2/2 - i\text{Im}(\alpha\beta^*) \right], \end{aligned} \quad (4.34)$$

and the fact that they form an overcomplete basis is due to [compare with the completeness relation (1.12)]

$$\boxed{\frac{1}{\pi} \int d^2\alpha |\alpha\rangle \langle \alpha| = \mathbb{1}}, \quad (4.35)$$

where $d^2\alpha = d\text{Re}(\alpha) d\text{Im}(\alpha)$. As a consequence, any state $|\psi\rangle$ can be expanded in terms of the coherent states as

$$|\psi\rangle = \frac{1}{\pi} \int d^2\alpha \langle\alpha|\psi\rangle |\alpha\rangle. \quad (4.36)$$

Let us prove (4.35): Consider, for all $m, n \in \mathbb{N}_0$,

$$\begin{aligned} \langle m| \int d^2\alpha |\alpha\rangle\langle\alpha| |n\rangle &\stackrel{(4.16)}{=} \int d^2\alpha e^{-|\alpha|^2} \frac{\alpha^m (\alpha^*)^n}{\sqrt{m!n!}} \\ &= \int_0^\infty dr e^{-r^2} \frac{r^{m+n+1}}{\sqrt{m!n!}} \int_0^{2\pi} d\theta e^{i\theta(m-n)}, \end{aligned} \quad (4.37)$$

where we transformed into polar coordinates $\alpha = re^{i\theta}$, with $d^2\alpha = r dr d\theta$. The second integral gives

$$\int_0^{2\pi} d\theta e^{i\theta(m-n)} = \begin{cases} \frac{e^{i\theta(m-n)}}{i(m-n)} \Big|_{\theta=0}^{2\pi} & \text{if } m \neq n \\ 2\pi & \text{if } m = n \end{cases} = 2\pi\delta_{m,n}, \quad (4.38)$$

such that

$$\langle m| \int d^2\alpha |\alpha\rangle\langle\alpha| |n\rangle = 2\pi\delta_{m,n} \frac{1}{n!} \int_0^\infty dr e^{-r^2} r^{2n+1} \quad (4.39)$$

$$= \pi\delta_{m,n} \frac{1}{n!} \underbrace{\int_0^\infty d\rho e^{-\rho} \rho^n}_{\Gamma(n+1)=n!}. \quad (4.40)$$

where we changed the variable to $\rho = r^2$, with $d\rho = 2r dr$. Here we can identify the Gamma function $\Gamma(n+1) = \int_0^\infty d\rho e^{-\rho} \rho^n = n!$, such that

$$\langle m| \int d^2\alpha |\alpha\rangle\langle\alpha| |n\rangle = \pi\delta_{m,n}, \quad (4.41)$$

which proves the relation from Eq. (4.35).

Review Questions 4.2.

- What is a coherent state?
- What is the action of a displacement operator on creation and annihilation operators. How do we interpret this action in the complex-plane representation of phase space?
- In the Schrödinger picture, the time evolved coherent state is $|\alpha(t)\rangle$. How does the phase space representation change in time?
- Consider now the time evolution in the Heisenberg picture. What happens now with the complex-plane representation of phase space?
- What would be the corresponding phasor diagram of a Fock number state? Does it change in time? Why? Why not?
- Why is there a factor of $\frac{1}{\pi}$ in the completeness relation (4.35) of the coherent state basis?

4.3 Thermal states

The thermal state When light is emitted by thermal sources of some inverse temperature $\beta = 1/k_B T$ (don't confuse it with a complex parameter), i.e., objects that are in thermal equilibrium such as a black body, e.g. the sun, we speak of thermal light. Thermal light is in a stationary state that does not evolve in time. It must thus be a statistical mixture of the eigenstates of the Hamiltonian. To this end, recall that in Sec. 4.1 we found that the Fock states $|n\rangle$ are the eigenstates of the single mode Hamiltonian $H = H_{\text{single mode}} = \hbar\omega(a^\dagger a + \frac{1}{2})$, and from statistical mechanics we know that the probability for a thermal excitation of the n -th level with energy $E_n = \hbar\omega(n + 1/2)$ is

$$P_n = \frac{e^{-\beta E_n}}{Z}, \quad (4.42)$$

with $Z = \sum_{n=0}^{\infty} e^{-\beta E_n}$ the partition function. Accordingly, the single-mode thermal state is

$$\rho_\beta = \sum_{n=0}^{\infty} P_n |n\rangle\langle n| = \frac{1}{Z} e^{-\beta H}. \quad (4.43)$$

Using that

$$Z = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+1/2)} = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}, \quad (4.44)$$

the thermal state can also be written as

$$\rho_\beta = (1 - e^{-\beta\hbar\omega}) \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} |n\rangle\langle n|. \quad (4.45)$$

The average photon number Let us now calculate the average photon number $\bar{n}$ of a thermal state, which is the expectation value of the number operator $N = \sum_{n=0}^{\infty} n |n\rangle\langle n|$,

$$\bar{n} = \langle N \rangle_{\rho_\beta} = \text{Tr}(N \rho_\beta) \stackrel{(4.43)}{=} \sum_{n=0}^{\infty} n P_n = \frac{e^{-\beta\hbar\omega/2}}{Z} \sum_{n=0}^{\infty} n e^{-\beta\hbar\omega n} = \dots = \frac{1}{e^{\beta\hbar\omega} - 1}. \quad (4.46)$$

This is nothing but the Bose-Einstein distribution, which is in accordance with our understanding of black-body radiation. From this we obtain $e^{-\beta\hbar\omega} = \bar{n}/(1 + \bar{n})$, which allows us to write the thermal state (4.45) as

$$\rho_\beta = \frac{1}{1 + \bar{n}} \sum_{n=0}^{\infty} \left(\frac{\bar{n}}{1 + \bar{n}} \right)^n |n\rangle\langle n| = \frac{1}{1 + \bar{n}} \left(\frac{\bar{n}}{1 + \bar{n}} \right)^N. \quad (4.47)$$

To get the fluctuations in the average photon number, $(\Delta N)^2 = \langle N^2 \rangle - \langle N \rangle^2$, we have to calculate the expectation value of N^2 ,

$$\langle N^2 \rangle_{\rho_\beta} = \text{Tr}(N^2 \rho_\beta) = \sum_{n=0}^{\infty} n^2 P_n = \dots = \bar{n} + 2\bar{n}^2. \quad (4.48)$$

Accordingly, for a thermal state, the variance

$$(\Delta N)^2 = \bar{n}(\bar{n} + 1) \quad (4.49)$$

is larger than the average $\bar{n}$, which is called a super-Poissonian statistics.

Uncertainties Let us consider the uncertainties of a single-mode thermal state in the complex-plane representation. To this end, we note that thermal states ρ_β are diagonal in the Fock basis [see Eq. (4.43)]. Hence, they yield vanishing expectation values for creation and annihilation operators, and, accordingly, also for the fields as well as the quadrature operators X and P . From the expectation values

$$\langle X \rangle_{\rho_\beta} = 0, \quad \langle X^2 \rangle_{\rho_\beta} = \frac{\bar{n}}{2} + \frac{1}{4}, \quad (4.50a)$$

$$\langle P \rangle_{\rho_\beta} = 0, \quad \langle P^2 \rangle_{\rho_\beta} = \frac{\bar{n}}{2} + \frac{1}{4}, \quad (4.50b)$$

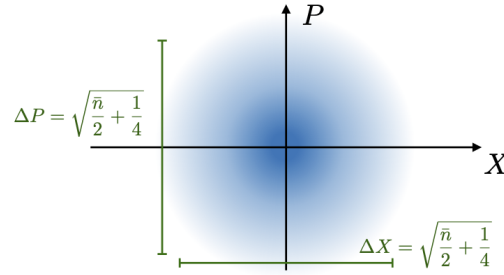
we then find the uncertainties in the quadrature operators to be

$$\Delta X = \Delta P = \sqrt{\frac{\bar{n}}{2} + \frac{1}{4}}, \quad (4.51)$$

giving rise to the uncertainty relation

$$\Delta X \Delta P = \frac{1}{4} + \frac{\bar{n}}{2}. \quad (4.52)$$

This exceeds the Heisenberg limit by $\bar{n}/2$, which is due to thermal fluctuations. A sketch of the thermal state in the complex plane is as follows:



Energy density For a single mode, as considered above, the average energy of the photons is $\hbar\omega\bar{n}$. If we consider now a set of modes, all in a thermal state, we have to take the density of modes into account to get the average energy. To this end, we calculate the density of modes of a cubic cavity of volume L^3 ,

$$\rho(\omega) = \frac{1}{L^3} \sum_{\kappa} \delta(\omega_{\kappa} - \omega), \quad (4.53)$$

and take the limit $L \rightarrow \infty$. Using that $\kappa = (p, \mathbf{k})$, with two polarization modes p , the limit $L \rightarrow \infty$ as taken in Eq. (2.89) gives

$$\rho(\omega) = \frac{2}{L^3} \sum_{\mathbf{k}} \delta(\omega_{\mathbf{k}} - \omega) \rightarrow \frac{1}{4\pi^3} \int d^3k \delta(c\mathbf{k} - \omega) = \frac{1}{\pi^2} \int dk k^2 \delta(c\mathbf{k} - \omega) = \frac{\omega^2}{\pi^2 c^3}. \quad (4.54)$$

The spectral energy density of thermal radiation is then given by

$$u(T, \omega) = \hbar\omega\bar{n}\rho(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\beta\hbar\omega} - 1}, \quad (4.55)$$

which is Planck's law. Integrating over ω then gives the total energy density, known as the Stefan-Boltzmann law,

$$u(T) = \int_0^\infty d\omega u(T, \omega) = \frac{\pi^2}{15\hbar^3 c^3 \beta^4}. \quad (4.56)$$

Review Questions 4.3.

- What does a thermal state describe?
- Why is a thermal state a statistical mixture of Fock number states?
- What is a super-Poissonian statistics?
- How would the phasor diagram of the thermal state look like. Discuss the outcome of a measurement of the electric field.

4.4 Squeezed states

The squeezing operator So far, we have only considered states with equal quadrature uncertainties $\Delta X = \Delta P$ [see Eqs. (4.10), (4.30), and (4.51)], and found that coherent states (including the vacuum state $|0\rangle$) are minimal uncertainty states, satisfying $\Delta X \Delta P = 1/4$. However, minimal uncertainty states are not necessarily required to have equal quadrature uncertainties, since $\Delta X \Delta P = 1/4$ can hold even if $\Delta X \neq \Delta P$. Such minimal uncertainty states with not necessarily equal quadrature uncertainties are called *squeezed states*. Their naming originates from their complex-plane representation of phase space, since, unlike the roundish coherent states (for which $\Delta X = \Delta P$), they look *squeezed* (due to $\Delta X \neq \Delta P$).

Squeezing a state can be performed by acting with the squeezing operator

$$S(\xi) = \exp\left(\frac{\xi^*}{2}a^2 - \frac{\xi}{2}(a^\dagger)^2\right), \quad \xi = re^{i\theta} \in \mathbb{C}, \quad (4.57)$$

which is a function of the complex squeezing parameter ξ , whose amplitude r determines the squeezing strength, and whose argument θ determines the angle, i.e., the direction of the squeezing. In order to see this, let us first note that the squeezing operator is unitary and satisfies

$$S(-\xi) = S(\xi)^\dagger = S(\xi)^{-1}. \quad (4.58)$$

Its action on the creation and annihilation operators is found (similar to the action of the displacement operator) using (4.20),

$$S(\xi)^\dagger a S(\xi) = a \cosh r - a^\dagger e^{i\theta} \sinh r, \quad (4.59a)$$

$$S(\xi)^\dagger a^\dagger S(\xi) = a^\dagger \cosh r - a e^{-i\theta} \sinh r. \quad (4.59b)$$

By rotating the quadratures by an angle $\theta/2$ (which entails a rotation of the complex plane spanned by X and P) through the definition of the rotated quadrature operators

$$X_{\theta/2} = \frac{ae^{-i\theta/2} + a^\dagger e^{i\theta/2}}{2} = X \cos\left(\frac{\theta}{2}\right) + P \sin\left(\frac{\theta}{2}\right), \quad (4.60a)$$

$$P_{\theta/2} = \frac{ae^{-i\theta/2} - a^\dagger e^{i\theta/2}}{2i} = P \cos\left(\frac{\theta}{2}\right) - X \sin\left(\frac{\theta}{2}\right), \quad (4.60b)$$

we see, using (4.59) as well as $\cosh r \pm \sinh r = e^{\pm r}$, that

$$S(\xi)^\dagger X_{\theta/2} S(\xi) = e^{-r} X_{\theta/2}, \quad (4.61a)$$

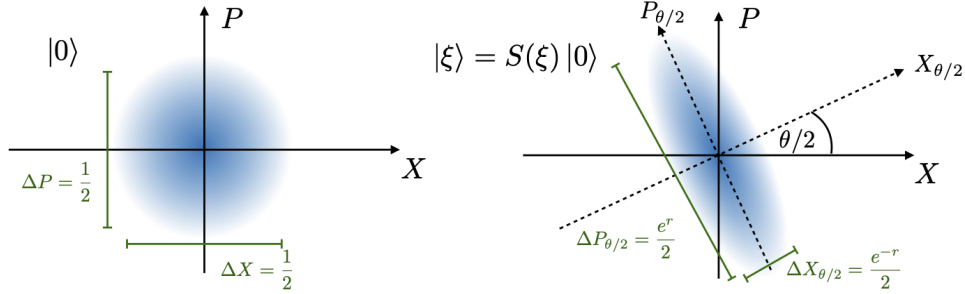
$$S(\xi)^\dagger P_{\theta/2} S(\xi) = e^r P_{\theta/2}. \quad (4.61b)$$

That is, the squeezing operator $S(\xi)$ *squeezes* the rotated quadrature $X_{\theta/2}$ by e^{-r} , and, at the same time, *stretches* the rotated quadrature $P_{\theta/2}$ by e^r .

Squeezed vacuum An important example is the squeezed vacuum state (note that we reserve greek letters from the beginning of the alphabet for coherent states, and use greek letters from the end of the alphabet to denote squeezed states)

$$|\xi\rangle = S(\xi) |0\rangle, \quad (4.62)$$

whose complex-plane representation is as follows:



With the help of Eqs. (4.60), (4.61), and the expectation values for $|0\rangle$ from Eq. (4.9), we find the expectation values of the squeezed vacuum to be

$$\langle X_{\theta/2} \rangle_{|\xi\rangle} = 0, \quad \langle X_{\theta/2}^2 \rangle_{|\xi\rangle} = \frac{1}{4} e^{-2r} \quad (4.63a)$$

$$\langle P_{\theta/2} \rangle_{|\xi\rangle} = 0, \quad \langle P_{\theta/2}^2 \rangle_{|\xi\rangle} = \frac{1}{4} e^{2r}, \quad (4.63b)$$

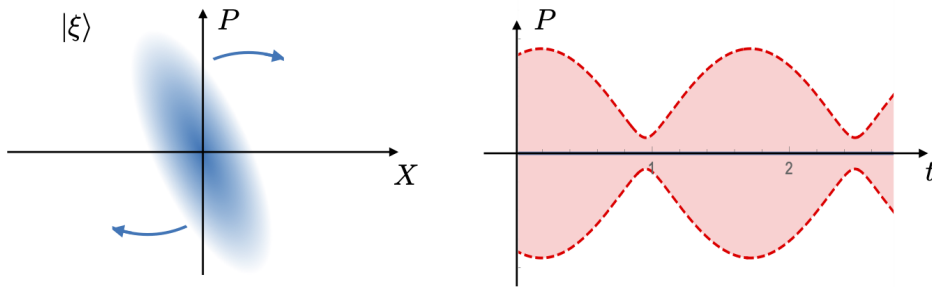
such that

$$\Delta X_{\theta/2} = \frac{1}{2} e^{-r}, \quad \Delta P_{\theta/2} = \frac{1}{2} e^r. \quad (4.64)$$

This shows that the squeezed vacuum state is a state of minimal uncertainty, with

$$\Delta X_{\theta/2} \Delta P_{\theta/2} = \frac{1}{4}. \quad (4.65)$$

What we see from the above illustration is that the noise of the squeezed vacuum state is phase dependent. The projection on the P axis, i.e., the electric field $E_{\perp} \propto P$, thus has a time dependent uncertainty as can be seen in the corresponding phasor diagram:



Consider now the number statistics. While for the vacuum state $|0\rangle$, we clearly have $\langle N \rangle_{|0\rangle} = 0$ and $\langle N^2 \rangle_{|0\rangle} = 0$, such that $\Delta N = 0$, this changes significantly once we squeeze the vacuum. To see this, let us first calculate the mean photon number for the squeezed vacuum. Using that $S(\xi)$ is unitary, we get

$$\begin{aligned} \bar{n} &= \langle \xi | N | \xi \rangle \\ &= \langle 0 | [S(\xi)^\dagger a^\dagger S(\xi)] [S(\xi)^\dagger a S(\xi)] | 0 \rangle \\ &\stackrel{(4.59)}{=} \langle 0 | [a^\dagger \cosh r - a e^{-i\theta} \sinh r] [a \cosh r - a^\dagger e^{i\theta} \sinh r] | 0 \rangle \\ &= \sinh^2 r, \end{aligned} \quad (4.66)$$

and, similarly, we get (making use of $\cosh^2 x - \sinh^2 x = 1$),

$$\begin{aligned}
\langle \xi | N^2 | \xi \rangle &= \langle 0 | \left(\left[a^\dagger \cosh r - a e^{-i\theta} \sinh r \right] \left[a \cosh r - a^\dagger e^{i\theta} \sinh r \right] \right)^2 | 0 \rangle \\
&= \langle 0 | a e^{-i\theta} \sinh r \left[a \cosh r - a^\dagger e^{i\theta} \sinh r \right] \left[a^\dagger \cosh r - a e^{-i\theta} \sinh r \right] a^\dagger e^{i\theta} \sinh r | 0 \rangle \\
&= \sinh^2 r \langle 1 | \left[a \cosh r - a^\dagger e^{i\theta} \sinh r \right] \left[a^\dagger \cosh r - a e^{-i\theta} \sinh r \right] | 1 \rangle \\
&= \sinh^2 r (2 \cosh^2 r + \sinh^2 r) \\
&= \bar{n}(3\bar{n} + 2).
\end{aligned} \tag{4.67}$$

Accordingly, the squeezed vacuum has a non-vanishing mean photon number with variance

$$(\Delta N)^2 = 2\bar{n}(\bar{n} + 1), \tag{4.68}$$

which is twice as large as the photon number variance (4.49) of the thermal state, corresponding to highly super-Poissonian statistics.

Finally, let us expand the squeezed vacuum in the Fock state basis. To this end note that the squeezing operator only contains squared creation and annihilation operators. The squeezed vacuum state can thus only have contributions from Fock states corresponding to an even number of photons. Keeping this in mind and considering $a|0\rangle = 0$, such that

$$0 = S(\xi) a S(\xi)^\dagger | \xi \rangle = S(-\xi)^\dagger a S(-\xi) | \xi \rangle \stackrel{(4.59)}{=} \left(a \cosh r + a^\dagger e^{i\theta} \sinh r \right) | \xi \rangle, \tag{4.69}$$

we get a recursion relation by projecting this on $|2n-1\rangle$:

$$\begin{aligned}
\sqrt{2n} \cosh r \langle 2n | \xi \rangle + \sqrt{2n-1} e^{i\theta} \sinh r \langle 2(n-1) | \xi \rangle &= 0. \\
\Leftrightarrow \langle 2n | \xi \rangle &= \sqrt{\frac{2n-1}{2n}} (-e^{i\theta} \tanh r) \langle 2(n-1) | \xi \rangle.
\end{aligned} \tag{4.70}$$

By iteration, we get, using $(2n-1)!! = (2n)!/(2^n n!)$ and $(2n)!! = 2^n n!$,

$$\langle 2n | \xi \rangle = \sqrt{\frac{(2n-1)!!}{(2n)!!}} (-e^{i\theta} \tanh r)^n \langle 0 | \xi \rangle = \frac{\sqrt{(2n)!}}{2^n n!} (-e^{i\theta} \tanh r)^n \langle 0 | \xi \rangle, \tag{4.71}$$

such that

$$| \xi \rangle = \langle 0 | \xi \rangle \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} (-e^{i\theta} \tanh r)^n | 2n \rangle. \tag{4.72}$$

Finally, the factor $\langle 0 | \xi \rangle$ is obtained from normalization,

$$1 = \langle \xi | \xi \rangle = |\langle 0 | \xi \rangle|^2 \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n} (n!)^2} \tanh^{2n} r = |\langle 0 | \xi \rangle|^2 \frac{1}{\sqrt{1 - \tanh^2 r}} = |\langle 0 | \xi \rangle|^2 \cosh r. \tag{4.73}$$

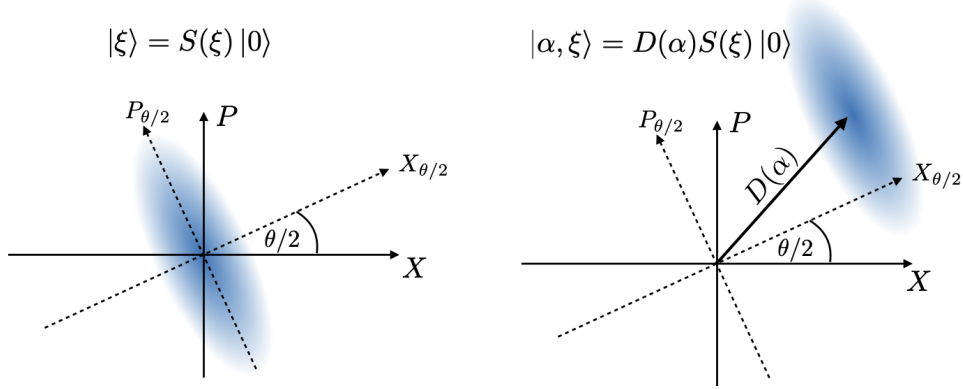
We can thus choose $\langle 0 | \xi \rangle = 1/\sqrt{\cosh r}$, such that the expansion of the squeezed vacuum state in the Fock state basis becomes

$$\boxed{| \xi \rangle = \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} (-e^{i\theta} \tanh r)^n | 2n \rangle.} \tag{4.74}$$

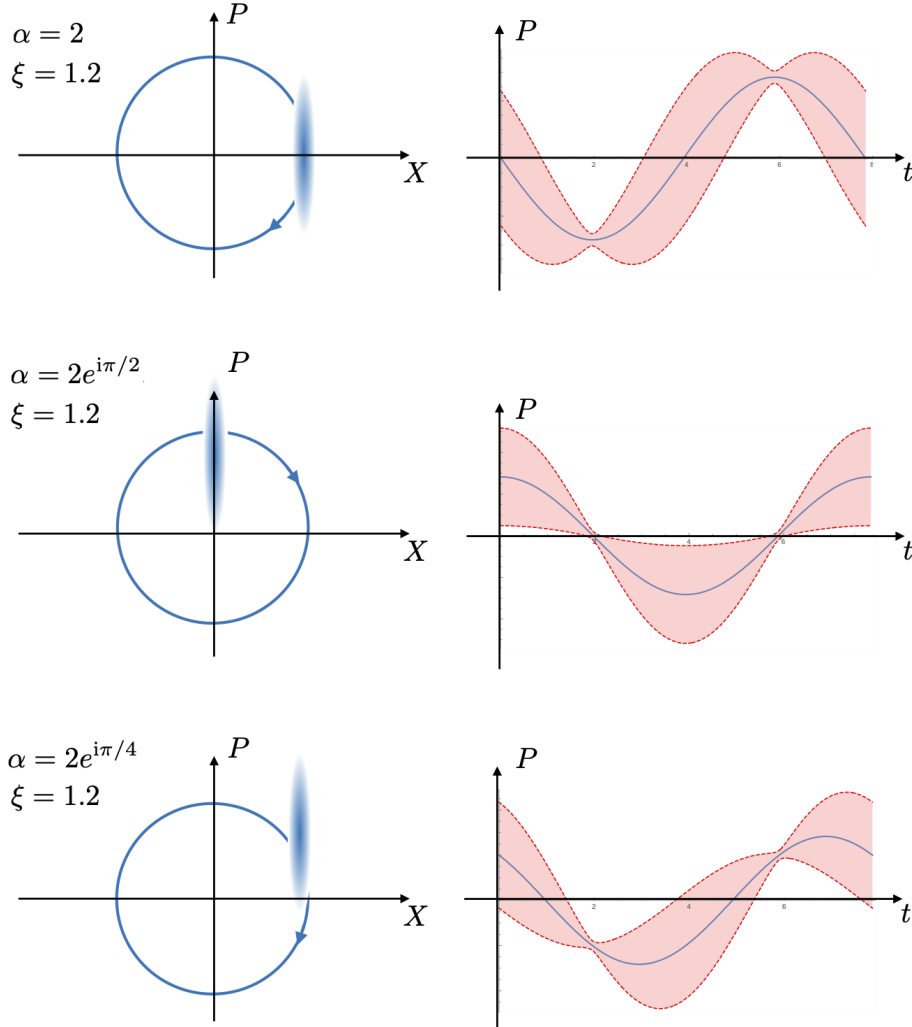
Displaced squeezed state When we act with a displacement operator $D(\alpha)$ on the squeezed vacuum $S(\xi) |0\rangle$, we get the so-called displaced squeezed vacuum

$$|\alpha, \xi\rangle = D(\alpha)S(\xi) |0\rangle, \quad (4.75)$$

whose quadratures have the same variance as the squeezed vacuum but with non-vanishing average value.



From this phase-space diagram, we directly see that the electric field $E_{\perp} \propto P$ of a displaced squeezed vacuum state that evolves freely has a time dependent noise that strongly depends on ξ and α . This is shown for some examples in the following:



Review Questions 4.4.

- Are there states with an infinitesimal uncertainty of the electric field? What would be the consequence for such a state?
- Explain the action of the squeezing operator via the complex-plane representation of phase space of the squeezed vacuum.
- Explain the phasor diagram of the squeezed vacuum state.
- We discussed the displaced squeezed state $D(\alpha)S(\xi)|0\rangle$. What happens if we first displace the vacuum and then squeeze, $S(\xi)D(\alpha)|0\rangle$? Discuss this via the complex-plane representation of phase space. How would we have to displace and squeeze to get the same states as illustrated above?

4.5 Phase-space representations

Above we illustrated quantum states of the electromagnetic field via a complex-plane representation of phase-space. However, we did not provide a rigorous formal description of these phase-space representations. This is the aim of the following.

4.5.1 Characteristic function of a classical random variable

Let us start with a brief recap of the concept of characteristic functions of a classical random variable. Let $\rho(x) \geq 0$ be the classical probability density associated with the outcome x of the random variable (note that x is a real number, it is not an operator). The n -th moment of the random variable is then

$$\langle x^n \rangle = \int dx x^n \rho(x). \quad (4.76)$$

If all moments $\langle x^n \rangle$ are known, then $\rho(x)$ is completely specified. To see this, let us introduce the characteristic function

$$C(k) = \langle e^{ikx} \rangle = \int dx e^{ikx} \rho(x) = \sum_{n=0}^{\infty} \frac{(ik)^n}{n!} \langle x^n \rangle, \quad (4.77)$$

which is obtained from the moments $\langle x^n \rangle$, and related to $\rho(x)$ through Fourier transformation,

$$\rho(x) = \frac{1}{2\pi} \int dk e^{-ikx} C(k). \quad (4.78)$$

Accordingly, knowing all moments $\langle x^n \rangle$ implies knowing $C(k)$, which implies knowing $\rho(x)$. Also note that if we know the characteristic function $C(k)$, we can calculate the moments via

$$\langle x^n \rangle = \frac{1}{i^n} \left. \frac{d^n C(k)}{dk^n} \right|_{k=0}. \quad (4.79)$$

4.5.2 Characteristic functions

Let us now consider single-mode quantum states ρ , which can fully be characterized by the expectation values of all possible products of creation and annihilation operators. Analogously to the classical case, we can extract these expectation values from characteristic functions. However,

we have to be a bit careful, since the ordering of the creation and annihilation operators in the expectation value matters. To this end, let us consider three important orderings of operators (note that expectation values of a product of creation and annihilation operators can be brought into different orderings using the operators' commutation relations).

- **Normal ordering:** A product of creation and annihilation operators is in normal ordering if all creation operators are to the left of all annihilation operators, e.g., $(a^\dagger)^m a^n$ is in normal ordering. Normal ordering is usually indicated by colons, for example,

$$: a^\dagger a a^\dagger a a :_N = (a^\dagger)^2 a^3. \quad (4.80)$$

For normal ordering, we usually drop the subscript and write $: a^\dagger a a^\dagger a a : = a^\dagger a a^\dagger a a :$.

- **Weyl ordering:** A product of creation and annihilation operators is in Weyl ordering if it is symmetrized with respect to all possible symmetric products. For example, $\frac{1}{2}(a^\dagger a + a a^\dagger)$ is in Weyl ordering, and

$$\begin{aligned} : a^\dagger a a^\dagger a a :_W = & \frac{1}{10} (a^\dagger a^\dagger a a a + a^\dagger a a^\dagger a a + a^\dagger a a a^\dagger a + a^\dagger a a a a^\dagger + a a^\dagger a^\dagger a a \\ & + a a^\dagger a a^\dagger a + a a^\dagger a a a^\dagger + a a a^\dagger a^\dagger a + a a a^\dagger a a^\dagger + a a a a^\dagger a^\dagger). \end{aligned} \quad (4.81)$$

- **Antinormal ordering:** A product of creation and annihilation operators is in antinormal ordering if all annihilation operators are to the left of all creation operators. For example, $a^n (a^\dagger)^m$ is in antinormal ordering, and we write, for example,

$$: a^\dagger a a^\dagger a a :_{AN} = a^3 (a^\dagger)^2. \quad (4.82)$$

The characteristic functions for these orderings are then given in terms of a complex variable β (which stands for two real variables, its real and imaginary part, such that we can treat β and β^* as independent variables),

$$\chi_\rho^N(\beta) = \text{Tr} \left(e^{\beta a^\dagger} e^{-\beta^* a} \rho \right), \quad (4.83a)$$

$$\chi_\rho^W(\beta) = \text{Tr} \left(e^{\beta a^\dagger - \beta^* a} \rho \right), \quad (4.83b)$$

$$\chi_\rho^{AN}(\beta) = \text{Tr} \left(e^{-\beta^* a} e^{\beta a^\dagger} \rho \right). \quad (4.83c)$$

They give rise to the expectation values, for $s \in \{N, W, AN\}$,

$$\langle : (a^\dagger)^m a^n :_s \rangle = \frac{d^{n+m}}{d\beta^m d(-\beta^*)^n} \chi_\rho^s(\beta) \Big|_{\beta=0}. \quad (4.84)$$

We directly see that the characteristic functions (4.83) are related to the expectation value of the displacement operator $D(\beta) = \exp(\beta a^\dagger - \beta^* a)$. In particular, using (4.21), we have $D(\beta) = e^{\beta a^\dagger} e^{-\beta^* a} e^{-|\beta|^2/2} = e^{-\beta^* a} e^{\beta a^\dagger} e^{|\beta|^2/2}$, and, thus,

$$\boxed{\chi_\rho^s(\beta) = \text{Tr} (D(\beta) \rho) e^{\frac{s|\beta|^2}{2}}}, \quad (4.85)$$

with

$$\begin{aligned} s = 1 & \Leftrightarrow \text{normal ordering } (N) \\ s = 0 & \Leftrightarrow \text{Weyl ordering } (W) \\ s = -1 & \Leftrightarrow \text{antinormal ordering } (AN). \end{aligned} \quad (4.86)$$

4.5.3 Quasiprobability distributions

Defining quasiprobability distributions Analogously to the classical random variable, where the classical probability density is obtained from the characteristic function via Fourier transform, we are interested in the distributions obtained from the Fourier transform of the characteristic functions (4.85). To this end, note that the two-dimensional Fourier transform of a function $f(\beta)$ of a complex variable β can be written, using $\beta = x' + ip'$ and $\alpha = x + ip$ (note that x and p are real variables, no operators),

$$\tilde{f}(\alpha) = \frac{1}{\pi^2} \int d^2\beta f(\beta) e^{\alpha\beta^* - \alpha^*\beta} = \frac{1}{\pi^2} \int dx' dp' f(x', p') e^{2ipx' - 2ixp'} = \tilde{f}(x, p), \quad (4.87)$$

with, in particular,

$$\frac{1}{\pi^2} \int d^2\beta e^{\alpha\beta^* - \alpha^*\beta} = \frac{1}{\pi^2} \int dx' dp' e^{2ipx' - 2ixp'} = \frac{1}{(2\pi)^2} \int du dv e^{ipu - ixdv} = \delta(x)\delta(p) = \delta(\alpha). \quad (4.88)$$

Now, taking the two-dimensional Fourier transform of the characteristic function (4.85) defines the corresponding quasiprobability distribution

$$W_\rho^s(\alpha) = \frac{1}{\pi^2} \int d^2\beta \chi_\rho^s(\beta) e^{\alpha\beta^* - \alpha^*\beta}, \quad (4.89)$$

whose inverse Fourier transform gives back the characteristic function,

$$\chi_\rho^s(\beta) = \int d^2\alpha W_\rho^s(\alpha) e^{\beta\alpha^* - \beta^*\alpha}. \quad (4.90)$$

Calculating the expectation values (4.84) by taking the derivatives then gives

$$\langle : (a^\dagger)^m a^n :_s \rangle = \frac{d^{n+m}}{d\beta^m d(-\beta^*)^n} \chi_\rho^s(\beta) \Big|_{\beta=0} \stackrel{(4.90)}{=} \int d^2\alpha (\alpha^*)^m \alpha^n W_\rho^s(\alpha). \quad (4.91)$$

Accordingly, the expectation value of any suitably ordered function $f_s(a, a^\dagger)$ of annihilation and creation operators can be expressed through the integral of the complex function $f_s(\alpha, \alpha^*)$, weighted by the quasiprobability distribution,

$$\langle f_s(a, a^\dagger) \rangle_\rho = \text{Tr} \left(f_s(a, a^\dagger) \rho \right) = \int d^2\alpha f_s(\alpha, \alpha^*) W_\rho^s(\alpha). \quad (4.92)$$

This is the so-called optical equivalence theorem, which strengthens our interpretation of $W_\rho^s(\alpha)$ as a distribution function over phase-space.

Relations between quasiprobability distributions Let us now briefly discuss how two quasiprobability distributions $W_\rho^s(\alpha)$ and $W_\rho^{s'}(\alpha)$ are related. To this end, we first note that by Eq. (4.85), the characteristic functions are related by

$$\chi_\rho^{s'}(\beta) = \chi_\rho^s(\beta) e^{\frac{(s'-s)|\beta|^2}{2}}. \quad (4.93)$$

By taking the Fourier transform, we then get

$$W_\rho^{s'}(\alpha) = \frac{1}{\pi^2} \int d^2\beta \chi_\rho^s(\beta) e^{\frac{(s'-s)|\beta|^2}{2}} e^{\alpha\beta^* - \alpha^*\beta}. \quad (4.94)$$

The integral converges for $s' < s$. In this case, we can apply the convolution theorem for the product of $\chi_\rho^s(\beta)$ and $\exp(-(s-s')|\beta|^2/2)$, yielding

$$W_\rho^{s'}(\alpha) = \frac{2}{\pi(s-s')} \int d^2\alpha' W_\rho^s(\alpha') e^{-\frac{|\alpha-\alpha'|^2}{s-s'}}. \quad (4.95)$$

We thus see that when going from s to $s' < s$, there is a smoothing of the quasiprobability distribution as described by the convolution of $W_\rho^s(\alpha')$ with a Gaussian (i.e., for $s' < s$, $W_\rho^{s'}(\alpha)$ is a smeared out version of $W_\rho^s(\alpha')$).

Properties of the quasiprobability distributions We now want to understand $W_\rho^s(\alpha)$ in more detail. To this end, we first note that due to the uncertainty principle, in quantum mechanics, two points in phase-space cannot be associated with mutually exclusive events. Instead, any event must be smeared out in phase-space. Accordingly, phase-space cannot be treated as a probability space. We thus see that the density function in phase-space, $W_\rho^s(\alpha)$, cannot be a probability density. However, it shares some properties with a probability function (it is real and normalized; as we will see), but also exhibits clear differences to a probability distribution (most prominently, it can be negative; as we will see) highlighting its “quantum nature”, hence the name *quasiprobability distribution*.

To see that the quasiprobability distributions are normalized, consider

$$\int d^2\alpha W_\rho^s(\alpha) \stackrel{(4.89)}{=} \frac{1}{\pi^2} \int d^2\alpha d^2\beta \chi_\rho^s(\beta) e^{\alpha\beta^* - \alpha^*\beta} \stackrel{(4.88)}{=} \int d^2\beta \chi_\rho^s(\beta) \delta(\beta) = \chi_\rho^s(0) \stackrel{(4.85)}{=} \text{Tr}(\rho) = 1, \quad (4.96)$$

and to see that it takes real values, consider, by Eqs. (4.89) and (4.85),

$$\begin{aligned} W_\rho^s(\alpha) &= \frac{1}{\pi^2} \int d^2\beta \text{Tr}(D(\beta)\rho) e^{\frac{s|\beta|^2}{2}} e^{\alpha\beta^* - \alpha^*\beta} \\ &\stackrel{(4.19)}{=} \frac{1}{\pi^2} \int d^2\beta \text{Tr}\left(e^{\beta(a^\dagger - \alpha^*) - \beta^*(a - \alpha)} \rho\right) e^{\frac{s|\beta|^2}{2}} \\ &\stackrel{(4.25)}{=} \frac{1}{\pi^2} \int d^2\beta \text{Tr}\left(D(\alpha)D(\beta)D^\dagger(\alpha)\rho\right) e^{\frac{s|\beta|^2}{2}} \\ &= \text{Tr}\left(\underbrace{\frac{1}{\pi^2} \int d^2\beta D(\beta) e^{\frac{s|\beta|^2}{2}} D^\dagger(\alpha)\rho D(\alpha)}_{T(s)}\right) \\ &= \text{Tr}\left(T(s)D^\dagger(\alpha)\rho D(\alpha)\right), \end{aligned} \quad (4.97)$$

where

$$T(s) = \frac{1}{\pi^2} \int d^2\beta D(\beta) e^{\frac{s|\beta|^2}{2}} \quad (4.98)$$

is Hermitian, $T(s) = T^\dagger(s)$. Accordingly, $W_\rho^s(\alpha)$ is the expectation value of $T(s)$ with respect to the displaced state $D^\dagger(\alpha)\rho D(\alpha)$. And since the expectation value of a Hermitian operator is always real, we see that $W_\rho^s(\alpha)$ is real.

For $s < 1$, the operator $T(s)$ can be nicely expressed in the Fock state basis. To see this, let us first plug $D(\beta)$ from Eq. (4.22) into (4.98),

$$\begin{aligned} T(s) &= \frac{1}{\pi^2} \int d^2\beta e^{\beta a^\dagger} e^{-\beta^* a} e^{\frac{(s-1)|\beta|^2}{2}} \\ &= \frac{1}{\pi^2} \int d^2\beta \sum_{m,n} (-1)^n \frac{\beta^m (a^\dagger)^m}{m!} \frac{(\beta^*)^n a^n}{n!} e^{\frac{(s-1)|\beta|^2}{2}}. \end{aligned} \quad (4.99)$$

Using polar coordinates, $\beta = re^{i\theta}$, with $d^2\beta = r dr d\theta$, this yields

$$\begin{aligned} T(s) &= \frac{1}{\pi^2} \sum_{m,n} \frac{(-1)^n (a^\dagger)^m a^n}{m!n!} \int_0^\infty dr r^{m+n+1} e^{\frac{(s-1)r^2}{2}} \underbrace{\int_0^{2\pi} d\theta e^{i\theta(m-n)}}_{2\pi\delta_{m,n}} \\ &= \frac{2}{\pi} \sum_n \frac{(-1)^n (a^\dagger)^n a^n}{(n!)^2} \int_0^\infty dr r^{2n+1} e^{\frac{(s-1)r^2}{2}}. \end{aligned} \quad (4.100)$$

For $s < 1$ the integral over r converges, yielding $2^n n! (1-s)^{-n-1}$, and we obtain

$$T(s) = \frac{2}{\pi} \sum_n \frac{(-2)^n (a^\dagger)^n a^n}{n!} (1-s)^{-n-1} = \frac{2}{\pi(1-s)} \sum_n \frac{(a^\dagger)^n a^n}{n!} \left(\frac{2}{s-1} \right)^n. \quad (4.101)$$

When acting with $T(s)$ on the Fock state $|m\rangle$, the sum truncates at $n = m$, and we get

$$\begin{aligned} T(s) |m\rangle &= \frac{2}{\pi(1-s)} \sum_{n=0}^m \frac{m!}{n!(m-n)!} \left(\frac{2}{s-1} \right)^n |m\rangle \\ &= \frac{2}{\pi(1-s)} \frac{1}{(s-1)^m} \underbrace{\sum_{n=0}^m \binom{m}{n} 2^n (s-1)^{m-n}}_{(2+s-1)^m} |m\rangle \\ &= \frac{2}{\pi(1-s)} \left(\frac{s+1}{s-1} \right)^m |m\rangle. \end{aligned} \quad (4.102)$$

We thus see that $T(s)$ is diagonal in the Fock state basis, and we can write, for $s < 1$,

$$T(s) = \frac{2}{\pi(1-s)} \sum_n \left(\frac{s+1}{s-1} \right)^n |n\rangle\langle n| = \frac{2}{\pi(1-s)} \left(\frac{s+1}{s-1} \right)^N, \quad (4.103)$$

with, in particular, $T(0) = \frac{2}{\pi}(-1)^N$, and $T(-1) = \frac{1}{\pi}|0\rangle\langle 0|$.

Review Questions 4.5.

- Why are there different orderings of creation and annihilation operators? How can we transform between different orderings? Transform $a^\dagger a a^\dagger$ into normal ordering.
- Why can characteristic functions fully characterize a single-mode quantum state ρ ?
- What is the optical equivalence theorem?
- Which properties have quasiprobability distributions?

4.5.4 Glauber-Sudarshan P-representation

It turns out that the diagonal matrix elements in the coherent state basis (remember that this basis is overcomplete) provides a complete description of the single-mode quantum state. To make this explicit, considering two operators, A and B , for which

$$\langle\alpha| A |\alpha\rangle = \langle\alpha| B |\alpha\rangle \quad \forall \alpha \in \mathbb{C}. \quad (4.104)$$

For $C = A - B$, we thus have,

$$0 = \langle\alpha| C |\alpha\rangle \stackrel{(4.16)}{=} \sum_{m,n} \frac{\langle m| C |n\rangle}{\sqrt{m!n!}} (\alpha^*)^m \alpha^n \quad \forall \alpha \in \mathbb{C}. \quad (4.105)$$

This can only hold for all $\alpha \in \mathbb{C}$ if $\langle m|C|n\rangle = 0$ for all $m, n \in \mathbb{N}_0$, such that $C = 0$, and, thus, $A = B$. Accordingly, any operator A is fully described by the diagonal matrix elements $\langle \alpha|A|\alpha\rangle$ in the coherent state basis.

Given that the diagonal matrix elements in the coherent state basis are sufficient for a complete description of a single-mode quantum state, it is suggestive to write the density operator as

$$\rho = \int d^2\alpha P(\alpha) |\alpha\rangle\langle\alpha|, \quad (4.106)$$

where $P(\alpha)$ is the so-called Glauber-Sudarshan P-representation. As we will show now, $P(\alpha)$ corresponds to the quasiprobability distribution for $s = 1$, i.e., $W_\rho^1(\alpha) = P(\alpha)$. To see this, consider the characteristic function for $s = 1$ from Eq. (4.83a),

$$\chi_\rho^1(\beta) = \text{Tr} \left(e^{\beta a^\dagger} e^{-\beta^* a} \rho \right) \quad (4.107)$$

$$\stackrel{(4.106)}{=} \int d^2\alpha P(\alpha) \langle \alpha| e^{\beta a^\dagger} e^{-\beta^* a} |\alpha\rangle \quad (4.108)$$

$$\stackrel{(4.14)}{=} \int d^2\alpha P(\alpha) e^{\beta\alpha^* - \beta^*\alpha}. \quad (4.109)$$

By comparing this with (4.90), we indeed see that $P(\alpha) = W_\rho^1(\alpha)$, which justifies that the density matrix representation (4.106) via the Glauber-Sudarshan P-representation provides a complete representation of the single-mode density operator ρ .

Note that if the field is in a quasi-classical state, such as a coherent state or a statistical mixture thereof, then $P(\alpha)$ is positive and no more singular than a Dirac delta function. For example, for a coherent state $\rho = |\alpha_0\rangle\langle\alpha_0|$,

$$P(\alpha) = \delta(\alpha - \alpha_0), \quad (4.110)$$

and for a thermal state ρ_β with average photon number $\bar{n}$,

$$P(\alpha) = \frac{1}{\pi\bar{n}} \exp\left(-\frac{|\alpha|^2}{\bar{n}}\right). \quad (4.111)$$

However, for non-classical states of light, such as Fock number states or squeezed states, $P(\alpha)$ is not a regular function – it becomes a highly singular distribution, e.g., involving derivatives of delta functions. In this case $P(\alpha)$ only makes sense as a distribution.

Note that since $P(\alpha)$ is not a regular function for non-classical states, it cannot straightforwardly be illustrated as a function of α (we can only understand it in a distributional sense). To plot $P(\alpha)$, one would have to regularize it, for example through the convolution with a Gaussian function. However, in (4.95) we have just seen that the quasiprobability distributions $W_\rho^s(\alpha)$ for $s < 1$ correspond to such convolutions, and are, thus, regularized versions of $P(\alpha) = W_\rho^1(\alpha)$. We thus proceed with studying $W_\rho^s(\alpha)$ for $s < 1$. These quasiprobability distributions will then be illustrated for some examples, including non-classical states such as Fock states and squeezed states.

4.5.5 Wigner function

The Wigner function For $s = 0$ (i.e., characteristic functions $\chi_\rho^0(\beta)$ which give rise to expectation values in Weyl ordering), the quasiprobability distribution is the well-known Wigner

function $W_\rho^0(\alpha) = W(\alpha)$. By inserting $T(0)$ from Eq. (4.98) into (4.97), we get

$$\begin{aligned} W(\alpha) &= \frac{2}{\pi} \text{Tr} \left((-1)^N D^\dagger(\alpha) \rho D(\alpha) \right) \\ &= \frac{2}{\pi} \text{Tr} \left(\sum_n (-1)^n |n\rangle \langle n| D^\dagger(\alpha) \rho D(\alpha) \right) \\ &= \frac{2}{\pi} \left(\sum_n \langle 2n| D^\dagger(\alpha) \rho D(\alpha) |2n\rangle - \sum_n \langle 2n+1| D^\dagger(\alpha) \rho D(\alpha) |2n+1\rangle \right). \end{aligned} \quad (4.112)$$

The expression in parentheses is simply the difference between the probability to measure an even number of photons versus an odd number of photons in the displaced state $D^\dagger(\alpha) \rho D(\alpha)$. As this difference can range from -1 to 1 , we see that $-2/\pi \leq W(\alpha) \leq 2/\pi$. The Wigner function can, thus, take negative values. This property is associated with a quantum mechanical character, since it stands in strong contrast to classical probability distributions, which can only take positive values. As an example, for Fock number states $|n\rangle$, the Wigner function at the origin is $W(0) = (-1)^n 2/\pi$ (compare with the plots provided in Sec. 4.5.7 below).

In order to write the Wigner function in terms of overlaps with coherent states, let us insert the completeness relation of coherent states, $\mathbb{1} = \frac{1}{\pi} \int d^2\beta |\beta\rangle \langle \beta|$, in the first line of (4.112), and use $(-1)^N |\beta\rangle = |-\beta\rangle$,

$$W(\alpha) = \frac{2}{\pi^2} \int d\beta^2 \langle -\beta| D^\dagger(\alpha) \rho D(\alpha) |\beta\rangle = \frac{2}{\pi^2} \int d\beta^2 \langle 0| D^\dagger(-\beta) D^\dagger(\alpha) \rho D(\alpha) D(\beta) |0\rangle. \quad (4.113)$$

From Eq. (4.24), we have $D(\alpha)D(\beta) = D(\alpha + \beta)e^{(\alpha\beta^* - \alpha^*\beta)/2}$ and $D^\dagger(-\beta)D^\dagger(\alpha) = D^\dagger(\alpha - \beta)e^{(-\beta\alpha^* + \beta^*\alpha)/2}$, such that the Wigner function becomes

$$W(\alpha) = \frac{2}{\pi^2} \int d\beta^2 \langle \alpha - \beta| \rho |\alpha + \beta\rangle e^{\alpha\beta^* - \alpha^*\beta}. \quad (4.114)$$

Rewriting the Wigner function $W(x, p)$ as a function of the phase-space variables x and p In the following, let us resolve the connection to the Wigner function as it is often presented in the quantum mechanics literature. To this end, we note that with $\alpha = x + ip$, the displacement operator $D(\alpha)$ can be written in terms of the quadrature operators $X = (a + a^\dagger)/2$ and $P = (a - a^\dagger)/2i$, which satisfy $[X, P] = i/2$,

$$D(x + ip) = e^{(x+ip)a^\dagger - (x-ip)a} = e^{2i(pX - xP)} \stackrel{(4.21)}{=} e^{2ipX} e^{-2ixP} e^{-ixp} \quad (4.115a)$$

$$= e^{-2ixP} e^{2ipX} e^{ixp}. \quad (4.115b)$$

Now, when taking the trace in (4.112) with respect to the eigenstates $|x\rangle$ of X , i.e., $X|x\rangle = x|x\rangle$, and using (4.115a), we get

$$W(x, p) = \frac{2}{\pi} \int dx'' \langle x''| (-1)^N e^{2ixP} e^{-2ipX} \rho e^{2ipX} e^{-2ixP} |x''\rangle. \quad (4.116)$$

Since quadrature operators always add or remove one photon from the field, they change the parity of the photon number. Accordingly, they satisfy $(-1)^N X (-1)^N = -X$ and $(-1)^N P (-1)^N = -P$, and, thus, $(-1)^N |x\rangle = |-x\rangle$. Using this together with $e^{-2ixP} |x''\rangle = |x'' + x\rangle$ in (4.116) gives

$$\begin{aligned} W(x, p) &= \frac{2}{\pi} \int dx'' \langle x - x''| e^{-2ipX} \rho e^{2ipX} |x + x''\rangle \\ &= \frac{2}{\pi} \int dx'' \langle x - x''| \rho |x + x''\rangle e^{4ipx''}. \end{aligned} \quad (4.117)$$

Finally, by substituting $x'' = x'/2$, we arrive at the common expression for the Wigner function,

$$W(x, p) = \frac{1}{\pi} \int dx' \left\langle x - \frac{x'}{2} \right| \rho \left| x + \frac{x'}{2} \right\rangle e^{2ipx'}. \quad (4.118)$$

Similarly, when taking the trace in (4.112) with respect to the eigenstates $|p\rangle$ of P , i.e., $P|p\rangle = p|p\rangle$, one finds

$$W(x, p) = \frac{1}{\pi} \int dp' \left\langle p - \frac{p'}{2} \right| \rho \left| p + \frac{p'}{2} \right\rangle e^{-2ip'x}. \quad (4.119)$$

Marginal distributions With the help of the above expressions for $W(x, p)$, we can now find the so-called marginal distribution on a quadrature, which is the probability distribution $W(x)$ for x and $W(p)$ for p , obtained by integrating $W(x, p)$ over p and over x , respectively. In particular, we find

$$W(x) = \int dp W(x, p) \stackrel{(4.118)}{=} \int dx' \left\langle x - \frac{x'}{2} \right| \rho \left| x + \frac{x'}{2} \right\rangle \underbrace{\frac{1}{\pi} \int dp e^{2ipx'}}_{\delta(x')} = \langle x | \rho | x \rangle, \quad (4.120)$$

and, similarly,

$$W(p) = \int dx W(x, p) \stackrel{(4.119)}{=} \int dp' \left\langle p - \frac{p'}{2} \right| \rho \left| p + \frac{p'}{2} \right\rangle \underbrace{\frac{1}{\pi} \int dx e^{-2ip'x}}_{\delta(p')} = \langle p | \rho | p \rangle. \quad (4.121)$$

Note that while the Wigner function $W(x, p)$ can take negative values, $W(x)$ and $W(p)$ are necessarily positive, they describe probability distributions.

4.5.6 Husimi Q-function

The quasiprobability function corresponding to the characteristic function $\chi_\rho^{-1}(\beta)$ for antinormal ordering, $s = -1$, is known as the Husimi Q-function, and denoted by $W_\rho^{-1}(\alpha) = Q(\alpha)$. Using $T(-1) = \frac{1}{\pi}|0\rangle\langle 0|$ from (4.98) in Eq. (4.97) and $D(\alpha)|0\rangle = |\alpha\rangle$, we find that

$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \rho | \alpha \rangle, \quad (4.122)$$

which are the diagonal matrix elements in the coherent state basis, with $0 \leq Q(\alpha) \leq 1/\pi$.

By the fact that a single-mode quantum state can equivalently well be described by the density operator ρ or by the quasiprobability distribution $W_\rho^s(\alpha)$ (recall that $W_\rho^s(\alpha)$ gives rise to the expectation values of all possible products of creation and annihilation operators), Eq. (4.122) implies once more that a density matrix ρ is fully determined by the diagonal elements in the coherent state basis.

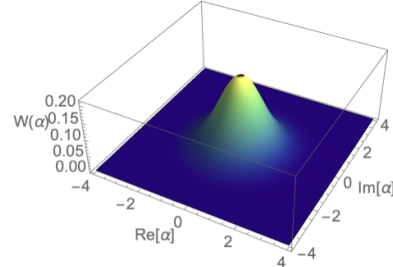
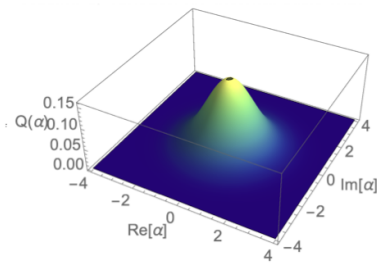
Review Questions 4.6.

- Why is (4.106) not a spectral decomposition of ρ ?
- Why cannot the Glauber-Sudarshan P -representation be drawn without problems?
- Can the Wigner function be seen as a regularization of the Glauber-Sudarshan P -representation?
- What are the marginal distributions of the Wigner function, and why are they important?
- What is the Husimi Q -function $Q(\alpha)$ for the coherent state $|\beta\rangle$? How does it simplify when restricting to $\alpha, \beta \in \mathbb{R}$?

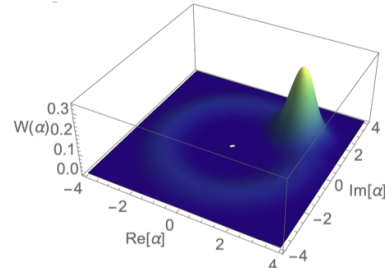
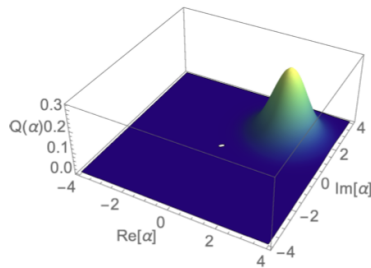
4.5.7 Illustration of some examples

Let us plot the Husimi Q -function (left) and the Wigner function (right) for various states:

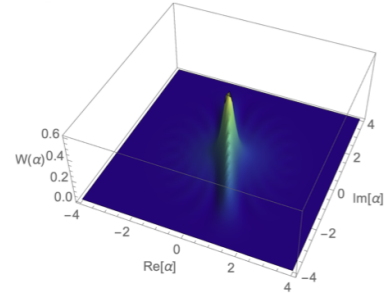
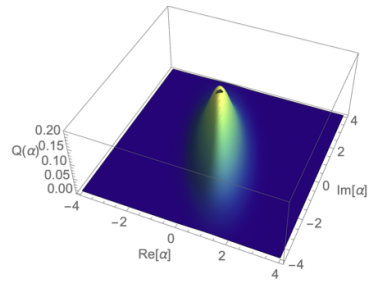
- The thermal state ρ_β with $\bar{n} = 1$:



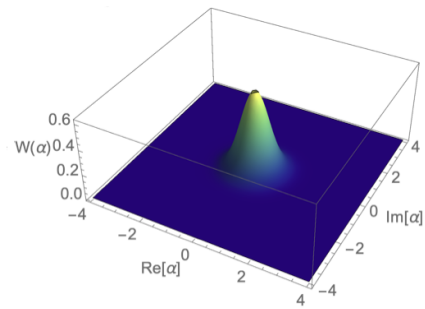
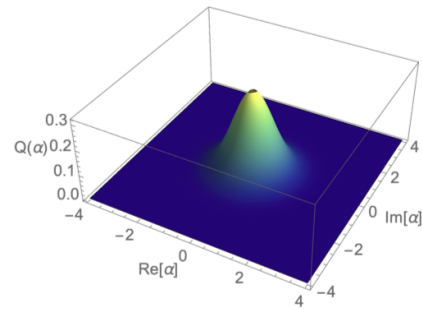
- The coherent state $|\alpha\rangle$ with $\alpha = 2 + i2$ (the ring in the Wigner function is a numerical artefact):



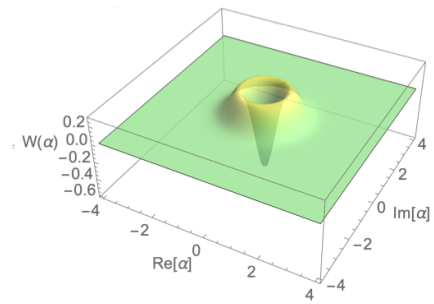
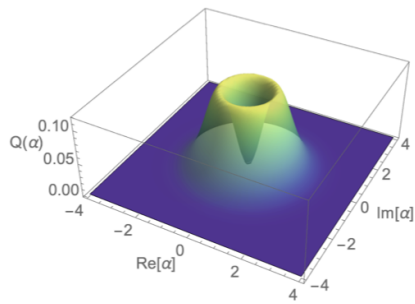
- The squeezed vacuum state $|\xi\rangle$ from Eq. (4.62) with $\xi = re^{i\theta}$, $r = 1$, $\theta = \frac{\pi}{4}$:



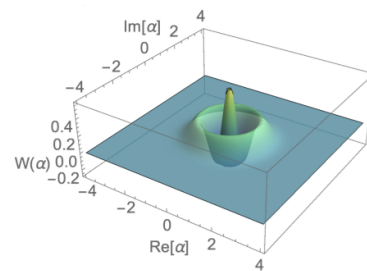
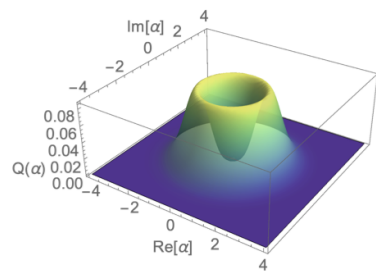
- The vacuum state $|0\rangle$:



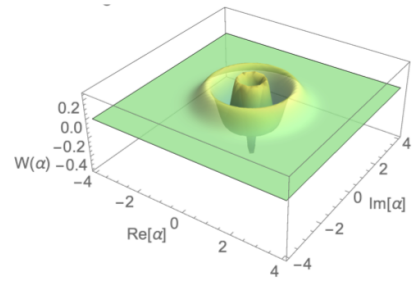
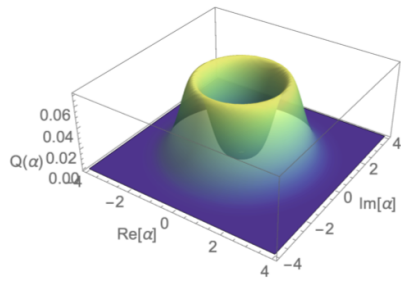
- The Fock number state $|1\rangle$:



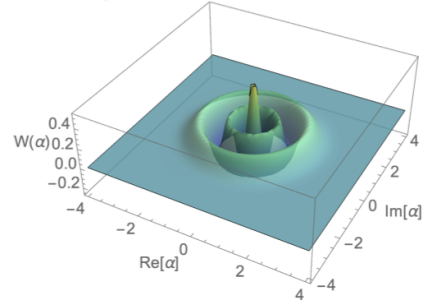
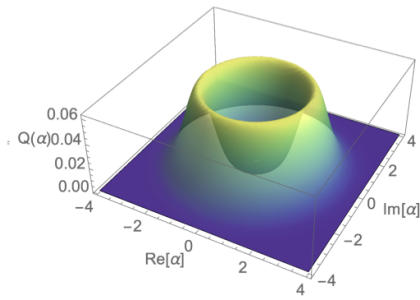
- The Fock number state $|2\rangle$:



- The Fock number state $|3\rangle$:



- The Fock number state $|4\rangle$:



Review Questions 4.7.

- *Explain the properties of the above illustrated states.*

Chapter 5

Linear optics

5.1 Single-photon states

Above we discussed states of a single optical mode, which (by their description in the Fock basis) are composed of different photon number states, with all photons living in the same mode. While such states are central to quantum optics, many quantum optical experiments deal with a single photon only (or a well defined number of photons). We therefore turn now to the description of a single photon, and allow the photon to be spread across multiple modes.

States of single photons Let us consider the single photon state $|\kappa\rangle = a_\kappa^\dagger |\Omega\rangle$, with $|\Omega\rangle = |0\rangle \otimes \cdots \otimes |0\rangle$ the multi-mode vacuum state. It describes a single excitation of the field mode κ and lives in the Fock space $\mathcal{H}_{\text{rad}} = \bigotimes_\kappa \mathcal{H}_\kappa$. The set of all possible single-photon states $\{|\kappa\rangle\}_\kappa$ spans the subspace of the Fock space with exactly one field excitation, which is therefore called the single-photon Hilbert space $\mathcal{H}_1$. A state living in $\mathcal{H}_1$, i.e., a single-photon state, is thus a superposition of the basis states of $\mathcal{H}_1$ (i.e., a single photon spread over multiple modes),

$$|\psi\rangle = \sum_\kappa \psi_\kappa |\kappa\rangle = \sum_\kappa \psi_\kappa a_\kappa^\dagger |\Omega\rangle, \quad (5.1)$$

where $\psi_\kappa \in \mathbb{C}$ is the amplitude of the single photon to be in mode κ , and $\sum_\kappa |\psi_\kappa|^2 = 1$. Since the state (5.1) describes a single photon in a *coherent* superposition over several modes, we can define the creation and annihilation operators associated with this photon,

$$a_{|\psi\rangle}^\dagger = \sum_\kappa \psi_\kappa a_\kappa^\dagger, \quad \text{and} \quad a_{|\psi\rangle} = \sum_\kappa \psi_\kappa^* a_\kappa. \quad (5.2)$$

Their action creates and annihilates a photon in state $|\psi\rangle$, respectively,

$$a_{|\psi\rangle}^\dagger |\Omega\rangle = |\psi\rangle, \quad \text{and} \quad a_{|\psi\rangle} |\psi\rangle = |\Omega\rangle. \quad (5.3)$$

For two arbitrary single-photon states $|\psi\rangle$ and $|\phi\rangle$, the commutator

$$[a_{|\phi\rangle}, a_{|\psi\rangle}^\dagger] \stackrel{(5.2)}{=} \sum_{\kappa, \kappa'} \phi_\kappa^* \psi_{\kappa'} \underbrace{[a_\kappa, a_{\kappa'}^\dagger]}_{\delta_{\kappa, \kappa'}} = \sum_\kappa \phi_\kappa^* \psi_\kappa = \langle \phi | \psi \rangle \quad (5.4)$$

reflects the orthogonality of the states.

Single photons in the free field In (quantum) optics, one often works with fields that are localized in space and propagate along a given direction (think of a laser beam or the wavepacket of a single photon traveling in a certain direction). To this end, let us describe the

allowed directions and moduli of the corresponding wave-vectors of such a field by the transverse vector field $\mathbf{w}(\mathbf{k})$ obeying the normalization condition

$$\int d^3k \mathbf{w}^*(\mathbf{k}) \cdot \mathbf{w}(\mathbf{k}) = 1. \quad (5.5)$$

In particular, let us use the transverse vector field $\mathbf{w}(\mathbf{k})$ to single out the directions of the allowed wave vectors via the dot product $\mathbf{e}_{\mathbf{k},p}^* \cdot \mathbf{w}(\mathbf{k})$, where $\mathbf{e}_{\mathbf{k},p}$ is the unit vector of the mode with wave vector $\mathbf{k}$ and polarization p , and to single out the spectrum of the allowed wave vectors via the amplitudes $|\mathbf{w}(\mathbf{k})|^2$. In analogy to (5.2), we can then define the creation operator associated with the transverse vector field $\mathbf{w}(\mathbf{k})$:

$$a_w^\dagger = \sum_p \int d^3k \mathbf{e}_{\mathbf{k},p}^* \cdot \mathbf{w}(\mathbf{k}) a_p^\dagger(\mathbf{k}). \quad (5.6)$$

Note that since the width of the frequency distribution fixes the spatial extension of the wave along the beam direction, quasi-monochromatic single photons have an extended spatial extension, while photons with a wide frequency distribution result in a localized wavepacket.

Review Questions 5.1.

- What is the single-photon Hilbert space, and what's its span?
- How can we describe optical modes corresponding to light that is localized in space and travels in a certain direction? Do you know examples?

5.2 Single-photon unitary transformations

Creation and annihilation operators under unitary transformations Let us consider now a unitary transformation u acting on the single-particle Hilbert space $\mathcal{H}_1$, which we suppose to be spanned by the discrete basis $\{|\kappa\rangle\}_\kappa$. Since unitary transformations preserve the inner product [see Eq. (1.21)], $\{u|\kappa\rangle\}_\kappa$ defines a new basis, with basis vectors

$$u|\kappa\rangle = \underbrace{\sum_{\kappa'} |\kappa'\rangle \langle \kappa'| u |\kappa\rangle}_{\mathbb{1}} = \sum_{\kappa'} \underbrace{\langle \kappa'| u |\kappa\rangle}_{u_{\kappa',\kappa}} |\kappa'\rangle = \underbrace{\sum_{\kappa'} u_{\kappa',\kappa} a_{\kappa'}^\dagger |\Omega\rangle}_{b_\kappa^\dagger} = b_\kappa^\dagger |\Omega\rangle. \quad (5.7)$$

In the new basis, the corresponding creation and annihilation operators are, thus,

$$b_\kappa^\dagger = a_{u|\kappa}^\dagger = \sum_{\kappa'} u_{\kappa',\kappa} a_{\kappa'}^\dagger \quad \text{and} \quad b_\kappa = a_{u|\kappa} = \sum_{\kappa'} u_{\kappa',\kappa}^* a_{\kappa'}, \quad (5.8)$$

which, by (5.4) and the orthogonality $\langle \kappa| u^\dagger u |\kappa'\rangle = \langle \kappa|\kappa'\rangle = \delta_{\kappa,\kappa'}$, obey the canonical commutation relation

$$[b_\kappa, b_{\kappa'}^\dagger] = \delta_{\kappa,\kappa'}. \quad (5.9)$$

Note that similar to Eq. (5.8), we can express the operators in the old basis in terms of the operators in the new basis through the inverse unitary transformation $u^\dagger$,

$$a_\kappa^\dagger = \sum_{\kappa'} u_{\kappa',\kappa}^\dagger b_{\kappa'}^\dagger \quad \text{and} \quad a_\kappa = \sum_{\kappa'} (u_{\kappa',\kappa}^\dagger)^* b_{\kappa'}. \quad (5.10)$$

If you don't see this, show this by inserting (5.8) and using that u is unitary.

It is interesting to note that unitary transformations, which transform modes κ to modes κ' with equal frequency, i.e., $\omega_\kappa = \omega_{\kappa'}$, simply correspond to a different choice of the normal-mode functions. To see this, consider the vector field potential from Eq. (3.8),

$$\mathbf{A}^+(\mathbf{r}) = \sum_{\kappa} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} \mathbf{u}_{\kappa}(\mathbf{r}) a_{\kappa} \stackrel{(5.10)}{=} \sum_{\kappa, \kappa'} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} (u_{\kappa', \kappa}^{\dagger})^* \mathbf{u}_{\kappa}(\mathbf{r}) b_{\kappa'}. \quad (5.11)$$

Using that $\omega_{\kappa} = \omega_{\kappa'}$ and $(u_{\kappa', \kappa}^{\dagger})^* = u_{\kappa, \kappa'}$ then gives

$$\mathbf{A}^+(\mathbf{r}) = \sum_{\kappa'} \sqrt{\frac{\hbar}{2\omega_{\kappa'}\epsilon_0}} \underbrace{\left(\sum_{\kappa} u_{\kappa, \kappa'} \mathbf{u}_{\kappa}(\mathbf{r}) \right)}_{\mathbf{v}_{\kappa'}(\mathbf{r})} b_{\kappa'} = \sum_{\kappa'} \sqrt{\frac{\hbar}{2\omega_{\kappa'}\epsilon_0}} \mathbf{v}_{\kappa'}(\mathbf{r}) b_{\kappa'}, \quad (5.12)$$

where we introduced the transformed normal-mode functions

$$\mathbf{v}_{\kappa'}(\mathbf{r}) = \sum_{\kappa} u_{\kappa, \kappa'} \mathbf{u}_{\kappa}(\mathbf{r}). \quad (5.13)$$

That is, under the action of such a single-photon unitary transformation that doesn't mix different frequencies, the fields keep the same form. The same can be shown for the Hamiltonian (3.11):

$$\mathbf{A}^+(\mathbf{r}) = \sum_{\kappa} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} \mathbf{u}_{\kappa}(\mathbf{r}) a_{\kappa} = \sum_{\kappa} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} \mathbf{v}_{\kappa}(\mathbf{r}) b_{\kappa}, \quad (5.14a)$$

$$H_{\perp} = \sum_{\kappa} \hbar\omega_{\kappa} \left(a_{\kappa}^{\dagger} a_{\kappa} + \frac{1}{2} \right) = \sum_{\kappa} \hbar\omega_{\kappa} \left(b_{\kappa}^{\dagger} b_{\kappa} + \frac{1}{2} \right). \quad (5.14b)$$

Dynamics generated by a single-particle Hamiltonian In general, unitary transformations are generated by time evolution under a Hamiltonian. In order to generate a single-photon unitary which sends single-photon states to single-photon states, the Hamiltonian is only allowed to contain terms composed of a single creation and a single annihilation operator (which defines a single-particle operator),

$$H = \sum_{\kappa, \kappa'} h_{\kappa\kappa'} a_{\kappa}^{\dagger} a_{\kappa'}. \quad (5.15)$$

The dynamics generated by such a single-photon Hamiltonian is called linear, since creation and annihilation operators appear linear, respectively, and obey linear equations of motion in the Heisenberg picture. In particular, using $a_{\kappa}(t) = U^{\dagger}(t) a_{\kappa} U(t)$ in the Heisenberg picture, with $U(t) = \exp(-iH(t - t_0)/\hbar)$, the equation of motion for $a_{\kappa}(t)$ is

$$i\hbar \frac{d}{dt} a_{\kappa}(t) = [a_{\kappa}(t), H] = \sum_{\kappa'', \kappa'} h_{\kappa''\kappa'} U^{\dagger}(t) \underbrace{\left[a_{\kappa}, a_{\kappa''}^{\dagger} a_{\kappa'} \right]}_{\delta_{\kappa, \kappa''} a_{\kappa'}} U(t) = \sum_{\kappa'} h_{\kappa\kappa'} a_{\kappa'}(t), \quad (5.16)$$

which integrates to

$$a_{\kappa}(t) = \sum_{\kappa'} u_{\kappa\kappa'}(t) a_{\kappa'}, \quad \Rightarrow \quad a_{\kappa}^{\dagger}(t) = \sum_{\kappa'} u_{\kappa\kappa'}^*(t) a_{\kappa'}^{\dagger}, \quad (5.17)$$

where $u(t) = \exp(-iht/\hbar)$, and h has elements $h_{\kappa\kappa'}$. Using the above notation from (5.2), for any single-photon state $|\psi\rangle$, we can thus write

$$a_{|\psi\rangle}(t) = a_{u^{\dagger}|\psi\rangle}, \quad \text{and} \quad a_{|\psi\rangle}^{\dagger}(t) = a_{u^{\dagger}|\psi\rangle}^{\dagger}. \quad (5.18)$$

For completeness, let us also consider the time evolution in the Schrödinger picture. Here, we consider the evolution of any initial state $|\Psi\rangle$ (note that $|\Psi\rangle$ describes any pure state of, in general, many photons distributed across multiple modes), which we express through an arbitrary function $F(\{a_\kappa, a_\kappa^\dagger\})$ of creation and annihilation operators acting on the vacuum,

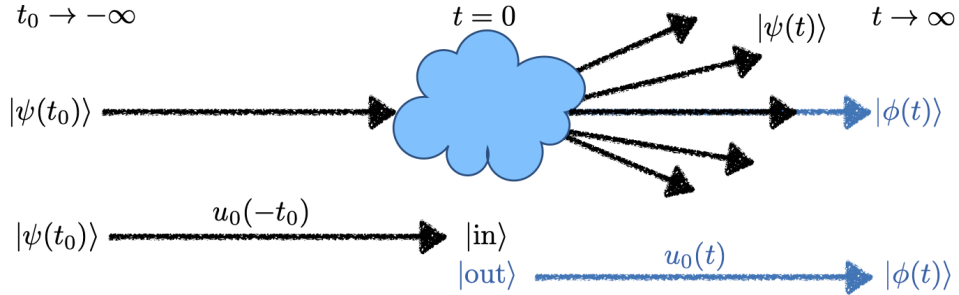
$$|\Psi\rangle = F(\{a_\kappa, a_\kappa^\dagger\}) |\Omega\rangle. \quad (5.19)$$

For its time evolution generated by a single-particle Hamiltonian (5.15), we have

$$\begin{aligned} |\Psi(t)\rangle &= U(t) |\Psi\rangle \\ &= U(t) F(\{a_\kappa, a_\kappa^\dagger\}) U^\dagger(t) \underbrace{U(t) |\Omega\rangle}_{|\Omega\rangle} \\ &= F\left(\left\{U(t)a_\kappa U^\dagger(t), U(t)a_\kappa^\dagger U^\dagger(t)\right\}\right) |\Omega\rangle \\ &\stackrel{(5.18)}{=} F\left(\left\{a_{u(t)|\kappa}, a_{u(t)|\kappa}^\dagger\right\}\right) |\Omega\rangle \\ &\stackrel{(5.8)}{=} F(\{b_\kappa(t), b_\kappa^\dagger(t)\}) |\Omega\rangle. \end{aligned} \quad (5.20)$$

We thus see, that $|\Psi(t)\rangle$ can be obtained by evolving the single-particle states $|\kappa\rangle$ individually via $u(t)|\kappa\rangle$, resulting in the evolved creation and annihilation operators $\{b_\kappa(t), b_\kappa^\dagger(t)\}$, which then replace the initial ones. Linear dynamics thus corresponds to a time-dependent change of the single-particle basis.

Scattering matrix In (quantum) optics, one often deals with scattering scenarios where the electromagnetic field interacts with objects that are localized in space, such as a beam splitter or phase shifter. In this case, the field evolves freely, except within the region of space of the optical element. If one is not interested in the time-dependent evolution during the interaction process, it suffices to consider the relation between the ingoing and outgoing freely propagating waves, as typical for a scattering scenario. In the following we study such scattering scenarios:



As shown in the illustration, let us consider a single photon that is initially, i.e., long before its interaction with the object, at time $t_0 \rightarrow -\infty$, in the state $|\psi(t_0)\rangle$. We are interested in the transition of the single photon to the single-photon state $|\phi(t)\rangle$ at time $t \rightarrow \infty$, long after its interaction with the object. That is, we want to find the transition amplitude $\langle\phi(t)|u(t-t_0)|\psi(t_0)\rangle$, where u is the corresponding unitary time evolution operator that describes the interaction of the field with the object. Note that we assume a linear interaction with the object, such that u is a single-particle unitary that maps single-photon states to single-photon states.

First, we want to get rid of the time-dependence in the transition amplitude. To this end, let us define the input reference state $|\text{in}\rangle$ as the state to which $|\psi(t_0)\rangle$ would evolve to at time $t=0$, if there was no interaction. Hence, $|\text{in}\rangle = u_0(-t_0)|\psi(t_0)\rangle$, with $u_0(t)$ the single-particle unitary corresponding to a free propagation. We thus have

$$|\psi(t_0)\rangle = u_0(t_0) |\text{in}\rangle, \quad \text{with} \quad |\text{in}\rangle = a_{|\text{in}}^\dagger |\Omega\rangle. \quad (5.21)$$

Similarly, we define the time-independent reference state for the final state $|\phi(t)\rangle$ as the state $|\text{out}\rangle = u_0(-t) |\phi(t)\rangle$ from which $|\phi(t)\rangle$ would have originated if there was no interaction. We thus have

$$|\phi(t)\rangle = u_0(t) |\text{out}\rangle, \quad \text{with} \quad |\text{out}\rangle = a_{|\text{out}\rangle}^\dagger |\Omega\rangle. \quad (5.22)$$

The transition amplitude therefore becomes

$$\begin{aligned} \langle\phi(t)|\psi(t)\rangle &= \langle\phi(t)| u(t-t_0) |\psi(t_0)\rangle \\ &= \langle\text{out}| u_0^\dagger(t) u(t-t_0) u_0(t_0) |\text{in}\rangle \\ &= \langle\text{out}| \underbrace{u_0^\dagger(t) u(t-t_0) u^\dagger(t_0) u_0(t_0)}_s |\text{in}\rangle. \end{aligned} \quad (5.23)$$

We therefore obtain

$$\boxed{\langle\phi(t)|\psi(t)\rangle = \langle\text{out}| s |\text{in}\rangle}, \quad (5.24)$$

which defines the scattering matrix

$$s = \lim_{\substack{t_0 \rightarrow -\infty \\ t \rightarrow \infty}} u_0^\dagger(t) u(t) u^\dagger(t_0) u_0(t_0), \quad (5.25)$$

which relates the input and output reference states. These input/output reference states are referred to as input/output modes or channels. Given that we consider the limits $t \rightarrow \infty$ and $t_0 \rightarrow -\infty$, we can set $t_0 = -t$, and write the scattering matrix as

$$s = \lim_{t \rightarrow \infty} u_{\text{int}}(t) u_{\text{int}}^\dagger(-t), \quad (5.26)$$

where $u_{\text{int}}(t) = u_0^\dagger(t) u(t)$. Note that the scattering matrix is unitary. We say that it describes a passive transformation if the interaction does not change the frequency of the incoming field (think of a beam splitter or phase shifter). We will discuss some passive transformation below.

In order to translate the above described single-photon scattering scenario to many-photon states, we can adapt our considerations in the Schrödinger picture from Eq. (5.20): For a set of input channels $\{|\text{in}_j\rangle\}_j$, the initial state at time $t_0 \rightarrow -\infty$ is then

$$|\Psi(t_0)\rangle = U_0(t_0) |\Psi_{\text{in}}\rangle, \quad \text{with} \quad |\Psi_{\text{in}}\rangle = F(a_{|\text{in}_j\rangle}, a_{|\text{in}_j\rangle}^\dagger) |\Omega\rangle, \quad (5.27)$$

and, since the scattering scenario corresponds to the mapping $|\text{in}_j\rangle \rightarrow s |\text{in}_j\rangle$, the final state at time $t \rightarrow \infty$ is

$$|\Psi(t)\rangle = U_0(t) |\Psi_{\text{out}}\rangle, \quad \text{with} \quad |\Psi_{\text{out}}\rangle = F(a_{s|\text{in}_j\rangle}, a_{s|\text{in}_j\rangle}^\dagger) |\Omega\rangle. \quad (5.28)$$

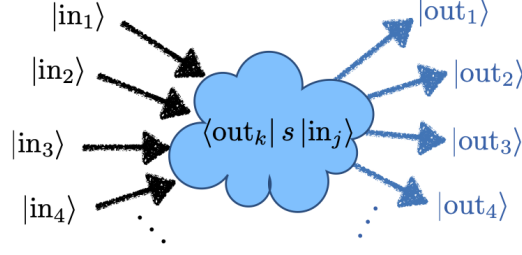
If $\{|\text{out}_k\rangle\}_k$ constitutes a complete set of output channels, then the annihilation operators of the input channels are mapped to those of the output by

$$a_{|\text{in}_j\rangle}^\dagger \rightarrow a_{s|\text{in}_j}^\dagger = \sum_k \langle\text{out}_k| s |\text{in}_j\rangle a_{|\text{out}_k\rangle}^\dagger. \quad (5.29)$$

In the following, to ease notation, we set $a_{|\text{in}_j\rangle}^\dagger = a_j^\dagger$, $a_{|\text{out}_k\rangle}^\dagger = b_k^\dagger$, and $\langle\text{out}_k| s |\text{in}_j\rangle = s_{kj}$, such that (5.29) becomes

$$\boxed{a_j^\dagger \rightarrow \sum_k s_{kj} b_k^\dagger}. \quad (5.30)$$

The multi-mode scattering scenarios can be illustrated as follows:

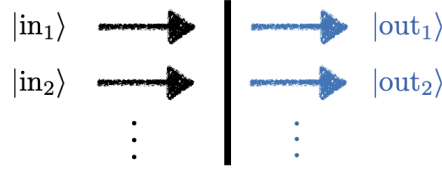


Review Questions 5.2.

- Show that Eq. (5.10) is true.
- What is the form of a single-particle operator? Why? What is the form of a n -particle operator?
- What does the scattering matrix describe?
- What is a passive transformation?
- How does the state $a_1^\dagger a_2^\dagger |\Omega\rangle$ transform in the multi-mode scattering scenario?

5.3 The phase shifter

Possibly the simplest optical element is the phase shifter.



It simply changes the phase of each mode. The scattering matrix acts as

$$s |\text{in}_j\rangle = e^{i\varphi_j} |\text{out}_j\rangle, \quad (5.31)$$

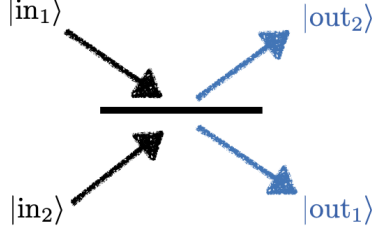
with $\varphi_j \in \mathbb{R}$, and is therefore diagonal,

$$s = \begin{pmatrix} e^{i\varphi_1} & 0 & \dots \\ 0 & e^{i\varphi_2} & \\ \vdots & & \ddots \end{pmatrix}. \quad (5.32)$$

Typical examples of phase shifters are wave plates. For example, a phase shifter with scattering matrix $s = \text{diag}(1, -1)$ acts on the single-photon states $|H\rangle = \frac{1}{\sqrt{2}}(|\text{in}_1\rangle + |\text{in}_2\rangle)$ and $|V\rangle = \frac{1}{\sqrt{2}}(|\text{in}_1\rangle - |\text{in}_2\rangle)$ as $|H\rangle \rightarrow |V\rangle$ and $|V\rangle \rightarrow |H\rangle$. Associating $|H\rangle$ and $|V\rangle$ with a horizontally and vertically polarized photon, respectively, this action corresponds to the action of a half-wave plate at an angle of 45° with respect to the linearly polarized photon.

5.4 The beam splitter

Scattering matrix of the beam splitter The beam splitter is one of the most central elements in (quantum) optical experiments. It corresponds to a partially reflective mirror that transmits and reflects some portion of the incoming light.



In particular, when light is incident in mode $|\text{in}_1\rangle$, it can either be transmitted to mode $|\text{out}_1\rangle$ or reflected to mode $|\text{out}_2\rangle$. The corresponding scattering matrix must therefore act as

$$s |\text{in}_1\rangle = t_1 |\text{out}_1\rangle + r_1 |\text{out}_2\rangle, \quad (5.33)$$

with t_1 and $r_1 \in \mathbb{C}$ the corresponding amplitudes for transmission and reflection. The same output modes can also be reached by a second input mode $|\text{in}_2\rangle$ from the other side of the beam splitter. Incoming light in mode $|\text{in}_2\rangle$ will then leave the beam splitter in mode $|\text{out}_1\rangle$ in the case of reflection, and in mode $|\text{out}_2\rangle$ in the case of transmission, i.e.,

$$s |\text{in}_2\rangle = r_2 |\text{out}_1\rangle + t_2 |\text{out}_2\rangle, \quad (5.34)$$

with $t_2, r_2 \in \mathbb{C}$. The creation operators of the input and output modes therefore transform as

$$\begin{pmatrix} a_1^\dagger \\ a_2^\dagger \end{pmatrix} \rightarrow \begin{pmatrix} t_1 b_1^\dagger + r_1 b_2^\dagger \\ r_2 b_1^\dagger + t_2 b_2^\dagger \end{pmatrix} = \underbrace{\begin{pmatrix} t_1 & r_1 \\ r_2 & t_2 \end{pmatrix}}_{\tilde{s}} \begin{pmatrix} b_1^\dagger \\ b_2^\dagger \end{pmatrix}. \quad (5.35)$$

Note that while a beam splitter can possibly have multiple input and output modes, it always connects two input modes with two output modes. Accordingly, the matrix $\tilde{s}$ in (5.35) corresponds to a submatrix of the corresponding scattering matrix s . If the beam splitter is lossless, then $\tilde{s}$ is unitary, such that its rows (and columns) must be orthonormal, imposing the constraints

$$|t_1|^2 + |r_1|^2 = 1, \quad |t_1|^2 + |r_2|^2 = 1 \quad (5.36a)$$

$$|r_2|^2 + |t_2|^2 = 1, \quad |r_1|^2 + |t_2|^2 = 1 \quad (5.36b)$$

$$r_2 t_1^* + t_2 r_1^* = 0, \quad t_1 r_1^* + r_2 t_2^* = 0. \quad (5.36c)$$

We can therefore express $\tilde{s}$ in terms of four real parameters $\alpha, \beta, \gamma, \delta$,

$$\tilde{s} = \begin{pmatrix} t_1 & r_1 \\ r_2 & t_2 \end{pmatrix} = \begin{pmatrix} e^{i(\alpha-\beta/2-\delta/2)} \cos(\gamma/2) & -e^{i(\alpha-\beta/2+\delta/2)} \sin(\gamma/2) \\ e^{i(\alpha+\beta/2-\delta/2)} \sin(\gamma/2) & e^{i(\alpha+\beta/2+\delta/2)} \cos(\gamma/2) \end{pmatrix}. \quad (5.37)$$

Note that the unitarity of $\tilde{s}$ ensure that the commutation relations $[a_j, a_k^\dagger] = \delta_{j,k}$ from the input modes are correctly transferred to the output modes, $[b_j, b_k^\dagger] = \delta_{j,k}$. For example,

$$\begin{aligned} [a_1, a_1^\dagger] &\rightarrow [t_1^* b_1 + r_1^* b_2, t_1 b_1^\dagger + r_1 b_2^\dagger] = |t_1|^2 + |r_1|^2 = 1, \\ [a_1, a_2^\dagger] &\rightarrow [t_1^* b_1 + r_1^* b_2, r_2 b_1^\dagger + t_2 b_2^\dagger] = r_2 t_1^* + t_2 r_1^* = 0. \end{aligned} \quad (5.38)$$

Finally note that the general properties of the beam splitter (i.e, the probabilities for reflection and transmission, and unitarity) do not depend on the choice of the phases, since the phases are simply a result of local phase operations in the input and output modes:

$$\tilde{s} = e^{i\alpha} \begin{pmatrix} e^{-i\beta/2} & 0 \\ 0 & e^{i\beta/2} \end{pmatrix} \begin{pmatrix} \cos(\gamma/2) & -\sin(\gamma/2) \\ \sin(\gamma/2) & \cos(\gamma/2) \end{pmatrix} \begin{pmatrix} e^{-i\delta/2} & 0 \\ 0 & e^{i\delta/2} \end{pmatrix}. \quad (5.39)$$

Common choices of the phases are

$$\alpha = \frac{\pi}{2}, \beta = 0, \delta = \pi \quad \tilde{s} = \begin{pmatrix} t & r \\ r & -t \end{pmatrix} \quad (5.40a)$$

$$\alpha = 0, \beta = \frac{\pi}{2}, \delta = -\frac{\pi}{2} \quad \tilde{s} = \begin{pmatrix} t & ir \\ ir & t \end{pmatrix}, \quad (5.40b)$$

where $t = \cos(\gamma/2)$ and $r = \sin(\gamma/2)$.

Photon indivisibility Let us see what the beam splitter does to a single photon. For a single-photon in the first input mode, $|\psi_{\text{in}}\rangle = a_1^\dagger |\Omega\rangle$, we obtain, by Eq. (5.30), $|\psi_{\text{out}}\rangle = (t_1 b_1^\dagger + r_1 b_2^\dagger) |\Omega\rangle$. A measurement of the photon numbers $N_j = b_j^\dagger b_j$ in the output modes then gives the expectation values

$$\langle N_1 \rangle = |t_1|^2, \quad \text{and} \quad \langle N_2 \rangle = |r_1|^2. \quad (5.41)$$

That is, the field mode $|\text{out}_1\rangle$ will be excited with probability $|t_1|^2$, and the field mode $|\text{out}_2\rangle$ with probability $|r_1|^2$. However, since

$$\langle N_1 N_2 \rangle = 0, \quad (5.42)$$

both modes will never be excited at the same time. Therefore, a single photon can never be detected in both outputs simultaneously. This property supports a corpuscular interpretation of a single photon (remember that, when we introduced Fock number states $|n\rangle$, we interpreted n as the number of excitations in the electromagnetic field, and called these excitations photons), which is in stark contrast to the behavior of classical waves. To see this, consider a coherent state in the first input mode,

$$|\Psi_{\text{in}}\rangle = \exp(\alpha a_1^\dagger - \alpha^* a_1) |\Omega\rangle = |\alpha\rangle_1, \quad (5.43)$$

for which we get the output

$$\begin{aligned} |\Psi_{\text{out}}\rangle &= \exp(\alpha(t_1 b_1^\dagger + r_1 b_2^\dagger) - \alpha^*(t_1^* b_1 + r_1^* b_2)) |\Omega\rangle \\ &= \exp(\alpha t_1 b_1^\dagger - \alpha^* t_1^* b_1) \exp(\alpha r_1 b_2^\dagger - \alpha^* r_1^* b_2) |\Omega\rangle \\ &= |\alpha t_1\rangle_1 |\alpha r_1\rangle_2, \end{aligned} \quad (5.44)$$

which is a coherent state in each of the modes. One then finds [recall Eq. (4.18)]

$$\langle N_1 \rangle = |\alpha t_1|^2, \quad \langle N_2 \rangle = |\alpha r_1|^2, \quad (5.45)$$

and

$$\langle N_1 N_2 \rangle = \langle N_1 \rangle \langle N_2 \rangle \neq 0, \quad (5.46)$$

showing that the coherent state got split into two uncorrelated coherent states, with a non-vanishing probability to detect an excited field in both output modes simultaneously. This is in agreement with a wave-like behavior.

Review Questions 5.3.

- Do you know optical elements that act as a phase shifter? Explain briefly?
- What is the scattering matrix of a beam splitter?
- Can the scattering matrix (5.37) describe any 2×2 unitary matrix? Why?
- Can a beam splitter divide a single field excitation?

Homodyne detection The beam splitter can be used to measure the expectation value of the quadrature operators for an unknown single-mode state $|\Psi\rangle$. The measurement protocol is called homodyne detection and is based on the interference of $|\Psi\rangle$ with a coherent state $|\alpha\rangle$ (often called the local oscillator) on the beam splitter. In particular, consider the input state (we denote by $|\Phi_j\rangle$ a state $|\Phi\rangle$ in mode j)

$$|\Psi_{\text{in}}\rangle = |\Psi_1\rangle |\alpha_2\rangle \quad (5.47)$$

to a balanced beam splitter (often called a 50/50 beam splitter) with scattering matrix

$$\tilde{s} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (5.48)$$

The creation and annihilation operators at the output are then $b_1^\dagger = (a_1^\dagger + a_2^\dagger)/\sqrt{2}$ and $b_2^\dagger = (a_1^\dagger - a_2^\dagger)/\sqrt{2}$. By measuring the difference between the intensities at the output, we thus get

$$\begin{aligned} D &= \langle b_1^\dagger b_1 - b_2^\dagger b_2 \rangle \\ &= \frac{1}{2} \langle (a_1^\dagger + a_2^\dagger)(a_1 + a_2) - (a_1^\dagger - a_2^\dagger)(a_1 - a_2) \rangle \\ &= \langle a_1^\dagger a_2 + a_2^\dagger a_1 \rangle. \end{aligned} \quad (5.49)$$

Given the input state from (5.47), this gives

$$\begin{aligned} D &= \langle \Psi | a_1^\dagger | \Psi \rangle \langle \alpha | a_2 | \alpha \rangle + \langle \Psi | a_1 | \Psi \rangle \langle \alpha | a_2^\dagger | \alpha \rangle \\ &= \alpha \langle \Psi | a_1^\dagger | \Psi \rangle + \alpha^* \langle \Psi | a_1 | \Psi \rangle. \end{aligned} \quad (5.50)$$

With $a_1 = X + iP$ and $\alpha = re^{i\theta} = r(\cos \theta + i \sin \theta)$, this becomes

$$\begin{aligned} D &= \langle \Psi | r(\cos \theta + i \sin \theta) (X - iP) | \Psi \rangle + \langle \Psi | r(\cos \theta - i \sin \theta) (X + iP) | \Psi \rangle \\ &= 2r \langle \Psi | (\cos \theta X + \sin \theta P) | \Psi \rangle \\ &\stackrel{(4.60a)}{=} 2r \langle \Psi | X_\theta | \Psi \rangle, \end{aligned} \quad (5.51)$$

where X_θ is the quadrature X rotated by an angle θ . Accordingly, using homodyne detection, we can measure the expectation value $\langle X_\theta \rangle$ of the rotated quadrature, where the angle θ can be controlled via the phase of the local oscillator.

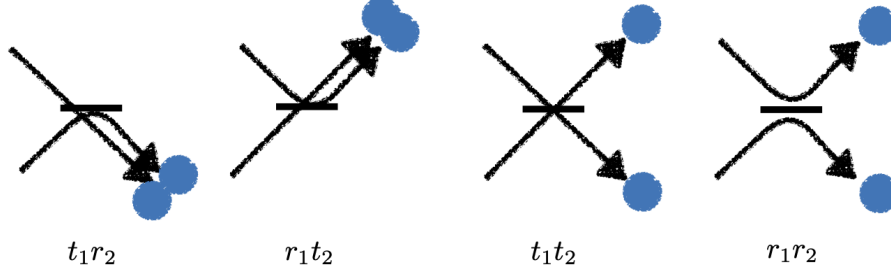
Hong-Ou-Mandel interference Let us see now what happens when we send two photons into the beam splitter, one in each input mode. The input state is then given by

$$|\Psi_{\text{in}}\rangle = a_1^\dagger a_2^\dagger |\Omega\rangle. \quad (5.52)$$

The output state is found to be

$$\begin{aligned} |\Psi_{\text{out}}\rangle &= (t_1 b_1^\dagger + r_1 b_2^\dagger) (r_2 b_1^\dagger + t_2 b_2^\dagger) |\Omega\rangle \\ &= [t_1 r_2 (b_1^\dagger)^2 + r_1 t_2 (b_2^\dagger)^2 + (t_1 t_2 + r_1 r_2) b_1^\dagger b_2^\dagger] |\Omega\rangle \\ &= \sqrt{2} t_1 r_2 |2_1\rangle |0_2\rangle + \sqrt{2} r_1 t_2 |0_1\rangle |2_2\rangle + (t_1 t_2 + r_1 r_2) |1_1\rangle |1_1\rangle, \end{aligned} \quad (5.53)$$

where $|n_j\rangle$ describes a Fock state in mode j . Here we see that four different two-particle transitions can occur, with corresponding transition amplitudes $t_1 r_2$, $r_1 t_2$, $t_1 t_2$, and $r_1 r_2$:



What we recognize is that two two-particle transitions result in the output state $|1_1\rangle|1_2\rangle$ with one photon in each output port, which is commonly called a coincidence: when both photons are transmitted, with amplitude t_1t_2 , and when both photons are reflected, with amplitude r_1r_2 . As a result, these two two-particle transitions interfere (since there is no information through which we can distinguish which of the two transitions had happened). The probability to find the output state $|1_1\rangle|1_2\rangle$ is therefore given by a coherent sum of two two-particle transition amplitudes:

$$\begin{aligned} p_{11} &= |\langle 1_1 | \langle 1_2 | \Psi_{\text{out}} \rangle|^2 = |t_1t_2 + r_1r_2|^2 \\ &= |t_1t_2|^2 + |r_1r_2|^2 + 2\text{Re}(t_1t_2r_1^*r_2^*). \end{aligned} \quad (5.54)$$

Note that by inserting the expressions for t_1, t_2, r_1, r_2 from Eq. (5.37), we see that the interference is always destructive, since the interference term (the last term in (5.54)) is always negative, and we get

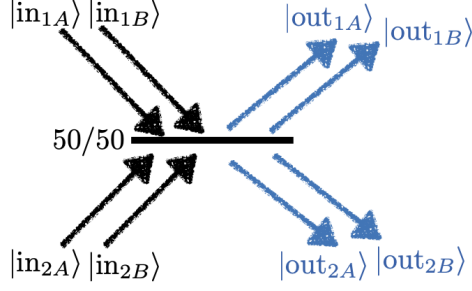
$$p_{11} = \cos^4\left(\frac{\gamma}{2}\right) + \sin^4\left(\frac{\gamma}{2}\right) - 2\cos^2\left(\frac{\gamma}{2}\right)\sin^2\left(\frac{\gamma}{2}\right) = \cos^2\gamma. \quad (5.55)$$

On the other hand, the probabilities $p_{20} = 2|t_1r_2|^2$ and $p_{02} = 2|r_1t_2|^2$ to observe the bunching of both photons in the same output mode is

$$p_{20} = p_{02} = 2\sin^2\left(\frac{\gamma}{2}\right)\cos^2\left(\frac{\gamma}{2}\right) = \frac{1}{2}\sin^2\gamma. \quad (5.56)$$

Indeed, for a 50/50 beam splitter, for which $\gamma = \pi/2$ (compare with Eq. (5.48)), the interference in (5.55) is totally destructive, such that the probability to observe one photon in each output mode is identically zero. Note that different to the interference between single-particle transition amplitudes (e.g. in the double-slit experiment), here two-particle transition amplitudes interfere. The interference results from the fact that photons are identical particles, such that by measuring one photon in state $|1_1\rangle$ and one photon in state $|1_2\rangle$, there is no information (i.e., no physically observable degree of freedom) through which we can distinguish whether the photons were transmitted or reflected at the beam splitter. This two-particle interference effect is commonly called Hong-Ou-Mandel effect, to the honor of the discoverers.

Hong-Ou-Mandel interference of partially distinguishable photons Above we discussed the interference of two indistinguishable photons on the beam splitter and found that the interference is perfectly destructive in the case of a 50/50 beam splitter. Let us now assume such a 50/50 beam splitter and see what happens to the output statistics if we make the particles partially distinguishable through some additional (often called internal) degrees of freedom. These internal degrees of freedom can be, for example, the photons' spectrum, their time of arrival at the beam splitter, or their polarization.



As illustrated, for each of the two input and two output directions of the 50/50 beam splitter we consider two orthogonal internal states, for example $|\text{in}_{1A}\rangle$ and $|\text{in}_{1B}\rangle$, where A and B label orthogonal internal states of the photons (such as orthogonal polarization states or orthogonal spectral properties). We suppose that the photon hitting the beam splitter from direction 1 is in the state

$$|\text{in}_{1A}\rangle = a_{1A}^\dagger |\Omega\rangle, \quad (5.57)$$

and the photon hitting the beam splitter from direction 2 is in a superposition of the internal states labeled by A and B ,

$$\alpha |\text{in}_{2A}\rangle + \sqrt{1 - |\alpha|^2} |\text{in}_{2B}\rangle = \left(\alpha a_{2A}^\dagger + \sqrt{1 - |\alpha|^2} a_{2B}^\dagger \right) |\Omega\rangle, \quad (5.58)$$

where $\alpha \in \mathbb{C}$ controls the contributions from the two orthogonal internal states, with $0 \leq |\alpha| \leq 1$, and $\langle \text{in}_{2A} | \text{in}_{2B} \rangle = 0$. The two-photon input state is therefore

$$|\Psi_{\text{in}}\rangle = \left(\alpha a_{1A}^\dagger a_{2A}^\dagger + \sqrt{1 - |\alpha|^2} a_{1A}^\dagger a_{2B}^\dagger \right) |\Omega\rangle. \quad (5.59)$$

Moreover, we assume that the 50/50 beam splitter doesn't change the internal states A and B and acts equally for both,

$$a_{jC}^\dagger \rightarrow \sum_k s_{kj} b_{kC}^\dagger, \quad C \in \{A, B\}. \quad (5.60)$$

Considering the scattering matrix

$$\tilde{s} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5.61)$$

for the 50/50 beam splitter, the input state (5.59) transforms to

$$\begin{aligned} |\Psi_{\text{out}}\rangle &= \frac{1}{2} \left(\alpha [b_{1A}^\dagger + b_{2A}^\dagger] [b_{1A}^\dagger - b_{2A}^\dagger] + \sqrt{1 - |\alpha|^2} [b_{1A}^\dagger + b_{2A}^\dagger] [b_{1B}^\dagger - b_{2B}^\dagger] \right) |\Omega\rangle \\ &= \frac{1}{2} \left(\alpha [(b_{1A}^\dagger)^2 - (b_{2A}^\dagger)^2] + \sqrt{1 - |\alpha|^2} \left[b_{1A}^\dagger b_{1B}^\dagger - b_{2A}^\dagger b_{2B}^\dagger + \underbrace{b_{2A}^\dagger b_{1B}^\dagger - b_{1A}^\dagger b_{2B}^\dagger}_{\text{coincidences}} \right] \right) |\Omega\rangle. \end{aligned} \quad (5.62)$$

We now further assume that the photon detectors at the output cannot resolve the internal states A and B . They can only resolve the output directions 1 and 2 (in experiments often so-called avalanche photodiodes are used, which record a “click” if a photon hits the detector, independently of the photon's polarization and exact spectral properties). We then find a coincidence, i.e., the detectors at output 1 and output 2 click at the same time, with probability

$$p_{11} = \sum_{C, C'} |\langle 1_{1C} | \langle 1_{2C'} | |\Psi_{\text{out}}\rangle|^2 = \frac{1}{2} (1 - |\alpha|^2). \quad (5.63)$$

We see that the probability for a coincidence strongly depends on $|\alpha|$. If $|\alpha| = 1$, we recover the case of two photons in the same internal state as discussed before, for which a perfect 50/50 beam splitter results in $p_{11} = 0$ due to perfectly destructive two-photon interference. On the other hand, with decreasing $|\alpha|$ the destructive interference effect decreases, such that the coincidence probability p_{11} increases. The coincidence probability finally reaches its maximum of $p_{11} = 1/2$ for $|\alpha| = 0$, in which case the two photons are in orthogonal internal states, and, thus, cannot interfere on the beam splitter.

Let us note that the parameter α quantifies the indistinguishability of the input photons with respect to their internal degrees of freedom. Although the photons are input to orthogonal input states (since they hit the beam splitter from different directions, 1 and 2), they are often said to be *indistinguishable* if $\alpha = 0$, and *partially distinguishable* for variable α . Indistinguishability is here meant to be with respect to the photons' internal degrees of freedom.

Review Questions 5.4.

- *How can you measure the expectation values of the quadrature operators?*
- *What interferes in the Hong-Ou-Mandel effect?*
- *Explain how two-photon interference on the beam splitter depends on the splitting ration of the beam splitter (i.e., the transmission and reflection coefficients), and how it depends on the photons' input states!*
- *What would be the behavior of the coincidence probability in the case of a general beam splitter and general single photon states at input?*

Chapter 6

Photodetection and coherence

In our discussion of linear dynamics of single photons in the previous chapter, we mentioned the detecting of single photons without further discussion. We are now proceeding towards a more formal description of photodetection. Usually, the electromagnetic field is detected via its interaction with matter. Thereby, the absorption of photons results in the ionization of electrons, whose current can be measured (possibly after amplification). While we postpone our discussion of light-matter interaction to the next chapters, let us here place the emphasis on the electric field, which, as we will see in the next chapter, is typically responsible for the dominant processes in light-matter interaction. That is, we consider ideal detectors working on an absorption mechanism which is sensitive to the electric field at space-time event $(\mathbf{r}, t)$. To ease notation, we assume a fixed polarization, and write the space-time event as $x = (\mathbf{r}, t)$, such that we can describe the electric field through a scalar operator $E(x)$.

6.1 Photodetection rates

Photodetection rate Suppose that the state of the field is described by $|\phi_i\rangle$ before the measurement. The transition probability to absorb a photon at the space-time event x , with the field resulting in the final state $|\phi_f\rangle$ is then proportional to

$$T_{fi} = \left| \langle \phi_f | E^+(x) | \phi_i \rangle \right|^2, \quad (6.1)$$

with $E^+(x)$ the positive frequency part of the electric field operator, which involves only a single annihilation operator [cf. Eq. (3.16)], hence, describing the absorption of a single photon when acting on the initial state $|\phi_i\rangle$. Now note that the photodetector doesn't measure the final state of the field $|\phi_f\rangle$. Instead, it measures the photoelectron, which results from the absorption process. Since every possible final state of the field reached after the absorption of a single photon results in such a photoelectron, we have to sum over all final states $|\phi_f\rangle$ to describe the rate of our photodetector. Using the completeness relation $\sum_f |\phi_f\rangle \langle \phi_f| = \mathbb{1}$, we obtain the detection rate

$$\begin{aligned} I(x) &= \sum_f T_{fi} \\ &\stackrel{(6.1)}{=} \sum_f \langle \phi_f | E^+(x) | \phi_i \rangle \langle \phi_i | E^-(x) | \phi_f \rangle \\ &= \langle \phi_i | E^-(x) \left(\sum_f |\phi_f\rangle \langle \phi_f| \right) E^+(x) | \phi_i \rangle \\ &= \langle \phi_i | E^-(x) E^+(x) | \phi_i \rangle. \end{aligned} \quad (6.2)$$

The photodetection rate of our photodetector thus provides us with the expectation value

$$I(x) = \langle E^-(x)E^+(x) \rangle \quad (6.3)$$

of the operator $E^-(x)E^+(x)$, which corresponds to the field intensity at x . Note that we get the same result if the field is initially in a mixed state $\rho = \sum_i p_i |\phi_i\rangle\langle\phi_i|$, since then, the rate is $I(x) = \sum_i p_i \sum_f T_{fi} = \sum_i p_i \langle\phi_i| E^-(x)E^+(x) |\phi_i\rangle = \langle E^-(x)E^+(x) \rangle$.

Joint photodetection Consider now a joint detection of one photon at space-time event $x_1 = (\mathbf{r}_1, t_1)$, and one photon at $x_2 = (\mathbf{r}_2, t_2)$, with $t_2 > t_1$. Following our above considerations, the joint detection rate gets

$$\langle E^-(x_1)E^-(x_2)E^+(x_2)E^+(x_1) \rangle. \quad (6.4)$$

Note that the ordering of the field operators reflects the fact that first a photon at x_1 is absorbed, and then a photon at x_2 . We can generalize this to the detection of n photons at space-time events $x_1, \dots, x_n$, with $t_1 < \dots < t_n$, for which the joint detection rate gets

$$\langle E^-(x_1) \cdots E^-(x_n)E^+(x_n) \cdots E^+(x_1) \rangle. \quad (6.5)$$

Note that photodetection rates are always given by expectation values of operators in normal ordering, ensuring that the n -photon detection rate vanishes in a state with fewer than n photons.

Single-mode detection rates As an example, let us consider the detection rates for a single field mode. Using $E^+(x) = i\sqrt{\frac{\hbar\omega}{2\epsilon_0}}u(\mathbf{r})a(t)$, the detection rate at space-time event $x = (\mathbf{r}, t)$ becomes proportional to the number operator $N(t) = a^\dagger(t)a(t)$ of that mode,

$$I(x) = \langle E^-(x)E^+(x) \rangle = \frac{|u(\mathbf{r})|^2}{2\epsilon_0} \hbar\omega \langle a^\dagger(t)a(t) \rangle. \quad (6.6)$$

It is interesting to note that the joint detection of two photons at the same time $t_1 = t_2 = t$ is not proportional to $\langle N(t)^2 \rangle$, but to $\langle N(t)[N(t) - 1] \rangle$, since the same photon cannot be detected twice,

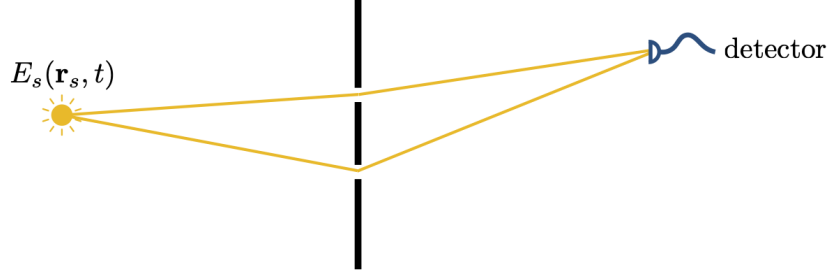
$$\begin{aligned} \langle E^-(\mathbf{r}_1, t)E^-(\mathbf{r}_2, t)E^+(\mathbf{r}_2, t)E^+(\mathbf{r}_1, t) \rangle &= \frac{|u(\mathbf{r}_1)u(\mathbf{r}_2)|^2}{4\epsilon_0^2} \hbar^2 \omega^2 \langle a^\dagger(t) \underbrace{a^\dagger(t)a(t)}_{a(t)a^\dagger(t)-1} a(t) \rangle \\ &= \frac{|u(\mathbf{r}_1)u(\mathbf{r}_2)|^2}{4\epsilon_0^2} \hbar^2 \omega^2 \langle N(t)[N(t) - 1] \rangle. \end{aligned} \quad (6.7)$$

Review Questions 6.1.

- What is the corresponding operator describing the photodetection rate.
- Why are photodetection rates always given by expectation values in normal ordering?

6.2 First-order coherence

The double-slit experiment In order to motivate the quantum mechanical description of first-order coherence, let us briefly discuss Young's double-slit experiment:



Here, a source field $E_s(x)$ is split into two components by a double-slit. Note that we ignore diffraction effects due to finite widths of the slits. Since the resulting two paths to the detector can have different lengths, the positive-frequency part of the field at the detector must read

$$E^+(\mathbf{r}, t) = c_1(\mathbf{r})E_s^+(\mathbf{r}_s, t - \tau_1(\mathbf{r})) + c_2(\mathbf{r})E_s^+(\mathbf{r}_s, t - \tau_2(\mathbf{r})), \quad (6.8)$$

where $\tau_j(\mathbf{r})$ is the traveling time for path $j \in \{1, 2\}$, and $c_j(\mathbf{r}) \in \mathbb{C}$ describes the scattering due to the j -th slit. The photodetection rate at the detector is therefore given by the intensity (to ease notation we drop the position arguments $\mathbf{r}$ and $\mathbf{r}_s$)

$$\begin{aligned} I(x) &= \langle E^-(x)E^+(x) \rangle \\ &= |c_1|^2 \langle E_s^-(t - \tau_1)E_s^+(t - \tau_1) \rangle + |c_2|^2 \langle E_s^-(t - \tau_2)E_s^+(t - \tau_2) \rangle \\ &\quad + 2\text{Re} \left[c_1 c_2^* \langle E_s^-(t - \tau_2)E_s^+(t - \tau_1) \rangle \right]. \end{aligned} \quad (6.9)$$

While the first two terms correspond to the detection rates associated with light traveling through the individual slits, the last term describes an interference of the source between different times, which is essentially determined by the two-time correlation function

$$\langle E_s^-(\mathbf{r}_s, t - \tau_2)E_s^+(\mathbf{r}_s, t - \tau_1) \rangle. \quad (6.10)$$

This correlation function provides a measure for the degree of temporal coherence of the source, which, by Eq. (6.9), gives rise to interference fringes in Young's double slit experiment. As a measure of the strength of interference, one usually uses the visibility

$$v = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}. \quad (6.11)$$

It is straightforward to check that in the case of balanced slit contributions, $|c_1| = |c_2|$, the visibility becomes

$$v = \left| \frac{\langle E_s^-(t - \tau_2)E_s^+(t - \tau_1) \rangle}{\sqrt{\langle E_s^-(t - \tau_1)E_s^+(t - \tau_1) \rangle \langle E_s^-(t - \tau_2)E_s^+(t - \tau_2) \rangle}} \right|. \quad (6.12)$$

First-order correlation and coherence function Let us now generalize the two-time correlation function to account for correlations of the electromagnetic field between two different space-time events x_1 and x_2 : To this end, we define the first-order correlation function

$$G^{(1)}(x_1; x_2) = \langle E^-(x_1)E^+(x_2) \rangle. \quad (6.13)$$

Note that the first-order correlation function evaluated at the same space-time events yields the photodetection rate, i.e., $G^{(1)}(x; x) = I(x)$. In analogy to the visibility (6.12), which provides a normalized correlation measure, let us also introduce a normalized version of (6.13). We define the first-order coherence function

$$g^{(1)}(x_1; x_2) = \frac{G^{(1)}(x_1; x_2)}{\sqrt{G^{(1)}(x_1; x_1)G^{(1)}(x_2; x_2)}}, \quad (6.14)$$

satisfying $0 \leq |g^{(1)}(x_1; x_2)| \leq 1$. Different coherence strengths are usually called as follows:

$$\begin{aligned} |g^{(1)}(x_1; x_2)| &= 1 && \text{complete coherence} \\ 0 < |g^{(1)}(x_1; x_2)| &< 1 && \text{partial coherence} \\ |g^{(1)}(x_1; x_2)| &= 0 && \text{incoherent.} \end{aligned}$$

Note that in the case of temporal correlations, for which $\mathbf{r}_2 = \mathbf{r}_1$ and $t_2 = t_1 + \tau$, the first-order coherence function is usually denoted by $g^{(1)}(\tau)$.

Examples Let us now consider the first-order coherence function for some examples. To this end, we write the positive frequency part of the electric field (which we suppose to be polarized along a certain direction) as $E^+(x) = \sum_{\mathbf{k}} K_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t + \pi/2)} a_{\mathbf{k}}$, with $K_{\mathbf{k}} = \sqrt{\frac{\hbar \omega_{\mathbf{k}}}{2\epsilon_0 L^3}}$.

- For any single-mode state $\rho \in \mathcal{H}_{\mathbf{k}}$, we get

$$\begin{aligned} G^{(1)}(x; x) &= K_{\mathbf{k}}^2 \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle, \\ G^{(1)}(x_1; x_2) &= K_{\mathbf{k}}^2 e^{i\mathbf{k} \cdot (\mathbf{r}_2 - \mathbf{r}_1) - i\omega_{\mathbf{k}}(t_2 - t_1)} \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle. \end{aligned}$$

The modulus of the first-order coherence function thus becomes

$$|g^{(1)}(x_1, x_2)| = 1, \quad (6.15)$$

showing that any single-mode state is completely coherent (this includes all single-mode states discussed in Ch. 4, such as Fock number states $|n\rangle$, coherent states $|\alpha\rangle$ or single-mode thermal states ρ_β).

- Consider now polarized, parallel multi-mode light along a certain direction, for which we choose the z -direction. That is, we consider wave vectors $\mathbf{k}$ along the z -direction, and write the positive frequency part of the electric field as $E^+(x) = \sum_{\mathbf{k}} K_{\mathbf{k}} e^{i(kz - \omega_{\mathbf{k}} t + \pi/2)} a_{\mathbf{k}}$. For the multi-mode chaotic light

$$\rho = \sum_{\{n_{\mathbf{k}}\}} P(\{n_{\mathbf{k}}\}) |\{n_{\mathbf{k}}\}\rangle \langle \{n_{\mathbf{k}}\}|, \quad (6.16)$$

with

$$P(\{n_{\mathbf{k}}\}) = \prod_{\mathbf{k}} P_{\mathbf{k}}(n_{\mathbf{k}}) = \prod_{\mathbf{k}} \frac{\bar{n}_{\mathbf{k}}^{n_{\mathbf{k}}}}{(1 + \bar{n}_{\mathbf{k}})^{n_{\mathbf{k}}+1}}, \quad (6.17)$$

where $|\{n_{\mathbf{k}}\}\rangle = |n_{\mathbf{k}_1}\rangle |n_{\mathbf{k}_2}\rangle \dots$ are multi-mode Fock states, one can show that

$$g^{(1)}(\tau) = \frac{\sum_{\mathbf{k}} \omega_{\mathbf{k}} \bar{n}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}} \tau}}{\sum_{\mathbf{k}} \omega_{\mathbf{k}} \bar{n}_{\mathbf{k}}}. \quad (6.18)$$

Since the intensity at frequency $\omega_{\mathbf{k}}$ is proportional to $\omega_{\mathbf{k}} \bar{n}_{\mathbf{k}}$, for chaotic light with a Lorentzian spectrum of width γ around ω_0 , we have

$$\omega_{\mathbf{k}} \bar{n}_{\mathbf{k}} \propto \frac{\gamma \pi}{(\omega_0 - \omega_{\mathbf{k}})^2 + \gamma^2}. \quad (6.19)$$

Using this in (6.18) and assuming a narrow frequency width $\gamma \ll \omega_0$, one can show that the first-order coherence function results in

$$g^{(1)}(\tau) = e^{-i\omega_0 \tau - \gamma |\tau|}. \quad (6.20)$$

Accordingly, the temporal coherence decays as $|g^{(1)}(\tau)| = e^{-\gamma |\tau|}$, illustrating the limited coherence of multi-mode chaotic light.

Review Questions 6.2.

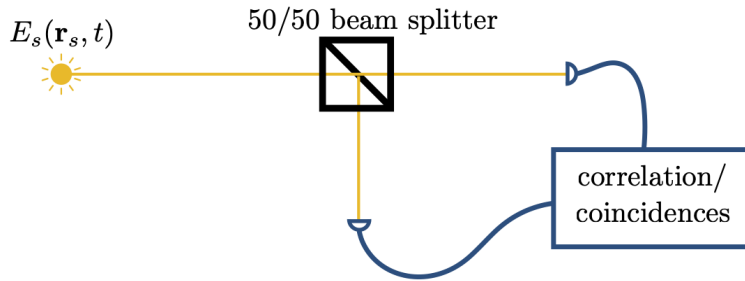
- What is quantified by the first-order coherence function?
- Explain the double-slit experiment in terms of first-order coherence.
- Do single-mode states differ in their coherence? Why?
- What happens to the first order coherence if the light has multiple frequencies?

6.3 Second-order coherence

In the first example above, we have seen that any single-mode light state is completely coherent. This is due to the fact that first-order coherence quantifies the degree to which a light source is monochromatic. Such states, however, can strongly differ in their photon number statistics. First-order coherence is therefore unable to distinguish between different photon number distributions.

6.3.1 Second-order correlation and coherence function

The photon number distributions become relevant if one correlates intensities instead of fields (this was initially introduced by Hanbury Brown and Twiss to measure the angular diameter of distant stars [20]). To this end consider a light source $E_s(\mathbf{r}_s, t)$, which is split into two parts by a 50/50 beam splitter, such that the fields in the output arms are proportional to $E_s(\mathbf{r}_s, t - \tau_1)$ and $E_s(\mathbf{r}_s, t - \tau_2)$, with τ_j the time-delay corresponding to the j -th output arm. Now suppose we place photon detectors in the individual outputs of the beam splitter, and measure coincidence counts (or intensity correlations in the case of photodiodes).



Transforming $t - \tau_1 \rightarrow t$, and $t - \tau_2 \rightarrow t + \tau$, with $\tau = \tau_1 - \tau_2$, the rate of coincidence counts is then proportional to

$$G^{(2)}(\tau) = \langle E_s^-(t) E_s^-(t + \tau) E_s^+(t + \tau) E_s^+(t) \rangle, \quad (6.21)$$

which, after normalization, yields the second-order temporal coherence function

$$g^{(2)}(\tau) = \frac{\langle E_s^-(t) E_s^-(t + \tau) E_s^+(t + \tau) E_s^+(t) \rangle}{\langle E_s^-(t) E_s^+(t) \rangle \langle E_s^-(t + \tau) E_s^+(t + \tau) \rangle}. \quad (6.22)$$

This can be interpreted as the conditional probability that if a photon is detected at time t , another photon is also detected at time $t + \tau$. Hence, this measurement is sensitive to the photon number distribution of the light source $E_s(t)$.

Before we investigate $g^{(2)}(\tau)$ further, let us generalize (6.21) and (6.22): We define the second-order correlation function by

$$G^{(2)}(x_1, x_2; x_3, x_4) = \langle E^-(x_1) E^-(x_2) E^+(x_3) E^+(x_4) \rangle, \quad (6.23)$$

and the second-order coherence function by

$$g^{(2)}(x_1, x_2; x_3, x_4) = \frac{G^{(2)}(x_1, x_2; x_3, x_4)}{\sqrt{G^{(1)}(x_1; x_1)G^{(1)}(x_2; x_2)G^{(1)}(x_3; x_3)G^{(1)}(x_4; x_4)}}. \quad (6.24)$$

By evaluating this function at a single spatial point $\mathbf{r}$ and for two distinct times $t_1 = t_4 = t$ and $t_2 = t_3 = t + \tau$, we obtain second-order temporal coherence function $g^{(2)}(\tau)$ from Eq. (6.22).

6.3.2 Distinguishing quantum and classical light

In order to see how the second order temporal coherence function can be used to distinguish quantum from classical light, let us see how $g^{(2)}(\tau)$ is bounded by classical light.

Classical light The second-order temporal coherence function for a classical electric field $E(t)$ with complex conjugate $E^*(t)$ is obtained by replacing the field operators in (6.22) with their corresponding classical fields,

$$g^{(2)}(\tau) = \frac{\langle E^*(t)E^*(t+\tau)E(t+\tau)E(t) \rangle}{\langle E^*(t)E(t) \rangle \langle E^*(t+\tau)E(t+\tau) \rangle} = \frac{\langle I(t)I(t+\tau) \rangle}{\langle I(t) \rangle^2}, \quad (6.25)$$

where $I(t) = \frac{1}{2}\epsilon_0 c |E(t)|^2$ is the intensity, $\langle I(t) \rangle = \frac{1}{T} \int dt I(t)$ the average intensity, and $\langle I(t)I(t+\tau) \rangle = \frac{1}{T} \int dt I(t)I(t+\tau)$. For $\tau = 0$, we therefore have

$$g^{(2)}(0) = \frac{\langle I(t)^2 \rangle}{\langle I(t) \rangle^2}. \quad (6.26)$$

Using the Cauchy-Schwarz inequality,

$$\langle I(t) \rangle^2 = \left(\int dt \frac{1}{T} I(t) \right)^2 \leq \int dt' \frac{1}{T^2} \int dt I(t)^2 = \frac{1}{T} \int dt I(t)^2 = \langle I(t)^2 \rangle, \quad (6.27)$$

we find

$$1 \leq g^{(2)}(0). \quad (6.28)$$

We thus see that classical light establishes a lower bound on the second-order temporal coherence function at $\tau = 0$. Moreover, using again the Cauchy-Schwarz inequality,

$$\langle I(t)I(t+\tau) \rangle^2 = \left(\frac{1}{T} \int dt I(t)I(t+\tau) \right)^2 \leq \frac{1}{T} \int dt' I(t')^2 \frac{1}{T} \int dt I(t+\tau)^2 = \langle I(t)^2 \rangle^2, \quad (6.29)$$

we get $\langle I(t)I(t+\tau) \rangle \leq \langle I(t)^2 \rangle$, and, by Eq. (6.25),

$$g^{(2)}(\tau) \leq g^{(2)}(0). \quad (6.30)$$

For classical light, the degree of second-order temporal coherence can therefore never exceed its value for zero time delay. We now show that the quantized electromagnetic field can take states that break the classical bounds (6.28) and (6.30), highlighting that a measurement of $g^{(2)}(\tau)$ can demonstrate the “quantum nature” of light.

Quantum light Let us, for simplicity, consider a single mode. From Eq. (6.6), we see that in this case the second-order temporal coherence function (6.22) becomes

$$g^{(2)}(\tau) = \frac{\langle a^\dagger a^\dagger a a \rangle}{\langle a^\dagger a \rangle^2} = \frac{\langle a^\dagger (a a^\dagger - 1) a \rangle}{\langle a^\dagger a \rangle^2} = \frac{\langle N^2 \rangle - \langle N \rangle}{\langle N \rangle^2} = 1 + \frac{\langle N^2 \rangle - \langle N \rangle^2 - \langle N \rangle}{\langle N \rangle^2}, \quad (6.31)$$

which is independent of τ but strongly depends on the number distribution of the light. Using $(\Delta N)^2 = \langle N^2 \rangle - \langle N \rangle^2$, this becomes

$$g^{(2)}(\tau) = 1 + Q, \quad (6.32)$$

where we defined the Mandel parameter

$$Q = \frac{(\Delta N)^2 - \langle N \rangle}{\langle N \rangle^2}, \quad (6.33)$$

which measures the deviation of the photon-number distribution from Poissonian statistics (recall that for Poissonian statistics $(\Delta N)^2 = \langle N \rangle$). In particular, $Q > 0$ indicates super-Poissonian statistics (as, for example, in the case of thermal light; recall (4.49)), and $Q < 0$ sub-Poissonian statistics, which can only appear in the case of “quantum light”. For example, in the case of a Fock state $|n\rangle$, one has $\langle N \rangle = n$ and $\langle N^2 \rangle = n^2$, such that $Q = -1/n$. Accordingly, $g^{(2)}(\tau) = 1 - \frac{1}{n} < 1$, which breaks the classical bound (6.28), hence, indicating the quantum nature of Fock states.

Review Questions 6.3.

- What does the second-order temporal coherence function $g^{(2)}(\tau)$ quantify? How does it differ from the first order coherence function $g^{(1)}(\tau)$?
- How can $g^{(2)}(\tau)$ be used to distinguish quantum from classical light?
- What are the bounds on $g^{(2)}(\tau)$ imposed by classical light?
- What is $g^{(2)}(\tau)$ for Fock states?
- Discuss how $g^{(2)}(0)$ can be used to check whether a light source is a single-photon emitter (emitting one photon at a time).

Chapter 7

Interaction between quantum matter and a classical external electromagnetic field

Let us now come back to a classical treatment of the electromagnetic field. In particular, in Chapter 2, we described the interaction between the classical electromagnetic field and a set of classical point particles with charges q_α . We now want to additionally include an external field (or externally applied field) whose sources [the charge density $\rho_e(\mathbf{r}, t)$ and current density $\mathbf{j}_e(\mathbf{r}, t)$] are fixed independently of the field generated by the particles. That is, the particles evolve under the simultaneous action of the external field and the rest of the electromagnetic field.

7.1 Classical Hamiltonian in the presence of an external field

We describe the external field by the classical potentials $\mathbf{A}_e(\mathbf{r}, t)$ and $\phi_e(\mathbf{r}, t)$. The potentials $\mathbf{A}_t(\mathbf{r}, t)$ and $\phi_t(\mathbf{r}, t)$ of the total classical electromagnetic field are then

$$\mathbf{A}_t(\mathbf{r}, t) = \mathbf{A}_e(\mathbf{r}, t) + \mathbf{A}(\mathbf{r}, t) \quad (7.1a)$$

$$\phi_t(\mathbf{r}, t) = \phi_e(\mathbf{r}, t) + \phi(\mathbf{r}, t). \quad (7.1b)$$

Now we require that the equations of motion for the potentials $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$ are governed by the sources $\rho(\mathbf{r}, t)$ and $\mathbf{j}(\mathbf{r}, t)$ of the particles alone, while the dynamics of the particles (i.e., the Lorentz equations) are governed by the total field corresponding to the potentials $\mathbf{A}_t(\mathbf{r}, t)$ and $\phi_t(\mathbf{r}, t)$. To this end, we consider the electromagnetic field in the Coulomb gauge (i.e., $\mathbf{A}_\perp = \mathbf{A}$), while keeping the gauge of the external field unspecified. As shown in detail in [4] (see Complement C_{II} there), the corresponding total energy is given by [c.f. Eq. (2.52)]

$$\begin{aligned} H = & \sum_{\alpha} \frac{1}{2m_{\alpha}} (\mathbf{p}_{\alpha} - q_{\alpha} [\mathbf{A}_{\perp}(\mathbf{r}_{\alpha}) + \mathbf{A}_e(\mathbf{r}_{\alpha}, t)])^2 \\ & + \epsilon_{\text{Coul}} + \frac{1}{8\pi\epsilon_0} \sum_{\alpha \neq \beta} \frac{q_{\alpha} q_{\beta}}{|\mathbf{r}_{\alpha} - \mathbf{r}_{\beta}|} \\ & + \sum_{\alpha} q_{\alpha} \phi_e(\mathbf{r}_{\alpha}, t) \\ & + \int \left(\frac{\epsilon_0 \mathbf{E}_{\perp}^2}{2} + \frac{\mathbf{B}^2}{2\mu_0} \right) d^3r. \end{aligned} \quad (7.2)$$

In Appendix A.1, we show that this Hamiltonian indeed leads to the Lorentz equations with respect to the total fields for the dynamics of the charged particles. Let us now decompose the

Hamiltonian (7.2) as

$$H = H_P + H_R + H_I, \quad (7.3)$$

where

$$H_P = \sum_{\alpha} \frac{[\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_e(\mathbf{r}_{\alpha}, t)]^2}{2m_{\alpha}} + \underbrace{\epsilon_{\text{Coul}} + \frac{1}{8\pi\epsilon_0} \sum_{\alpha \neq \beta} \frac{q_{\alpha} q_{\beta}}{|\mathbf{r}_{\alpha} - \mathbf{r}_{\beta}|}}_{V_{\text{Coul}}=H_{\parallel}} + \sum_{\alpha} q_{\alpha} \phi_e(\mathbf{r}_{\alpha}, t) \quad (7.4)$$

is the Hamiltonian of the charged particles in the Coulomb field and the classical external field,

$$H_R = H_{\perp} = \int \left(\frac{\epsilon_0 \mathbf{E}_{\perp}^2}{2} + \frac{\mathbf{B}^2}{2\mu_0} \right) d^3r \quad (7.5)$$

is the Hamiltonian of the free electromagnetic field, and

$$H_I = - \sum_{\alpha} \frac{q_{\alpha}}{m_{\alpha}} [\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_e(\mathbf{r}_{\alpha}, t)] \cdot \mathbf{A}_{\perp}(\mathbf{r}_{\alpha}) + \sum_{\alpha} \frac{q_{\alpha}^2}{2m_{\alpha}} \mathbf{A}_{\perp}^2(\mathbf{r}_{\alpha}) \quad (7.6)$$

is the interaction Hamiltonian between the particles and the electromagnetic field.

7.1.1 Coulomb interacting particles

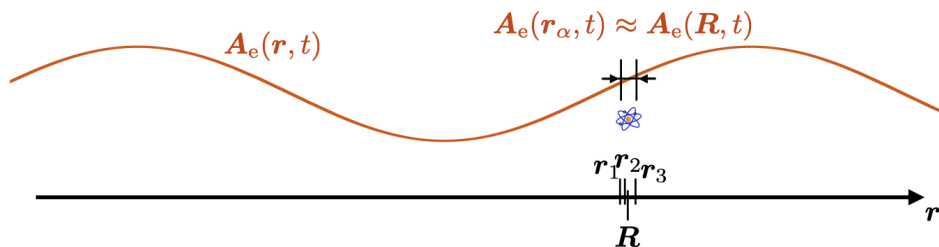
Suppose now that the particles are sufficiently near to one other, such that the Coulomb interaction due to the longitudinal component of the electromagnetic field (described by $V_{\text{Coul}} = H_{\parallel}$) is a very good approximation to their real interaction. For the dynamical evolution of the system, we can then neglect the particles interaction with the transverse component of the electromagnetic field as described by H_I from Eq. (7.6). Then, the Hamiltonian $H_R = H_{\perp}$ of the transverse component of the electromagnetic field can likewise be neglected. We are thus left with the Hamiltonian

$$H = \sum_{\alpha} \frac{[\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_e(\mathbf{r}_{\alpha}, t)]^2}{2m_{\alpha}} + V_{\text{Coul}} + \sum_{\alpha} q_{\alpha} \phi_e(\mathbf{r}_{\alpha}, t), \quad (7.7)$$

which is a function of the dynamical variables of the charged particles in the presence of an external field.

7.1.2 Dipole approximation

We assume now that the system of particles forms a globally neutral system, i.e. $\sum_{\alpha} q_{\alpha} = 0$, and that the spatial extension of the system of charged particles is much smaller than the wavelength of the radiation with which it interacts. In this case, we can Taylor expand the spatial variation of the electromagnetic field at the position $\mathbf{R}$ of the system. The Taylor expansion to first non-vanishing order is called the long-wavelength approximation or dipole approximation. It is justified, for example, by considering the interaction of an atom or small molecule whose spatial extension is in the order of Angströms (i.e., 10^{-10} m) with radiation whose wavelength is in the visible range (i.e., 10^{-7} m).



Let us now expand the field potentials around $\mathbf{r} = \mathbf{R}$ to their first non-vanishing order. For the vector potential, we therefore set $\mathbf{A}_e(\mathbf{r}_\alpha, t) \approx \mathbf{A}_e(\mathbf{R}, t)$, that is, we consider it to be spatially constant in the vicinity of the system. Now consider the scalar potential. Since $\sum_\alpha q_\alpha = 0$, by setting the scalar potential spatially constant, we would get $\sum_\alpha q_\alpha \phi_e(\mathbf{R}, t) = 0$. However, we want the first *non-vanishing* order. Accordingly, we must include the next order and expand $\phi_e(\mathbf{r}, t)$ in (7.7) around $\mathbf{R}$ by

$$\phi_e(\mathbf{r}, t) = \phi_e(\mathbf{R}, t) + \nabla \phi_e(\mathbf{R}, t) \cdot (\mathbf{r} - \mathbf{R}) + \dots, \quad (7.8)$$

where $\nabla \phi_e(\mathbf{R}, t)$ is short for $\nabla \phi_e(\mathbf{r}, t)|_{\mathbf{r}=\mathbf{R}}$. Using this together with $\sum_\alpha q_\alpha = 0$, we thus have

$$\begin{aligned} \sum_\alpha q_\alpha \phi_e(\mathbf{r}_\alpha, t) &= \sum_\alpha q_\alpha [\phi_e(\mathbf{R}, t) + \nabla \phi_e(\mathbf{R}, t) \cdot (\mathbf{r}_\alpha - \mathbf{R}) + \dots] \\ &= \underbrace{\sum_\alpha q_\alpha \mathbf{r}_\alpha \cdot \nabla \phi_e(\mathbf{R}, t)}_{\mathbf{d}} + \underbrace{\sum_\alpha q_\alpha \phi_e(\mathbf{R}, t)}_{=0} - \underbrace{\sum_\alpha q_\alpha \nabla \phi_e(\mathbf{R}, t) \cdot \mathbf{R}}_{=0} + \dots \\ &= \mathbf{d} \cdot \nabla \phi_e(\mathbf{R}, t) + \dots, \end{aligned} \quad (7.9)$$

where we defined the dipole moment

$$\mathbf{d} = \sum_\alpha q_\alpha \mathbf{r}_\alpha. \quad (7.10)$$

Here “...” are higher order terms, with the next non-vanishing order corresponding to the quadrupole moment. Under consideration of the dipole approximation, the Hamiltonian (7.7) of the charged particles then becomes

$$H = \sum_\alpha \frac{[\mathbf{p}_\alpha - q_\alpha \mathbf{A}_e(\mathbf{R}, t)]^2}{2m_\alpha} + V_{\text{Coul}} + \mathbf{d} \cdot \nabla \phi_e(\mathbf{R}, t). \quad (7.11)$$

Review Questions 7.1.

- What distinguishes the external field from the usual electromagnetic field?
- What governs the equations of motion of the particles and fields in the presence of an external field.
- What is the approximation behind Coulomb interacting particles? How does it simplify the corresponding Hamiltonian?
- What is the dipole approximation? Under which conditions is it valid?

7.1.3 Length gauge under dipole approximation

Now recall that we didn't set the gauge of the external electromagnetic field yet. Accordingly, we can simplify the Hamiltonian (7.11) using the gauge transformation (2.29) for the external field. To this end, we want to achieve the transformation $\mathbf{p}_\alpha - q_\alpha \mathbf{A}_e(\mathbf{R}, t) \rightarrow \mathbf{p}_\alpha$, which would significantly simplify the Hamiltonian. As we show now, this can be achieved via the gauge transformation

$$\chi(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{A}_e(\mathbf{R}, t). \quad (7.12)$$

With the help of (2.29), the external field then transforms as

$$\mathbf{A}_e(\mathbf{r}, t) \rightarrow \mathbf{A}_e(\mathbf{r}, t) + \nabla \chi(\mathbf{r}, t) = \mathbf{A}_e(\mathbf{r}, t) - \mathbf{A}_e(\mathbf{R}, t) \quad (7.13a)$$

$$\phi_e(\mathbf{r}, t) \rightarrow \phi_e(\mathbf{r}, t) - \frac{\partial \chi(\mathbf{r}, t)}{\partial t} = \phi_e(\mathbf{r}, t) + \mathbf{r} \cdot \frac{\partial \mathbf{A}_e(\mathbf{R}, t)}{\partial t}. \quad (7.13b)$$

Accordingly, the vector potential in the Hamiltonian (7.11) transforms as $\mathbf{A}_e(\mathbf{R}, t) \rightarrow 0$, and the gradient of the scalar potential as

$$\nabla\phi_e(\mathbf{R}, t) \rightarrow \nabla\phi_e(\mathbf{R}, t) + \frac{\partial\mathbf{A}_e(\mathbf{R}, t)}{\partial t} \stackrel{(2.25)}{=} -\mathbf{E}_e(\mathbf{R}, t), \quad (7.14)$$

where we used $\mathbf{E}(\mathbf{r}, t) = -\partial\mathbf{A}(\mathbf{r}, t)/\partial t - \nabla\phi(\mathbf{r}, t)$ from Eq. (2.25) in the last line. Accordingly, in the new gauge of the external field, the Hamiltonian (7.11) reads

$$H = \sum_{\alpha} \frac{\mathbf{p}_{\alpha}^2}{2m_{\alpha}} + V_{\text{Coul}} - \mathbf{d} \cdot \mathbf{E}_e(\mathbf{R}, t). \quad (7.15)$$

Here we explicitly see the electric dipole interaction between the dipole moment $\mathbf{d}$ and the classical external electric field $\mathbf{E}_e$. This gauge is often called the length gauge.

7.1.4 Velocity gauge

Let us briefly mention another often used gauge, the so-called velocity gauge. It is usually applied to the Hamiltonian [c.f. Eq. (7.11)]

$$H = \frac{[\mathbf{p}^2 - q\mathbf{A}_e(\mathbf{R}, t)]^2}{2m} + V_{\text{Coul}}(\mathbf{r}) + q\phi_e(\mathbf{R}, t) \quad (7.16)$$

under dipole approximation of a single charged particle (e.g. an electron) with mass m and charge q in a Coulomb potential $V_{\text{Coul}}(\mathbf{r})$, and in the presence of an external electromagnetic field with vector potential $\mathbf{A}_e(\mathbf{R}, t)$. Often the scalar potential $\phi_e(\mathbf{r}, t)$ is assumed to be zero. Using (2.29), the gauge transformation according to

$$\chi(t) = \frac{q}{2m} \int^t ds \mathbf{A}_e^2(\mathbf{R}, s) \quad (7.17)$$

becomes

$$\mathbf{A}_e(\mathbf{r}, t) \rightarrow \mathbf{A}_e(\mathbf{r}, t) + \nabla\chi(t) = \mathbf{A}_e(\mathbf{r}, t) \quad (7.18a)$$

$$\phi_e(\mathbf{r}, t) \rightarrow \phi_e(\mathbf{r}, t) - \frac{\partial\chi(t)}{\partial t} = \phi_e(\mathbf{r}, t) - \frac{q}{2m} \mathbf{A}_e^2(\mathbf{R}, t). \quad (7.18b)$$

In the velocity gauge, the Hamiltonian of the single charged particle therefore becomes

$$H = \frac{\mathbf{p}^2}{2m} - \frac{q}{m} \mathbf{p} \cdot \mathbf{A}_e(\mathbf{R}, t) + V_{\text{Coul}}(\mathbf{r}) + q\phi_e(\mathbf{R}, t) \quad (7.19)$$

Here $V_{\text{Coul}}(\mathbf{r})$ describes the Coulomb potential energy, and $\mathbf{p}^2/2m$ the energy of the particle in the external field. The second term describes the interaction between the particle and the external field. It couples the velocity $\mathbf{p}/m$ to the vector potential $\mathbf{A}_e(\mathbf{R}, t)$, hence the name velocity gauge. The last term corresponds to the potential energy of the charge in the external scalar field. The scalar field is often assumed to be zero, resulting in the Hamiltonian

$$H = \frac{\mathbf{p}^2}{2m} - \frac{q}{m} \mathbf{p} \cdot \mathbf{A}_e(\mathbf{R}, t) + V_{\text{Coul}}(\mathbf{r}). \quad (7.20)$$

Review Questions 7.2.

- What is the Hamiltonian in the length gauge under dipole approximation?
- What is the velocity gauge, and where does the name come from?

7.2 Quantum Hamiltonian in the presence of a classical external field

7.2.1 Quantization of the charged particles

Now we quantize the charged particles while keeping the external electromagnetic field classical. For the quantization of the particles, we replace the position of the α th particle by the quantum mechanical position operator $\mathbf{r}_\alpha$, and the momentum of the α th particle by the quantum mechanical momentum operator $\mathbf{p}_\alpha$ (recall that for the position and momentum operator we use the same notation as for the classical position and momentum; accordingly, in the following $\mathbf{r}_\alpha$ and $\mathbf{p}_\alpha$ always refer to operators). These operators obey the commutation relation

$$[r_{\alpha j}, p_{\alpha' j'}] = i\hbar \delta_{\alpha, \alpha'} \delta_{j, j'}. \quad (7.21)$$

The quantum Hamiltonian of the charged particles is therefore obtained by replacing the position and momentum with the corresponding *quantum mechanical operators* $\mathbf{r}_\alpha$ and $\mathbf{p}_\alpha$. For Coulomb interacting particles with classical Hamiltonian (7.7) we thus get the quantum Hamiltonian

$$H = \sum_{\alpha} \frac{[\mathbf{p}_\alpha - q_\alpha \mathbf{A}_e(\mathbf{r}_\alpha, t)]^2}{2m_\alpha} + V_{\text{Coul}} + \sum_{\alpha} q_\alpha \phi_e(\mathbf{r}_\alpha, t), \quad (7.22)$$

where $\mathbf{A}_e$ and ϕ_e are the potentials of the *classical* external field. Similarly, the quantum Hamiltonian under dipole approximation is obtained from (7.11), and its expression in the length gauge from (7.15).

Particles with spin Note that we only considered charged particles without further internal degrees of freedom. If we want to include the particles' spin degree of freedom, we must complement the Hamiltonian (7.22) [or the quantized version of (7.2)] by

$$H_1^S = - \sum_{\alpha} \mathbf{M}_\alpha^S \cdot \mathbf{B}_e(\mathbf{r}_\alpha, t), \quad (7.23)$$

which couples the magnetic spin moments

$$\mathbf{M}_\alpha^S = g_\alpha \frac{q_\alpha}{2m_\alpha} \mathbf{S}_\alpha \quad (7.24)$$

of the particles with the external magnetic field. In Eq. (7.24), g_α is the Lande factor and $\mathbf{S}_\alpha$ the spin operator of the α th particle. Note that the spin-dependent term (7.23) is introduced heuristically. It can be justified for electrons by the non-relativistic limit of the Dirac equation.

If not mentioned otherwise, we neglect the particles' spin degrees of freedom. This is a good approximation if the photon energy is small as compared to the particles' momenta, i.e., $\hbar k \ll p$.

7.2.2 Gauge transformations of quantum Hamiltonians

Let us now ask whether there is a quantum counterpart to a gauge transformation (2.29) of the classical field potentials. As we will see, the quantum counterpart to a gauge transformation is a unitary transformation of the form

$$T^\dagger(t) = \exp \left[\frac{i}{\hbar} \sum_{\alpha} q_\alpha \chi(\mathbf{r}_\alpha, t) \right], \quad (7.25)$$

where $\chi(\mathbf{r}, t)$ is now an operator (it is a function of operators). In order to see this, consider the Schrödinger equation $i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle$. The time evolution of the transformed state $|\psi_T(t)\rangle = T^\dagger |\psi(t)\rangle$ thus reads (to ease notation we don't write the time argument of $T^\dagger$)

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\psi_T(t)\rangle &= i\hbar \frac{\partial}{\partial t} [T^\dagger |\psi(t)\rangle] \\ &= \left(i\hbar \frac{\partial T^\dagger}{\partial t} + T^\dagger i\hbar \frac{\partial}{\partial t} \right) |\psi(t)\rangle \\ &= \left(i\hbar \frac{\partial T^\dagger}{\partial t} + T^\dagger H \right) T T^\dagger |\psi(t)\rangle \\ &= \underbrace{\left(i\hbar \frac{\partial T^\dagger}{\partial t} T + T^\dagger H T \right)}_{H_T} |\psi_T(t)\rangle. \end{aligned} \quad (7.26)$$

The Hamiltonian must therefore transform as

$$H_T = T^\dagger H T + i\hbar \frac{\partial T^\dagger}{\partial t} T. \quad (7.27)$$

We can now apply this transformation to the Hamiltonian (7.22). Let us start with the unitary transformation according to the first term in (7.27). Since the second and third term of the Hamiltonian (7.22) are functions of the position operators $\mathbf{r}_\alpha$ only, they commute with $T^\dagger$, and, hence, remain invariant. Next, consider the first term of the Hamiltonian (7.22). To this end, consider $T^\dagger \mathbf{p}_\alpha T$: Using that the operators x, p , which commute with their commutator, obey $[f(x), p] = [x, p] \frac{\partial f(x)}{\partial x}$, and $[x, p] = i\hbar$, we have $e^{f(x)} p e^{-f(x)} = p + [f(x), p] + \frac{1}{2}[f(x), [f(x), p]] + \dots = p + [x, p] \frac{\partial f(x)}{\partial x} = p + i\hbar \frac{\partial f(x)}{\partial x}$. Accordingly, we get

$$T^\dagger \mathbf{p}_\alpha T = \mathbf{p}_\alpha + \frac{i}{\hbar} \sum_{\alpha'} q_{\alpha'} [\chi(\mathbf{r}_{\alpha'}, t), \mathbf{p}_\alpha] = \mathbf{p}_\alpha - q_\alpha \nabla \chi(\mathbf{r}_\alpha, t). \quad (7.28)$$

Altogether, the unitary transformation of the Hamiltonian then becomes

$$T^\dagger H T = \sum_\alpha \frac{[\mathbf{p}_\alpha - q_\alpha [\mathbf{A}_e(\mathbf{r}_\alpha, t) + \nabla \chi(\mathbf{r}_\alpha, t)]]^2}{2m_\alpha} + V_{\text{Coul}} + \sum_\alpha q_\alpha \phi_e(\mathbf{r}_\alpha, t). \quad (7.29)$$

Finally, the second term in (7.27) is

$$i\hbar \frac{\partial T^\dagger}{\partial t} T = - \sum_\alpha q_\alpha \frac{\partial \chi(\mathbf{r}_\alpha, t)}{\partial t} T^\dagger T = - \sum_\alpha q_\alpha \frac{\partial \chi(\mathbf{r}_\alpha, t)}{\partial t}. \quad (7.30)$$

The transformation (7.27) of the Hamiltonian (7.22) therefore reads

$$H_T = \sum_\alpha \frac{[\mathbf{p}_\alpha - q_\alpha [\mathbf{A}_e(\mathbf{r}_\alpha, t) + \nabla \chi(\mathbf{r}_\alpha, t)]]^2}{2m_\alpha} + V_{\text{Coul}} + \sum_\alpha q_\alpha \left[\phi_e(\mathbf{r}_\alpha, t) - \frac{\partial \chi(\mathbf{r}_\alpha, t)}{\partial t} \right]. \quad (7.31)$$

By comparing with H from (7.22), we recognise that the transformation (7.27) corresponds to the quantum mechanical counterpart of the gauge transformation (2.29), now with $\chi(\mathbf{r}_\alpha, t)$ being an operator. That is, the unitary transformation (7.25) led to a Hamiltonian obtained by exchanging the operators

$$\mathbf{A}_e(\mathbf{r}_\alpha, t) \rightarrow \mathbf{A}_e(\mathbf{r}_\alpha, t) + \nabla \chi(\mathbf{r}_\alpha, t) \quad (7.32a)$$

$$\phi_e(\mathbf{r}_\alpha, t) \rightarrow \phi_e(\mathbf{r}_\alpha, t) - \frac{\partial \chi(\mathbf{r}_\alpha, t)}{\partial t}, \quad (7.32b)$$

in analogy to the classical gauge transformation (2.29).

Review Questions 7.3.

- *What is the quantum mechanical counterpart of a gauge transformation?*
- *Recall that unitary transformations of the total quantum system simply correspond to a change of the “reference frame”. Discuss why your answer to the first question is thus not surprising.*

7.3 Interaction between electrons and the classical external electromagnetic field

We now want to study the interaction between atoms or molecules with laser pulses described by a classical external electromagnetic field. To simplify the problem, we suppose that only a single electron (of an atom or a molecule) is active at a time, known as the single active electron approximation (SAE). We suppose that the single electron is initially in its ground state $|g\rangle$, i.e., it is bound to the parent ion, which we assume to be localised at $\mathbf{R}$, i.e., we neglect the motion of the ion. To ease notation, we therefore denote by $\mathbf{E}_e(t) = \mathbf{E}_e(\mathbf{R}, t)$ the external electromagnetic field at the position of the electron. We further suppose no further interaction of the parent ion with the electromagnetic field than the contribution of a *static* Coulomb potential $V_{\text{Coul}}(\mathbf{r})$ seen by the electron. We can therefore neglect the parent ion in our Hamiltonian (except for the static Coulomb potential seen by the electron). Under consideration of the dipole approximation, in the length gauge the Hamiltonian therefore reads [c.f. Eq. (7.15)]

$$H(t) = \underbrace{\frac{\mathbf{p}^2}{2m} + V_{\text{Coul}}(\mathbf{r})}_{H_0} - \underbrace{\mathbf{d} \cdot \mathbf{E}_e(t)}_{V(t)}, \quad (7.33)$$

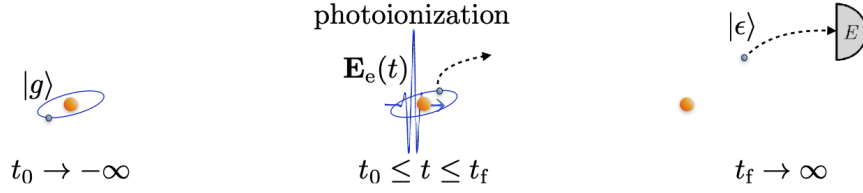
where m is the electron’s mass, $q = -e$ the electron’s charge, and $\mathbf{d} = q\mathbf{r} = -e\mathbf{r}$ the dipole operator. The above Hamiltonian decomposes into a static “field-free” Hamiltonian H_0 , and a time dependent interaction Hamiltonian $V(t)$. While solving the electronic dynamics under the Hamiltonian H_0 is already challenging (but can be done analytically, for example, for hydrogen-like atoms), an analytic solution to the electronic dynamics under the full Hamiltonian $H(t) = H_0 + V(t)$ requires further approximations. To this end, we suppose that the interaction is weak (i.e., the interaction described by $V(t)$ is weak compared to the energies of H_0) and solve the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = [H_0 + V(t)] |\psi(t)\rangle \quad (7.34)$$

for electronic states $|\psi(t)\rangle$ via a perturbative treatment of the interaction Hamiltonian $V(t)$.

7.3.1 Solving electron-field interaction via perturbation theory

As a typical example of electron-field interaction, we consider in the following the process of photoionization, i.e., the ionization of an electron due to its interaction with light. (Note that the following calculations are not restricted to photoionization, and also hold, for example, for transitions between bound states of the electron).



At time $t_0 \rightarrow -\infty$ the electron is prepared in the ground state $|g\rangle$. Between $t_0 \leq t \leq t_f$ it interacts with a finite light pulse described by the external field $\mathbf{E}_e(t)$. The light pulse is assumed to have sufficient high energy components, such that the electron can get ionized into the continuum state $|\epsilon\rangle$. This is typically achieved with femto- and/or atto-second laser pulses in the XUV range. At time $t_f \rightarrow \infty$, i.e., after the interaction with the laser pulse, a projective measurement on the continuum state $|\epsilon\rangle$ is performed (we measure the kinetic energy of the electron, possibly together with other quantum numbers of the electron, such as orbital angular momentum). Our aim is to express the outcome of this measurement, i.e., the photoelectron spectrum. To this end, we assume weak electron-field interactions, as typical for weak and moderate field intensities. We can therefore consider the interaction Hamiltonian $V(t)$ in (7.34) as a perturbation.

Interaction picture and perturbative expansion Let us first transform into the interaction picture with respect to H_0 via the unitary transformation $U_0^\dagger(t) = \exp[\frac{i}{\hbar}H_0t]$ ¹. Accordingly, the state of the electron in the interaction picture becomes

$$|\psi_I(t)\rangle = U_0^\dagger(t) |\psi(t)\rangle, \quad (7.35)$$

and the Schrödinger equation in the interaction picture reads [cf. Eq. (1.60)]

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle, \quad (7.36)$$

with

$$V_I(t) = U_0^\dagger(t) V(t) U_0(t) \quad (7.37)$$

the Hamiltonian in the interaction picture. Note that $V_I(t)$ describes the interaction between the electron and the external electromagnetic field. In the case of a light pulse of finite temporal width, there is no interaction before and after the light pulse, i.e., $\lim_{t \rightarrow \pm\infty} V_I(t) = 0$. Accordingly, in the interaction picture the electron is in a stationary state before and after the light pulse.

The formal solution to the Schrödinger equation (7.36) is obtained by acting with the unitary time evolution operator [see Eq. (1.47)]

$$U(t, t_0) = \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_{t_0}^t ds V_I(s) \right) \quad (7.38)$$

on the initial state $|\psi_I(t_0)\rangle$. We assume the electron to be initially in the bound ground state $|\psi(t_0)\rangle = e^{-i\omega_g t_0} |g\rangle$. Note that the global phase of the initial state is random, and can be chosen freely. Our choice becomes obvious by considering the initial state in the interaction picture, $|\psi_I(t_0)\rangle = U_0^\dagger(t_0) e^{-i\omega_g t_0} |g\rangle = |g\rangle$, where we used that $H_0 |g\rangle = \hbar\omega_g |g\rangle$. The time evolved state thus becomes

$$|\psi_I(t)\rangle = U(t, t_0) |g\rangle. \quad (7.39)$$

¹Note that in Sec. 1.5.4 we defined the interaction picture by transforming according to $U_0^\dagger(t) = \exp[\frac{i}{\hbar}H_0(t - t_0)]$, which includes the initial time t_0 . However, since we can transform into any picture via any unitary of our choice, we here choose $U_0^\dagger(t) = \exp[\frac{i}{\hbar}H_0t]$ in order to avoid too much phase terms of the form $\exp(i\omega t_0)$

Using the expression for $U(t, t_0)$ from Eq. (1.42), the solution (7.39) can be written as

$$|\psi_I(t)\rangle = \underbrace{|g\rangle}_{|\psi_I^{(0)}(t)\rangle} + \sum_{n=1}^{\infty} \underbrace{\left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n V_I(t_1) V_I(t_2) \cdots V_I(t_n)}_{|\psi_I^{(n)}(t)\rangle} |g\rangle. \quad (7.40)$$

This defines a perturbative expansion, with n the perturbation order,

$$|\psi_I(t)\rangle = \sum_{n=0}^{\infty} |\psi_I^{(n)}(t)\rangle. \quad (7.41)$$

Up to second order, the expansion reads

$$|\psi_I(t)\rangle = |g\rangle - \frac{i}{\hbar} \int_{t_0}^t dt_1 V_I(t_1) |g\rangle - \frac{1}{\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 V_I(t_1) V_I(t_2) |g\rangle + \mathcal{O}(V_I^3). \quad (7.42)$$

Since the perturbation order n corresponds to the number of interaction terms appearing in the corresponding expansion term, we recognize that approximating the electron's state via the first few orders is only valid in the case of weak interactions as compared to the unperturbed Hamiltonian. As already mentioned above, we assume such weak interactions, and continue with considering the perturbation expansion of $|\psi_I(t)\rangle$ up to second order as provided in Eq. (7.42).

Let us now consider the interaction Hamiltonian in the interaction picture from Eq. (7.37). Using the definition of $V(t)$ from Eq. (7.33), it reads

$$V_I(t) = - \underbrace{U_0^\dagger(t) \mathbf{d} U_0(t)}_{\mathbf{d}_I(t)} \cdot \mathbf{E}_e(t) = -\mathbf{d}_I(t) \cdot \mathbf{E}_e(t), \quad (7.43)$$

where $\mathbf{d}_I(t)$ is the dipole operator in the interaction picture.

Transition amplitudes Suppose now that we perform a measurement at a late time $t_f \rightarrow \infty$ when the light pulse has passed, such that there is no interaction between electron and electromagnetic field at the time of the measurement, i.e., $H(t_f) = H_0$. Further suppose that we perform a projective measurement [see Sec. 1.4.2] that can resolve the eigenstates $|\epsilon\rangle$ of the Hamiltonian H_0 at the time of the measurement. These states satisfy $H_0 |\epsilon\rangle = \hbar\omega_\epsilon |\epsilon\rangle$, with the spectral decomposition of H_0 reading

$$H_0 = \oint d\epsilon \hbar\omega_\epsilon |\epsilon\rangle\langle\epsilon|. \quad (7.44)$$

Note that ϵ can be a set of quantum numbers, which can include both continuous and discrete values, i.e., $\epsilon = (E, \alpha)$, with E the energy, and α all other quantum numbers such as the spin or orbital angular momentum quantum numbers of the electron.

The corresponding observable [see Eq. (1.30)] of the considered measurement reads

$$M = \oint d\epsilon \epsilon |\epsilon\rangle\langle\epsilon|. \quad (7.45)$$

The outcome ϵ therefore occurs at time t_f with probability

$$p(\epsilon) = |\langle\epsilon|\psi(t_f)\rangle|^2 = \left| \langle\epsilon| U_0(t_f) U_0^\dagger(t_f) |\psi(t_f)\rangle \right|^2 \stackrel{(7.35)}{=} |\langle\epsilon| U_0(t_f) |\psi_I(t_f)\rangle|^2 = |\langle\epsilon|\psi_I(t_f)\rangle|^2, \quad (7.46)$$

where the last equality is due to $|\epsilon\rangle$ being an eigenstate of H_0 , such that $U_0^\dagger(t_f) |\epsilon\rangle = e^{i\omega_\epsilon t_f} |\epsilon\rangle$. Note that if we can only resolve the photoelectron's energy E but are blind to all other quantum numbers α [recall that $\epsilon = (E, \alpha)$ describes a set of quantum numbers], then the probability to

measure a photoelectron with energy E is given by $p(E) = \sum_{\alpha} p(E, \alpha) = \sum_{\alpha} |\langle E\alpha | \psi(t_f) \rangle|^2$, i.e., we must sum over all probabilities $p(\epsilon) = p(E, \alpha)$ corresponding to the same energy E .

In order to describe our measurement outcome, we must consider the transition amplitudes

$$A(\epsilon) = \langle \epsilon | \psi_I(t_f) \rangle, \quad p(\epsilon) = |A(\epsilon)|^2. \quad (7.47)$$

Using the perturbative expansion of the electron's state from Eq. (7.41), the perturbative expansion of the transition amplitudes becomes

$$A(\epsilon) = \sum_{n=0}^{\infty} \underbrace{\langle \epsilon | \psi_I^{(n)}(t_f) \rangle}_{A^{(n)}(\epsilon)} = \sum_{n=0}^{\infty} A^{(n)}(\epsilon). \quad (7.48)$$

The amplitudes up to second order thus read [using (7.42)]

$$A^{(0)}(\epsilon) = \langle \epsilon | g \rangle = 0, \quad (7.49a)$$

$$A^{(1)}(\epsilon) = -\frac{i}{\hbar} \int_{t_0}^{t_f} dt_1 \langle \epsilon | V_I(t_1) | g \rangle, \quad (7.49b)$$

$$A^{(2)}(\epsilon) = -\frac{1}{\hbar^2} \int_{t_0}^{t_f} dt_1 \int_{t_0}^{t_1} dt_2 \langle \epsilon | V_I(t_1) V_I(t_2) | g \rangle. \quad (7.49c)$$

Note that $A^{(0)}(\epsilon)$ corresponds to the process where no photon has been absorbed or emitted, $A^{(1)}(\epsilon)$ to the process where a single photon is absorbed or emitted, and $A^{(2)}(\epsilon)$ corresponds to processes involving the absorption or emission of two photons. That is, the perturbation order n corresponds to the number of photons involved in the transition process. We now examine the first and second order processes separately, where this becomes more evident.

First order transition amplitudes Using Eq. (7.43), as well as $H_0 |g\rangle = \hbar\omega_g |g\rangle$ and $H_0 |\epsilon\rangle = \hbar\omega_{\epsilon} |\epsilon\rangle$, the first order transition amplitude (7.49b) becomes

$$\begin{aligned} A^{(1)}(\epsilon) &= \frac{i}{\hbar} \int_{t_0}^{t_f} dt_1 \mathbf{E}_e(t_1) \cdot \langle \epsilon | U_0^\dagger(t_1) d U_0(t_1) | g \rangle, \\ &= \frac{i}{\hbar} \int_{t_0}^{t_f} dt_1 e^{i(\omega_{\epsilon} - \omega_g)t_1} \mathbf{E}_e(t_1) \cdot \underbrace{\langle \epsilon | \mathbf{d} | g \rangle}_{\mathbf{d}_{\epsilon,g}}, \\ &= \frac{i}{\hbar} \mathbf{d}_{\epsilon,g} \cdot \int_{t_0}^{t_f} dt_1 e^{i\omega_{\epsilon g} t_1} \mathbf{E}_e(t_1), \end{aligned} \quad (7.50)$$

where we introduced the dipole matrix elements $\mathbf{d}_{\epsilon,g} = \langle \epsilon | \mathbf{d} | g \rangle$, and $\omega_{\epsilon g} = \omega_{\epsilon} - \omega_g$. Since we consider the limits $t_f \rightarrow \infty$ and $t_0 \rightarrow -\infty$, the first order transition amplitude (7.50) is proportional to the Fourier transform

$$\tilde{\mathbf{E}}_e(\omega_{\epsilon g}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt_1 e^{i\omega_{\epsilon g} t_1} \mathbf{E}_e(t_1) \quad (7.51)$$

of the external electromagnetic field (i.e., the laser pulse). As a result, we obtain

$$\boxed{A^{(1)}(\epsilon) = \frac{i}{\hbar} \sqrt{2\pi} \mathbf{d}_{\epsilon,g} \cdot \tilde{\mathbf{E}}_e(\omega_{\epsilon g})}. \quad (7.52)$$

When the electron gets ionized, with $|\epsilon\rangle$ describing a continuum state, in first order approximation, the spectrum of the photoelectron is thus a replica of the laser pulse spectrum that is weighted by the dipole matrix elements $\mathbf{d}_{\epsilon,g}$, which depend on the atom (or molecule), and

quantify the likelihood of the transition from the electronic ground state $|g\rangle$ to the continuum state $|\epsilon\rangle$.

We can express the transition amplitude (7.52) more compactly by considering the Fourier transform

$$\tilde{V}(\omega) = -\mathbf{d} \cdot \tilde{\mathbf{E}}_e(\omega), \quad (7.53)$$

of the interaction Hamiltonian $V(t) = -\mathbf{d} \cdot \mathbf{E}_e(t)$ [see Eq. (7.33)], with matrix elements

$$\tilde{V}_{\epsilon,g}(\omega) = \langle \epsilon | \tilde{V}(\omega) | g \rangle = -\langle \epsilon | \mathbf{d} | g \rangle \cdot \tilde{\mathbf{E}}_e(\omega) = -\mathbf{d}_{\epsilon,g} \cdot \tilde{\mathbf{E}}_e(\omega). \quad (7.54)$$

The first order transition amplitude (7.52) then reads

$$A^{(1)}(\epsilon) = -\frac{i}{\hbar} \sqrt{2\pi} \tilde{V}_{\epsilon,g}(\omega_{\epsilon g}). \quad (7.55)$$

We can thus associate $\tilde{V}_{\epsilon,g}(\omega_{\epsilon g})$ with a single-photon dipole transition from the ground state $|g\rangle$ to the continuum state $|\epsilon\rangle$ mediated by the external electric field with frequency $\omega_{\epsilon g}$.

Second order transition amplitudes By proceeding similar as above, for the second order transition amplitude (7.49c), we obtain

$$\begin{aligned} A^{(2)}(\epsilon) &= -\frac{1}{\hbar^2} \int_{t_0}^{t_f} dt_1 \int_{t_0}^{t_1} dt_2 \underbrace{\langle \epsilon | U_0^\dagger(t_1) V(t_1) U_0(t_1) U_0^\dagger(t_2) V(t_2) U_0(t_2) | g \rangle}_{e^{i\omega_\epsilon t_1} \langle \epsilon |} \underbrace{e^{-iH_0(t_1-t_2)/\hbar}}_{e^{-iH_0(t_1-t_2)/\hbar}} \underbrace{e^{-i\omega_g t_2} | g \rangle}_{e^{-i\omega_g t_2} | g \rangle} \\ &= -\frac{1}{\hbar^2} \int_{t_0}^{t_f} dt_1 \int_{t_0}^{t_1} dt_2 e^{i\omega_\epsilon t_1} e^{-i\omega_g t_2} \langle \epsilon | V(t_1) e^{-iH_0(t_1-t_2)/\hbar} V(t_2) | g \rangle. \end{aligned} \quad (7.56)$$

This expression has a nice interpretation: At time t_2 , the electron undergoes a transition from the ground state $|g\rangle$ to an intermediate state described by the interaction term $V(t_2)$. Between time t_2 and t_1 , the electron remains in the intermediate state and does not interact with the electromagnetic field, as accounted for by the free evolution operator $e^{-iH_0(t_1-t_2)/\hbar}$. Thereafter, at time t_1 , the electron again interacts with the electromagnetic field as described by $V(t_1)$, undergoing a transition from the intermediate state to the state $|\epsilon\rangle$.

For the first order transition amplitude we have seen that it is convenient to express it in terms of the electromagnetic field in the frequency domain [compare Eqs. (7.50) and (7.52)]. We thus aim at expressing the electric fields in (7.56) [contained in $V(t)$] also in the frequency domain. To do so, let us first lift the upper limit of the second integral to t_f , and compensate for that by the Heaviside function $\Theta(t_1 - t_2)$,

$$A^{(2)}(\epsilon) = -\frac{1}{\hbar^2} \int_{t_0}^{t_f} dt_1 \int_{t_0}^{t_f} dt_2 e^{i\omega_\epsilon t_1} e^{-i\omega_g t_2} \langle \epsilon | V(t_1) \underbrace{\Theta(t_1 - t_2) e^{-iH_0(t_1-t_2)/\hbar}}_{iG^+(t_1-t_2)} V(t_2) | g \rangle. \quad (7.57)$$

By introducing the retarded Green's function

$$G^+(t) = -i\Theta(t)e^{-iH_0 t/\hbar}, \quad (7.58)$$

the second order transition amplitude becomes

$$A^{(2)}(\epsilon) = -\frac{i}{\hbar^2} \int_{t_0}^{t_f} dt_1 \int_{t_0}^{t_f} dt_2 e^{i\omega_\epsilon t_1} e^{-i\omega_g t_2} \langle \epsilon | V(t_1) G^+(t_1 - t_2) V(t_2) | g \rangle. \quad (7.59)$$

We can now express the retarded Green's function in terms of its Fourier transform,

$$G^+(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega' \tilde{G}^+(\omega') e^{-i\omega' t}. \quad (7.60)$$

Therewith Eq. (7.59) becomes

$$A^{(2)}(\epsilon) = -\frac{i}{\sqrt{2\pi}\hbar^2} \int_{-\infty}^{\infty} d\omega' \int_{t_0}^{t_f} dt_1 dt_2 e^{i(\omega_\epsilon - \omega')t_1} e^{-i(\omega_g - \omega')t_2} \langle \epsilon | V(t_1) \tilde{G}^+(\omega') V(t_2) | g \rangle. \quad (7.61)$$

Substituting $\omega' = \omega_g + \omega$, and using $\omega_{\epsilon g} = \omega_\epsilon - \omega_g$, leads to

$$A^{(2)}(\epsilon) = -\frac{i}{\sqrt{2\pi}\hbar^2} \int_{-\infty}^{\infty} d\omega \int_{t_0}^{t_f} dt_1 dt_2 e^{i(\omega_{\epsilon g} - \omega)t_1} e^{i\omega t_2} \langle \epsilon | V(t_1) \tilde{G}^+(\omega_g + \omega) V(t_2) | g \rangle. \quad (7.62)$$

In the limit $t_0 \rightarrow -\infty$ and $t_f \rightarrow \infty$, we can consider the Fourier transform

$$\tilde{V}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt V(t) e^{i\omega t} \quad (7.63)$$

of the interaction operator: In Eq. (7.62), the integral over t_1 yields $\tilde{V}(\omega_{\epsilon g} - \omega)$, and the integral over t_2 yields $\tilde{V}(\omega)$. Accordingly, we have

$$A^{(2)}(\epsilon) = -\frac{i}{\hbar^2} \sqrt{2\pi} \int_{-\infty}^{\infty} d\omega \langle \epsilon | \tilde{V}(\omega_{\epsilon g} - \omega) \tilde{G}^+(\omega_g + \omega) \tilde{V}(\omega) | g \rangle. \quad (7.64)$$

As we show in Appendix A.2, the Fourier transform of the retarded Green's function can be expressed as

$$\tilde{G}^+(\omega) = \lim_{\eta \searrow 0} \frac{1}{\sqrt{2\pi}} \frac{1}{\omega - H_0/\hbar + i\eta}. \quad (7.65)$$

Since it is a function of H_0 , with spectral decomposition $H_0 = \oint d\epsilon' \hbar \omega_{\epsilon'} |\epsilon'\rangle \langle \epsilon'|$ [see Eq. (7.44)], we get

$$\tilde{G}^+(\omega) = \lim_{\eta \searrow 0} \frac{1}{\sqrt{2\pi}} \oint d\epsilon' \frac{1}{\omega - \omega_{\epsilon'} + i\eta} |\epsilon'\rangle \langle \epsilon'|. \quad (7.66)$$

Using this in Eq. (7.64),

$$A^{(2)}(\epsilon) = -\frac{i}{\hbar^2} \int_{-\infty}^{\infty} d\omega \lim_{\eta \searrow 0} \oint d\epsilon' \underbrace{\langle \epsilon | \tilde{V}(\omega_{\epsilon g} - \omega) | \epsilon' \rangle}_{\tilde{V}_{\epsilon, \epsilon'}(\omega_{\epsilon g} - \omega)} \frac{1}{\omega_g + \omega - \omega_{\epsilon'} + i\eta} \underbrace{\langle \epsilon' | \tilde{V}(\omega) | g \rangle}_{\tilde{V}_{\epsilon', g}(\omega)}. \quad (7.67)$$

and utilizing the notation introduced in (7.54), then results in

$$\boxed{A^{(2)}(\epsilon) = -\frac{i}{\hbar^2} \int_{-\infty}^{\infty} d\omega \lim_{\eta \searrow 0} \oint d\epsilon' \frac{\tilde{V}_{\epsilon, \epsilon'}(\omega_{\epsilon g} - \omega) \tilde{V}_{\epsilon', g}(\omega)}{\omega - \omega_{\epsilon'} + i\eta}.} \quad (7.68)$$

This equation has a nice interpretation: $\tilde{V}_{\epsilon', g}(\omega)$ describes a single-photon dipole transition from the initial ground state $|g\rangle$ to the intermediate eigenstate $|\epsilon'\rangle$ of H_0 (which can be a bound or continuum state) mediated by the external electric field with frequency ω . The transition is resonant if $\omega \approx \omega_{\epsilon' g}$, since then the denominator in (7.68) gets small, and otherwise off-resonant. As described by $\tilde{V}_{\epsilon, \epsilon'}(\omega_{\epsilon g} - \omega)$, this is followed by a second single-photon dipole transition from the intermediate state $|\epsilon'\rangle$ to the final state $|\epsilon\rangle$, which is mediated by the external electric field with frequency $\omega_{\epsilon g} - \omega$. The integration $\oint d\epsilon'$ states that the transition can be through all possible intermediate states $|\epsilon'\rangle$, and the integration $\int d\omega$ states that all frequency components of the external electric field can mediate the transition.

Review Questions 7.4.

- *What is the Hamiltonian of a single electron interacting with the external electromagnetic field under the dipole approximation? Explain the different terms.*
- *What is the perturbative expansion?*
- *What is an intuitive picture for the processes of the different perturbation orders?*
- *Consider the transition of a single electron from the ground state to a continuum state: What is the zeroth/first/second order transition amplitude. Explain the terms.*

Chapter 8

Interaction between quantum matter and the quantized electromagnetic field

So far we considered the quantization of the electromagnetic field in the absence of charges [Chapters 4, 5, and 6] and the interaction of quantized charged particles with a classical electromagnetic field [Chapter 7]. We now consider the interaction of charged particles with the electromagnetic field by quantizing both, the particles and the fields.

The Hamiltonian which describes the interactions of quantum particles with the quantized electromagnetic field can be obtained from the classical total energy (2.52) by quantizing both the electromagnetic field and the matter degrees of freedom (as discussed in Chapter 3). Using Eqs. (2.56) and (3.11), this yields

$$H = \sum_{\alpha} \frac{1}{2m_{\alpha}} (\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_{\perp}(\mathbf{r}_{\alpha}))^2 + \sum_{\kappa} \hbar \omega_{\kappa} \left(a_{\kappa}^{\dagger} a_{\kappa} + \frac{1}{2} \right) + V_{\text{Coul}}, \quad (8.1)$$

where

$$V_{\text{Coul}} = \epsilon_{\text{Coul}} + \frac{1}{8\pi\epsilon_0} \sum_{\alpha \neq \beta} \frac{q_{\alpha} q_{\beta}}{|\mathbf{r}_{\alpha} - \mathbf{r}_{\beta}|} = H_{\parallel} \quad (8.2)$$

is the Coulomb interaction Hamiltonian. Note that since we quantized both particles and fields, here $\mathbf{r}_{\alpha}$ is the position operator of the α -th particle, and $\mathbf{p}_{\alpha} = -i\hbar \nabla_{\alpha}$ the momentum operator.

8.1 Dipole approximation

Similar as in Sec. 7.1.2, we consider the dipole (or long-wavelength) approximation. That is, the wavelength of the radiation is considered to be much larger than the spatial extension of the system of charged particles (e.g, when an atom interacts with visible light). To this end, we make the approximation that all charged particles experience the same vector potential, $\mathbf{A}_{\perp}(\mathbf{r}_{\alpha}) \approx \mathbf{A}_{\perp}(\mathbf{R})$, where $\mathbf{R}$ is the position of the system of particles. The Hamiltonian (8.1) therefore reads

$$H_{\text{dip}} = \sum_{\alpha} \frac{1}{2m_{\alpha}} (\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_{\perp}(\mathbf{R}))^2 + \sum_{\kappa} \hbar \omega_{\kappa} \left(a_{\kappa}^{\dagger} a_{\kappa} + \frac{1}{2} \right) + V_{\text{Coul}}. \quad (8.3)$$

Here the first term suggests to translate each momentum operator $\mathbf{p}_{\alpha}$ by $q_{\alpha} \mathbf{A}_{\perp}(\mathbf{R})$ via the unitary transformation (note that the following unitary transformation leads to expressions similar to the length gauge in the case of classical fields)

$$T^{\dagger} \mathbf{p}_{\alpha} T = \mathbf{p}_{\alpha} + q_{\alpha} \mathbf{A}_{\perp}(\mathbf{R}). \quad (8.4)$$

This is accomplished by the unitary

$$T^\dagger = \exp \left[-\frac{i}{\hbar} \sum_{\alpha} q_{\alpha} \mathbf{r}_{\alpha} \cdot \mathbf{A}_{\perp}(\mathbf{R}) \right], \quad (8.5)$$

which can be seen using $e^A B e^{-A} = B + [A, B]$ for A, B commuting with their commutator $[A, B]$,

$$T^\dagger \mathbf{p}_{\alpha} T = \mathbf{p}_{\alpha} + \left[-\frac{i}{\hbar} \sum_{\alpha'} q_{\alpha'} \mathbf{r}_{\alpha'} \cdot \mathbf{A}_{\perp}(\mathbf{R}), \mathbf{p}_{\alpha} \right] = \mathbf{p}_{\alpha} + -\frac{i}{\hbar} q_{\alpha} \mathbf{A}_{\perp}(\mathbf{R}) \underbrace{[\mathbf{r}_{\alpha}, \mathbf{p}_{\alpha}]}_{i\hbar} = \mathbf{p}_{\alpha} + q_{\alpha} \mathbf{A}_{\perp}(\mathbf{R}). \quad (8.6)$$

As we will see in a bit, it is helpful to rewrite the unitary (8.5) with the help of the dipole operator $\mathbf{d} = \sum_{\alpha} q_{\alpha} \mathbf{r}_{\alpha}$, and using the vector potential from Eq. (3.8),

$$\mathbf{A}_{\perp}(\mathbf{R}) = \sum_{\kappa} \sqrt{\frac{\hbar}{2\omega_{\kappa}\epsilon_0}} \mathbf{u}_{\kappa}(\mathbf{R}) [a_{\kappa} + a_{\kappa}^{\dagger}], \quad (8.7)$$

yielding

$$T = \exp \left[\frac{i}{\hbar} \mathbf{d} \cdot \mathbf{A}_{\perp}(\mathbf{R}) \right] = \exp \left[\sum_{\kappa} \left(\lambda_{\kappa} a_{\kappa}^{\dagger} - \lambda_{\kappa}^* a_{\kappa} \right) \right], \quad (8.8)$$

where

$$\lambda_{\kappa} = \frac{i}{\sqrt{2\epsilon_0\hbar\omega_{\kappa}}} \mathbf{d} \cdot \mathbf{u}_{\kappa}(\mathbf{R}). \quad (8.9)$$

Our aim is now to transform the Hamiltonian (8.3) via the unitary (8.5), i.e., $H'_{\text{dip}} = T^\dagger H_{\text{dip}} T$. The first term in (8.3), which motivated this transformation, simply becomes

$$T^\dagger \left[\sum_{\alpha} \frac{1}{2m_{\alpha}} (\mathbf{p}_{\alpha} - q_{\alpha} \mathbf{A}_{\perp}(\mathbf{R}))^2 \right] T = \sum_{\alpha} \frac{\mathbf{p}_{\alpha}^2}{2m_{\alpha}}, \quad (8.10)$$

since $\mathbf{A}_{\perp}(\mathbf{R})$ commutes with T , and $\mathbf{p}_{\alpha}$ transforms according to (8.4). The third term in (8.3), the Coulomb interaction term from Eq. (8.2), remains invariant, since it is a function of $\mathbf{r}_{\alpha}$, and, thus, commutes with T ,

$$T^\dagger V_{\text{Coul}} T = V_{\text{Coul}}. \quad (8.11)$$

We are left with the second term in (8.3). Considering its transformation, we notice that T , as written in Eq. (8.8), constitutes a unitary displacement operator of the form (4.19). Accordingly, we can use (4.25), yielding

$$T^\dagger a_{\kappa} T = a_{\kappa} + \lambda_{\kappa}, \quad (8.12a)$$

$$T^\dagger a_{\kappa}^{\dagger} T = a_{\kappa}^{\dagger} + \lambda_{\kappa}^*. \quad (8.12b)$$

Therewith, we find

$$\begin{aligned} T^\dagger H_{\perp} T &= T^\dagger \left[\sum_{\kappa} \hbar\omega_{\kappa} \left(a_{\kappa}^{\dagger} a_{\kappa} + \frac{1}{2} \right) \right] T \\ &= \sum_{\kappa} \hbar\omega_{\kappa} \left(T^\dagger a_{\kappa}^{\dagger} T T^\dagger a_{\kappa} T + \frac{1}{2} \right) \\ &= \sum_{\kappa} \hbar\omega_{\kappa} \left((a_{\kappa}^{\dagger} + \lambda_{\kappa}^*)(a_{\kappa} + \lambda_{\kappa}) + \frac{1}{2} \right) \\ &= \underbrace{\sum_{\kappa} \hbar\omega_{\kappa} \left(a_{\kappa}^{\dagger} a_{\kappa} + \frac{1}{2} \right)}_{H_{\perp}} + \underbrace{\sum_{\kappa} \hbar\omega_{\kappa} \lambda_{\kappa}^* \lambda_{\kappa}}_{\epsilon_{\text{dip}}} + \underbrace{\sum_{\kappa} \hbar\omega_{\kappa} \left(\lambda_{\kappa} a_{\kappa}^{\dagger} + \lambda_{\kappa}^* a_{\kappa} \right)}_{H'_{\text{dip,I}}}. \end{aligned} \quad (8.13)$$

Here, the first term is $H_\perp$ from Eq. (3.11). Using the definition of λ_κ from (8.9), the second term gives

$$\epsilon_{\text{dip}} = \sum_{\kappa} \hbar \omega_{\kappa} \lambda_{\kappa}^* \lambda_{\kappa} = \sum_{\kappa} \frac{1}{2\epsilon_0} |\mathbf{d} \cdot \mathbf{u}_{\kappa}(\mathbf{R})|^2. \quad (8.14)$$

It only depends on the dipole operator $\mathbf{d}$ (which remains invariant under the unitary transformation, i.e., $T^\dagger \mathbf{d} T = \mathbf{d}$), and represents a dipole self-energy of the system. The dipole self-energy can be seen as the energy associated with the interaction of a dipole with its own electromagnetic field. On the other hand, the third term in (8.13) gives the dipole interaction Hamiltonian

$$H'_{\text{dip},\text{I}} = \sum_{\kappa} \hbar \omega_{\kappa} (\lambda_{\kappa} a_{\kappa}^\dagger + \lambda_{\kappa}^* a_{\kappa}) = - \sum_{\kappa} i \sqrt{\frac{\hbar \omega_{\kappa}}{2\epsilon_0}} \mathbf{d} \cdot \mathbf{u}_{\kappa}(\mathbf{R}) [a_{\kappa} - a_{\kappa}^\dagger] \stackrel{(3.10)}{=} -\mathbf{d} \cdot \mathbf{E}_\perp(\mathbf{R}), \quad (8.15)$$

where we identified the transverse electric field operator $\mathbf{E}_\perp(\mathbf{R})$ from the *previous representation*, i.e., before the unitary transformation. In fact, under the unitary transformation according to T , the transverse electric field operator changes to $\mathbf{E}'_\perp(\mathbf{r}) = T^\dagger \mathbf{E}_\perp(\mathbf{r}) T = \mathbf{E}_\perp(\mathbf{r}) - \frac{1}{\epsilon_0} \mathbf{P}_\perp(\mathbf{r})$, where $\mathbf{P}_\perp(\mathbf{r})$ is the transverse component of the polarization density $\mathbf{P}(\mathbf{r}) = \mathbf{d} \delta(\mathbf{r} - \mathbf{R})$ corresponding to a dipole $\mathbf{d}$ localised at $\mathbf{r} = \mathbf{R}$. By defining the displacement $\mathbf{D}(\mathbf{r}) = \epsilon_0 \mathbf{E}(\mathbf{r}) + \mathbf{P}(\mathbf{r})$, one can show that the displacement operator in the *new representation* corresponds to the transverse electric field in the *previous representation*, $\mathbf{D}'(\mathbf{r}) = T^\dagger \mathbf{D}(\mathbf{r}) T = \epsilon_0 \mathbf{E}_\perp(\mathbf{r})$. Hence, correctly speaking, the dipole interaction Hamiltonian (8.15) must read $-\mathbf{d} \cdot \mathbf{D}'(\mathbf{R})/\epsilon_0$ (for a detailed discussion, see Complement A_{IV} 2c in Ref. [4]). However, we continue with the “sloppy” notation from Eq. (8.15), which is often used in the literature.

Altogether, in the new representation, the Hamiltonian (8.3) becomes

$$H'_{\text{dip}} = \sum_{\alpha} \frac{\mathbf{p}_{\alpha}^2}{2m_{\alpha}} + \sum_{\kappa} \hbar \omega_{\kappa} \left(a_{\kappa}^\dagger a_{\kappa} + \frac{1}{2} \right) - \mathbf{d} \cdot \mathbf{E}_\perp(\mathbf{R}) + \epsilon_{\text{dip}} + V_{\text{Coul}}. \quad (8.16)$$

In contrast to the exact Hamiltonian (8.1), the dipole approximation led to an Hamiltonian where the interaction between the electromagnetic field and the charged quantum particles is purely linear in the field, i.e., in $H'_{\text{dip},\text{I}}$ creation and annihilation operators of the field are only involved linearly. This allows for rather simple calculations, justifying the relevance of the dipole approximation for light-matter interaction.

8.2 Dipole approximation with electromagnetic field in free space

Recall that for the electromagnetic field in free space, as discussed in Secs. 2.3.2 and 3.3, we consider the mode indexing $\kappa = (\mathbf{k}, p)$, where $\mathbf{k}$ is continuous, and $p \in \{1, 2\}$ represents the polarization index. The dipole approximation Hamiltonian (8.16) is then easily seen to be

$$H'_{\text{dip}} = \sum_{\alpha} \frac{\mathbf{p}_{\alpha}^2}{2m_{\alpha}} + \sum_p \int d^3k \hbar \omega_{\mathbf{k}} \left(a_p^\dagger(\mathbf{k}) a_p(\mathbf{k}) + \frac{1}{2} \right) - \mathbf{d} \cdot \mathbf{E}_\perp(\mathbf{R}) + \epsilon_{\text{dip}} + V_{\text{Coul}}, \quad (8.17)$$

where (c.f. Eq. (3.25))

$$\mathbf{E}_\perp(\mathbf{R}) = i \sum_p \int d^3k \sqrt{\frac{\hbar \omega_{\mathbf{k}}}{2\epsilon_0 (2\pi)^3}} \left[\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k} \cdot \mathbf{R}} a_p(\mathbf{k}) - \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k} \cdot \mathbf{R}} a_p^\dagger(\mathbf{k}) \right]. \quad (8.18)$$

The dipole self energy from Eq. (8.14) becomes

$$\epsilon_{\text{dip}} = \sum_p \int d^3k \frac{1}{2\epsilon_0 (2\pi)^3} \left| \mathbf{d} \cdot \mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k} \cdot \mathbf{R}} \right|^2, \quad (8.19)$$

and V_{Coul} from Eq. (8.2) remains unchanged.

Review Questions 8.1.

- Explain the dipole approximation. Under which condition can it be applied?
- Explain why it is expected that the unitary transformation (8.5) leads to expressions similar to the length gauge in the case of classical fields (compare with Eqs. (7.12) and (7.25))
- What is the dipole self-energy?
- Explain the individual terms of the Hamiltonian (8.16).

8.3 Transitions in first- and second-order perturbation theory

Let us now consider the dipole approximation Hamiltonian from Eq. (8.16), and decompose it into a matter Hamiltonian H_{mat} , a radiation Hamiltonian $H_{\text{rad}} = H_{\perp}$, and an interaction Hamiltonian H_{int} ,

$$H'_{\text{dip}} = H_{\text{mat}} + H_{\text{rad}} + H_{\text{int}}, \quad (8.20)$$

where

$$H_{\text{mat}} = \sum_{\alpha} \frac{\mathbf{p}_{\alpha}^2}{2m_{\alpha}} + \epsilon_{\text{dip}} + V_{\text{Coul}}, \quad (8.21a)$$

$$H_{\text{rad}} = H_{\perp} = \sum_{\kappa} \hbar\omega_{\kappa} \left(a_{\kappa}^{\dagger} a_{\kappa} + \frac{1}{2} \right), \quad (8.21b)$$

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}_{\perp}(\mathbf{R}). \quad (8.21c)$$

8.3.1 Coupling coefficients

Let us suppose now that the spectral decomposition of the matter Hamiltonian H_{mat} is known (e.g. when considering the hydrogen atom),

$$H_{\text{mat}} = \int d\epsilon E_{\text{mat},\epsilon} |\epsilon\rangle\langle\epsilon|, \quad (8.22)$$

where $E_{\text{mat},\epsilon} = \hbar\omega_{\epsilon}$. Moreover, in Chapter 4, we found that the multi-mode Fock states $|\{n_{\kappa}\}\rangle = |n_1, n_2, n_3, \dots\rangle$ are the eigenstates of the radiation Hamiltonian $H_{\text{rad}} = H_{\perp}$, whose spectral decomposition is

$$H_{\text{rad}} = \sum_{\{n_{\kappa}\}} \left[\sum_{\kappa} \hbar\omega_{\kappa} \left(n_{\kappa} + \frac{1}{2} \right) \right] |\{n_{\kappa}\}\rangle\langle\{n_{\kappa}\}|. \quad (8.23)$$

The eigenstates of the uncoupled matter-radiation system with Hamiltonian $H_0 = H_{\text{rad}} + H_{\text{mat}}$ is, thus, given by the states $|\epsilon, \{n_{\kappa}\}\rangle$, and the spectral decomposition reads

$$\begin{aligned} H_0 &= \mathbb{1}_{\text{mat}} \otimes H_{\text{rad}} + H_{\text{mat}} \otimes \mathbb{1}_{\text{rad}} \\ &= \sum_{\{n_{\kappa}\}} \int d\epsilon \underbrace{\left[E_{\text{mat},\epsilon} + \sum_{\kappa} \hbar\omega_{\kappa} \left(n_{\kappa} + \frac{1}{2} \right) \right]}_{E_{\alpha}} \underbrace{|\epsilon, \{n_{\kappa}\}\rangle\langle\epsilon, \{n_{\kappa}\}|}_{|\psi_{\alpha}\rangle\langle\psi_{\alpha}|} \\ &\quad \int d\alpha \\ &= \int d\alpha E_{\alpha} |\psi_{\alpha}\rangle\langle\psi_{\alpha}|. \end{aligned} \quad (8.24)$$

Note that since we additionally consider an interaction between matter and radiation according to the interaction Hamiltonian H_{int} from Eq. (8.21c), the states $|\epsilon, \{n_\kappa\}\rangle = |\psi_\alpha\rangle$ are not the eigenstates of the full Hamiltonian H'_{dip} .

The action of the interaction Hamiltonian H_{int} can be understood by expressing it through the ON-basis states $\{|\epsilon\rangle\}_\epsilon$ of the matter degrees of freedom and considering its action on the basis states $|\epsilon, \{n_\kappa\}\rangle$ of the uncoupled system. To this end, we expand the dipole operator $\mathbf{d}$ in this basis,

$$\mathbf{d} = \underbrace{\sum_{\epsilon'} |\epsilon'\rangle\langle\epsilon'|}_{\mathbb{1}} \mathbf{d} \underbrace{\sum_{\epsilon} |\epsilon\rangle\langle\epsilon|}_{\mathbb{1}} = \sum_{\epsilon, \epsilon'} \langle\epsilon'| \mathbf{d} |\epsilon\rangle |\epsilon'\rangle\langle\epsilon| = \sum_{\epsilon, \epsilon'} \mathbf{d}_{\epsilon', \epsilon} |\epsilon'\rangle\langle\epsilon|, \quad (8.25)$$

with $\mathbf{d}_{\epsilon', \epsilon} = \langle\epsilon'| \mathbf{d} |\epsilon\rangle$. Using this together with $\mathbf{E}_\perp(\mathbf{r}) = \sum_\kappa i\sqrt{\frac{\hbar\omega_\kappa}{2\epsilon_0}} \mathbf{u}_\kappa(\mathbf{r})(a_\kappa - a_\kappa^\dagger)$ from Eq. (3.10), the interaction Hamiltonian (8.21c) becomes

$$H_{\text{int}} = \sum_{\epsilon, \epsilon'} \sum_\kappa g_{\kappa}^{\epsilon', \epsilon} |\epsilon'\rangle\langle\epsilon| (a_\kappa - a_\kappa^\dagger), \quad (8.26)$$

where we defined the coupling coefficients

$$g_{\kappa}^{\epsilon', \epsilon} = -i\sqrt{\frac{\hbar\omega_\kappa}{2\epsilon_0}} \mathbf{d}_{\epsilon', \epsilon} \cdot \mathbf{u}_\kappa(\mathbf{R}). \quad (8.27)$$

Since $\mathbf{u}_\kappa(\mathbf{R})$ is real, and $\mathbf{d}_{\epsilon', \epsilon}^* = \mathbf{d}_{\epsilon, \epsilon'}$, we have that $(g_{\kappa}^{\epsilon', \epsilon})^* = -g_{\kappa}^{\epsilon, \epsilon'}$. Using this in (8.26) yields

$$H_{\text{int}} = \sum_{\epsilon, \epsilon'} \sum_\kappa \left[g_{\kappa}^{\epsilon', \epsilon} a_\kappa |\epsilon'\rangle\langle\epsilon| + (g_{\kappa}^{\epsilon, \epsilon'})^* a_\kappa^\dagger |\epsilon\rangle\langle\epsilon'| \right]. \quad (8.28)$$

By defining

$$V_\kappa^+ = \sum_{\epsilon', \epsilon} g_{\kappa}^{\epsilon', \epsilon} a_\kappa |\epsilon'\rangle\langle\epsilon| \quad \text{and} \quad V_\kappa^- = (V_\kappa^+)^\dagger, \quad (8.29)$$

the interaction Hamiltonian can then be written in the compact form

$$H_{\text{int}} = \sum_\kappa \left[V_\kappa^+ + V_\kappa^- \right]. \quad (8.30)$$

8.3.2 Transition amplitudes under weak coupling

Let us now assume the limit of weak coupling coefficients (8.27), such that we can treat the interaction Hamiltonian H_{int} as a perturbation. To this end, we transform the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \underbrace{(H_{\text{mat}} + H_{\text{rad}})}_{H_0} |\psi(t)\rangle + H_{\text{int}} |\psi(t)\rangle \quad (8.31)$$

through the unitary transformation $U_0^\dagger(t) = e^{\frac{i}{\hbar} H_0 t}$ into the interaction picture with respect to $H_0 = H_{\text{mat}} + H_{\text{rad}}$, yielding

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = H_{I, \text{int}}(t) |\psi_I(t)\rangle, \quad (8.32)$$

where $|\psi_I(t)\rangle = U_0^\dagger(t) |\psi(t)\rangle$, and $H_{I,\text{int}}(t) = U_0^\dagger(t) H_{\text{int}} U_0(t)$. Up to second order in perturbation, the unitary time evolution operator in the interaction picture is then [using (1.42)],

$$U_I(t, t_0) = \mathbb{1} - \frac{i}{\hbar} \int_{t_0}^t dt_1 H_{I,\text{int}}(t_1) - \frac{1}{\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H_{I,\text{int}}(t_1) H_{I,\text{int}}(t_2) + \mathcal{O}(g^3). \quad (8.33)$$

We are interested in the transition amplitude between two eigenstates $e^{-iE_i t/\hbar} |\psi_i\rangle$ and $e^{-iE_f t/\hbar} |\psi_f\rangle$ of H_0 , with $H_0 |\psi_i\rangle = E_i |\psi_i\rangle$ and $H_0 |\psi_f\rangle = E_f |\psi_f\rangle$, which becomes

$$\begin{aligned} \mathcal{A}_{fi}(t) &= e^{-i(E_i - E_f)t/\hbar} \langle \psi_f | U(t, t_0) | \psi_i \rangle \\ &= e^{-i(E_i - E_f)t/\hbar} \langle \psi_f | U_0(t) U_I(t, t_0) U_0^\dagger(t) | \psi_i \rangle \\ &= \langle \psi_f | U_I(t, t_0) | \psi_i \rangle. \end{aligned} \quad (8.34)$$

With the help of the perturbative expansion (8.33), the transition amplitude can be decomposed according to the perturbation order,

$$\mathcal{A}_{fi}(t) = \mathcal{A}_{fi}^{(0)}(t) + \mathcal{A}_{fi}^{(1)}(t) + \mathcal{A}_{fi}^{(2)}(t) + \mathcal{O}(g^3), \quad (8.35)$$

where, by Eq. (8.33), the zero-, first-, and second-order transition amplitudes are given by

$$\mathcal{A}_{fi}^{(0)}(t) = \langle \psi_f | \psi_i \rangle \quad (8.36a)$$

$$\mathcal{A}_{fi}^{(1)}(t) = -\frac{i}{\hbar} \int_{t_0}^t dt_1 \langle \psi_f | H_{I,\text{int}}(t_1) | \psi_i \rangle \quad (8.36b)$$

$$\mathcal{A}_{fi}^{(2)}(t) = -\frac{1}{\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \langle \psi_f | H_{I,\text{int}}(t_1) H_{I,\text{int}}(t_2) | \psi_i \rangle. \quad (8.36c)$$

First-order perturbation theory The transition amplitude (8.36b) in first order perturbation theory is

$$\mathcal{A}_{fi}^{(1)}(t) = -\frac{i}{\hbar} \int_{t_0}^t dt_1 \langle \psi_f | U_0^\dagger(t_1) H_{\text{int}} U_0(t_1) | \psi_i \rangle. \quad (8.37)$$

Using that $U_0(t_1) |\psi_i\rangle = e^{-iE_i t_1/\hbar} |\psi_i\rangle$, and, similarly, $\langle \psi_f | U_0^\dagger(t_1) = \langle \psi_f | e^{iE_f t_1/\hbar}$, this becomes

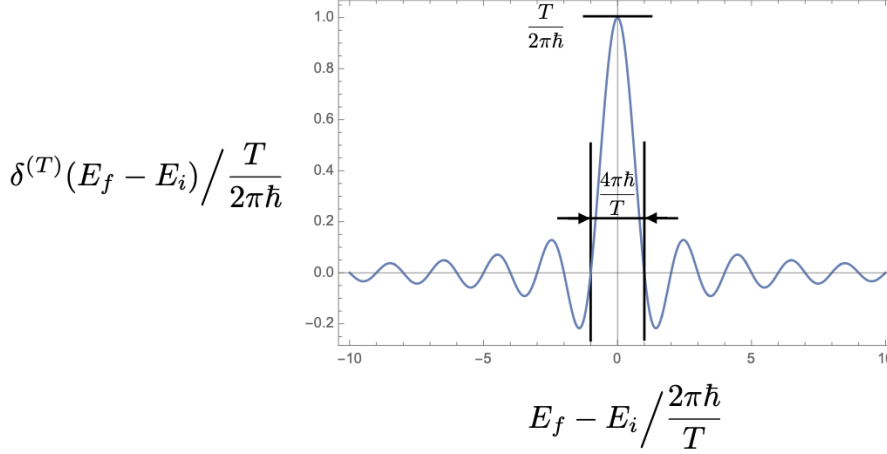
$$\begin{aligned} \mathcal{A}_{fi}^{(1)}(t) &= -\frac{i}{\hbar} \int_{t_0}^t dt_1 \langle \psi_f | H_{\text{int}} | \psi_i \rangle e^{i(E_f - E_i)t_1/\hbar}, \\ &= -2\pi i \langle \psi_f | H_{\text{int}} | \psi_i \rangle \left(\frac{1}{2\pi} \int_{t_0}^t \frac{dt_1}{\hbar} e^{i(E_f - E_i)t_1/\hbar} \right). \end{aligned} \quad (8.38)$$

Let us set

$$\boxed{t = \frac{T}{2} \quad \text{and} \quad t_0 = -\frac{T}{2}}, \quad (8.39)$$

such that the total interaction time is given by T . The integral in parentheses then defines the function

$$\begin{aligned} \delta^{(T)}(E_f - E_i) &= \frac{1}{2\pi} \int_{-T/2}^{T/2} \frac{dt_1}{\hbar} e^{i(E_f - E_i)t_1/\hbar} \\ &= \frac{1}{\pi} \frac{e^{i(E_f - E_i)T/2\hbar} - e^{-i(E_f - E_i)T/2\hbar}}{2i(E_f - E_i)} \\ &= \frac{1}{\pi} \frac{\sin\left((E_f - E_i)\frac{T}{2\hbar}\right)}{E_f - E_i}. \end{aligned} \quad (8.40)$$



From the first equality, we see that $\delta^{(T)}(E_f - E_i)$ tends to the Dirac delta function $\delta(E_f - E_i)$ in the limit $T \rightarrow \infty$. Indeed, $\delta^{(T)}(E_f - E_i)$ is peaked at $E_f - E_i = 0$ with peak height $\delta^{(T)}(0) = T/(2\pi\hbar)$, peak width on the order of $4\pi\hbar/T$, and it integrates to $\int_{-\infty}^{\infty} dE \delta^{(T)}(E) = 1$. It is, thus, an appropriate delta function that expresses the conservation of energy with an uncertainty $\hbar/T$ due to the finite duration of the interaction time.

Finally, we write the matrix elements of the interaction Hamiltonian as

$$V_{fi} = \langle \psi_f | H_{\text{int}} | \psi_i \rangle, \quad (8.41)$$

such that the first-order transition amplitude can compactly be written as

$$\boxed{\mathcal{A}_{fi}^{(1)}(T) = -2\pi i V_{fi} \delta^{(T)}(E_f - E_i).} \quad (8.42)$$

Second-order perturbation theory Let's now consider the second-order transition amplitude (8.36c),

$$\begin{aligned} \mathcal{A}_{fi}^{(2)}(t) &= -\frac{1}{\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \langle \psi_f | H_{\text{I,int}}(t_1) H_{\text{I,int}}(t_2) | \psi_i \rangle \\ &= -\frac{1}{\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \langle \psi_f | U_0^\dagger(t_1) H_{\text{int}} U_0(t_1) U_0^\dagger(t_2) H_{\text{int}} U_0(t_2) | \psi_i \rangle \\ &= -\frac{1}{\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 e^{-iE_i t_2/\hbar} e^{iE_f t_1/\hbar} \langle \psi_f | H_{\text{int}} U_0(t_1 - t_2) H_{\text{int}} | \psi_i \rangle \\ &= -\frac{1}{\hbar^2} \int_{t_0}^t dt_1 dt_2 e^{-iE_i t_2/\hbar} e^{iE_f t_1/\hbar} \langle \psi_f | H_{\text{int}} \underbrace{\Theta(t_1 - t_2) e^{-iH_0(t_1 - t_2)/\hbar}}_{iG^+(t_1 - t_2)} H_{\text{int}} | \psi_i \rangle, \end{aligned} \quad (8.43)$$

where we introduced the Heaviside function in the last step in order that both integrals get the same upper bound. We recognize the retarded Green's function $G^+(t) = -i\Theta(t)e^{-H_0 t/\hbar}$ [see Eq. (7.58)], which we can write by Eqs. (7.60) and (7.65), as

$$G^+(t_1 - t_2) = \lim_{\eta \searrow 0} \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega(t_1 - t_2)} \frac{1}{\omega - H_0/\hbar + i\eta}. \quad (8.44)$$

Turning the integral over ω into an integral over $E = \hbar\omega$, and using the spectral decomposition (8.24) of H_0 , this becomes

$$G^+(t_1 - t_2) = \lim_{\eta \searrow 0} \frac{1}{2\pi} \int_{-\infty}^{\infty} dE e^{-iE(t_1 - t_2)/\hbar} \oint d\alpha \frac{1}{E - E_\alpha + i\eta} |\psi_\alpha\rangle \langle \psi_\alpha|. \quad (8.45)$$

Plugging this into (8.43) gives for the second-order transition amplitude

$$\mathcal{A}_{fi}^{(2)}(t) = -\frac{i}{2\pi\hbar^2} \int_{-\infty}^{\infty} dE \left(\int_{t_0}^t dt_1 e^{-i(E-E_f)t_1/\hbar} \right) \left(\int_{t_0}^t dt_2 e^{i(E-E_i)t_2/\hbar} \right) \quad (8.46)$$

$$\times \lim_{\eta \searrow 0} \oint d\alpha \frac{1}{E - E_\alpha + i\eta} \langle \psi_f | H_{\text{int}} | \psi_\alpha \rangle \langle \psi_\alpha | H_{\text{int}} | \psi_i \rangle, \quad (8.47)$$

Similar as before, we use the shorthand notation $\langle \psi_\alpha | H_{\text{int}} | \psi_\beta \rangle = V_{\alpha\beta}$ [see Eq. (8.41)], and consider $t = T/2$ and $t_0 = -T/2$ [see Eq. (8.39)]. The expressions in parentheses then result in the delta function from Eq. (8.40). In particular, they can be identified with $2\pi\hbar\delta^{(T)}(E - E_f)$ and $2\pi\hbar\delta^{(T)}(E - E_i)$. Accordingly, we get

$$\mathcal{A}_{fi}^{(2)}(T) = -2\pi i \int_{-\infty}^{\infty} dE \delta^{(T)}(E - E_f) \delta^{(T)}(E - E_i) \underbrace{\lim_{\eta \searrow 0} \oint d\alpha \frac{V_{f\alpha} V_{\alpha i}}{E - E_\alpha + i\eta}}_{W_{fi}(E)}, \quad (8.48)$$

where

$$W_{fi}(E) = \lim_{\eta \searrow 0} \oint d\alpha \frac{V_{f\alpha} V_{\alpha i}}{E - E_\alpha + i\eta} \quad (8.49)$$

is a second-order transition element. Now let us recall that $\delta^{(T)}(E - E_f)$ and $\delta^{(T)}(E - E_i)$ is only non-vanishing at $E \approx E_f$ and $E \approx E_i$, respectively, with a width on in the order of $\hbar/T$. That is, the integral is only non-vanishing if E differs from E_f and E_i by no more than $\hbar/T$. The function $\delta^{(T)}(E - E_f)\delta^{(T)}(E - E_i)$ has thus a width on the order of $\hbar/T$ about $E_i \approx E_f$, and, thus, varies strongly as a function of E around $E_f \approx E_i$ for large T . Compared to this, in many cases the function $W_{fi}(E)$ varies only slightly with E over an interval of $\hbar/T$.¹ We can thus replace $W_{fi}(E)$ by $W_{fi}(E_i)$, and pull it out of the integral, which gives

$$\mathcal{A}_{fi}^{(2)}(T) = -2\pi i W_{fi}(E_i) \int_{-\infty}^{\infty} dE \delta^{(T)}(E - E_f) \delta^{(T)}(E - E_i). \quad (8.50)$$

We can now evaluate the integral over E ,

$$\begin{aligned} & \int_{-\infty}^{\infty} dE \delta^{(T)}(E - E_f) \delta^{(T)}(E - E_i) \\ & \stackrel{(8.40)}{=} \frac{1}{4\pi^2\hbar^2} \int_{-\infty}^{\infty} dE \int_{-T/2}^{T/2} dt_1 dt_2 e^{i(E-E_f)t_1/\hbar} e^{i(E-E_i)t_2/\hbar} \\ & = \frac{1}{2\pi\hbar} \int_{-T/2}^{T/2} dt_1 dt_2 e^{-iE_f t_1/\hbar} e^{-iE_i t_2/\hbar} \underbrace{\frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dE e^{iE(t_1+t_2)/\hbar}}_{\delta(t_1+t_2)} \\ & = \frac{1}{2\pi\hbar} \int_{-T/2}^{T/2} dt_1 e^{i(E_i-E_f)t_1/\hbar} \\ & \stackrel{(8.40)}{=} \delta^{(T)}(E_i - E_f). \end{aligned} \quad (8.51)$$

The second-order transition amplitude then results in

$$\boxed{\mathcal{A}_{fi}^{(2)}(T) = -2\pi i \left[\lim_{\eta \searrow 0} \oint d\alpha \frac{V_{f\alpha} V_{\alpha i}}{E_i - E_\alpha + i\eta} \right] \delta^{(T)}(E_i - E_f).} \quad (8.52)$$

¹Note that if H_0 has a discrete eigenstate $|\psi_k\rangle$ with eigenenergy E_k close to $E_i \approx E_f$, then $W_{fi}(E)$ changes rapidly with varying E near $E_i \approx E_f$, and the above approximation breaks down.

Here, the delta function $\delta^{(T)}(E_i - E_f)$ allows only those transitions, where the initial and final state is close in energy, within a width of $\hbar/T$. The term $V_{f\alpha}V_{\alpha i}$ describes a transition from the initial state $|\psi_i\rangle$ to an intermediate state $|\psi_\alpha\rangle$, followed by a transition from the intermediate state $|\psi_\alpha\rangle$ to the final state $|\psi_f\rangle$. The summation over α states that this transition can be through all intermediate states $|\psi_\alpha\rangle$, with the strength of the transition weighted by $1/(E_i - E_\alpha)$. That is, the closer the energy of the intermediate state to the energy of $E_i \approx E_f$, the larger the likelihood that the transition will go through this intermediate state.

Finally, let us combine the zero-, first-, and second-order transition amplitudes to get the total transition amplitude (8.35) up to second-order perturbation, reading

$$\mathcal{A}_{fi}(T) = \langle \psi_f | \psi_i \rangle - 2\pi i \delta^{(T)}(E_i - E_f) [V_{fi} + W_{fi}(E_i)] + \mathcal{O}(g^3). \quad (8.53)$$

Review Questions 8.2.

- What does the coupling coefficient $g_k^{\epsilon', \epsilon}$ describe?
- What is the transition amplitude in zeroth/first/second order perturbation theory? Describe the processes behind these terms.
- Explain the appearance of the function $\delta^{(T)}(E_i - E_f)$ in the transition amplitudes. What are its properties?

8.3.3 Transitions between different discrete states

Transition probability Suppose now, the initial state $|\psi_i\rangle$ and final state $|\psi_f\rangle$ are discrete and distinct, i.e., $\langle \psi_f | \psi_i \rangle = 0$. In this case, the transition probability up to second order perturbation becomes

$$p_{fi}(T) = |\mathcal{A}_{fi}(T)|^2 \stackrel{(8.53)}{=} 4\pi^2 \left[\delta^{(T)}(E_i - E_f) \right]^2 |V_{fi} + W_{fi}(E_i)|^2. \quad (8.54)$$

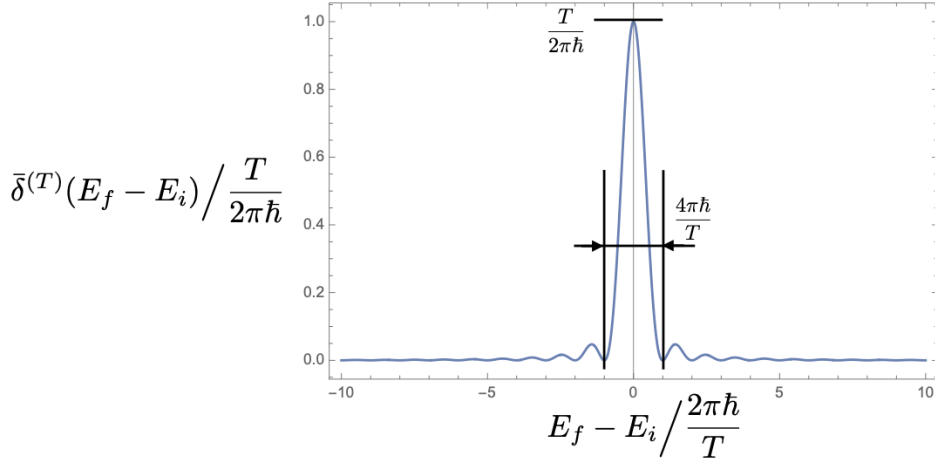
Here the squared delta function appears. By Eq. (8.40), it reads

$$\begin{aligned} \left[\delta^{(T)}(E_f - E_i) \right]^2 &= \frac{1}{\pi^2} \frac{\sin^2 \left((E_f - E_i) \frac{T}{2\hbar} \right)}{(E_f - E_i)^2} \\ &= \frac{T}{2\pi\hbar} \left[\frac{T}{2\pi\hbar} \frac{\sin^2 \left((E_f - E_i) \frac{T}{2\hbar} \right)}{\left((E_f - E_i) \frac{T}{2\hbar} \right)^2} \right] \\ &= \frac{T}{2\pi\hbar} \bar{\delta}^{(T)}(E_f - E_i), \end{aligned} \quad (8.55)$$

where

$$\bar{\delta}^{(T)}(E_f - E_i) = \frac{T}{2\pi\hbar} \frac{\sin^2 \left((E_f - E_i) \frac{T}{2\hbar} \right)}{\left((E_f - E_i) \frac{T}{2\hbar} \right)^2} \quad (8.56)$$

is yet another delta function, peaked at $E_f = E_i$, with a peak height of $T/(2\pi\hbar)$, a peak width scaling like $\hbar/T$, and unit integral, i.e., $\int_{-\infty}^{\infty} dE \bar{\delta}^{(T)}(E) = 1$. Accordingly, in the limit $T \rightarrow \infty$, it converges to the Dirac delta function.



Using (8.55) in the expression of the transition probability (8.54), we obtain

$$p_{fi}(T) = \frac{2\pi T}{\hbar} \bar{\delta}^{(T)}(E_f - E_i) |V_{fi} + W_{fi}(E_i)|^2. \quad (8.57)$$

First-order transition rate Since the transition probability is proportional to the duration of interaction T , we can define the transition rate in first-order perturbation theory to be

$$\Gamma_{i \rightarrow f} = \frac{p_{fi}(T)}{T} = \frac{2\pi}{\hbar} |V_{fi}|^2 \bar{\delta}^{(T)}(E_f - E_i), \quad (8.58)$$

and, therewith, the total decay rate

$$\Gamma = \sum_f \Gamma_{i \rightarrow f}. \quad (8.59)$$

8.3.4 Transitions towards a continuum of final states

Transition probability Suppose now that the final state belongs to an energy continuum, $|\psi_f\rangle = |E, \alpha\rangle$, with E characterizing the energy, and α all other quantum numbers. In this case $p_{fi}(T)$ is no longer a transition probability, but rather a transition probability density.² The quantity of interest, which has a physical meaning, is, thus, the probability that the system results in a region of final states within an interval δE_f and $\delta \alpha_f$,

$$\delta p(E_f, \alpha_f, T) = \int_{\substack{E \in \delta E_f \\ \alpha \in \alpha_f}} dE \, d\alpha \, \rho(E, \alpha) \left| \mathcal{A}_{(E, \alpha)i}(T) \right|^2, \quad (8.60)$$

where $\rho(E, \alpha)$ is the density of states in the continuum, and $\mathcal{A}_{(E, \alpha)i}(T) = \langle E, \alpha | U_I(T/2, -T/2) | \psi_i \rangle$. Using (8.53), up to second-order perturbation, we get

$$\delta p(E_f, \alpha_f, T) = 4\pi^2 \int_{\substack{E \in \delta E_f \\ \alpha \in \alpha_f}} dE \, d\alpha \, \rho(E, \alpha) \left| V_{(E, \alpha)i} + W_{(E, \alpha)i}(E_i) \right|^2 \left[\delta^{(T)}(E_i - E) \right]^2. \quad (8.61)$$

And using (8.55), we obtain

$$\begin{aligned} \delta p(E_f, \alpha_f, T) &= \frac{2\pi T}{\hbar} \int_{\substack{E \in \delta E_f \\ \alpha \in \alpha_f}} dE \, d\alpha \, \rho(E, \alpha) \left| V_{(E, \alpha)i} + W_{(E, \alpha)i}(E_i) \right|^2 \bar{\delta}^{(T)}(E_i - E) \\ &= \frac{2\pi T}{\hbar} \int_{\alpha \in \alpha_f} d\alpha \, \rho(E_i, \alpha) \left| V_{(E_i, \alpha)i} + W_{(E_i, \alpha)i}(E_i) \right|^2, \end{aligned} \quad (8.62)$$

²Note that since $\langle E, \alpha | E', \alpha \rangle = \delta(E - E')$, the norm of $|E, \alpha\rangle$ is infinite. The state $|E, \alpha\rangle$ alone has, thus, no physical meaning, and, instead, we must consider a region of final states within an interval δE_f and $\delta \alpha_f$ with density of states $\rho(E, \alpha)$.

where we assumed T to be sufficiently large, such that $\rho(E, \alpha)|V_{(E, \alpha)i}|^2$ varies much slower as a function of E as compared to $\bar{\delta}^{(T)}(E_i - E)$, which can, thus, be replaced by the Dirac delta function.

First-order transition rate Let us now restrict to lowest order in perturbation (i.e., we drop $W_{(E_i, \alpha)i}(E_i)$). Since the probability (8.62) is proportional to T , we get the transition rate

$$\Gamma_{i \rightarrow f} = \frac{\delta p(E_f, \alpha_f, T)}{T} = \frac{2\pi}{\hbar} \int_{\alpha \in \alpha_f} d\alpha \rho(E_i, \alpha) |V_{(E_i, \alpha)i}|^2, \quad (8.63)$$

which can also be written in terms of the individual transition rates,

$$\Gamma_{i \rightarrow f} = \int_{\substack{E \in \delta E_f \\ \alpha \in \alpha_f}} dE d\alpha \Gamma_{i \rightarrow (E, \alpha)}, \quad (8.64)$$

where

$$\Gamma_{i \rightarrow (E, \alpha)} = \frac{2\pi}{\hbar} \rho(E_i, \alpha) |V_{(E_i, \alpha)i}|^2 \bar{\delta}^{(T)}(E_i - E). \quad (8.65)$$

Fermi's Golden Rule Often one is not interested in the transition rate of a single transition $i \rightarrow f$, but in the total decay rate of the initial state, which is the sum of transition rates into all final states (e.g., think of the decay of an excited atom). In this case, the integral in (8.64) must be extended over all final states (E, α) ,

$$\Gamma = \int dE d\alpha \Gamma_{i \rightarrow (E, \alpha)} = \frac{2\pi}{\hbar} \int d\alpha \rho(E_i, \alpha) |V_{(E_i, \alpha)i}|^2. \quad (8.66)$$

Assuming that the transition matrix element $V_{(E_i, \alpha)i}$ is independent on α , i.e., $V_{(E_i, \alpha)i} = V_{fi}$, with $E_f \approx E_i$, it can be pulled out of the integral, resulting in Fermi's Golden Rule:

$$\boxed{\Gamma = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_i = E_f)}, \quad (8.67)$$

where $\rho(E) = \int d\alpha \rho(E, \alpha)$ is the density of states with energy E .

Note that by the delta function in (8.57) and (8.62), only transitions for which

$$E_i - E_f \approx 0 \quad (8.68)$$

are allowed. And since we assumed $|\psi_i\rangle$ and $|\psi_f\rangle$ to be eigenstates of $H_0 = H_{\text{mat}} + H_{\text{rad}}$, this is equivalent to the statement that the energy in the total system particles+field doesn't change. That is, a change of the energy in the matter state must come along with a change of the field excitations, which brings us to the discussion of absorption and emission processes.

Review Questions 8.3.

- What is the relation between the functions $\delta^{(T)}(E_f - E_i)$ and $\bar{\delta}^{(T)}(E_f - E_i)$? Compare their properties.
- What is the probability for a transition towards a continuum of final states? Why is the integration over final states needed?
- What is the transition rate? What is the total decay rate?
- Explain Fermi's Golden Rule.

8.4 Absorption and emission

Let us consider now the transition between two discrete matter states $|\epsilon\rangle$ and $|\epsilon'\rangle$ by considering the initial state $|\psi_i\rangle = |\epsilon, \{n_\kappa\}\rangle$ and final state $|\psi_f\rangle = |\epsilon', \{n'_\kappa\}\rangle$, where $|\{n_\kappa\}\rangle$ and $|\{n'_\kappa\}\rangle$ describe the initial and final radiation state, respectively. Above we have seen that the corresponding matrix element of the interaction Hamiltonian [c.f. Eq. (8.41)]

$$V_{fi} = V_{(\epsilon', \{n'_\kappa\})(\epsilon, \{n_\kappa\})} = \langle \epsilon', \{n'_\kappa\} | H_{\text{int}} | \epsilon, \{n_\kappa\} \rangle \quad (8.69)$$

is the essential ingredient for the calculation of the transition probabilities [see Eqs. (8.57) and (8.62)] and decay rates [see Eqs. (8.59) and (8.67)]. In the following, we are thus interested in investigating the matrix elements (8.69) in more detail. To this end, let us first recall from Eq. (8.30) that $H_{\text{int}} = \sum_\kappa [V_\kappa^+ + V_\kappa^-]$. In view of this, we investigate the action of V_κ^+ and V_κ^- separately.

Absorption of single-mode radiation The operator $V_\kappa^+ = \sum_{\epsilon, \epsilon'} g_{\kappa}^{\epsilon', \epsilon} a_\kappa |\epsilon'\rangle \langle \epsilon|$ describes the annihilation of a photon together with a change of the matter state from $|\epsilon\rangle$ to $|\epsilon'\rangle$, with the strength of this interaction being given by the coupling coefficient $g_{\kappa}^{\epsilon', \epsilon}$. Using $a_\kappa |n_\kappa\rangle = \sqrt{n_\kappa} |n_\kappa - 1\rangle$ [see Eq. (4.4a)], the transition matrix element reads

$$\langle \epsilon', n_\kappa - 1 | V_\kappa^+ | \epsilon, n_\kappa \rangle = g_{\kappa}^{\epsilon', \epsilon} \sqrt{n_\kappa}, \quad (8.70)$$

where the number of excitations in modes other than κ are not indicated, since they remain unaffected. Since this process corresponds to a change of the matter state into a higher energy state along with the annihilation of a single field excitation, it corresponds to the absorption of a photon. In particular, the condition (8.68) for transitions gives rise to the energy balance

$$E_{\epsilon'} + \hbar\omega_\kappa(n_\kappa - 1) = E_\epsilon + \hbar\omega_\kappa n_\kappa, \quad \Rightarrow \omega_\kappa = \frac{E_{\epsilon'} - E_\epsilon}{\hbar} \geq 0, \quad (8.71)$$

from which it follows that the frequency of the absorbed photon must match the energy difference between the final and initial matter states. The transition rate (8.58) corresponding to (8.70) is then

$$\Gamma_{(\epsilon, n_\kappa) \rightarrow (\epsilon', n_\kappa - 1)}^{\text{stim abs}} = \frac{2\pi}{\hbar} \left| g_{\kappa}^{\epsilon', \epsilon} \right|^2 n_\kappa \bar{\delta}^{(T)}(E_{\epsilon'} - \hbar\omega_\kappa - E_\epsilon). \quad (8.72)$$

It is proportional to the number n_κ of photons in mode κ , such that we speak of stimulated absorption. Note that absorption can only occur if the number of photons n_κ is non-vanishing. Further note that since by parity $\mathbf{d}_{\epsilon, \epsilon} = \langle \epsilon | \mathbf{d} | \epsilon \rangle = 0$, and, hence, $g_{\kappa}^{\epsilon, \epsilon} = 0$, absorption is always accompanied by a change of the matter state.

Emission of single mode-radiation Let's come back to (8.69), where the operator $V_\kappa^- = \sum_{\epsilon, \epsilon'} (g_{\kappa}^{\epsilon, \epsilon'})^* a_\kappa^\dagger |\epsilon'\rangle \langle \epsilon|$, on the other hand, describes the creation of a photon in mode κ together with a change of the matter state from $|\epsilon\rangle$ to $|\epsilon'\rangle$. Using $a_\kappa^\dagger |n_\kappa\rangle = \sqrt{n_\kappa + 1} |n_\kappa + 1\rangle$ [see Eq. (4.4b)], the non-zero transition matrix elements become

$$\langle \epsilon', n_\kappa + 1 | V_\kappa^- | \epsilon, n_\kappa \rangle = (g_{\kappa}^{\epsilon, \epsilon'})^* \sqrt{n_\kappa + 1}, \quad (8.73)$$

where we again drop the number of excitations in modes other than κ , which remain unaffected. Since this process changes the matter state into a lower energy state via the creation of a field excitation, we speak of the emission of a photon. Again, the energies must balance according to

$$E_{\epsilon'} + \hbar\omega_\kappa(n_\kappa + 1) = E_\epsilon + \hbar\omega_\kappa n_\kappa, \quad \Rightarrow \omega_\kappa = \frac{E_\epsilon - E_{\epsilon'}}{\hbar} \geq 0. \quad (8.74)$$

Similar to the absorption process, the frequency of the emitted photon must match the energy difference between the initial and final matter state. However, now the transition rate

$$\Gamma_{(\epsilon, n_\kappa) \rightarrow (\epsilon', n_\kappa + 1)}^{\text{em}} = \frac{2\pi}{\hbar} |g_{\kappa}^{\epsilon, \epsilon'}|^2 (n_\kappa + 1) \bar{\delta}^{(T)}(E_{\epsilon'} + \hbar\omega_\kappa - E_\epsilon) \quad (8.75)$$

is proportional to $n_\kappa + 1$, and doesn't vanish for an initial vacuum state for which $n_\kappa = 0$. We therefore distinguish between stimulated emission, whose rate is proportional to n_κ ,

$$\Gamma_{(\epsilon, n_\kappa) \rightarrow (\epsilon', n_\kappa + 1)}^{\text{stim em}} = \frac{2\pi}{\hbar} |g_{\kappa}^{\epsilon, \epsilon'}|^2 n_\kappa \bar{\delta}^{(T)}(E_{\epsilon'} + \hbar\omega_\kappa - E_\epsilon), \quad (8.76)$$

and spontaneous emission, which persists in the absence of photons, with rate

$$\Gamma_{(\epsilon, n_\kappa) \rightarrow (\epsilon', n_\kappa + 1)}^{\text{spon em}} = \frac{2\pi}{\hbar} |g_{\kappa}^{\epsilon, \epsilon'}|^2 \bar{\delta}^{(T)}(E_{\epsilon'} + \hbar\omega_\kappa - E_\epsilon). \quad (8.77)$$

Spontaneous emission in free space Let us consider now the total *spontaneous emission* rate for the initial matter state $|\epsilon\rangle$ decaying into the matter state $|\epsilon'\rangle$ regardless of the final radiation state. That is, we keep $|\epsilon'\rangle$ fixed, and sum over all final radiation states such that the energy $\hbar\omega_\kappa$ of the emitted photon equals the difference $\hbar\omega_0 = E_\epsilon - E_{\epsilon'}$ [see Eq. (8.74)] of the matter states,

$$\begin{aligned} \Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}} &= \sum_{\kappa} \Gamma_{(\epsilon, n_\kappa) \rightarrow (\epsilon', n_\kappa + 1)}^{\text{spon em}} \\ &\stackrel{(8.77)}{=} \frac{2\pi}{\hbar} \sum_{\kappa} |g_{\kappa}^{\epsilon, \epsilon'}|^2 \bar{\delta}^{(T)}(\hbar\omega_\kappa - \hbar\omega_0) \\ &\stackrel{(8.27)}{=} \frac{2\pi}{\hbar} \sum_{\kappa} \frac{\hbar\omega_\kappa}{2\epsilon_0} |\mathbf{d}_{\epsilon', \epsilon} \cdot \mathbf{u}_\kappa(\mathbf{R})|^2 \bar{\delta}^{(T)}(\hbar\omega_\kappa - \hbar\omega_0) \\ &= \frac{\pi}{\epsilon_0} \sum_{\kappa} \omega_\kappa |\mathbf{d}_{\epsilon', \epsilon} \cdot \mathbf{u}_\kappa(\mathbf{R})|^2 \bar{\delta}^{(T)}(\hbar\omega_\kappa - \hbar\omega_0). \end{aligned} \quad (8.78)$$

Now we want to place ourselves in free space (think of the emission process of an atom in free space). To do so, we consider the modes $\kappa = (\mathbf{k}, p)$, replace $\sum_{\kappa} \rightarrow \left(\frac{L}{2\pi}\right)^3 \int d^3k \sum_p$, use the mode functions $\mathbf{u}_\kappa(\mathbf{R}) \rightarrow \boldsymbol{\mu}_{\mathbf{k}, p}(\mathbf{R}) = \frac{1}{L^{3/2}} \exp(i\mathbf{k} \cdot \mathbf{R}) \mathbf{e}_{\mathbf{k}, p}$ from Eq. (2.83), and assume T to be large enough such that we can replace $\bar{\delta}^{(T)}(\hbar\omega_\kappa - \hbar\omega_0) \rightarrow \delta(\hbar\omega_\kappa - \hbar\omega_0)$, yielding

$$\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}} = \frac{\pi}{\epsilon_0} \frac{1}{(2\pi)^3} \int d^3k \omega_k \delta(\hbar\omega_k - \hbar\omega_0) \sum_p |\mathbf{d}_{\epsilon', \epsilon} \cdot \mathbf{e}_{\mathbf{k}, p}|^2. \quad (8.79)$$

For a fixed $\mathbf{k}$, we take θ to be the angle between $\mathbf{k}$ and $\mathbf{d}_{\epsilon', \epsilon}$. Since the unit polarization vectors $\mathbf{e}_{\mathbf{k}, p}$ are perpendicular to $\mathbf{k}$, we then get

$$\sum_p |\mathbf{d}_{\epsilon', \epsilon} \cdot \mathbf{e}_{\mathbf{k}, p}|^2 = |\mathbf{d}_{\epsilon', \epsilon}|^2 \sin^2 \theta. \quad (8.80)$$

Using this in the total spontaneous emission rate (8.79) together with $\delta(\hbar\omega_k - \hbar\omega_0) = \frac{1}{\hbar c} \delta(k - \omega_0/c)$, leads to

$$\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}} = \frac{\omega_0}{8\pi^2 \epsilon_0 \hbar c} |\mathbf{d}_{\epsilon', \epsilon}|^2 \underbrace{\int_0^\infty dk k^2 \delta(k - \omega_0/c)}_{\omega_0^2/c^2} \underbrace{\int_0^{2\pi} d\varphi}_{2\pi} \underbrace{\int_0^\pi d\theta \sin^3 \theta}_{4/3}, \quad (8.81)$$

from which we finally obtain

$$\boxed{\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}} = \frac{\omega_0^3}{3\pi \epsilon_0 \hbar c^3} |\mathbf{d}_{\epsilon', \epsilon}|^2.} \quad (8.82)$$

Note that since $\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}}$ goes with ω_0^3 , spontaneous emission is usually negligible in the radio frequency domain. It becomes important in the optical domain, and even more so for higher frequencies, such as in the ultraviolet and X-ray domain. The spontaneous emission rate $\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}}$ is also called the natural width of the excited matter state $|\epsilon\rangle$, and denoted by Γ . Its inverse $\tau_{\text{rad}} = 1/\Gamma$ is called the radiative lifetime of the excited state $|\epsilon\rangle$, which corresponds to the average time it takes to undergo the radiative decay.

Stimulated emission and absorption in free space To get the transition rates for stimulated absorption and stimulated emission in free space, recall that in the single-mode case, the transition rates (8.72) and (8.76) differ from the spontaneous emission rate (8.77) only by the photon number of the initial radiation mode $\langle n_\kappa | N_\kappa | n_\kappa \rangle = \langle N_\kappa \rangle = \bar{n}_\kappa = n_\kappa$. It thus becomes clear that the rates for stimulated absorption and stimulated emission in free space must be given by

$$\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{stim abs}} = \Gamma_{\epsilon \rightarrow \epsilon'}^{\text{stim em}} = \Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}} \bar{n}(\omega_0), \quad (8.83)$$

where $\bar{n}(\omega_0)$ is the average number of photons at the resonance frequency ω_0 .

Often, the transition rates are provided in terms of the spectral energy density $u(\omega)$. Using the density of modes in free space from Eq. (4.54), $\rho(\omega) = \frac{\omega^2}{\pi^2 c^3}$, it reads

$$u(\omega) = \hbar \omega \bar{n} \rho(\omega) \stackrel{(4.54)}{=} \frac{\hbar \omega^3}{\pi^2 c^3} \bar{n}. \quad (8.84)$$

By introducing the Einstein coefficients

$$A = \Gamma_{\epsilon \rightarrow \epsilon'}^{\text{spon em}} \quad \text{and} \quad B = \frac{\pi^2 c^3}{\hbar \omega_0^3} A, \quad (8.85)$$

the transition rates (8.83) can equally well be expressed as

$$\Gamma_{\epsilon \rightarrow \epsilon'}^{\text{stim abs}} = \Gamma_{\epsilon \rightarrow \epsilon'}^{\text{stim em}} = B u(\omega_0). \quad (8.86)$$

Review Questions 8.4.

- State and explain the transition matrix elements corresponding to the absorption and emission of a photon, respectively.
- What distinguishes spontaneous and stimulated emission?
- How is stimulated emission and absorption in free space related to spontaneous emission in free space?

8.5 Wigner-Weisskopf model of spontaneous emission

In the following, we want to discuss spontaneous emission non-perturbatively. To do so, we must employ some more approximations to the interaction Hamiltonian (8.28). In particular, we approximate the atom by a two-level system with ground state $|g\rangle$ and excited state $|e\rangle$, and employ the rotating wave approximation (we will see in a bit what that is).

Evolution equations in the interaction picture Let us start with the interaction Hamiltonian under dipole approximation, reading $H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}_\perp(\mathbf{R})$. First, we write the dipole operator under the assumption of a two-level atom as

$$\mathbf{d} = \mathbf{1} \mathbf{d} \mathbf{1} = (|g\rangle\langle g| + |e\rangle\langle e|) \mathbf{d} (|g\rangle\langle g| + |e\rangle\langle e|) = \mathbf{d}_{e,g} |e\rangle\langle g| + \mathbf{d}_{g,e} |g\rangle\langle e|, \quad (8.87)$$

where $\mathbf{d}_{e,g} = \langle e | \mathbf{d} | g \rangle$, and $\mathbf{d}_{g,g} = \mathbf{d}_{e,e} = 0$. Using this together with the electric field operator in free space from Eq. (8.18) gives

$$\begin{aligned} H_{\text{int}} &= -\mathbf{d} \cdot \mathbf{E}_\perp(\mathbf{R}) \\ &= -[\mathbf{d}_{e,g}|e\rangle\langle g| + \mathbf{d}_{g,e}|g\rangle\langle e|] \cdot i \sum_p \int d^3k \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3}} [\mathbf{e}_{\mathbf{k},p} e^{i\mathbf{k}\cdot\mathbf{R}} a_p(\mathbf{k}) - \mathbf{e}_{\mathbf{k},p}^* e^{-i\mathbf{k}\cdot\mathbf{R}} a_p^\dagger(\mathbf{k})] \\ &= \sum_p \int d^3k \left[g_{\mathbf{k},p}^{e,g} a_p(\mathbf{k}) |e\rangle\langle g| + g_{\mathbf{k},p}^{g,e} a_p(\mathbf{k}) |g\rangle\langle e| + (g_{\mathbf{k},p}^{e,g})^* a_p^\dagger(\mathbf{k}) |g\rangle\langle e| + (g_{\mathbf{k},p}^{g,e})^* a_p^\dagger(\mathbf{k}) |e\rangle\langle g| \right], \end{aligned} \quad (8.88)$$

with the coupling constant (cf. Eq. (6.18))

$$g_{\mathbf{k},p}^{e,g} = -i \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3}} e^{i\mathbf{k}\cdot\mathbf{R}} \mathbf{d}_{e,g} \cdot \mathbf{e}_{\mathbf{k},p}. \quad (8.89)$$

Now let us transform via the unitary $U_0^\dagger(t) = e^{iH_0 t/\hbar}$ into the interaction picture with respect to $H_0 = H_{\text{mat}} + H_{\text{rad}}$, where $H_{\text{mat}} = \hbar\omega_e |e\rangle\langle e| + \hbar\omega_g |g\rangle\langle g|$ and $H_{\text{rad}} = \sum_p \int d^3k \hbar\omega_{\mathbf{k}} (a_p^\dagger(\mathbf{k}) a_p(\mathbf{k}) + 1/2)$. The interaction Hamiltonian in the interaction picture $H_{\text{int,I}}(t) = U_0^\dagger(t) H_{\text{int}} U_0(t)$ then becomes

$$\begin{aligned} H_{\text{int,I}}(t) &= \sum_p \int d^3k \left[g_{\mathbf{k},p}^{e,g} a_p(\mathbf{k}) |e\rangle\langle g| e^{i(\omega_0 - \omega_{\mathbf{k}})t} + g_{\mathbf{k},p}^{g,e} a_p(\mathbf{k}) |g\rangle\langle e| e^{-i(\omega_0 + \omega_{\mathbf{k}})t} \right. \\ &\quad \left. + (g_{\mathbf{k},p}^{e,g})^* a_p^\dagger(\mathbf{k}) |g\rangle\langle e| e^{-i(\omega_0 - \omega_{\mathbf{k}})t} + (g_{\mathbf{k},p}^{g,e})^* a_p^\dagger(\mathbf{k}) |e\rangle\langle g| e^{i(\omega_0 + \omega_{\mathbf{k}})t} \right], \end{aligned} \quad (8.90)$$

where $\omega_0 = \omega_e - \omega_g$ corresponds to the energy difference of the matter states. Now we employ the rotating wave approximation: we neglect the rapidly oscillating terms with $e^{\pm i(\omega_0 + \omega_{\mathbf{k}})t}$, since they average to zero over relevant timescales. As a result, we are left only with terms that conserve the total number of excitations of matter and field,

$$\boxed{H_{\text{int,I}}(t) = \sum_p \int d^3k \left[g_{\mathbf{k},p}^{e,g} a_p(\mathbf{k}) |e\rangle\langle g| e^{i(\omega_0 - \omega_{\mathbf{k}})t} + (g_{\mathbf{k},p}^{e,g})^* a_p^\dagger(\mathbf{k}) |g\rangle\langle e| e^{-i(\omega_0 - \omega_{\mathbf{k}})t} \right]}. \quad (8.91)$$

Our aim is to solve the Schrödinger equation in the interaction picture,

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = H_{\text{int,I}}(t) |\psi_I(t)\rangle, \quad (8.92)$$

for the initial state $|e, \Omega\rangle$ of an excited atom with the field in the vacuum state at time $t_0 = 0$. To this end, we note that the interaction Hamiltonian (8.91) only couples the initial state $|e, \Omega\rangle$ with the states $|g, 1_{\mathbf{k},p}\rangle$ corresponding to the atom in the ground state, a single excitation of the field mode $(\mathbf{k}, p)$, and all other field modes in the vacuum state. For the state $|\psi_I(t)\rangle$, we can thus make the ansatz

$$\boxed{|\psi_I(t)\rangle = c_e(t) |e, \Omega\rangle + \sum_p \int d^3k c_{g,\mathbf{k},p}(t) |g, 1_{\mathbf{k},p}\rangle}, \quad (8.93)$$

with complex amplitudes $c_e(t)$ and $c_{g,\mathbf{k},p}(t)$. Plugging this ansatz into the Schrödinger equation (8.92) and using the action of the interaction Hamiltonian (8.91) gives

$$\begin{aligned} i\hbar \frac{d}{dt} c_e(t) |e, \Omega\rangle + \sum_p \int d^3k i\hbar \frac{d}{dt} c_{g,\mathbf{k},p}(t) |g, 1_{\mathbf{k},p}\rangle \\ = \sum_p \int d^3k \left[g_{\mathbf{k},p}^{e,g} e^{i(\omega_0 - \omega_{\mathbf{k}})t} c_{g,\mathbf{k},p}(t) |e, \Omega\rangle + (g_{\mathbf{k},p}^{e,g})^* e^{-i(\omega_0 - \omega_{\mathbf{k}})t} c_e(t) |g, 1_{\mathbf{k},p}\rangle \right]. \end{aligned} \quad (8.94)$$

Since the states $|e, \Omega\rangle$ and $|g, 1_{\mathbf{k}, p}\rangle$ form an ON-basis of the considered subspace, we can equate their coefficients on the left and right hand side of the equality, yielding a set of coupled differential equations for $c_e(t)$ and $c_{g, \mathbf{k}, p}(t)$:

$$i\hbar \frac{d}{dt} c_e(t) = \sum_p \int d^3k g_{\mathbf{k}, p}^{e, g} e^{i(\omega_0 - \omega_{\mathbf{k}})t} c_{g, \mathbf{k}, p}(t) \quad (8.95a)$$

$$i\hbar \frac{d}{dt} c_{g, \mathbf{k}, p}(t) = (g_{\mathbf{k}, p}^{e, g})^* e^{-i(\omega_0 - \omega_{\mathbf{k}})t} c_e(t). \quad (8.95b)$$

The spontaneous emission rate Integration of the second differential equation gives

$$c_{g, \mathbf{k}, p}(t) = \frac{1}{i\hbar} (g_{\mathbf{k}, p}^{e, g})^* \int_0^t dt' e^{-i(\omega_0 - \omega_{\mathbf{k}})t'} c_e(t'), \quad (8.96)$$

and by plugging this into the first differential equation, we get

$$\frac{d}{dt} c_e(t) = -\frac{1}{\hbar^2} \int d^3k \sum_p |g_{\mathbf{k}, p}^{e, g}|^2 \int_0^t dt' e^{i(\omega_0 - \omega_{\mathbf{k}})(t-t')} c_e(t'). \quad (8.97)$$

Similar as in our calculation for spontaneous emission in free space from the previous section [see Eqs. (8.79)–(8.82)], for a fixed wave vector $\mathbf{k}$, we take θ to be the angle between $\mathbf{k}$ and $\mathbf{d}_{e, g}$, and get

$$\sum_p |g_{\mathbf{k}, p}^{e, g}|^2 \stackrel{(8.89)}{=} \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3} \sum_p |\mathbf{d}_{e, g} \cdot \mathbf{e}_{\mathbf{k}, p}|^2 = \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3} |\mathbf{d}_{e, g}|^2 \sin^2 \theta. \quad (8.98)$$

Using this, and turning the integral over $\mathbf{k}$ in (8.97) into spherical coordinates then yields

$$\begin{aligned} \frac{d}{dt} c_e(t) &= -\frac{1}{\hbar^2} \int dk d\varphi d\theta k^2 \sin \theta \sum_p |g_{\mathbf{k}, p}^{e, g}|^2 \int_0^t dt' e^{i(\omega_0 - \omega_{\mathbf{k}})(t-t')} c_e(t') \\ &= -\frac{1}{16\pi^3 \epsilon_0 \hbar} |\mathbf{d}_{e, g}|^2 \int_0^\infty dk k^2 \omega_{\mathbf{k}} \int_0^t dt' e^{i(\omega_0 - \omega_{\mathbf{k}})(t-t')} c_e(t') \underbrace{\int_0^{2\pi} d\varphi}_{2\pi} \underbrace{\int_0^\pi d\theta \sin^3 \theta}_{4/3} \quad (8.99) \\ &= -\frac{1}{6\pi^2 \epsilon_0 \hbar} |\mathbf{d}_{e, g}|^2 \int_0^\infty dk k^2 \omega_{\mathbf{k}} \int_0^t dt' e^{i(\omega_0 - \omega_{\mathbf{k}})(t-t')} c_e(t'). \end{aligned}$$

By substituting $k = \omega_{\mathbf{k}}/c$ with $dk = d\omega_{\mathbf{k}}/c$, this turns into

$$\frac{d}{dt} c_e(t) = -\frac{1}{6\pi^2 \epsilon_0 \hbar c^3} |\mathbf{d}_{e, g}|^2 \int_0^\infty d\omega_{\mathbf{k}} \underbrace{\omega_{\mathbf{k}}^3}_{I} \int_0^t dt' e^{i(\omega_0 - \omega_{\mathbf{k}})(t-t')} c_e(t'). \quad (8.100)$$

We now consider the time integral: We make the assumption that the excited state population varies with a rate $\Gamma \ll \omega_0$. And since the exponential oscillates with a frequency proportional to ω_0 , the oscillations lead to a cancellation except around $t' \sim t$, such that we can set $c_e(t') = c_e(t)$ and put it out of the integral (sometimes called the Wigner-Weißkopf approximation),

$$I \approx c_e(t) \int_0^t dt' e^{i(\omega_0 - \omega_{\mathbf{k}})(t-t')}. \quad (8.101)$$

Substituting $\tau = t - t'$, with $d\tau = -dt'$, gives

$$I \approx -c_e(t) \int_t^0 d\tau e^{i(\omega_0 - \omega_{\mathbf{k}})\tau} = c_e(t) \int_0^t d\tau e^{i(\omega_0 - \omega_{\mathbf{k}})\tau}. \quad (8.102)$$

Given that the excited state population varies with a rate $\Gamma \ll \omega_0$, the time of interests are $t \gg 1/\omega_0$. We can therefore take the upper limit of the integral to infinity,

$$I \approx c_e(t) \int_0^\infty d\tau e^{i(\omega_0 - \omega_{\mathbf{k}})\tau}. \quad (8.103)$$

To perform the integral, we multiply with a factor $\exp(-\epsilon\tau)$, where $\epsilon > 0$, and take the limit $\lim_{\epsilon \searrow 0}$,

$$\begin{aligned} I &\approx c_e(t) \lim_{\epsilon \searrow 0} \int_0^\infty d\tau e^{i(\omega_0 - \omega_{\mathbf{k}} + i\epsilon)\tau} \\ &= c_e(t) \lim_{\epsilon \searrow 0} \left. \frac{e^{i(\omega_0 - \omega_{\mathbf{k}} + i\epsilon)\tau}}{i(\omega_0 - \omega_{\mathbf{k}} + i\epsilon)} \right|_{\tau=0}^\infty \\ &= c_e(t) \lim_{\epsilon \searrow 0} \left[0 - \frac{1}{i(\omega_0 - \omega_{\mathbf{k}} + i\epsilon)} \right] \\ &= i c_e(t) \lim_{\epsilon \searrow 0} \frac{1}{\omega_0 - \omega_{\mathbf{k}} + i\epsilon}. \end{aligned} \quad (8.104)$$

From principle part integrals, it is known that

$$\lim_{\epsilon \searrow 0} \frac{1}{\omega_0 - \omega_{\mathbf{k}} + i\epsilon} = -i\pi\delta(\omega_0 - \omega_{\mathbf{k}}) + \mathcal{P} \left(\frac{1}{\omega_0 - \omega_{\mathbf{k}}} \right), \quad (8.105)$$

where $\mathcal{P}$ is the Cauchy principle part (note that here it is assumed that the equality holds when the quantities appear within an integral; further recall that in the Cauchy principle part of the integral, an epsilon-region around the divergence is excluded, such that the integral may converges). Accordingly, we obtain

$$I \approx c_e(t) \left[\pi\delta(\omega_0 - \omega_{\mathbf{k}}) + i\mathcal{P} \left(\frac{1}{\omega_0 - \omega_{\mathbf{k}}} \right) \right]. \quad (8.106)$$

Using this in the differential equation (8.100), we obtain

$$\begin{aligned} \frac{d}{dt} c_e(t) &= -\frac{1}{6\pi^2\epsilon_0\hbar c^3} |\mathbf{d}_{e,g}|^2 c_e(t) \int_0^\infty d\omega_{\mathbf{k}} \omega_{\mathbf{k}}^3 \left[\pi\delta(\omega_0 - \omega_{\mathbf{k}}) + i\mathcal{P} \left(\frac{1}{\omega_0 - \omega_{\mathbf{k}}} \right) \right] \\ &= -c_e(t) \left[\underbrace{\frac{1}{2} \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3} |\mathbf{d}_{e,g}|^2}_{\Gamma = \Gamma_{e \rightarrow g}^{\text{spon em}}} + i \underbrace{\frac{1}{6\pi^2\epsilon_0\hbar c^3} |\mathbf{d}_{e,g}|^2 \mathcal{P} \int_0^\infty d\omega_{\mathbf{k}} \frac{\omega_{\mathbf{k}}^3}{\omega_0 - \omega_{\mathbf{k}}}}_{\delta\omega} \right] \end{aligned} \quad (8.107)$$

where we identify the same spontaneous emission rate $\Gamma = \Gamma_{e \rightarrow g}^{\text{spon em}}$ as in the perturbative treatment [see Eq. (8.82)], and defined the frequency (i.e., energy) shift

$$\delta\omega = \frac{\Gamma}{2\pi\omega_0^3} \mathcal{P} \int_0^\infty d\omega_{\mathbf{k}} \frac{\omega_{\mathbf{k}}^3}{\omega_0 - \omega_{\mathbf{k}}}. \quad (8.108)$$

The differential equation thus becomes

$$\frac{d}{dt} c_e(t) = - \left(\frac{\Gamma}{2} + i\delta\omega \right) c_e(t). \quad (8.109)$$

Note that the second term is imaginary such that it doesn't contribute to the decay rate, but rather to an energy shift. It turns out that this energy shift is related to the Lamb shift. It appears due to the coupling of the excited state with the continuum, such that the energy of the discrete excited state shifts from $\hbar\omega_e$ to $\hbar\omega_e + \hbar\delta\omega$.

Solving this differential equation for the initial condition of an excited atom, $c_e(0) = 1$, yields

$$\boxed{c_e(t) = e^{-(\Gamma/2 + i\delta\omega)t}}. \quad (8.110)$$

The probability to find the atom in the excited state therefore decays exponentially with rate Γ ,

$$\boxed{|c_e(t)|^2 = e^{-\Gamma t}}. \quad (8.111)$$

Spontaneous emission lineshape To find the coefficient $c_{g,\mathbf{k},p}(t)$, we can plug $c_e(t)$ from Eq. (8.110) into (8.96),

$$\begin{aligned} c_{g,\mathbf{k},p}(t) &= \frac{(g_{\mathbf{k},p}^{e,g})^*}{i\hbar} \int_0^t dt' e^{-i(\omega_0 - \omega_{\mathbf{k}})t'} e^{-(\Gamma/2 + i\delta\omega)t'} \\ &= \frac{(g_{\mathbf{k},p}^{e,g})^*}{i\hbar} \frac{e^{-i(\omega_0 - \omega_{\mathbf{k}} + \delta\omega)t - \Gamma t/2} - 1}{-i(\omega_0 - \omega_{\mathbf{k}} + \delta\omega) - \Gamma/2}. \end{aligned} \quad (8.112)$$

After long times $\Gamma t \gg 1$, the asymptotic state of the system therefore becomes [combining Eqs. (8.93), (8.110) and (8.112) for the limit $t \rightarrow \infty$]

$$|\psi_I(\infty)\rangle = \sum_p \int d^3k \frac{(g_{\mathbf{k},p}^{e,g})^*}{\hbar} \frac{1}{(\omega_{\mathbf{k}} - \omega_0 - \delta\omega) + i\Gamma/2} |g, 1_{\mathbf{k},p}\rangle. \quad (8.113)$$

The probability for an emission into the mode $(\mathbf{k}, p)$ is, therefore,

$$\boxed{p_{\mathbf{k},p} = \frac{|g_{\mathbf{k},p}^{e,g}|^2 / \hbar^2}{(\omega_{\mathbf{k}} - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2}}. \quad (8.114)$$

Finally, in order to get the probability for an emission at frequency ω , we sum these probabilities over the polarizations, and integrate over $\mathbf{k}$ with respect to the delta function $\delta(\omega_{\mathbf{k}} - \omega)$, which gives

$$\begin{aligned} p(\omega) &= \int d^3k \delta(\omega_{\mathbf{k}} - \omega) \sum_p p_{\mathbf{k},p} \\ &= \int d^3k \delta(\omega_{\mathbf{k}} - \omega) \frac{1}{(\omega_{\mathbf{k}} - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2} \frac{1}{\hbar^2} \sum_p |g_{\mathbf{k},p}^{e,g}|^2 \end{aligned} \quad (8.115)$$

Transforming into spherical coordinates and using $\sum_p |g_{\mathbf{k},p}^{e,g}|^2 = \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3} |\mathbf{d}_{e,g}|^2 \sin^2 \theta$ from Eq. (8.98) gives

$$\begin{aligned} p(\omega) &= \int dk k^2 \delta(\omega_{\mathbf{k}} - \omega) \frac{1}{(\omega_{\mathbf{k}} - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2} \frac{\omega_{\mathbf{k}}}{2\epsilon_0(2\pi)^3 \hbar} |\mathbf{d}_{e,g}|^2 \underbrace{\int_0^\pi d\theta \sin^3 \theta \int_0^{2\pi} d\varphi}_{8\pi/3} \\ &= \int dk k^2 \delta(\omega_{\mathbf{k}} - \omega) \frac{1}{(\omega_{\mathbf{k}} - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2} \frac{\omega_{\mathbf{k}}}{6\pi^2 \epsilon_0 \hbar} |\mathbf{d}_{e,g}|^2 \end{aligned} \quad (8.116)$$

Substituting $k = \omega_{\mathbf{k}}/c$ and $dk = d\omega_{\mathbf{k}}/c$ yields

$$\begin{aligned} p(\omega) &= \frac{1}{6\pi^2 \epsilon_0 \hbar c^3} |\mathbf{d}_{e,g}|^2 \int d\omega_{\mathbf{k}} \delta(\omega_{\mathbf{k}} - \omega) \frac{\omega_{\mathbf{k}}^3}{(\omega_{\mathbf{k}} - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2} \\ &= \frac{1}{6\pi^2 \epsilon_0 \hbar c^3} |\mathbf{d}_{e,g}|^2 \frac{\omega^3}{(\omega - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2} \end{aligned} \quad (8.117)$$

In the assumed limit $\Gamma \ll \omega_0$, the denominator of the last fraction is always large except around $\omega \approx \omega_0 + \delta\omega$. For small $\delta\omega$ one can therefore approximate the fraction by replacing ω^3 in the numerator by ω_0^3 , yielding

$$p(\omega) = \underbrace{\frac{\omega_0^3}{6\pi^2\epsilon_0\hbar c^3}}_{\Gamma/2\pi} |d_{e,g}|^2 \frac{1}{(\omega - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2}, \quad (8.118)$$

and, accordingly,

$$p(\omega) = \frac{1}{\pi} \frac{\Gamma/2}{(\omega - \omega_0 - \delta\omega)^2 + (\Gamma/2)^2} \quad (8.119)$$

For the frequency spectrum of spontaneous emission, we thus found a Lorentzian line shape centered around $\omega_0 + \delta\omega$ with FWHM Γ . This is why Γ is called the natural width of the excited state. Since by the coupling of the excited state to the continuum, its energy is shifted from $\hbar\omega_e$ to $\hbar\omega_e + \hbar\delta\omega$, the peak of the lineshape is centered around $\omega_0 + \delta\omega$ (instead of the bare resonance frequency ω_0 if there was no coupling to the continuum).

Review Questions 8.5.

- *What are the assumptions made in the Wigner-Weisskopf model of spontaneous emission?*
- *What is the spontaneous emission rate?*
- *What is the Wigner-Weißkopf approximation? Can you interpret this approximation in terms of memory effects? (It is similar to the Markov approximation, if you heard about that before)*
- *What is the spontaneous emission lineshape? What assumptions were made?*
- *Why is the lineshape not centered around the frequency $\omega_0 = \omega_e - \omega_g$?*

Chapter 9

Cavity quantum electrodynamics

In the previous chapter, we considered the interaction of matter with the electromagnetic field in free space. Now we impose additional boundary conditions to the electromagnetic field, such as reflective walls. To this end, we place a quantum object (think of an atom with well-defined energy levels) into a cavity and study its interaction with the electromagnetic field, which, by the boundary conditions imposed by the cavity, is confined, and composed of a discrete set of radiation modes. Such scenarios of light-matter interaction go under the name of cavity quantum electrodynamics (or cavity QED). As we will see, the additional boundary conditions can drastically affect the radiative properties of the system, such as the absorption and spontaneous emission rates.

9.1 Jaynes-Cummings model

The simplest model in cavity QED is the coupling of a two-level system (also called “atom” in the following) to a single mode of a cavity, known as the Jaynes-Cummings model. Since the cavity only hosts a single radiation mode, the radiation Hamiltonian reads

$$H_{\text{rad}} = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right). \quad (9.1)$$

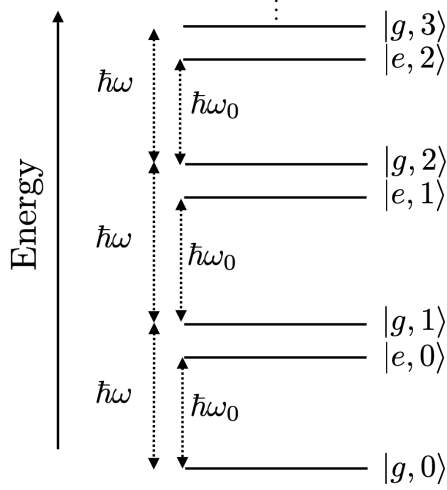
The atom is supposed to be placed inside the cavity with its two levels, $|g\rangle$ and $|e\rangle$, separated in energy by $\hbar\omega_0$. The matter Hamiltonian therefore reads

$$H_{\text{mat}} = \frac{\hbar\omega_0}{2} (|e\rangle\langle e| - |g\rangle\langle g|). \quad (9.2)$$

The eigenstates of $H_0 = H_{\text{rad}} + H_{\text{mat}}$, i.e., the uncoupled system of atom and field (sometimes called bare states) are thus $\{|g, n\rangle, |e, n\rangle\}_n$, where $n \in \mathbb{N}_0$ labels the Fock number state of the single radiation mode. The corresponding eigenenergies are

$$\hbar\omega \left(n + \frac{1}{2} \right) \pm \frac{\hbar\omega_0}{2}. \quad (9.3)$$

The energy spectrum of the bare states is as follows (shown for $\omega > \omega_0$):



For the interaction of atom and field, we consider the dipole approximation with interaction Hamiltonian $H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}_{\perp}(\mathbf{R})$ [cf. Eq. (8.16)]. Using $\mathbf{E}_{\perp}(\mathbf{R}) = i\sqrt{\frac{\hbar\omega}{2\epsilon_0}}\mathbf{u}(\mathbf{R})(a - a^{\dagger})$ and $\mathbf{d} = \mathbf{d}_{e,g}|e\rangle\langle g| + \mathbf{d}_{g,e}|g\rangle\langle e|$, and assuming that the dipole transition matrix elements $\mathbf{d}_{e,g} = \langle e|\mathbf{d}|g\rangle$ are real, such that $\mathbf{d}_{e,g} = \mathbf{d}_{g,e}$, the interaction Hamiltonian becomes

$$\begin{aligned} H_{\text{int}} &= -i\hbar \underbrace{\sqrt{\frac{\omega}{2\hbar\epsilon_0}}\mathbf{d}_{e,g} \cdot \mathbf{u}(\mathbf{R})}_g \left(\underbrace{|e\rangle\langle g|}_{\sigma_+} + \underbrace{|g\rangle\langle e|}_{\sigma_-} \right) (a - a^{\dagger}) \\ &= -i\hbar g (\sigma_+ + \sigma_-) (a - a^{\dagger}) \\ &= -i\hbar g (\sigma_+ a - \sigma_+ a^{\dagger} + \sigma_- a - \sigma_- a^{\dagger}), \end{aligned} \quad (9.4)$$

where we defined the real coupling constant $g = \sqrt{\frac{\omega}{2\hbar\epsilon_0}}\mathbf{d}_{e,g} \cdot \mathbf{u}(\mathbf{R})$, and the atomic raising and lowering operators

$$\sigma_+ = |e\rangle\langle g| \quad \text{and} \quad \sigma_- = |g\rangle\langle e|. \quad (9.5)$$

Let us transform into the interaction picture with respect to $H_0 = H_{\text{rad}} + H_{\text{mat}}$ via the unitary $U_0^{\dagger}(t) = e^{iH_0 t/\hbar}$. The interaction Hamiltonian in the interaction picture, $H_{\text{int,I}}(t) = U_0^{\dagger}(t)H_{\text{int}}U_0(t)$ then becomes, using $U_0^{\dagger}(t)aU_0(t) = ae^{-i\omega t}$ and $U_0^{\dagger}(t)\sigma_-U_0(t) = \sigma_-e^{-i\omega_0 t}$,

$$H_{\text{int,I}}(t) = -i\hbar g \left(\sigma_+ a e^{i(\omega_0 - \omega)t} - \sigma_+ a^{\dagger} e^{i(\omega_0 + \omega)t} + \sigma_- a e^{-i(\omega_0 + \omega)t} - \sigma_- a^{\dagger} e^{-i(\omega_0 - \omega)t} \right). \quad (9.6)$$

Assuming a small detuning $\delta = \omega_0 - \omega$, the second and third term with $e^{\pm i(\omega + \omega_0)t}$ oscillates rapidly as compared to the first and last term, and averages to zero over relevant timescales. We can thus employ the rotating wave approximation, and neglect the second and third term, yielding

$$H_{\text{int,I}}(t) \approx -i\hbar g \left(\sigma_+ a e^{-i(\omega - \omega_0)t} - \sigma_- a^{\dagger} e^{i(\omega - \omega_0)t} \right), \quad (9.7)$$

and, back in the Schrödinger picture,

$$\boxed{H_{\text{int}} \approx -i\hbar g (\sigma_+ a - \sigma_- a^{\dagger})}. \quad (9.8)$$

We now see that the rotating wave approximation left us with the terms that conserve the total number of excitations of atom and field. Accordingly, the state $|e, n\rangle$ is only coupled to $|g, n+1\rangle$. Moreover, we see that in the bare-state basis, the total Hamiltonian is block-diagonal, and can be diagonalized.

9.2 Diagonalization

First observe that the atom-field ground state $|g, 0\rangle$ is not coupled to any other state, and, thus, remains unaffected.

Now consider the subspace spanned by the states $\{|e, n\rangle, |g, n+1\rangle\}$, for any $n \in \mathbb{N}_0$. The corresponding block of the Hamiltonian in this subspace reads

$$\begin{aligned} H_n &= \begin{pmatrix} \langle e, n | H | e, n \rangle & \langle e, n | H | g, n+1 \rangle \\ \langle g, n+1 | H | e, n \rangle & \langle g, n+1 | H | g, n+1 \rangle \end{pmatrix} \\ &= \underbrace{\hbar\omega \begin{pmatrix} n + \frac{1}{2} & 0 \\ 0 & n + \frac{3}{2} \end{pmatrix}}_{H_{\text{rad}}} + \underbrace{\frac{\hbar\omega_0}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{H_{\text{mat}}} - \underbrace{i\hbar g \begin{pmatrix} 0 & \sqrt{n+1} \\ -\sqrt{n+1} & 0 \end{pmatrix}}_{H_{\text{int}}} \\ &= \begin{pmatrix} \hbar\omega(n+1) + \frac{\hbar}{2}(\omega_0 - \omega) & -i\hbar g\sqrt{n+1} \\ i\hbar g\sqrt{n+1} & \hbar\omega(n+1) - \frac{\hbar}{2}(\omega_0 - \omega) \end{pmatrix}. \end{aligned} \quad (9.9)$$

By introducing $\Omega_n = 2g\sqrt{n+1}$ and the detuning $\delta = \omega_0 - \omega$, this becomes

$$H_n = \hbar\omega(n+1)\mathbb{1} + \frac{\hbar}{2} \begin{pmatrix} \delta & -i\Omega_n \\ i\Omega_n & -\delta \end{pmatrix}. \quad (9.10)$$

We can now introduce $\theta_n \in [0, \pi]$, such that

$$\cos \theta_n = \frac{\delta}{\sqrt{\delta^2 + \Omega_n^2}} \quad \text{and} \quad \sin \theta_n = \frac{\Omega_n}{\sqrt{\delta^2 + \Omega_n^2}}, \quad (9.11)$$

and write

$$H_n = \hbar\omega(n+1)\mathbb{1} + \frac{\hbar\sqrt{\delta^2 + \Omega_n^2}}{2} \begin{pmatrix} \cos \theta_n & -i \sin \theta_n \\ i \sin \theta_n & -\cos \theta_n \end{pmatrix}. \quad (9.12)$$

Diagonalizing this two-by-two matrix yields the eigenvectors

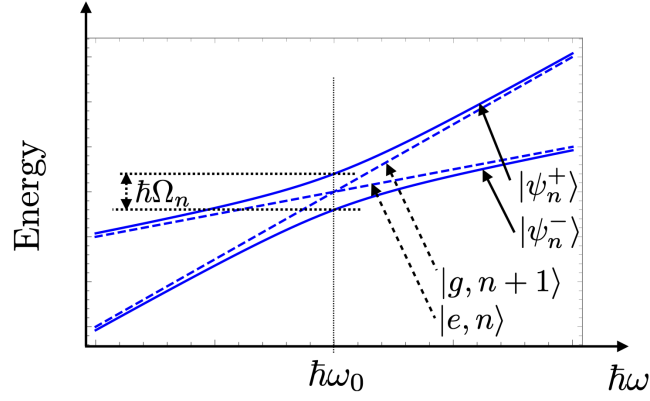
$$|\psi_n^+\rangle = \cos\left(\frac{\theta_n}{2}\right) |e, n\rangle + i \sin\left(\frac{\theta_n}{2}\right) |g, n+1\rangle, \quad (9.13a)$$

$$|\psi_n^-\rangle = \sin\left(\frac{\theta_n}{2}\right) |e, n\rangle - i \cos\left(\frac{\theta_n}{2}\right) |g, n+1\rangle, \quad (9.13b)$$

with corresponding eigenenergies

$$E_n^\pm = (n+1)\hbar\omega \pm \frac{\hbar\sqrt{\delta^2 + \Omega_n^2}}{2}. \quad (9.14)$$

The states $|\psi_n^\pm\rangle$ are often called dressed states (or dressed-atom states), since they may be viewed as describing the atom being “dressed” by the interaction with the radiation mode rather than the free atom. As characteristic for two coupled oscillators, this interaction leads to an anticrossing in the total energy spectrum (9.14), with the dressed states $|\psi_n^+\rangle$ and $|\psi_n^-\rangle$ separated in energy by $\hbar\sqrt{\delta^2 + \Omega_n^2}$:



Review Questions 9.1.

- Describe the Jaynes-Cummings model. What are the assumptions and approximations behind this model?
- What are the so-called dressed states?
- Why do the eigenenergies split on resonance? Explain the anticrossing.

9.3 Rabi oscillations

Suppose now that the system of atom+cavity mode is initially in the state corresponding to an excited atom together with n field excitations,

$$|\psi(0)\rangle = |e, n\rangle \stackrel{(9.13)}{=} \cos\left(\frac{\theta_n}{2}\right) |\psi_n^+\rangle + \sin\left(\frac{\theta_n}{2}\right) |\psi_n^-\rangle. \quad (9.15)$$

Since we expressed $|\psi(0)\rangle$ in terms of the eigenstates of the Hamiltonian H , the time evolved state is simply

$$\begin{aligned} |\psi(t)\rangle &= \cos\left(\frac{\theta_n}{2}\right) e^{-iE_n^+ t/\hbar} |\psi_n^+\rangle + \sin\left(\frac{\theta_n}{2}\right) e^{-iE_n^- t/\hbar} |\psi_n^-\rangle \\ &= \left[\cos^2\left(\frac{\theta_n}{2}\right) e^{-iE_n^+ t/\hbar} + \sin^2\left(\frac{\theta_n}{2}\right) e^{-iE_n^- t/\hbar} \right] |e, n\rangle \\ &\quad + i \sin\left(\frac{\theta_n}{2}\right) \cos\left(\frac{\theta_n}{2}\right) \left[e^{-iE_n^+ t/\hbar} - e^{-iE_n^- t/\hbar} \right] |g, n+1\rangle. \end{aligned} \quad (9.16)$$

The probability to find the atom in the excited state at time t is, thus,

$$\begin{aligned} p_n^e(t) &= \left| \cos^2\left(\frac{\theta_n}{2}\right) e^{-iE_n^+ t/\hbar} + \sin^2\left(\frac{\theta_n}{2}\right) e^{-iE_n^- t/\hbar} \right|^2 \\ &= \underbrace{\cos^4\left(\frac{\theta_n}{2}\right) + \sin^4\left(\frac{\theta_n}{2}\right)}_{\frac{1}{2}(1+\cos^2\theta_n)} + \underbrace{\sin^2\left(\frac{\theta_n}{2}\right) \cos^2\left(\frac{\theta_n}{2}\right)}_{\frac{1}{4}\sin^2\theta_n} \left[e^{-i(E_n^+ - E_n^-)t/\hbar} + e^{i(E_n^+ - E_n^-)t/\hbar} \right] \\ &= \frac{1}{2} \left[1 + \cos^2\theta_n + \sin^2\theta_n \cos\left(\frac{E_n^+ - E_n^-}{\hbar} t\right) \right] \end{aligned} \quad (9.17)$$

Using $\sin \theta_n$ and $\cos \theta_n$ from Eq. (9.11), as well as $E_n^+ - E_n^- = \hbar \sqrt{\delta^2 + \Omega_n^2}$ [see Eq. (9.14)], this becomes

$$p_n^e(t) = \frac{1}{2} \left[1 + \frac{\delta^2}{\delta^2 + \Omega_n^2} + \frac{\Omega_n^2}{\delta^2 + \Omega_n^2} \cos \left(\sqrt{\delta^2 + \Omega_n^2} t \right) \right], \quad (9.18)$$

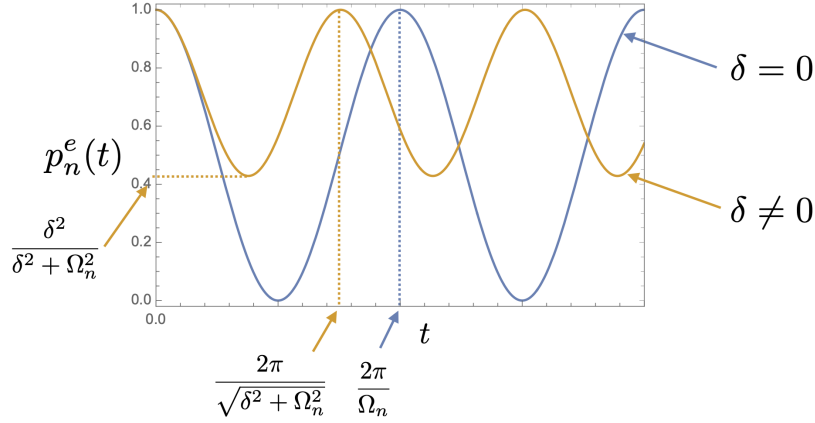
and, thus,

$$p_n^e(t) = \frac{1}{\delta^2 + \Omega_n^2} \left\{ \delta^2 + \frac{1}{2} \Omega_n^2 \left[1 + \cos \left(\sqrt{\delta^2 + \Omega_n^2} t \right) \right] \right\}. \quad (9.19)$$

One thus observes Rabi oscillations between the two coupled states with frequency $\sqrt{\delta^2 + \Omega_n^2}$. In particular, on resonance, $\delta = \omega_0 - \omega = 0$, the probability

$$p_n^e(t)|_{\delta=0} = \frac{1}{2} [1 + \cos(\Omega_n t)] \quad (9.20)$$

shows Rabi oscillations with unit amplitude. Note that these oscillations are also present if the cavity is initially in the vacuum state, $n = 0$, which is called vacuum Rabi flopping. The corresponding frequency (or vacuum Rabi splitting) is, on resonance, $\Omega_0 = 2g$, providing a measure for the coupling strength.



9.4 Decays and revivals

Let's still consider the atom initially in the excited state, however, suppose the field is in a superposition of several Fock number states $|n\rangle$. The probability to find the atom in the excited state must then be given by a superposition of signals that oscillate with frequency $\sqrt{\delta^2 + \Omega_n^2}$. Since these oscillations dephase with increasing time, the probability to find the atom in the excited state will decay, followed by a revival when the oscillations come into phase again.

Let us take a look at this phenomenon for an initial coherent state $|\alpha\rangle$ of the field, when the field mode is on resonance with the atomic transition, $\delta = 0$. Given the initial state

$$|\psi(0)\rangle = |e, \alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |e, n\rangle, \quad (9.21)$$

the probability to find the atom in the excited state is, thus, a superposition of the probabilities $p_n^e(t)|_{\delta=0}$ from Eq. (9.20), weighted by the modulus square of the amplitudes from (9.21),

$$p_\alpha^e(t)|_{\delta=0} = e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} p_n^e(t)|_{\delta=0} \quad (9.22)$$

$$= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \frac{1 + \cos(\Omega_n t)}{2}. \quad (9.23)$$

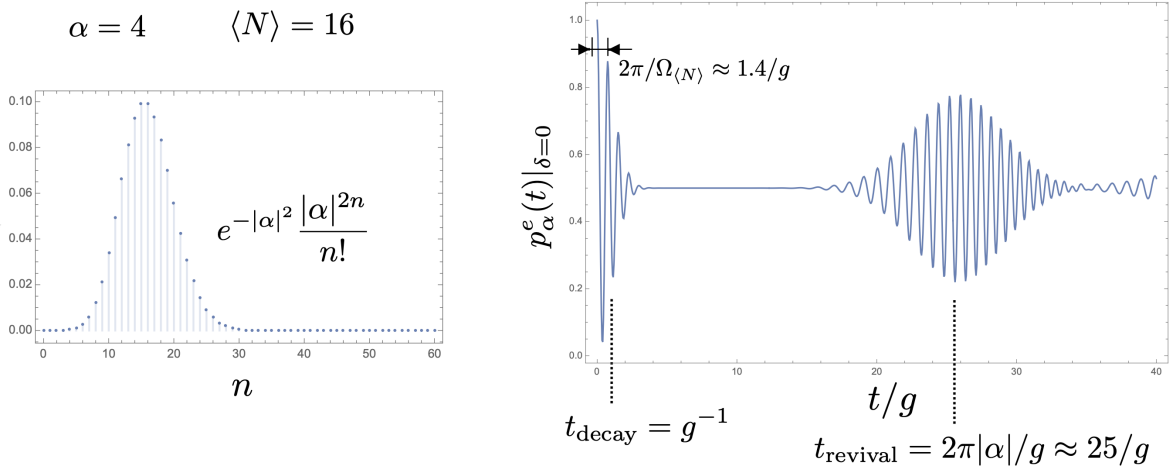
Now recall from our discussion of coherent states that the photon number is distributed around the mean value $\langle N \rangle = |\alpha|^2$. Accordingly, the short time behavior of $p_n^e(t)|_{\delta=0}$ must be given by oscillations with frequency $\Omega_{\langle N \rangle}$. Now recall that the variance is $(\Delta N)^2 = |\alpha|^2$. Thus, the distribution of the frequencies Ω_n has a width (assuming $|\alpha| \gg 1$)

$$\begin{aligned}
& \Omega_{\langle N \rangle + \Delta N} - \Omega_{\langle N \rangle - \Delta N} \\
&= 2g \left(\sqrt{|\alpha|^2 + |\alpha| + 1} - \sqrt{|\alpha|^2 - |\alpha| + 1} \right) \\
&= 2g \frac{\overbrace{\left(\sqrt{|\alpha|^2 + |\alpha| + 1} - \sqrt{|\alpha|^2 - |\alpha| + 1} \right) \left(\sqrt{|\alpha|^2 + |\alpha| + 1} + \sqrt{|\alpha|^2 - |\alpha| + 1} \right)}^{=2|\alpha|}}{\underbrace{\sqrt{|\alpha|^2 + |\alpha| + 1} + \sqrt{|\alpha|^2 - |\alpha| + 1}}_{\approx 2|\alpha|}} \quad (9.24) \\
&\approx 2g.
\end{aligned}$$

The decay of the oscillations therefore occurs on a timescale of the order $t_{\text{decay}} = g^{-1}$. When the frequencies at the center of the spectrum come back into phase, the oscillations revive. This happens on a timescale t_{revival} , for which $(\Omega_{\langle N \rangle} - \Omega_{\langle N \rangle - 1})t_{\text{revival}} = 2\pi$. We thus find

$$\begin{aligned}
t_{\text{revival}} &= \frac{2\pi}{2g \left(\sqrt{|\alpha|^2 + 1} - |\alpha| \right)} \\
&= \frac{2\pi}{2g \frac{(\sqrt{|\alpha|^2 + 1} - |\alpha|)(\sqrt{|\alpha|^2 + 1} + |\alpha|)}{\sqrt{|\alpha|^2 + 1} + |\alpha|}} \\
&\approx \frac{2\pi}{2g \frac{1}{2|\alpha|}} \\
&= \frac{2\pi|\alpha|}{g}.
\end{aligned} \tag{9.25}$$

Note that the revivals are a signature of the discreteness of the frequency spectrum. As an example, the probability to find the atom in the excited state is shown in the following for $\alpha = 4$:



Review Questions 9.2.

- *What are Rabi oscillations? Why do they occur in the Jaynes-Cummings model?*
- *Explain the behavior of the Rabi oscillations (Eq. (9.19))*
- *Why do we expect a decay and revival of Rabi oscillations if multiple photon numbers contribute to the initial state?*
- *What are the time scales of the decay and revival times for coherent states on resonance with the atomic transition?*
- *What changes if the frequency of the coherent state is off-resonant, i.e., $\delta \neq 0$?*

Appendices

Appendix A

Supplement to Ch. 7

A.1 Details to Eq. (7.2)

As mention in Sec. 7.1, the Hamiltonian (7.2) describes particles whose dynamics are governed by the total fields $\mathbf{A}_t = \mathbf{A}_e + \mathbf{A}$ and $\phi_t = \phi_e + \phi$ from Eq. (7.1), according to the Lorentz equations

$$m_\alpha \ddot{\mathbf{r}}_\alpha = q_\alpha \mathbf{E}_t + q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t. \quad (\text{A.1})$$

On the other hand, the equations of motion for the potentials $\mathbf{A}$ and ϕ are governed by the sources ρ and $\mathbf{j}$ of the particles alone, and given by the Maxwell equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad (\text{A.2a})$$

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right), \quad (\text{A.2b})$$

where $\mathbf{E} = -\nabla\phi - \partial\mathbf{A}/\partial t$ and $\mathbf{B} = \nabla \times \mathbf{A}$. That is, the evolution of $\mathbf{A}$ and ϕ is independently of the externally applied field.

We now show that Eq. (A.1) indeed follows from the Hamiltonian (7.2) (the validity of (A.2) is shown in [4], and will not be discussed here): Consider the Hamiltonian (7.2),

$$H = \underbrace{\sum_\alpha \frac{1}{2m_\alpha} (\mathbf{p}_\alpha - q_\alpha \mathbf{A}_t(\mathbf{r}_\alpha, t))^2}_{H_1} + \underbrace{V_{\text{Coul}}}_{H_\parallel} + \underbrace{\sum_\alpha q_\alpha \phi_e(\mathbf{r}_\alpha, t)}_{H_3} + H_\perp, \quad (\text{A.3})$$

where $\mathbf{A}_t(\mathbf{r}_\alpha, t) = \mathbf{A}_\perp(\mathbf{r}_\alpha) + \mathbf{A}_e(\mathbf{r}_\alpha, t)$ (since $\mathbf{A}_\perp = \mathbf{A}$ in the Coulomb gauge with respect to the electromagnetic field),

$$V_{\text{Coul}} = H_\parallel \stackrel{(2.56)}{=} \sum_{\alpha, \beta} \frac{q_\alpha q_\beta}{2\epsilon_0(2\pi)^3} \int d^3k \frac{e^{i\mathbf{k} \cdot (\mathbf{r}_\alpha - \mathbf{r}_\beta)}}{k^2}, \quad (\text{A.4})$$

and

$$H_\perp = \int \left(\frac{\epsilon_0 \mathbf{E}_\perp^2}{2} + \frac{\mathbf{B}^2}{2\mu_0} \right) d^3r. \quad (\text{A.5})$$

The particles' phase space coordinates are $(\mathbf{r}_\alpha, \mathbf{p}_\alpha)$. Accordingly we obtain the particles' equations of motion via Hamilton's equations

$$\frac{d(\mathbf{r}_\alpha)_i}{dt} = \frac{\partial H}{\partial (\mathbf{p}_\alpha)_i}, \quad (\text{A.6a})$$

$$\frac{d(\mathbf{p}_\alpha)_i}{dt} = -\frac{\partial H}{\partial (\mathbf{r}_\alpha)_i}. \quad (\text{A.6b})$$

Since $\mathbf{p}_\alpha$ only appears in the first term of the Hamiltonian (A.3), i.e., in H_1 , Eq. (A.6a) leads to

$$\begin{aligned} (\dot{\mathbf{r}}_\alpha)_i &= \frac{1}{2m_\alpha} \frac{\partial}{\partial (\mathbf{p}_\alpha)_i} \left[\mathbf{p}_\alpha^2 + q_\alpha^2 \mathbf{A}_t^2(\mathbf{r}_\alpha, t) - 2q_\alpha \mathbf{p}_\alpha \cdot \mathbf{A}_t(\mathbf{r}_\alpha, t) \right] \\ &= \frac{1}{2m_\alpha} [2(\mathbf{p}_\alpha)_i - 2q_\alpha (\mathbf{A}_t(\mathbf{r}_\alpha, t))_i]. \end{aligned} \quad (\text{A.7})$$

We thus arrive at

$$\boxed{\mathbf{p}_\alpha = m_\alpha \dot{\mathbf{r}}_\alpha + q_\alpha \mathbf{A}_t(\mathbf{r}_\alpha, t)}, \quad (\text{A.8})$$

which defines the particle's canonical momentum in terms of its kinetic momentum and the total vector potential [c.f. Eq. (2.51)].

Next, we consider Eq. (A.6b): To this end, we recognize that $\mathbf{r}_\alpha$ appears in the first three terms of the Hamiltonian (A.3). Let us consider these terms separately. For H_1 we have, using the notation $(\nabla_\alpha H_1)_i = \frac{\partial H_1}{\partial (\mathbf{r}_\alpha)_i}$, and dropping the arguments of $\mathbf{A}_t(\mathbf{r}_\alpha, t)$ to ease notation,

$$\begin{aligned} \nabla_\alpha H_1 &= \frac{1}{2m_\alpha} \nabla_\alpha \left[\mathbf{p}_\alpha^2 + q_\alpha^2 \mathbf{A}_t^2 - 2q_\alpha \mathbf{p}_\alpha \cdot \mathbf{A}_t \right] \\ &= \frac{1}{2m_\alpha} \left[q_\alpha^2 \nabla_\alpha \mathbf{A}_t^2 - 2q_\alpha \nabla_\alpha (\mathbf{p}_\alpha \cdot \mathbf{A}_t) \right]. \end{aligned} \quad (\text{A.9})$$

Now, we note that

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} + (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{B} \times (\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla \times \mathbf{B}), \quad (\text{A.10})$$

such that

$$\nabla_\alpha (\mathbf{p}_\alpha \cdot \mathbf{A}_t) = (\mathbf{p}_\alpha \cdot \nabla_\alpha) \mathbf{A}_t + \mathbf{p}_\alpha \times \underbrace{(\nabla_\alpha \times \mathbf{A}_t)}_{\mathbf{B}_t}. \quad (\text{A.11})$$

Using this in (A.9) leads to

$$\begin{aligned} \nabla_\alpha H_1 &= \frac{q_\alpha^2}{2m_\alpha} \nabla_\alpha \mathbf{A}_t^2 - \frac{q_\alpha}{m_\alpha} (\mathbf{p}_\alpha \cdot \nabla_\alpha) \mathbf{A}_t - \frac{q_\alpha}{m_\alpha} \mathbf{p}_\alpha \times \mathbf{B}_t \\ &\stackrel{(\text{A.8})}{=} \frac{q_\alpha^2}{2m_\alpha} \nabla_\alpha \mathbf{A}_t^2 - q_\alpha (\dot{\mathbf{r}}_\alpha \cdot \nabla_\alpha) \mathbf{A}_t - \frac{q_\alpha^2}{m_\alpha} (\mathbf{A}_t \cdot \nabla_\alpha) \mathbf{A}_t - q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t - \frac{q_\alpha^2}{m_\alpha} \mathbf{A}_t \times \mathbf{B}_t. \end{aligned} \quad (\text{A.12})$$

From (A.10), we have $\frac{1}{2} \nabla_\alpha \mathbf{A}_t^2 = \mathbf{A}_t \times (\nabla_\alpha \times \mathbf{A}_t) + (\mathbf{A}_t \cdot \nabla_\alpha) \mathbf{A}_t = \mathbf{A}_t \times \mathbf{B}_t + (\mathbf{A}_t \cdot \nabla_\alpha) \mathbf{A}_t$, and, thus,

$$\nabla_\alpha H_1 = -q_\alpha (\dot{\mathbf{r}}_\alpha \cdot \nabla_\alpha) \mathbf{A}_t - q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t. \quad (\text{A.13})$$

Next, for $H_\parallel$ from Eq. (A.4), we get

$$\begin{aligned} \nabla_\alpha H_\parallel &= \nabla_\alpha \sum_{\gamma, \beta} \frac{q_\gamma q_\beta}{2\epsilon_0 (2\pi)^3} \int d^3k \frac{e^{i\mathbf{k} \cdot (\mathbf{r}_\gamma - \mathbf{r}_\beta)}}{k^2}, \\ &= \sum_\gamma \frac{q_\gamma q_\alpha}{2\epsilon_0 (2\pi)^3} \int d^3k \frac{(-i\mathbf{k}) e^{i\mathbf{k} \cdot (\mathbf{r}_\gamma - \mathbf{r}_\alpha)}}{k^2} + \sum_\beta \frac{q_\alpha q_\beta}{2\epsilon_0 (2\pi)^3} \int d^3k \frac{(i\mathbf{k}) e^{i\mathbf{k} \cdot (\mathbf{r}_\alpha - \mathbf{r}_\beta)}}{k^2} \\ &= \sum_\beta \frac{q_\alpha q_\beta}{\epsilon_0 (2\pi)^3} \int d^3k \frac{(i\mathbf{k}) e^{i\mathbf{k} \cdot (\mathbf{r}_\alpha - \mathbf{r}_\beta)}}{k^2} \end{aligned} \quad (\text{A.14})$$

where we substituted $\mathbf{k} \rightarrow -\mathbf{k}$ in the first integral of the second line to arrive at the last line. Next, we use the Fourier transformation $\frac{1}{(2\pi)^{3/2}} \int d^3k \frac{-i\mathbf{k}}{(2\pi)^{3/2}k^2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{\mathbf{r}}{4\pi r^3}$, yielding

$$\begin{aligned} \nabla_\alpha H_\parallel &= -q_\alpha \frac{1}{4\pi\epsilon_0} \sum_\beta q_\beta \frac{\mathbf{r}_\alpha - \mathbf{r}_\beta}{|\mathbf{r}_\alpha - \mathbf{r}_\beta|^3} \\ &\stackrel{(2.20)}{=} -q_\alpha \mathbf{E}_\parallel(\mathbf{r}_\alpha, t). \end{aligned} \quad (\text{A.15})$$

Since, in the Coulomb gauge, $\mathbf{E}_\parallel(\mathbf{r}_\alpha, t) = -\nabla\phi(\mathbf{r}_\alpha, t)$, we arrive at

$$\nabla_\alpha H_\parallel = q_\alpha \nabla_\alpha \phi(\mathbf{r}_\alpha, t). \quad (\text{A.16})$$

Continuing with H_3 , we have

$$\nabla_\alpha H_3 = q_\alpha \nabla_\alpha \phi_e(\mathbf{r}_\alpha, t). \quad (\text{A.17})$$

Putting all together, for the right hand side of (A.6b), we have

$$\nabla_\alpha H = -q_\alpha (\dot{\mathbf{r}}_\alpha \cdot \nabla_\alpha) \mathbf{A}_t - q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t + q_\alpha \nabla_\alpha (\phi + \phi_e). \quad (\text{A.18})$$

On the other hand, for the left hand side of (A.6b), we have

$$\begin{aligned} \frac{d}{dt} \mathbf{p}_\alpha &\stackrel{(\text{A.8})}{=} \frac{d}{dt} (m_\alpha \dot{\mathbf{r}}_\alpha + q_\alpha \mathbf{A}_t) \\ &= m_\alpha \ddot{\mathbf{r}}_\alpha + q_\alpha \frac{d}{dt} \mathbf{A}_t \\ &= m_\alpha \ddot{\mathbf{r}}_\alpha + q_\alpha \frac{\partial}{\partial t} \mathbf{A}_t + q_\alpha (\dot{\mathbf{r}}_\alpha \cdot \nabla_\alpha) \mathbf{A}_t. \end{aligned} \quad (\text{A.19})$$

Since Eq. (A.6b) reads $\frac{d}{dt} \mathbf{p}_\alpha = -\nabla_\alpha H$, Eqs. (A.18) and (A.19) lead to

$$\begin{aligned} q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t - q_\alpha \nabla_\alpha (\underbrace{\phi + \phi_e}_{\phi_t}) &= m_\alpha \ddot{\mathbf{r}}_\alpha + q_\alpha \frac{\partial}{\partial t} \mathbf{A}_t \\ \Leftrightarrow m_\alpha \ddot{\mathbf{r}}_\alpha &= q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t + q_\alpha \underbrace{\left(-\nabla_\alpha \phi_t - \frac{\partial}{\partial t} \mathbf{A}_t \right)}_{\mathbf{E}_t}. \end{aligned} \quad (\text{A.20})$$

We thus found the the dynamics of the charged particles is governed by the Lorentz equations with respect to the total electric and magnetic field,

$$\boxed{m_\alpha \ddot{\mathbf{r}}_\alpha = q_\alpha \dot{\mathbf{r}}_\alpha \times \mathbf{B}_t(\mathbf{r}_\alpha, t) + q_\alpha \mathbf{E}_t(\mathbf{r}_\alpha, t).} \quad (\text{A.21})$$

A.2 Proof of Eq. (7.65)

We are now proving the expression of the Fourier transformed retarded Green's function provided in Eq. (7.65): We start with taking the Fourier transform of the retarded Green's function from Eq. (7.58),

$$\tilde{G}^+(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt e^{i\omega t} G^+(t) = \frac{-i}{\sqrt{2\pi}} \int_0^{\infty} dt e^{i\omega t} e^{-iH_0 t/\hbar}. \quad (\text{A.22})$$

The integral on the right hand side can be brought into the form of a Laplace transformation. To this end, recall that the Laplace transformation of a function $f(t)$ is defined, for $s \in \mathbb{C}$, by

$$\mathcal{L}(f(t))(s) := \int_0^{\infty} dt e^{-st} f(t). \quad (\text{A.23})$$

Under consideration of the integral in Eq. (A.22), we identify $f(t) = e^{-iH_0 t/\hbar} = \cos(H_0 t/\hbar) - i\sin(H_0 t/\hbar)$ and $s = -i\omega$. In order to use the known Laplace transformations

$$\mathcal{L}(\cos(H_0 t/\hbar))(s) = \frac{s}{s^2 + (H_0/\hbar)^2} \quad \text{for } \operatorname{Re}(s) > 0, \quad (\text{A.24a})$$

$$\mathcal{L}(\sin(H_0 t/\hbar))(s) = \frac{H_0/\hbar}{s^2 + (H_0/\hbar)^2} \quad \text{for } \operatorname{Re}(s) > 0, \quad (\text{A.24b})$$

we must ensure that $\operatorname{Re}(s) > 0$. To this end, we introduce $\eta > 0$, and consider $s = \eta - i\omega$ in the limit $\eta \searrow 0$. Equation (A.22) can then be written as

$$\begin{aligned} \tilde{G}^+(\omega) &= \lim_{\eta \searrow 0} \frac{-i}{\sqrt{2\pi}} \int_0^\infty dt e^{-(\eta-i\omega)t} e^{-iH_0 t/\hbar} \\ &= \lim_{\eta \searrow 0} \frac{-i}{\sqrt{2\pi}} \mathcal{L}(e^{-iH_0 t/\hbar})(\eta - i\omega) \\ &= \lim_{\eta \searrow 0} \frac{-i}{\sqrt{2\pi}} [\mathcal{L}(\cos(H_0 t/\hbar))(\eta - i\omega) - i\mathcal{L}(\sin(H_0 t/\hbar))(\eta - i\omega)] \\ &= \lim_{\eta \searrow 0} \frac{-i}{\sqrt{2\pi}} \left[\frac{\eta - i\omega}{(\eta - i\omega)^2 + (H_0/\hbar)^2} - i \frac{H_0/\hbar}{(\eta - i\omega)^2 + (H_0/\hbar)^2} \right] \\ &= \lim_{\eta \searrow 0} \frac{-i}{\sqrt{2\pi}} \left[\frac{\eta - i\omega - iH_0/\hbar}{(\eta - i\omega + iH_0/\hbar)(\eta - i\omega - iH_0/\hbar)} \right] \\ &= \lim_{\eta \searrow 0} \frac{-i}{\sqrt{2\pi}} \frac{1}{\eta - i\omega + iH_0/\hbar}. \end{aligned} \quad (\text{A.25})$$

We thus obtain

$$\tilde{G}^+(\omega) = \lim_{\eta \searrow 0} \frac{1}{\sqrt{2\pi}} \frac{1}{\omega - H_0/\hbar + i\eta},$$

which coincides with the sought-after relation from Eq. (7.65).

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